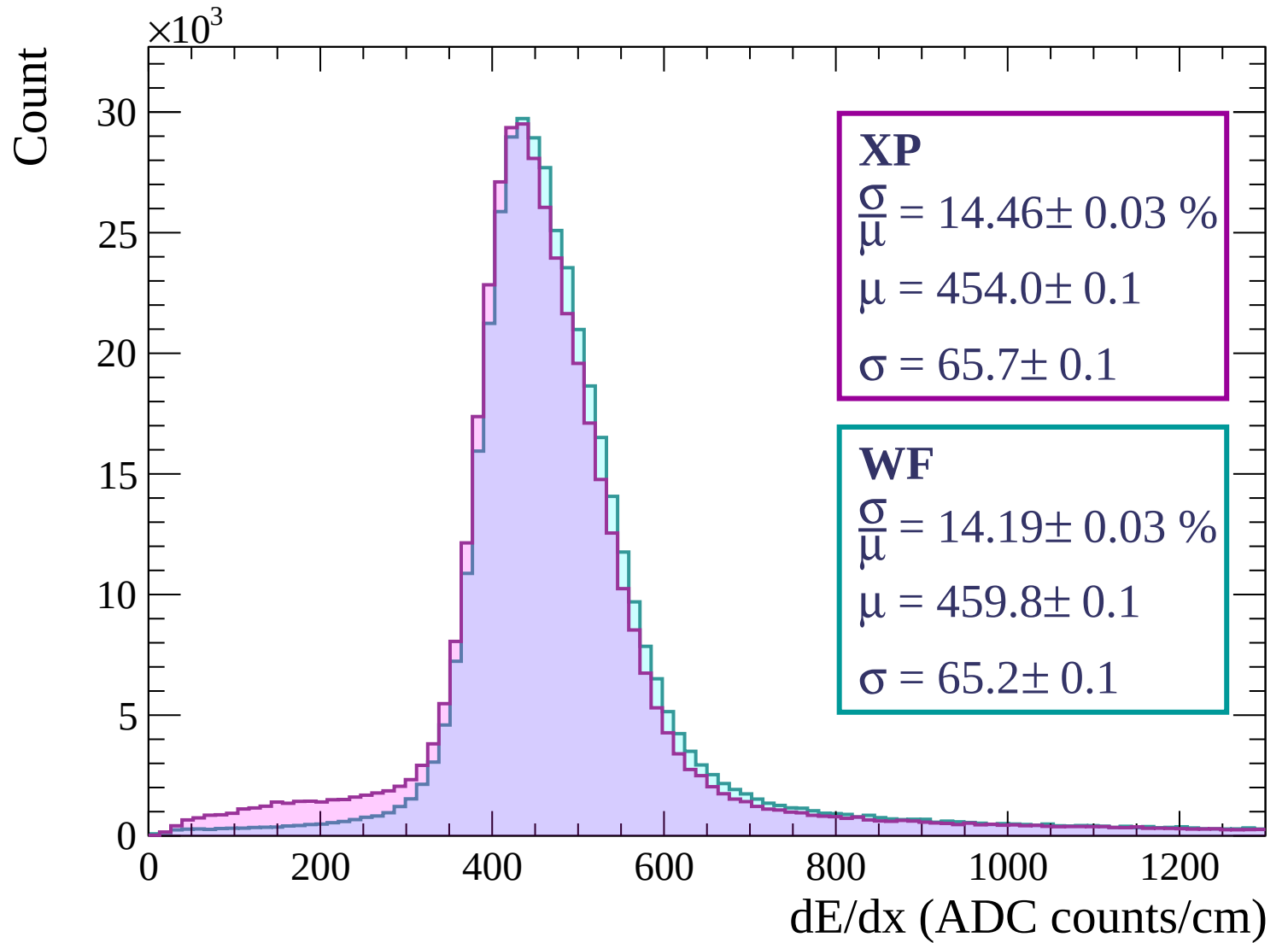
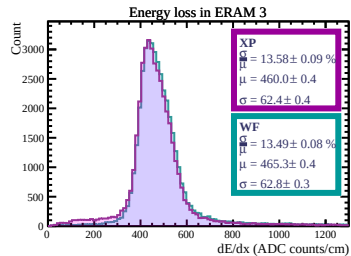
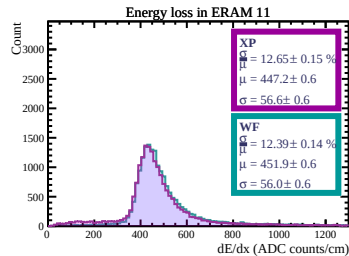
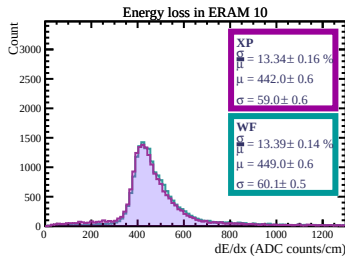
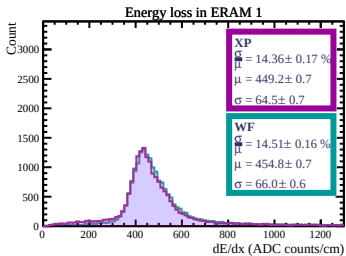
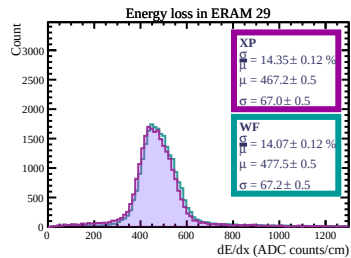
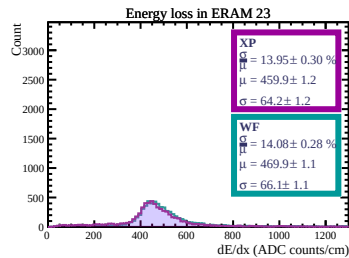
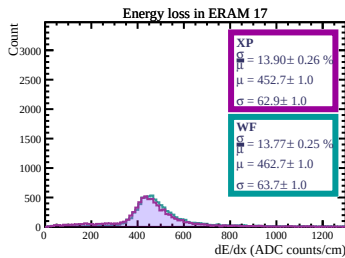
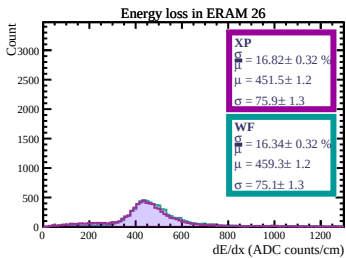
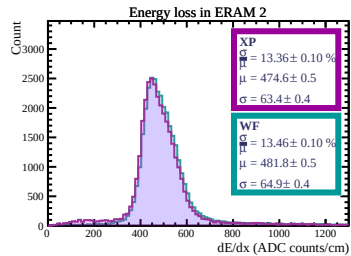
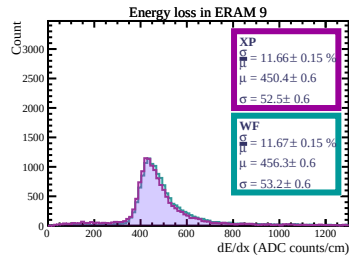
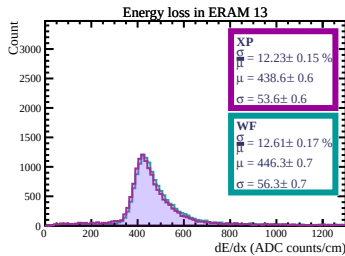
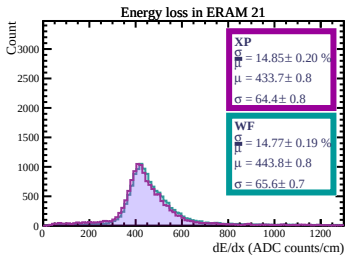
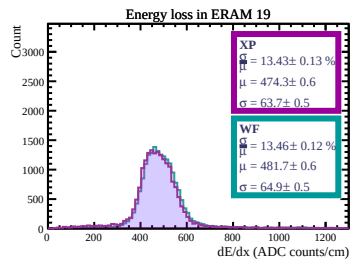
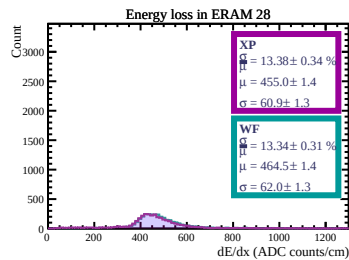
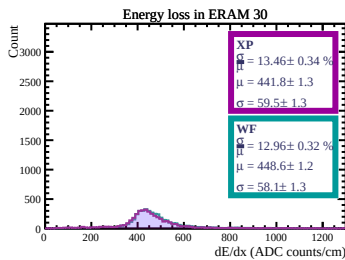
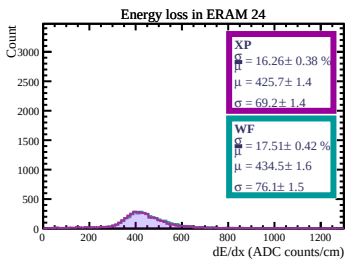
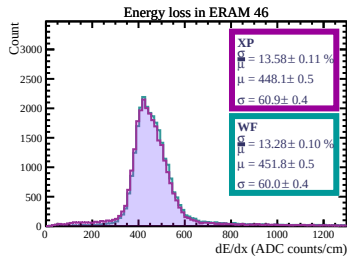
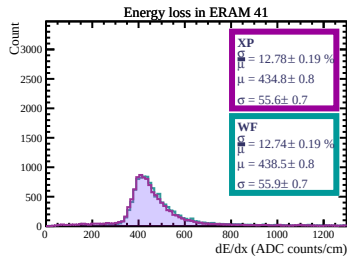
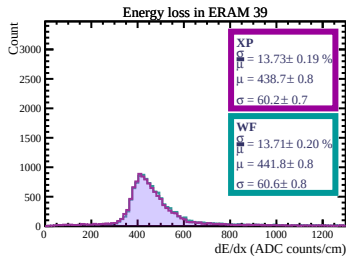
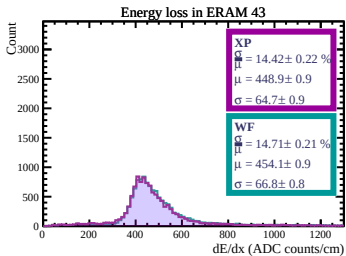
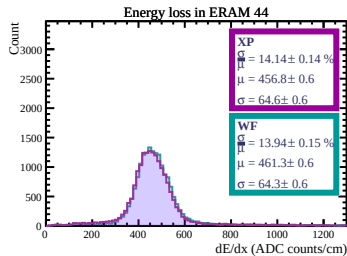
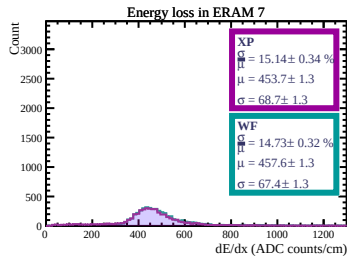
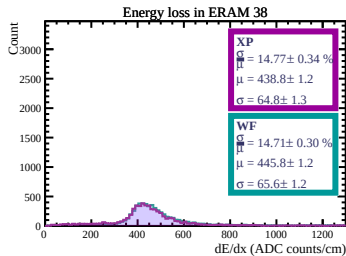
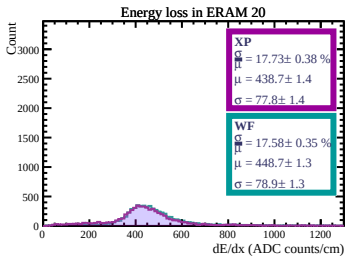
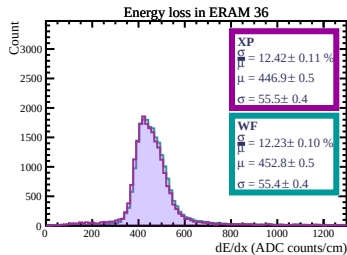
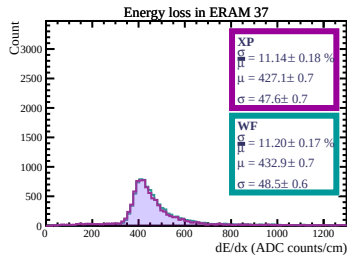
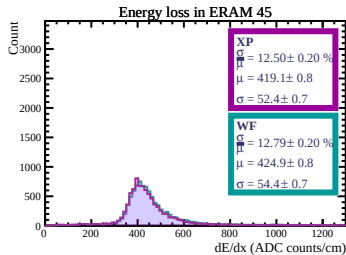
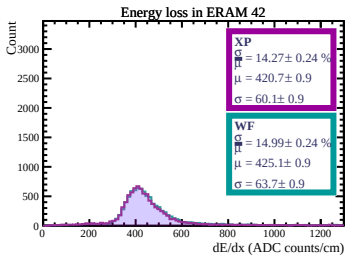
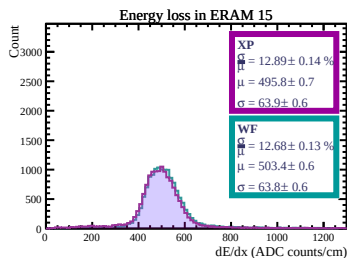
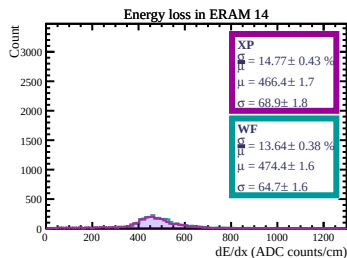
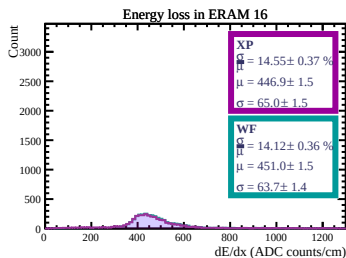
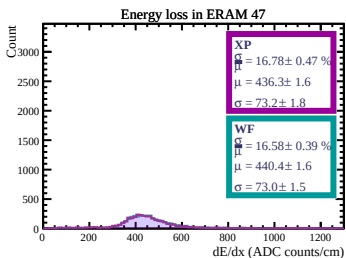
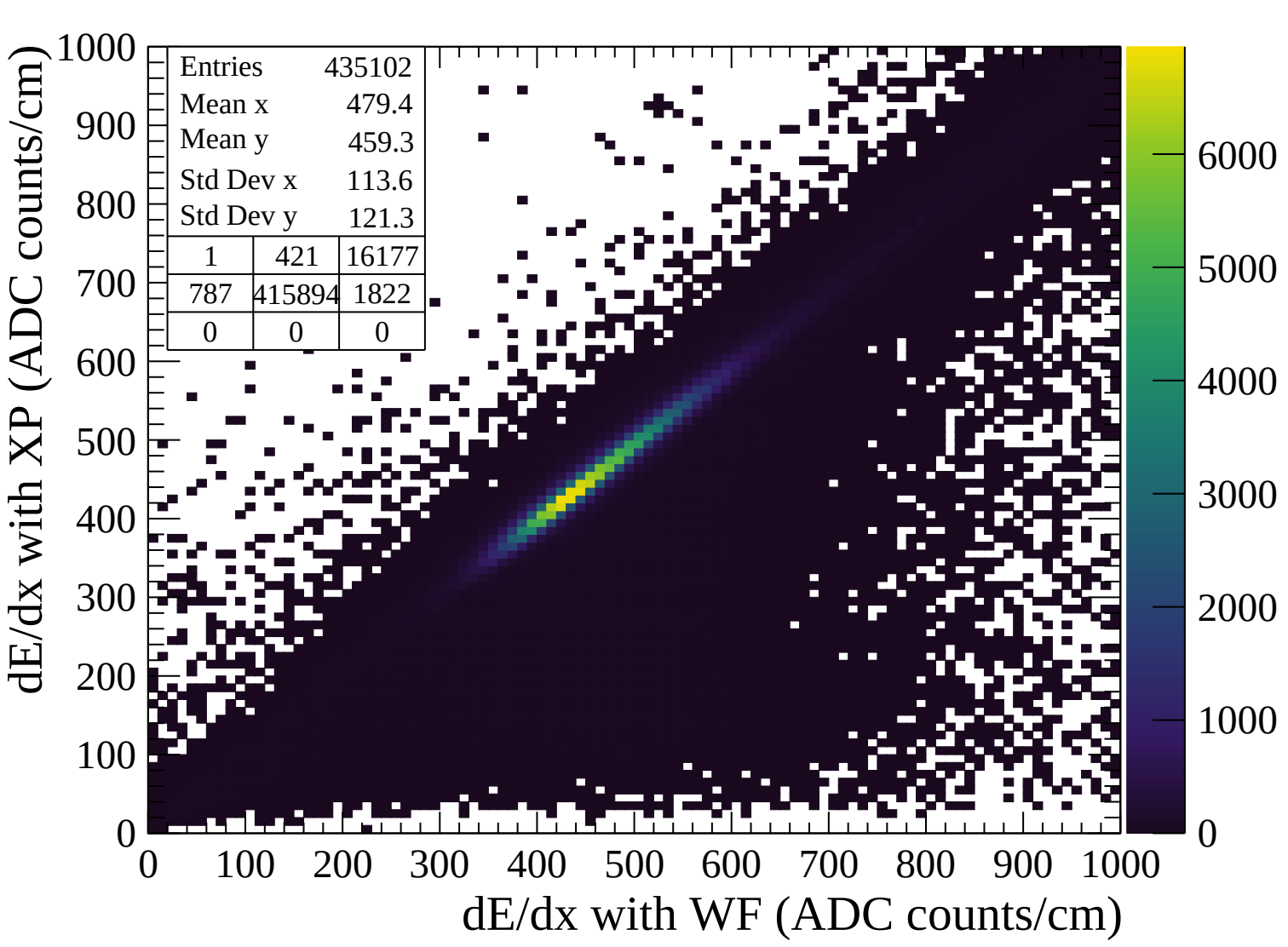
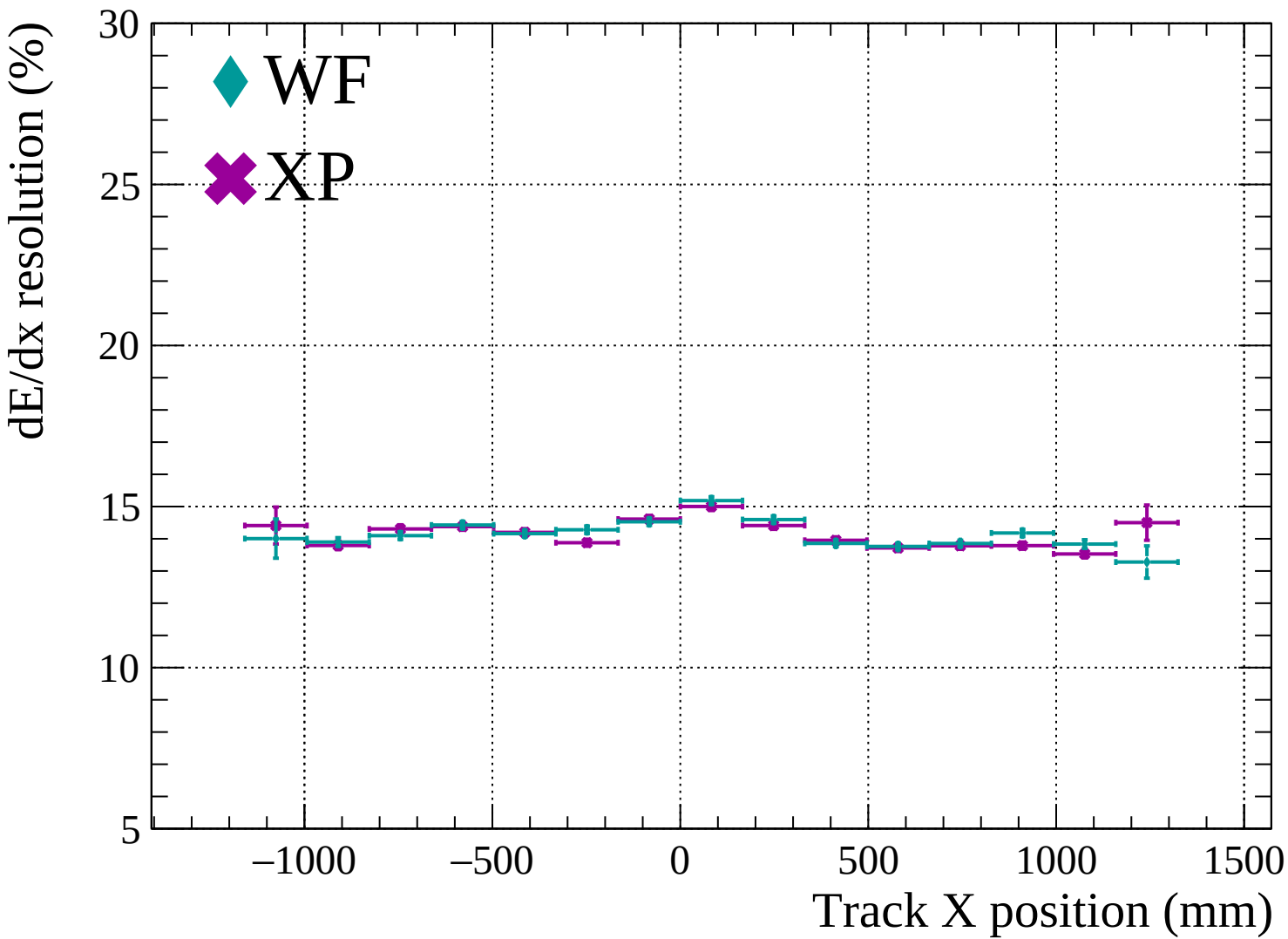


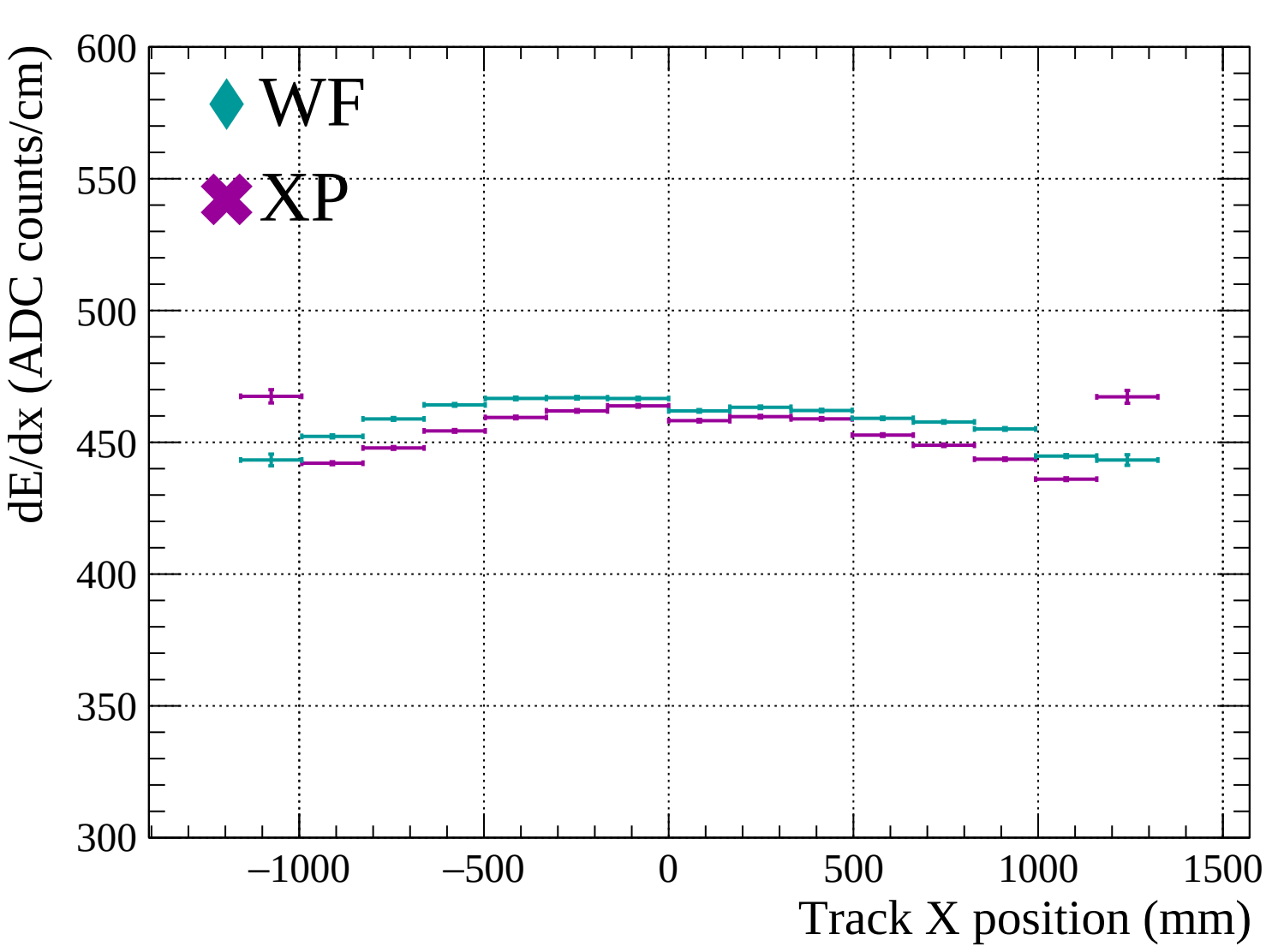


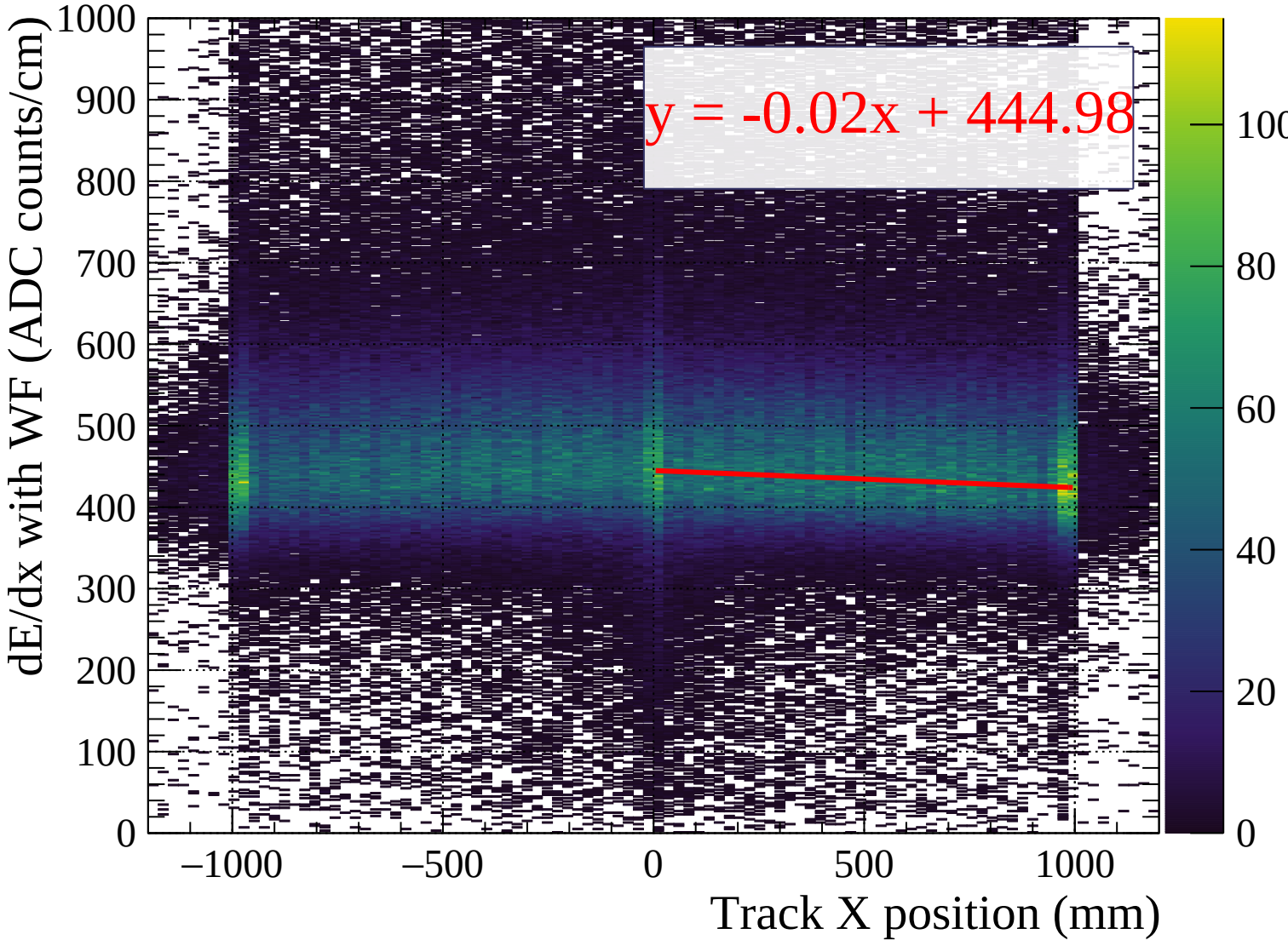


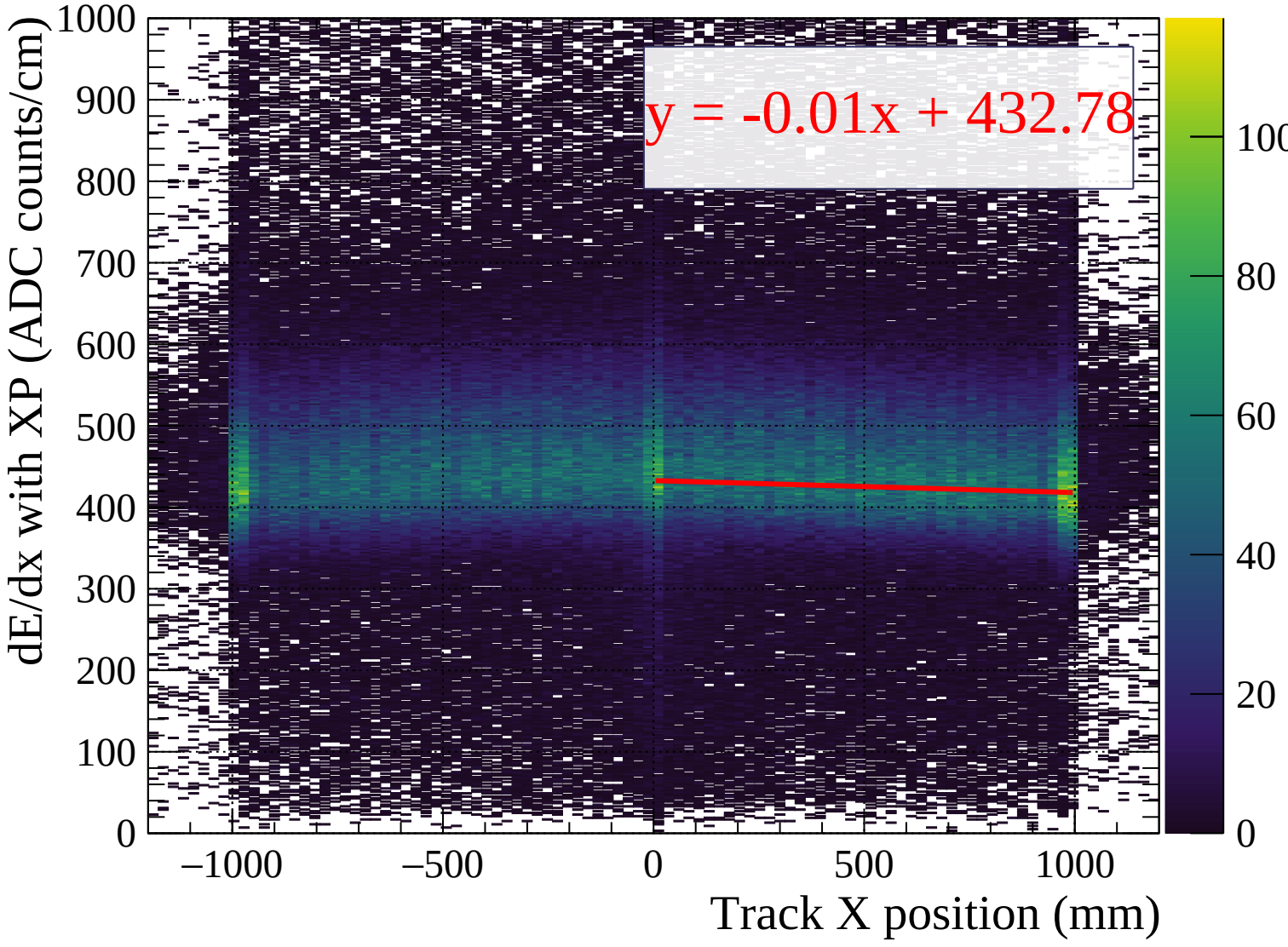




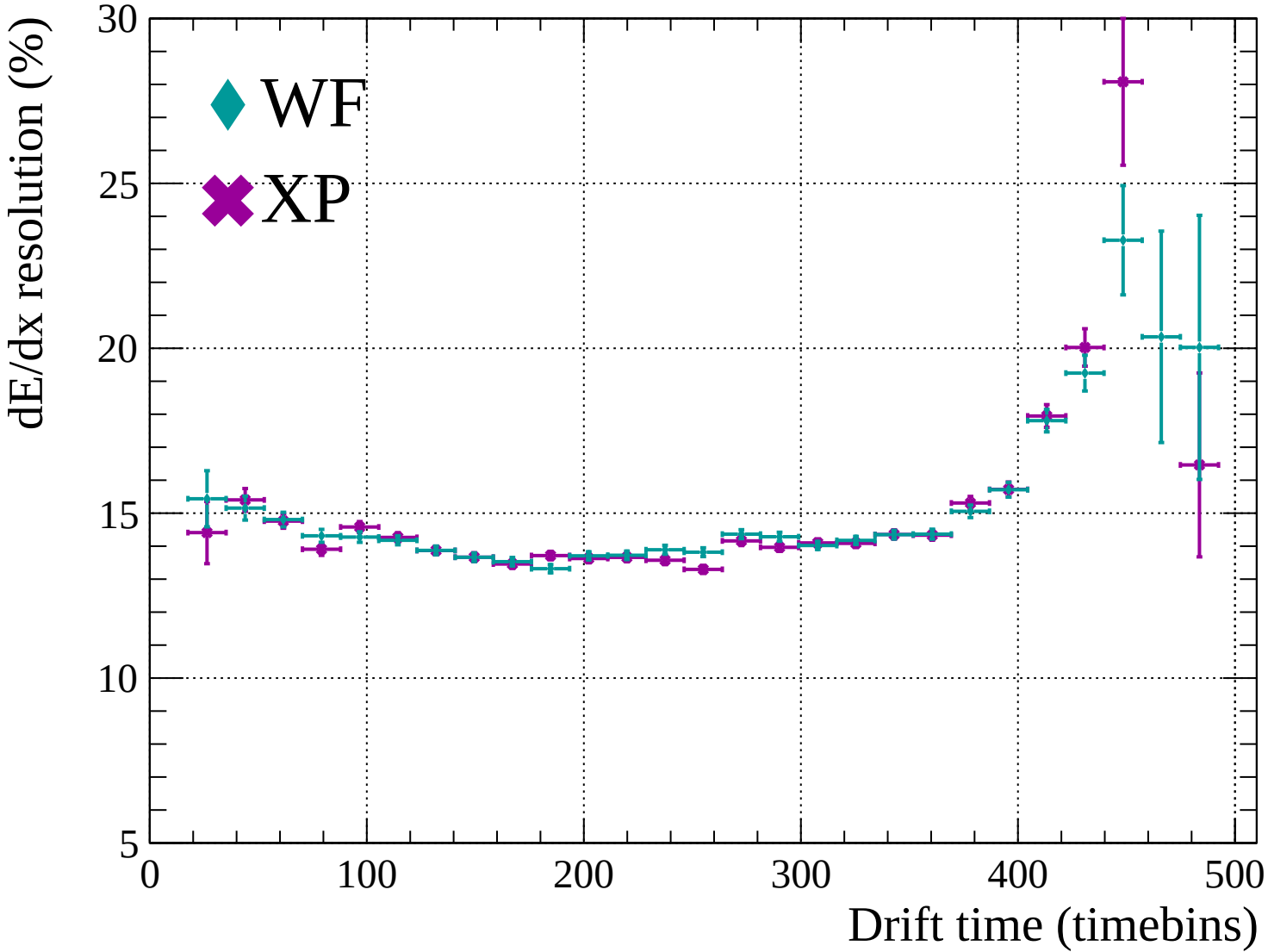


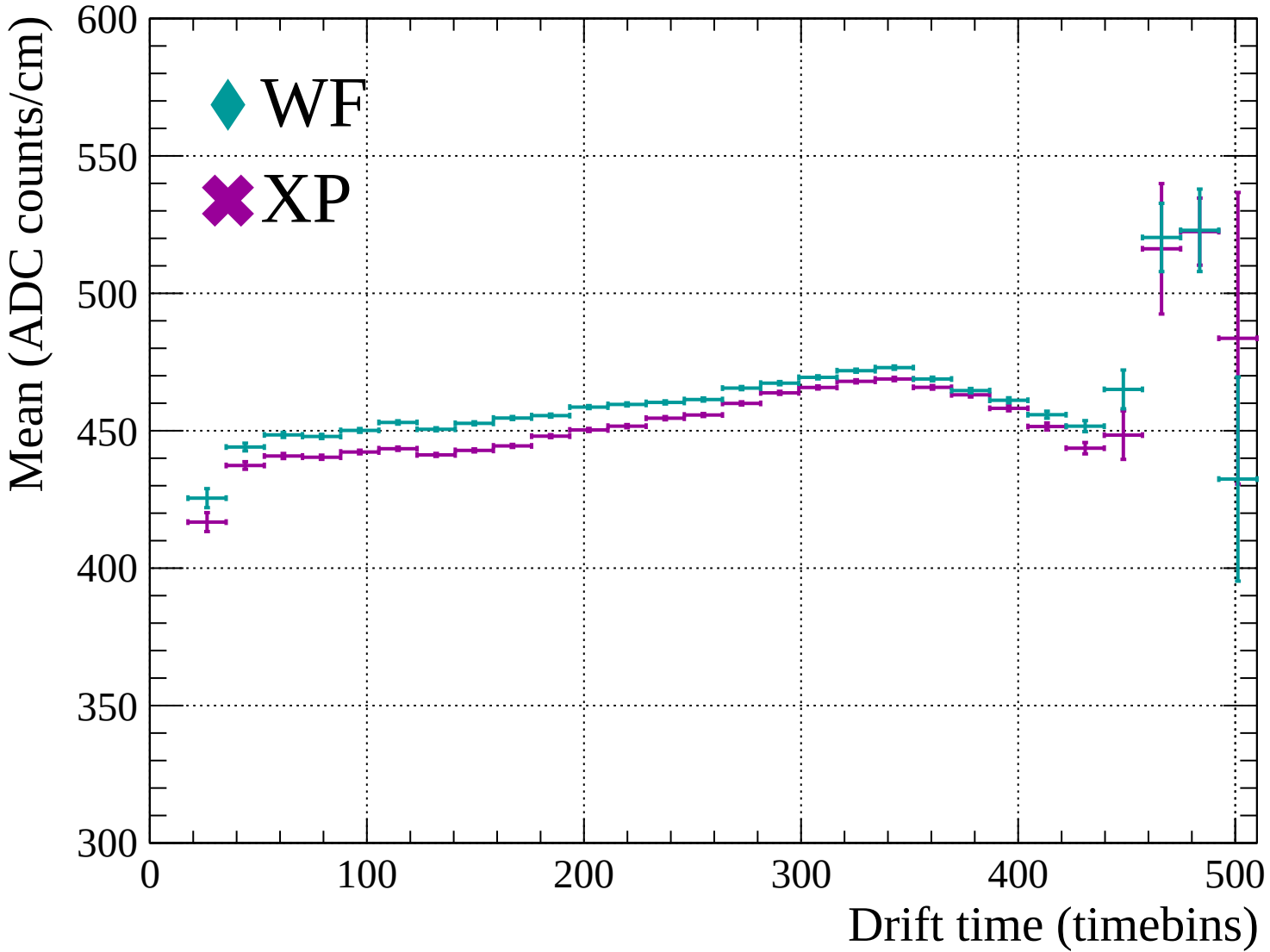


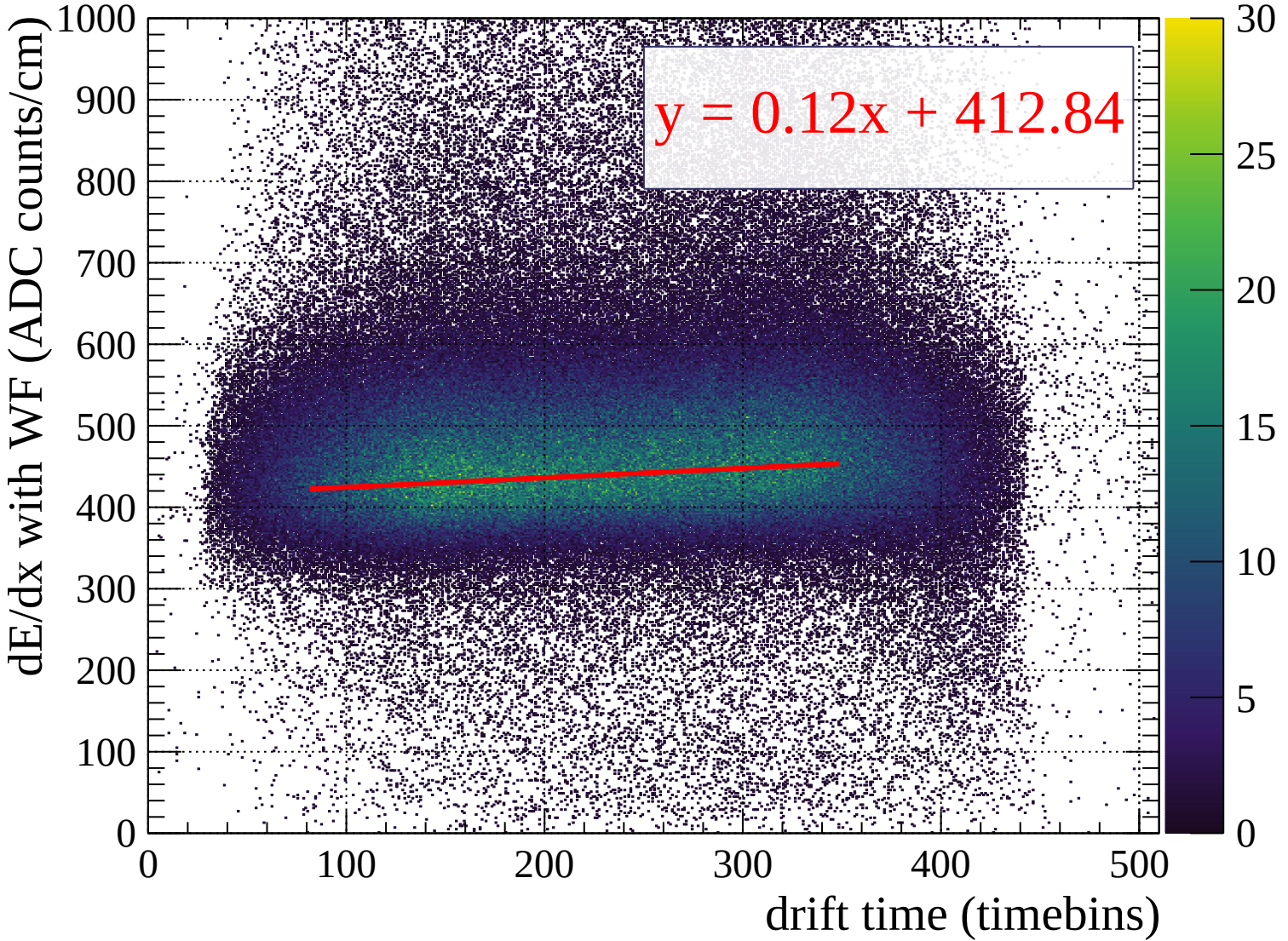


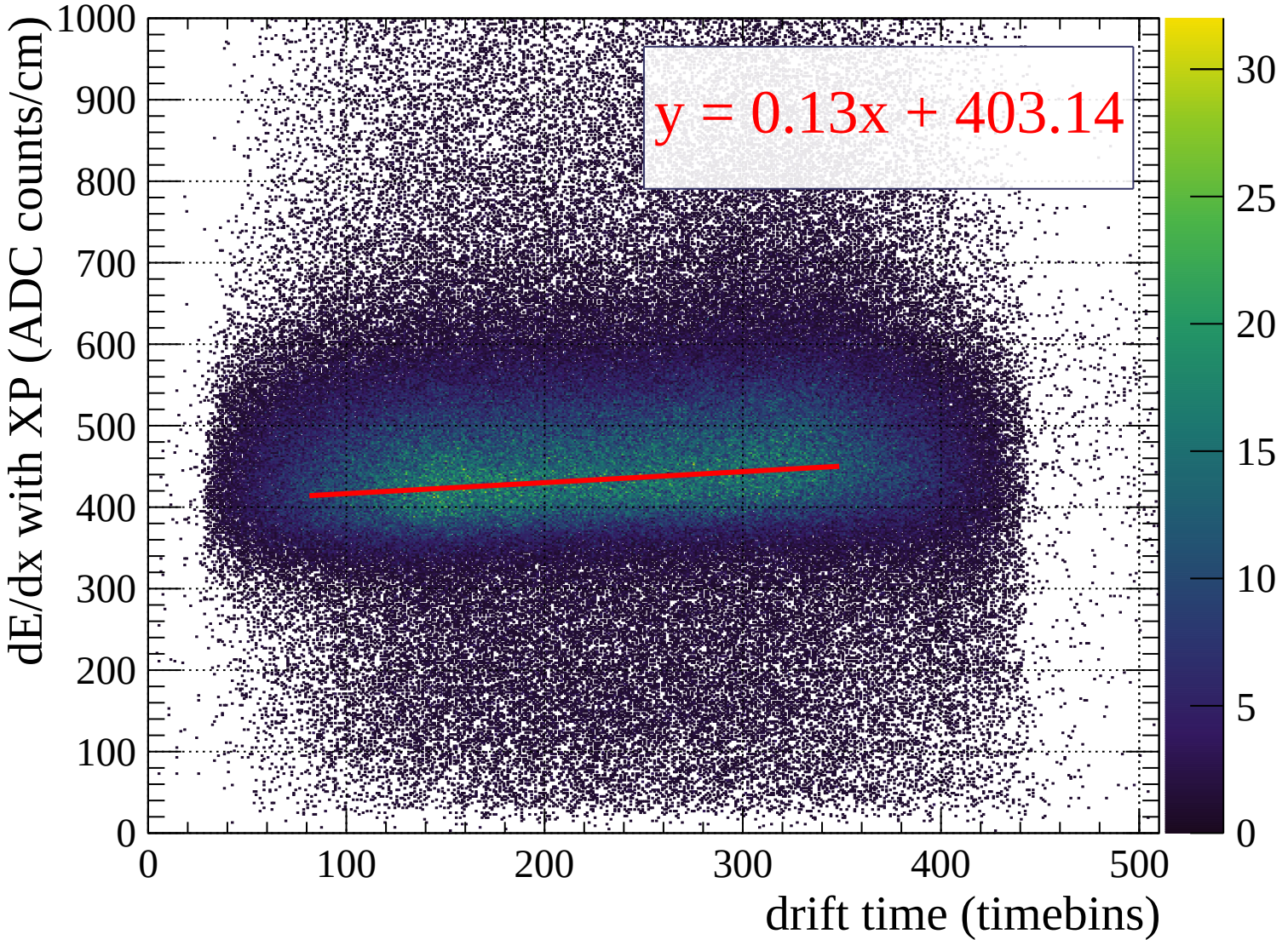




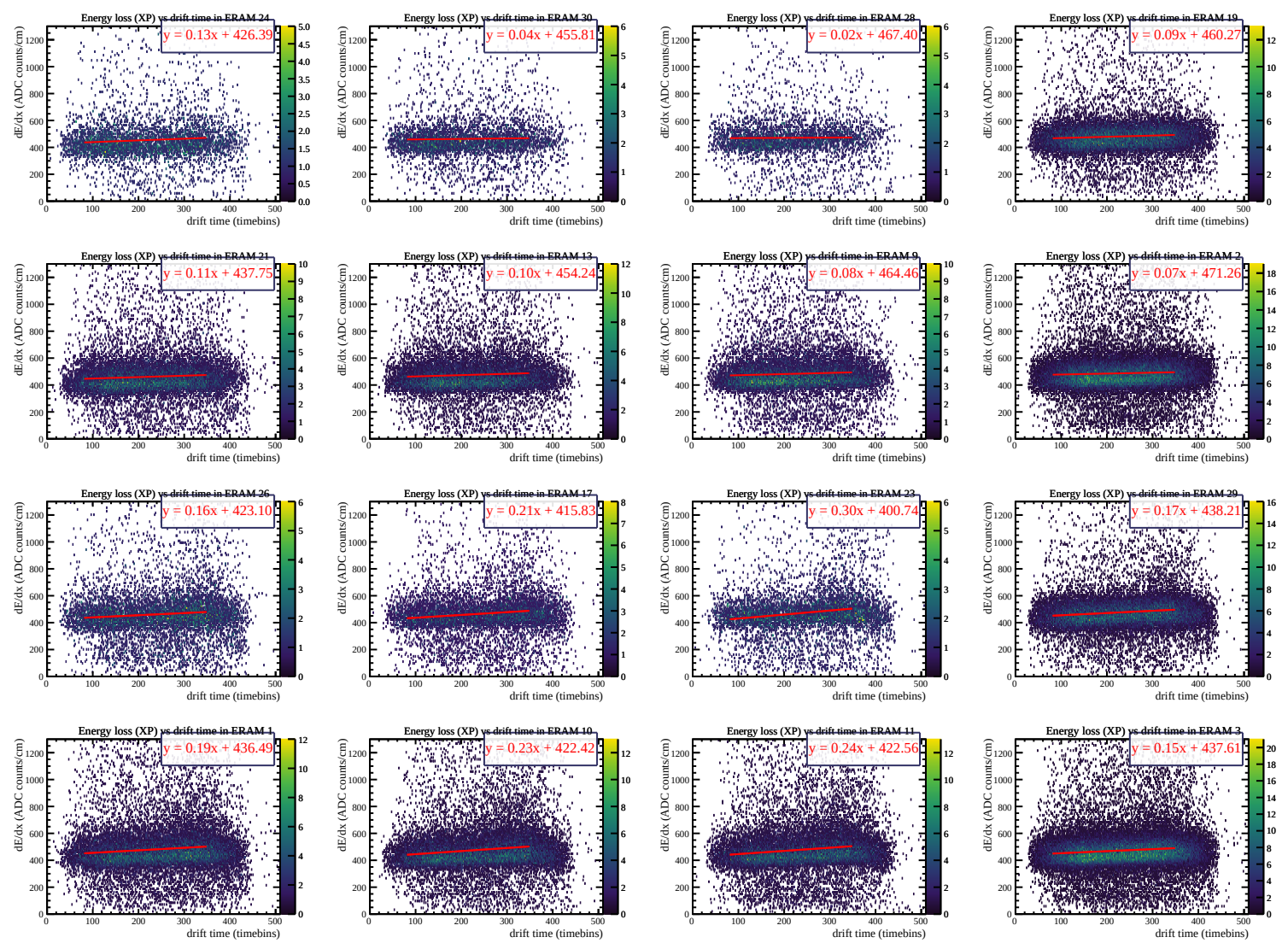


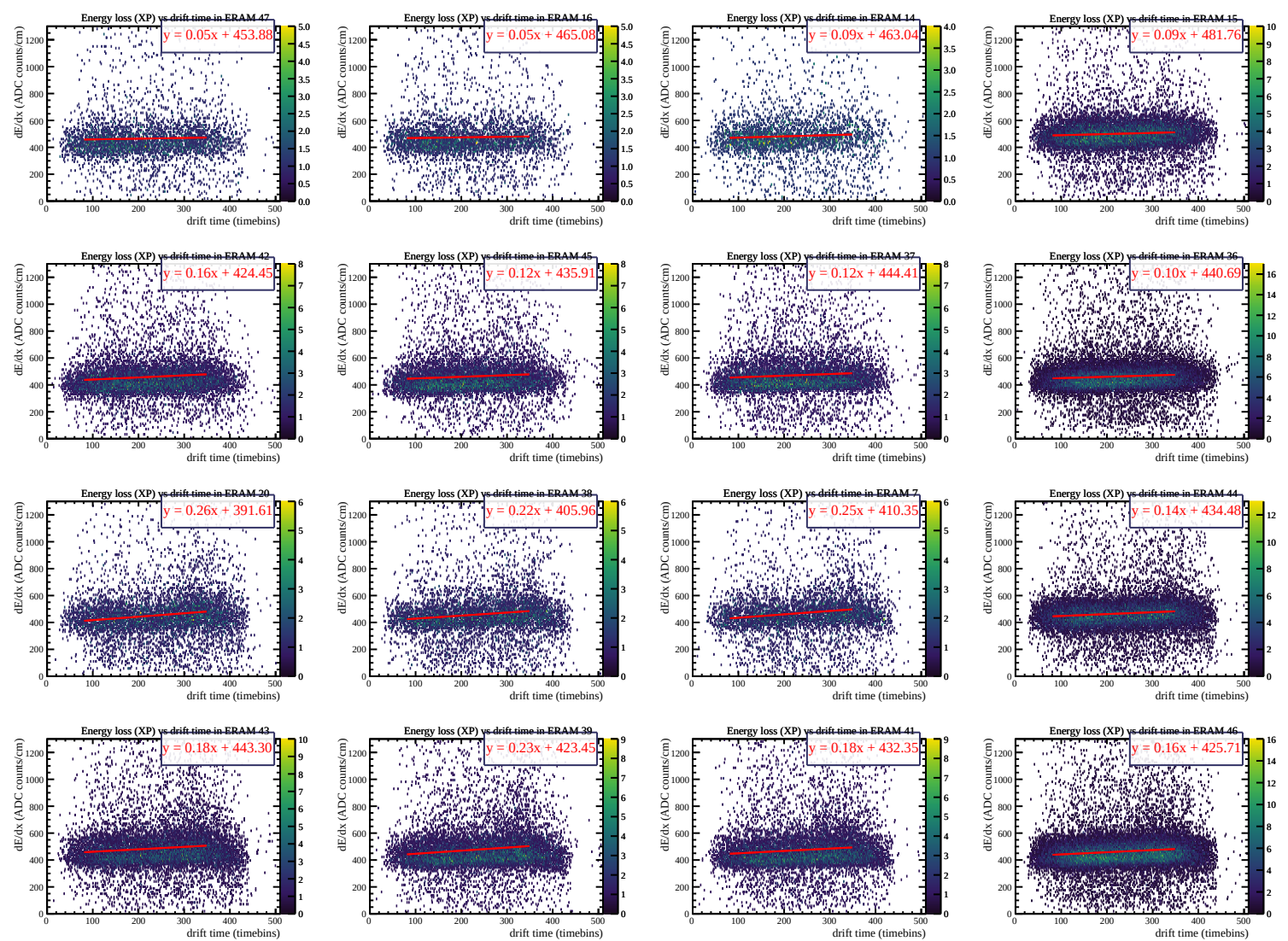


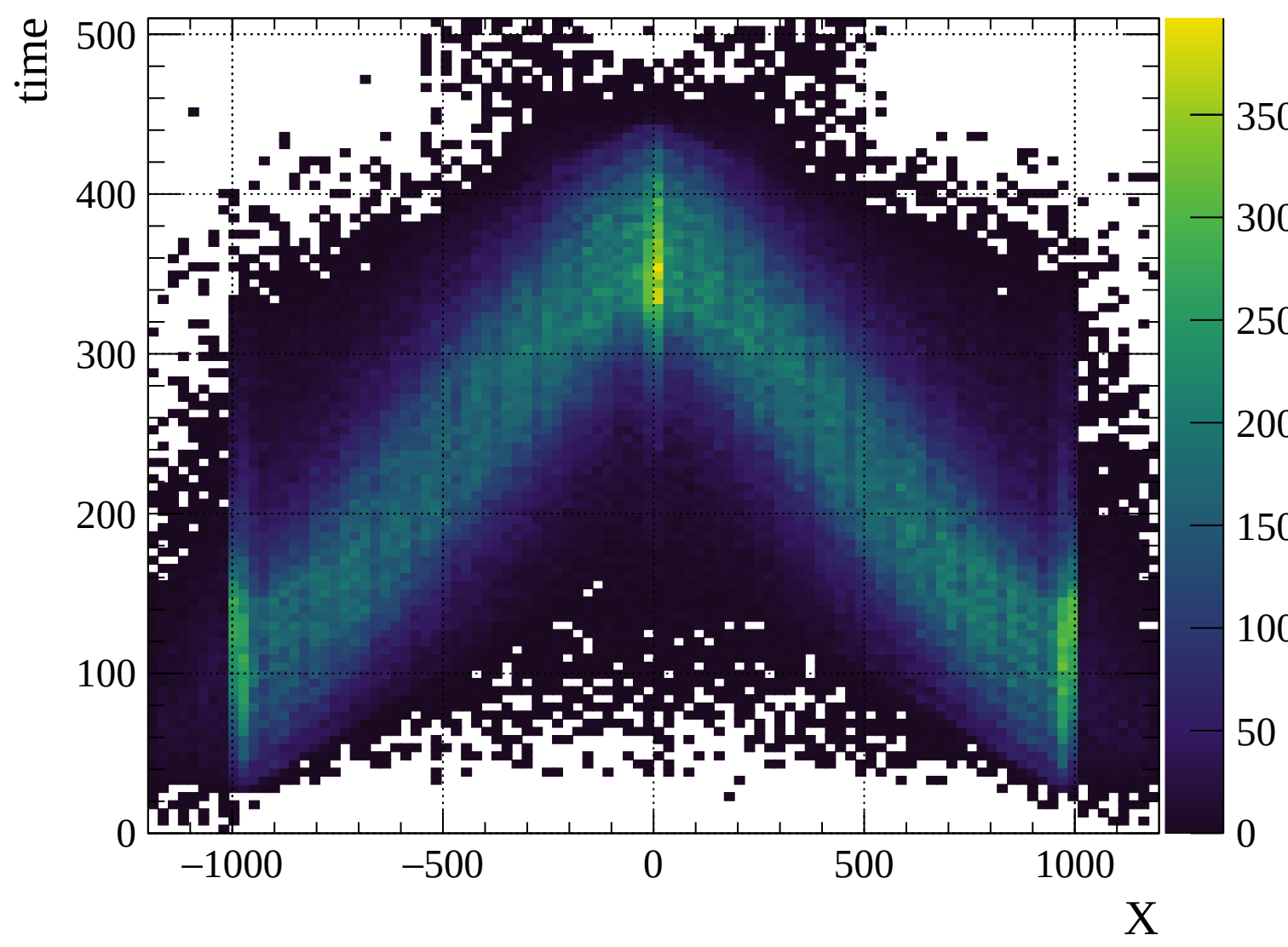


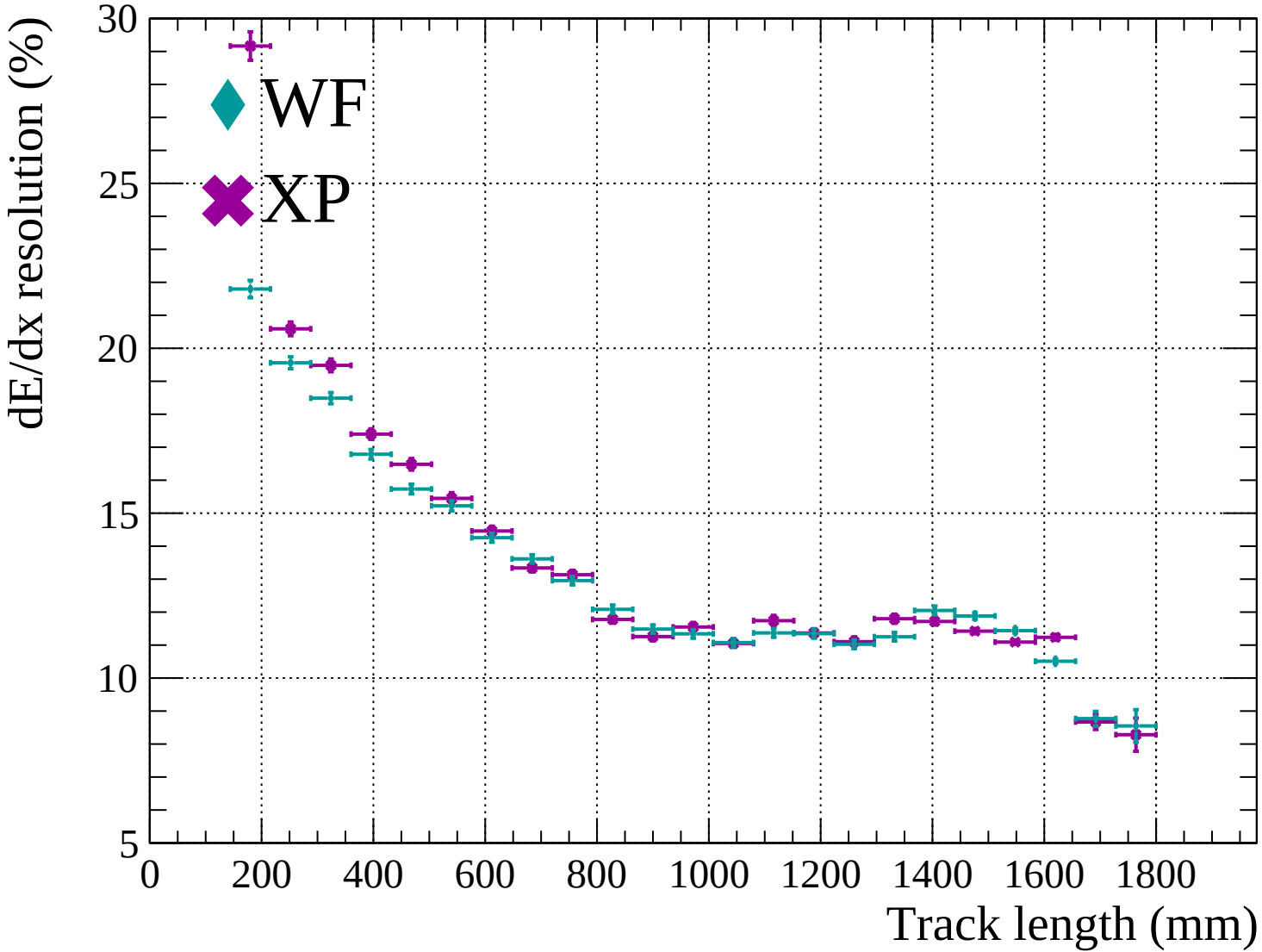




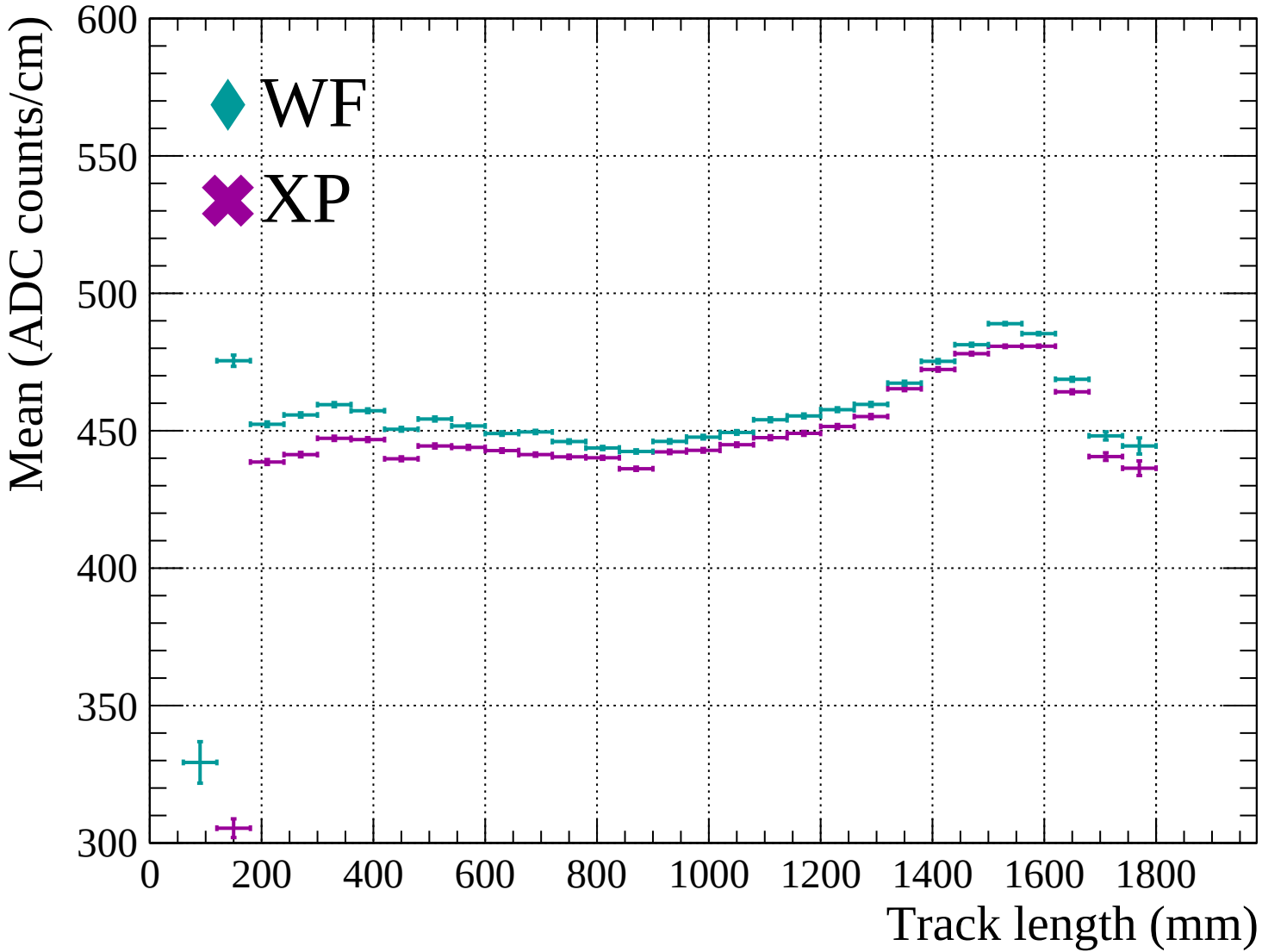


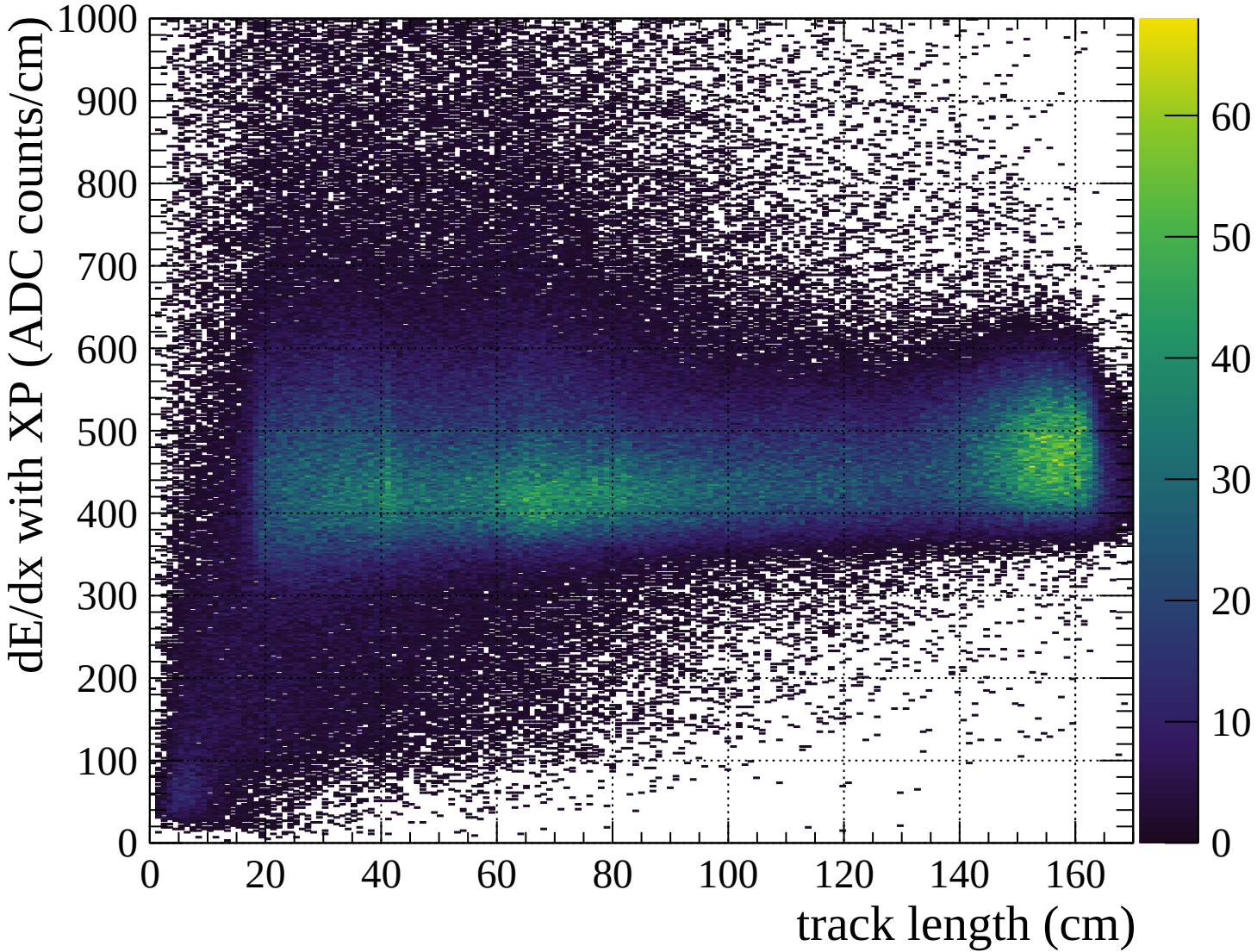


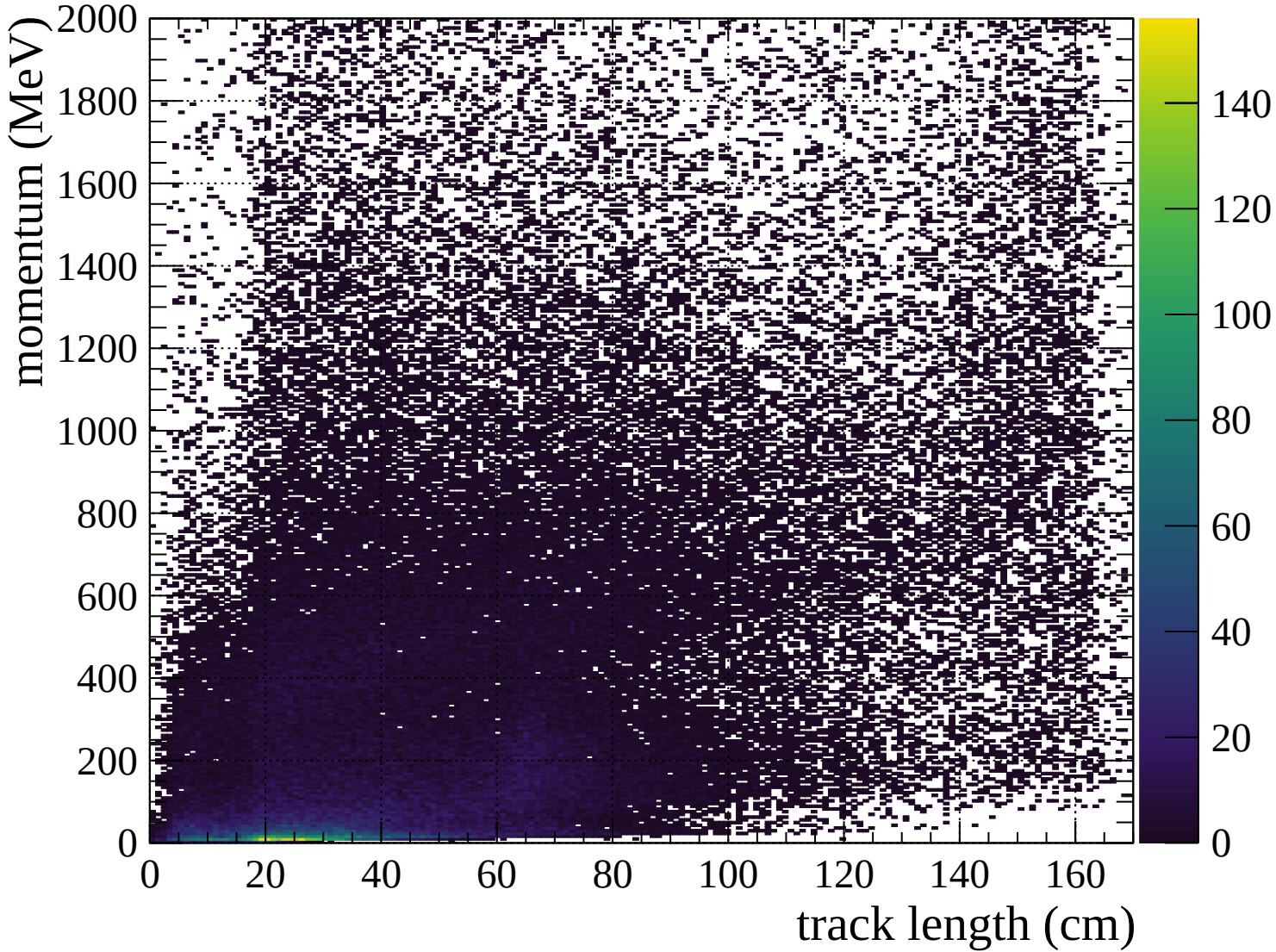


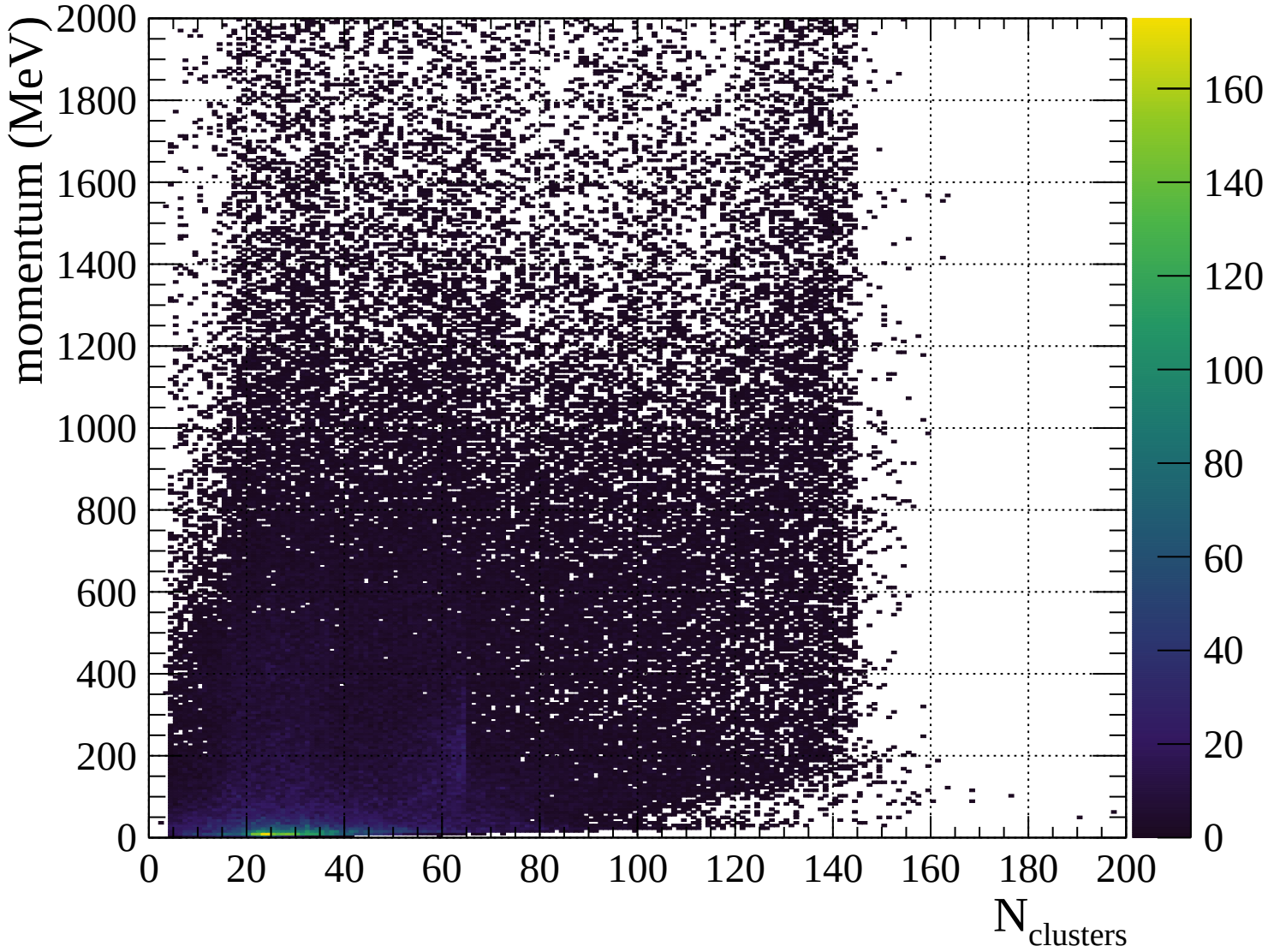


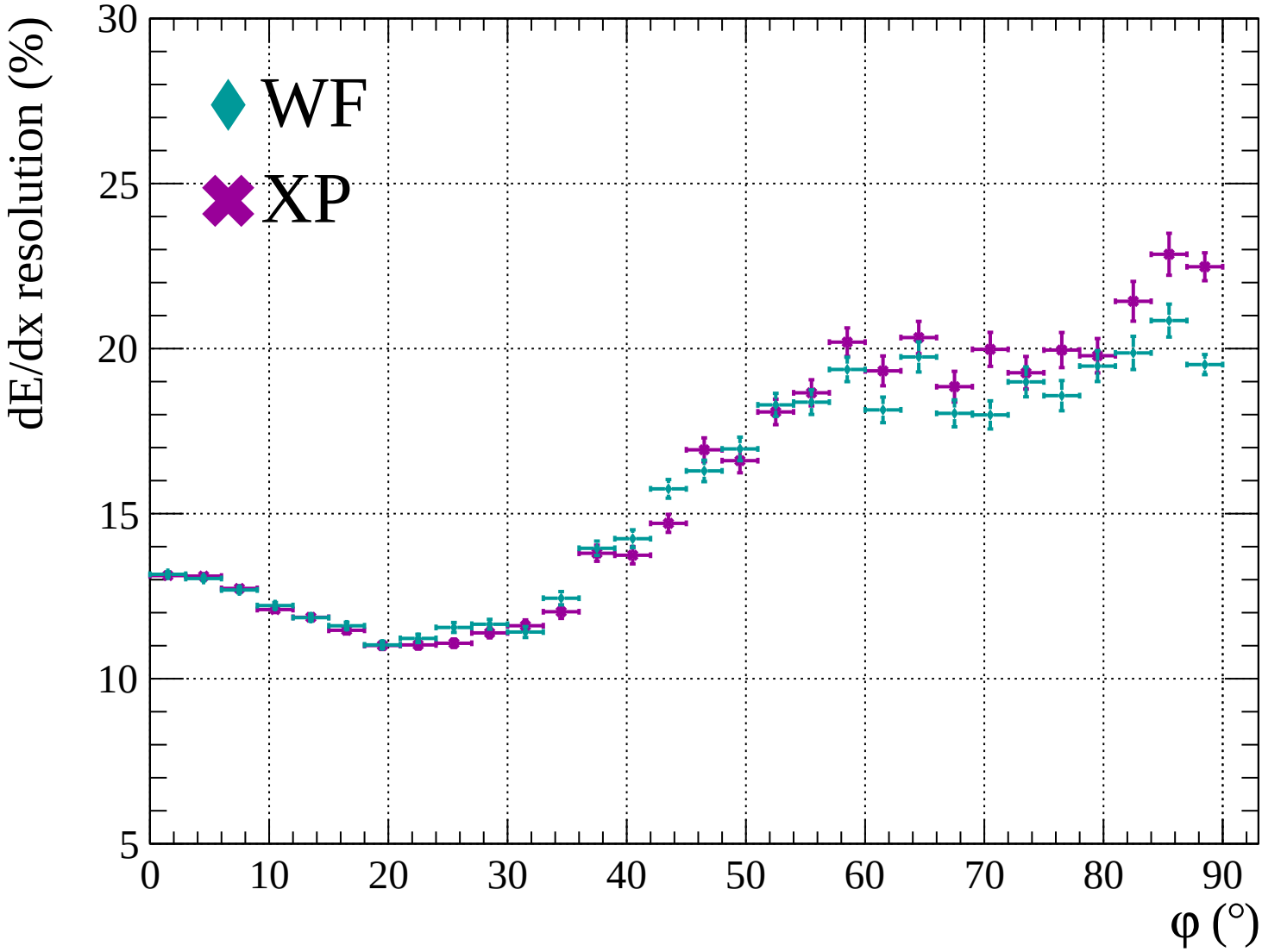


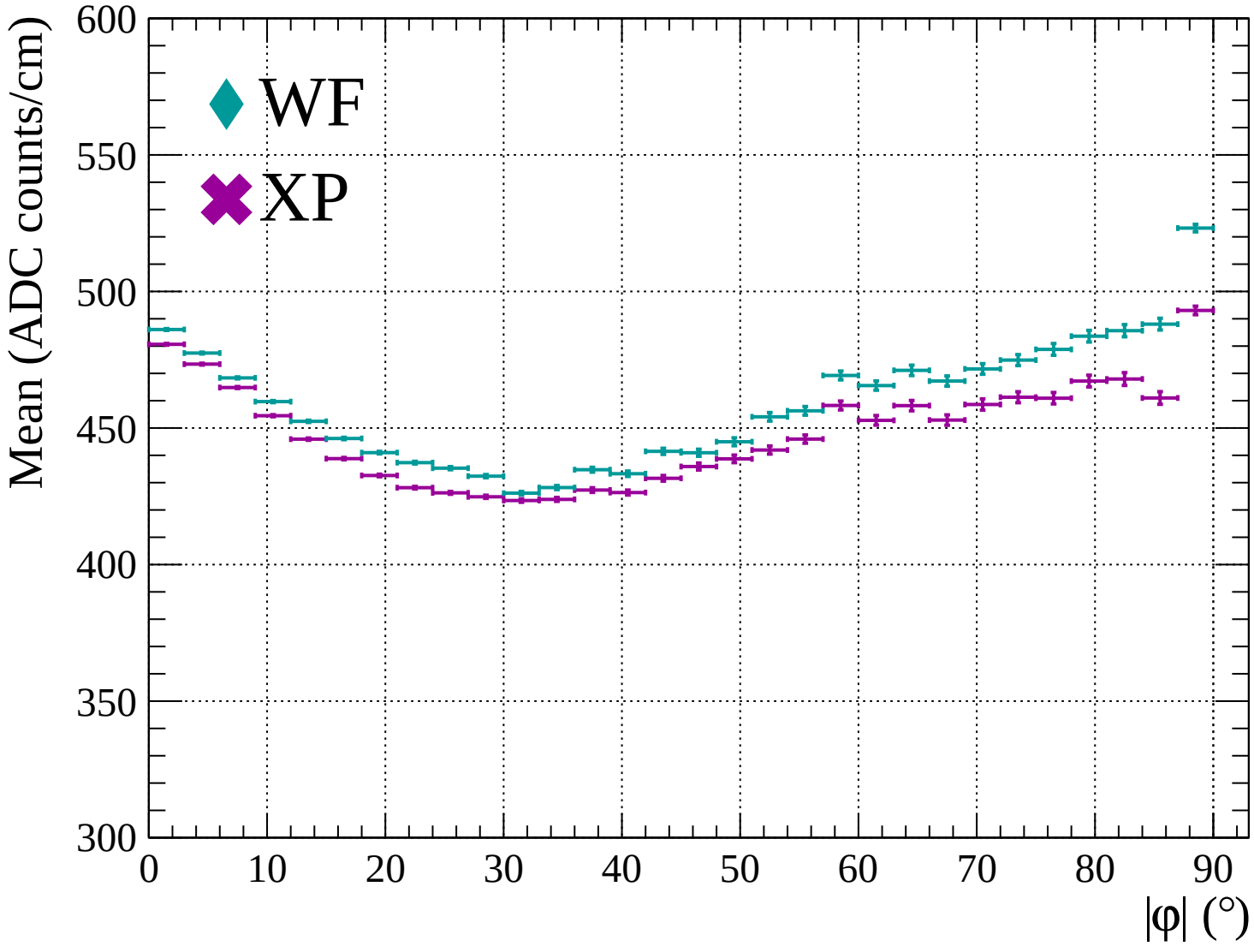


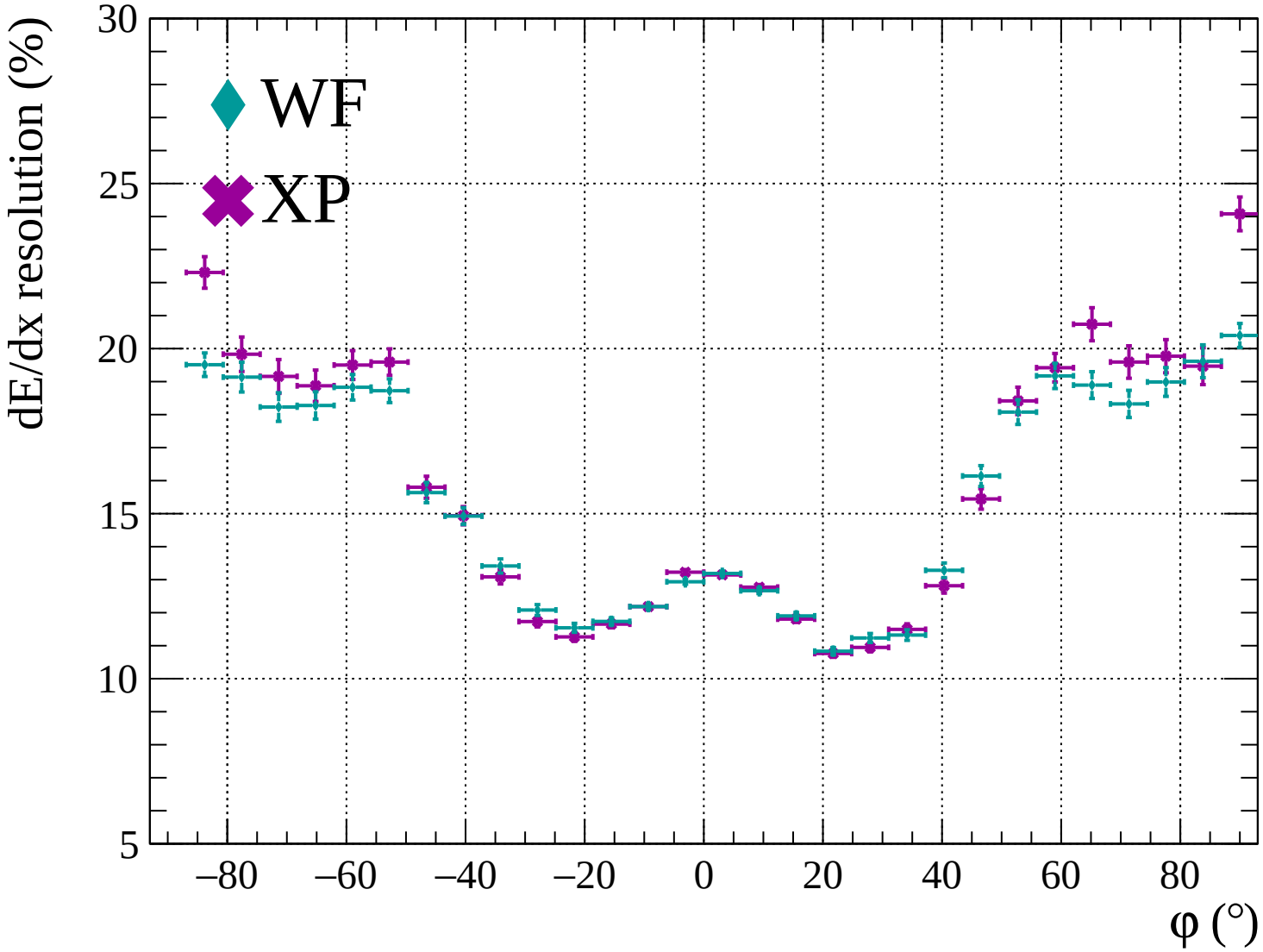




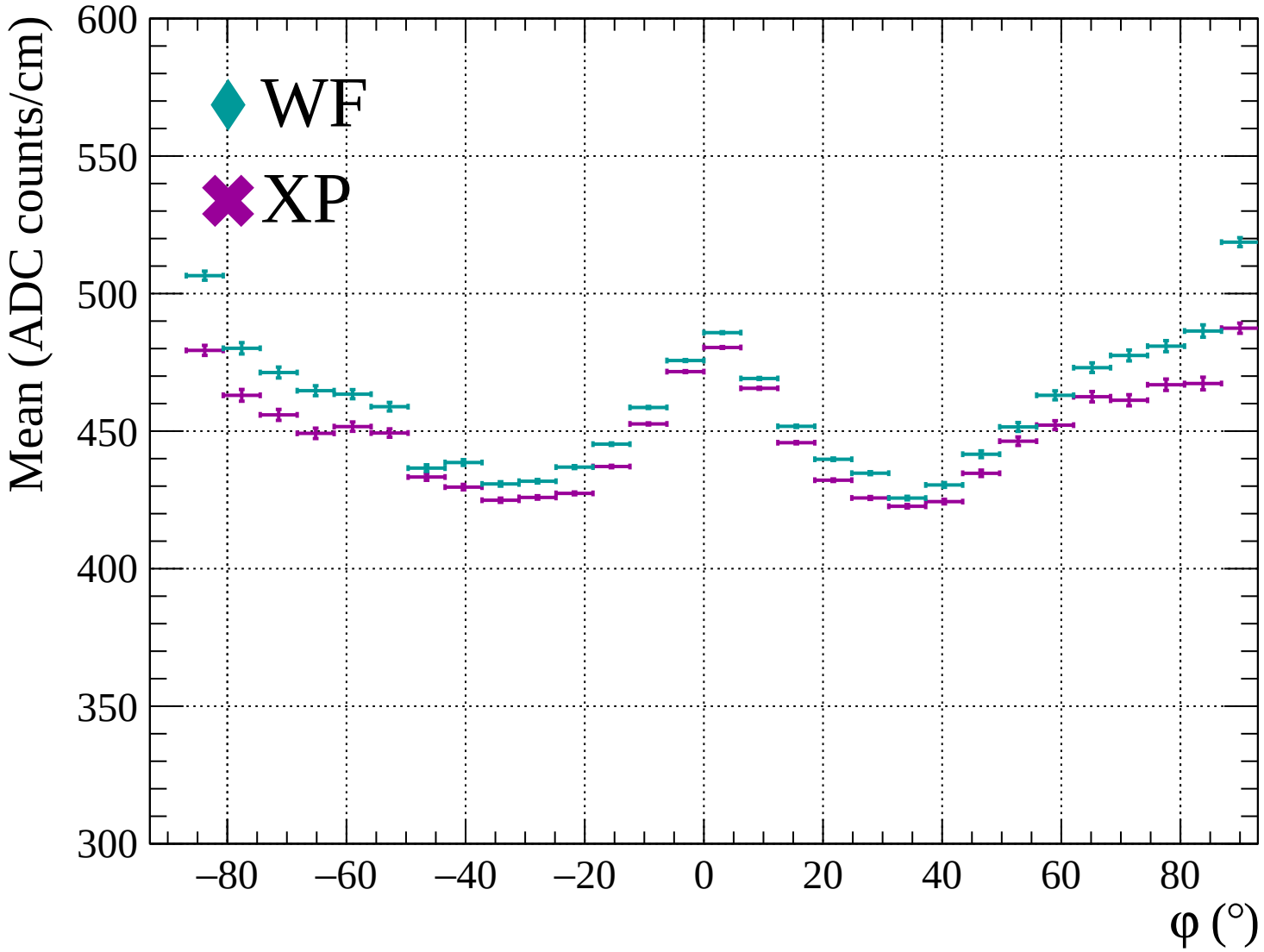


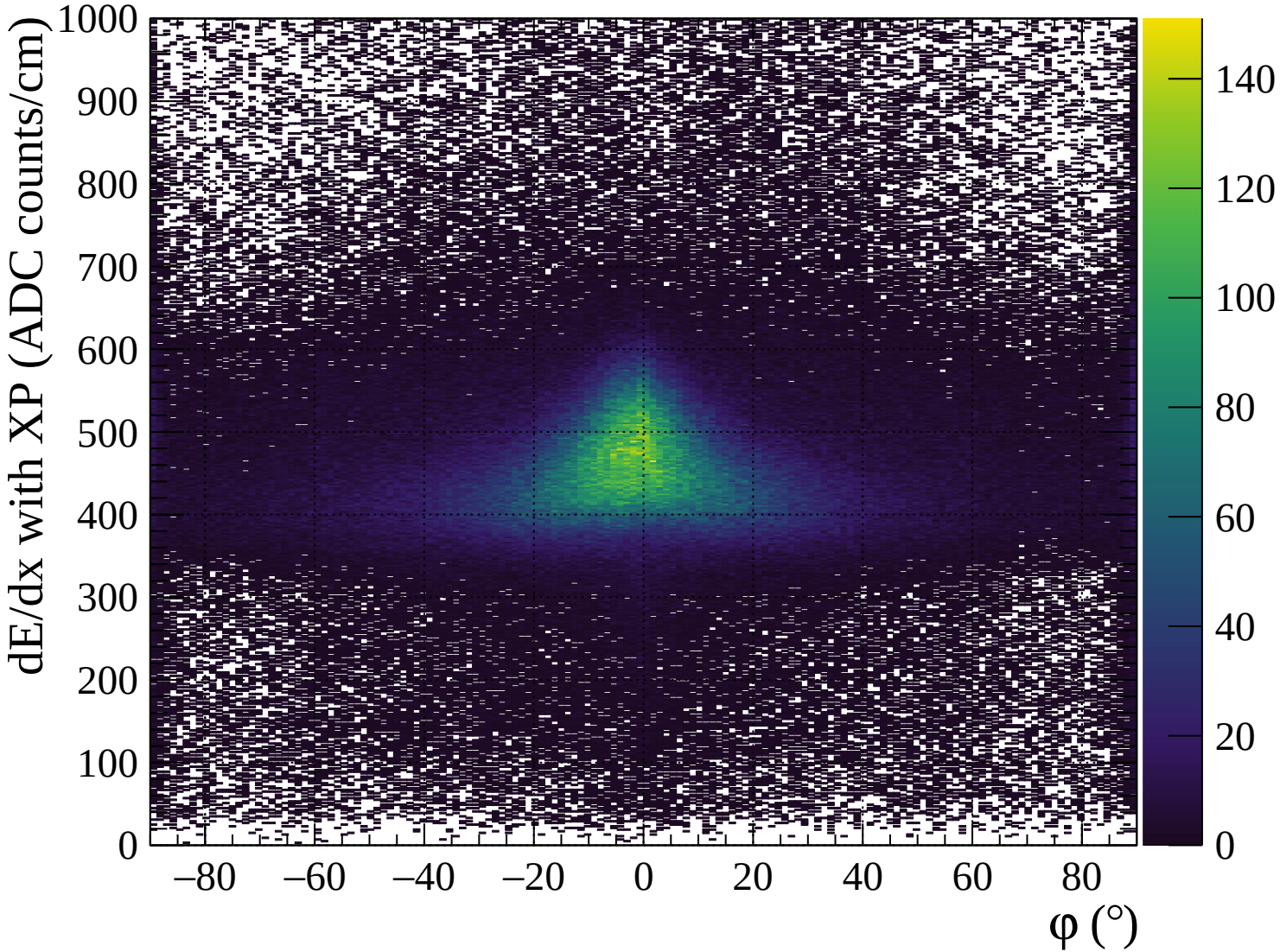


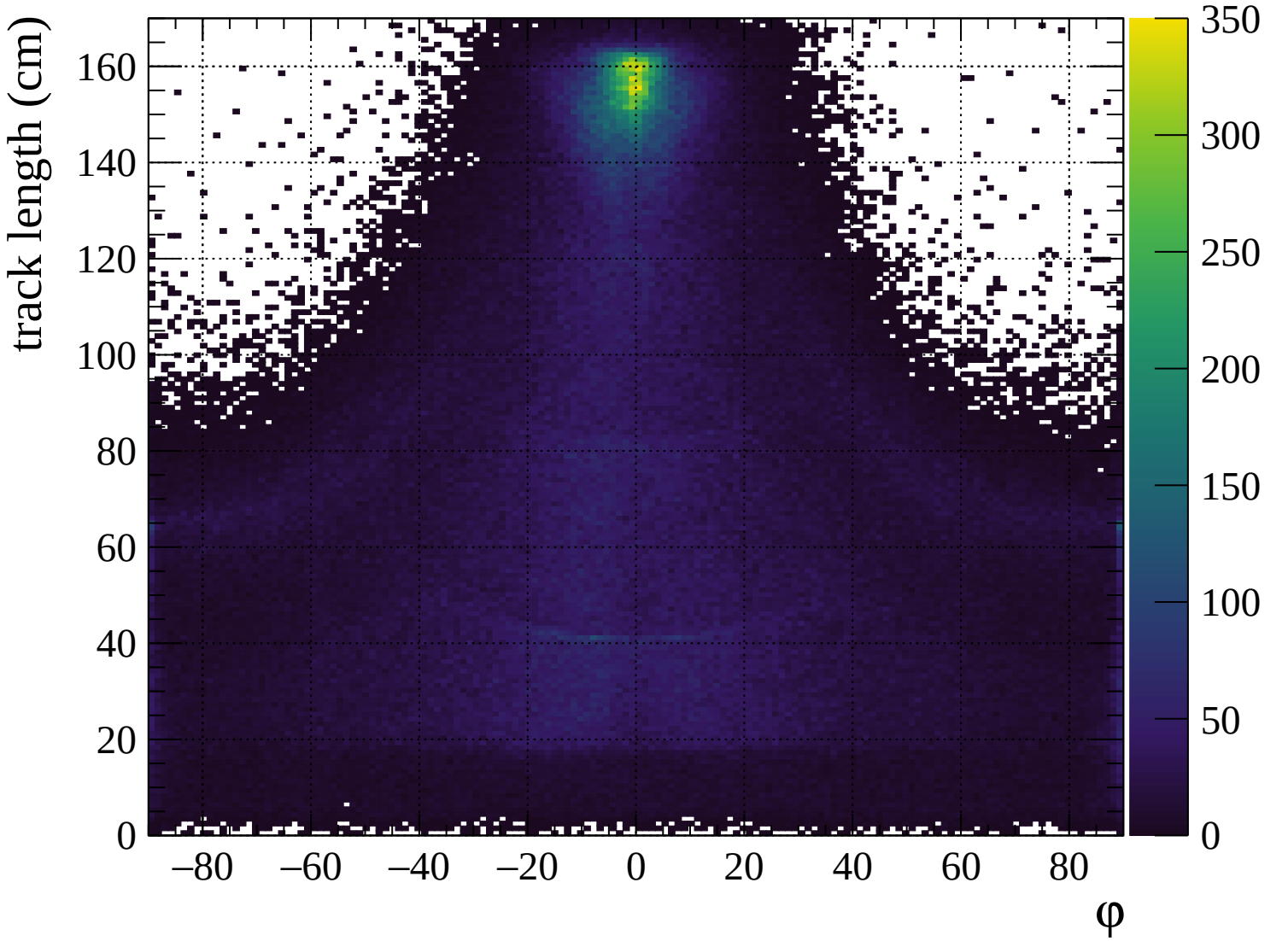


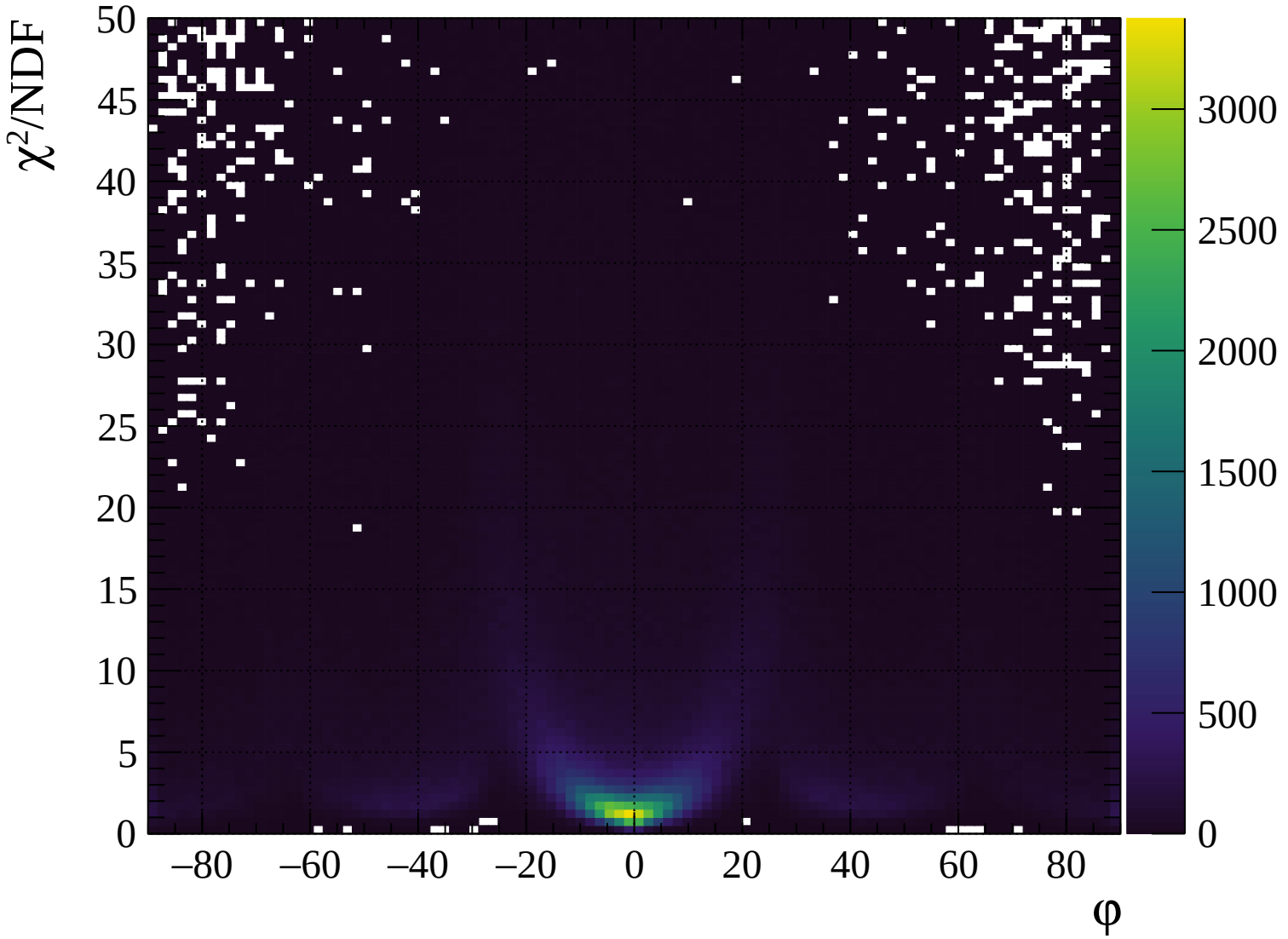


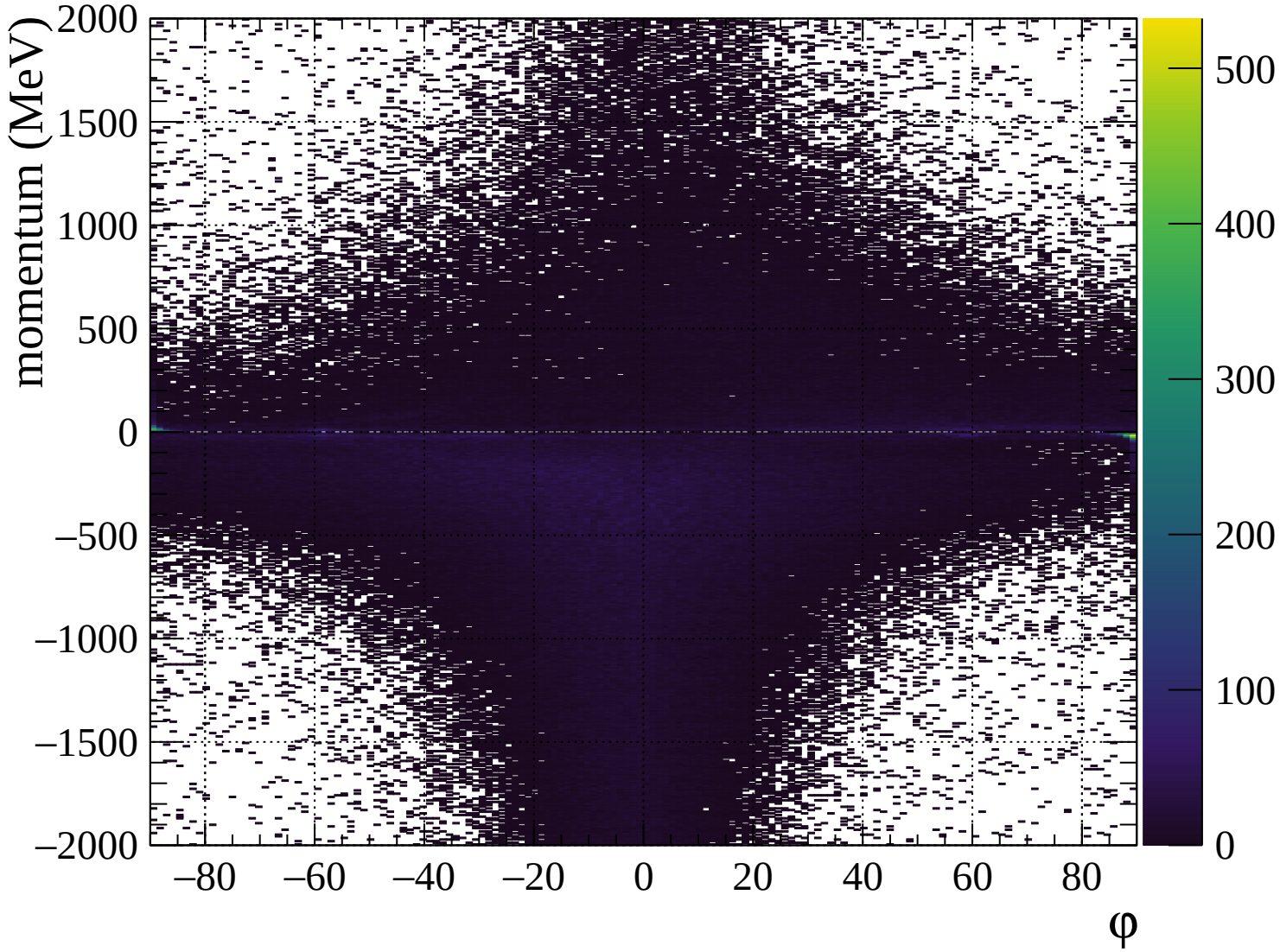


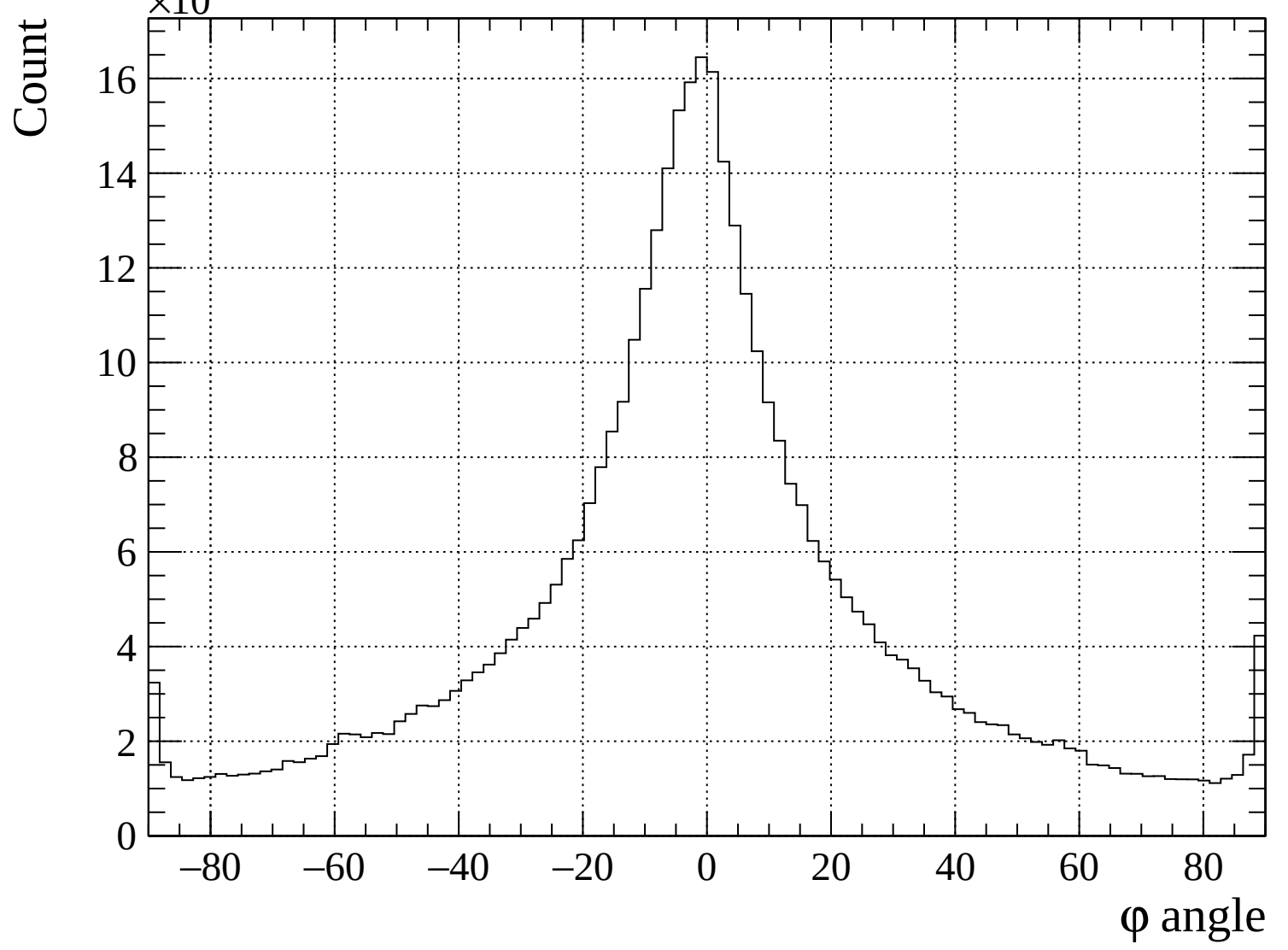


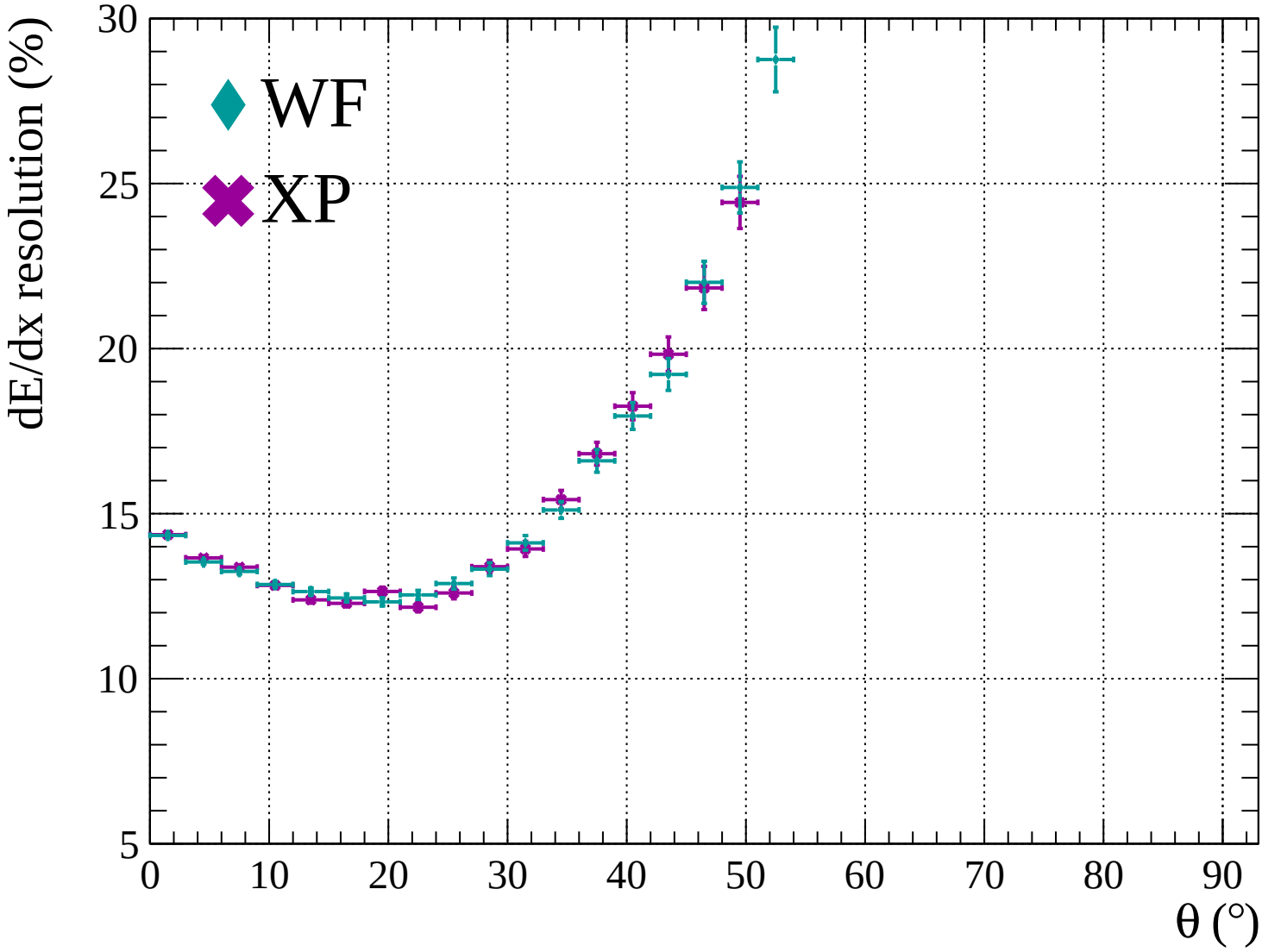


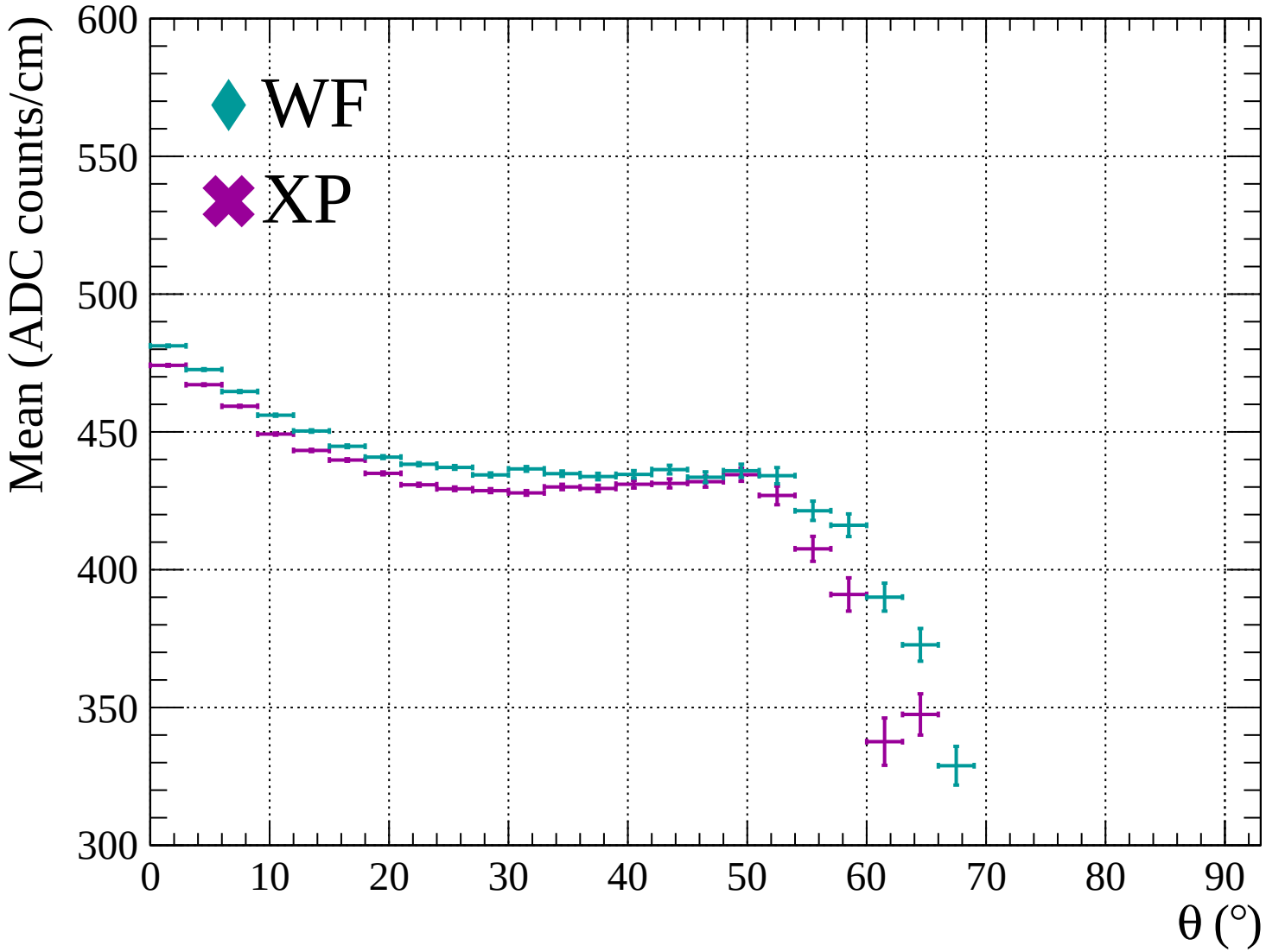




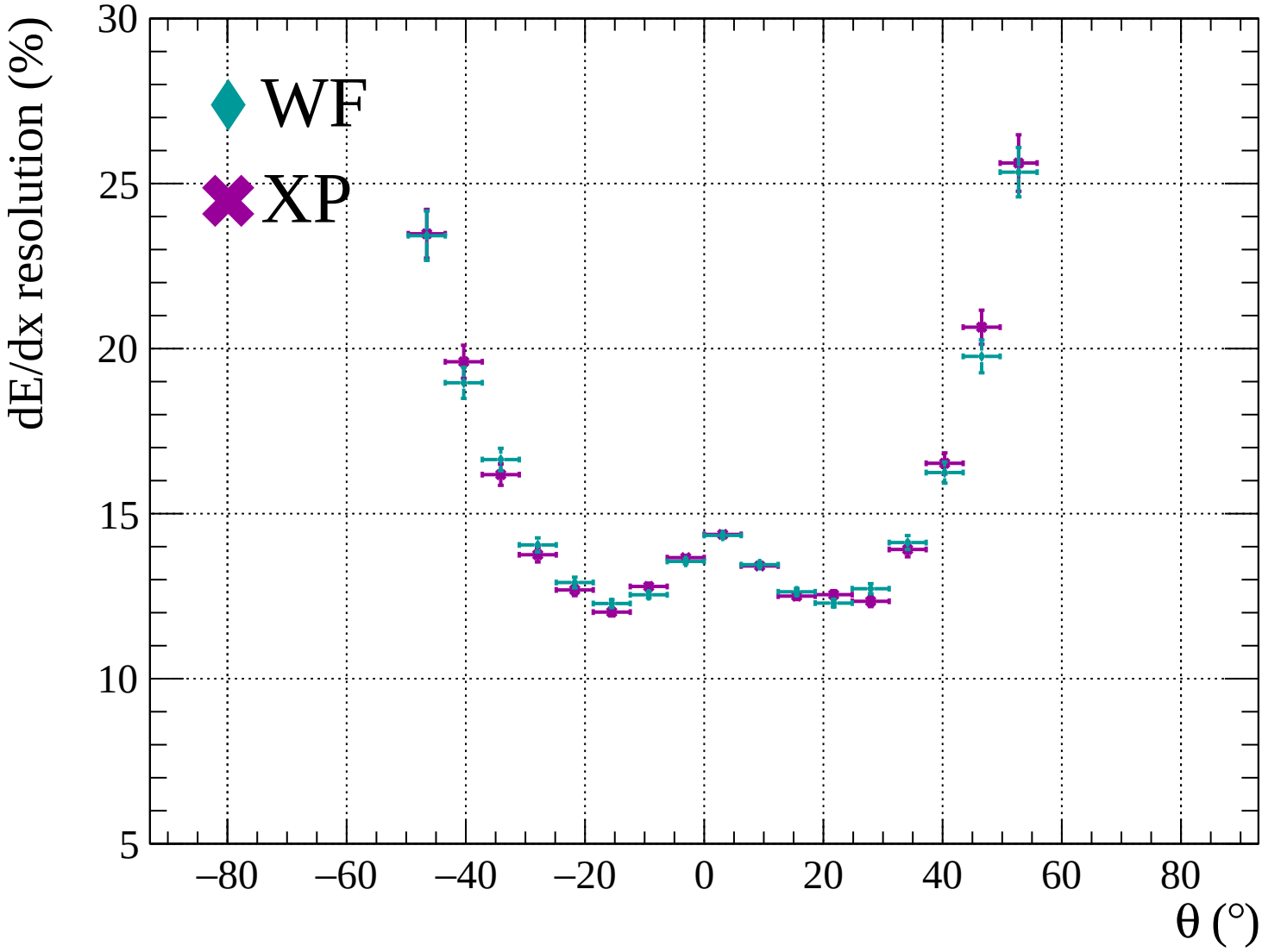


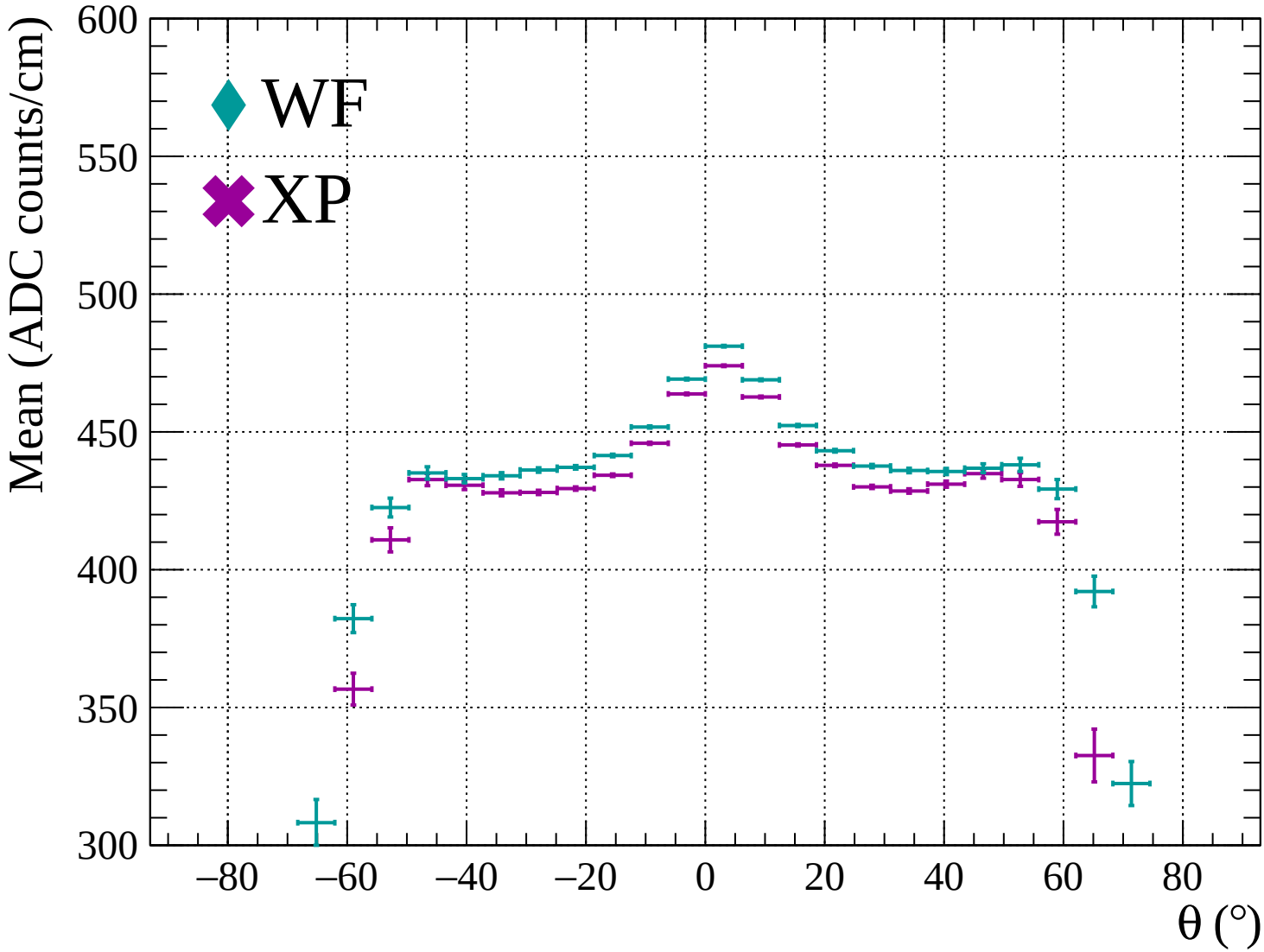


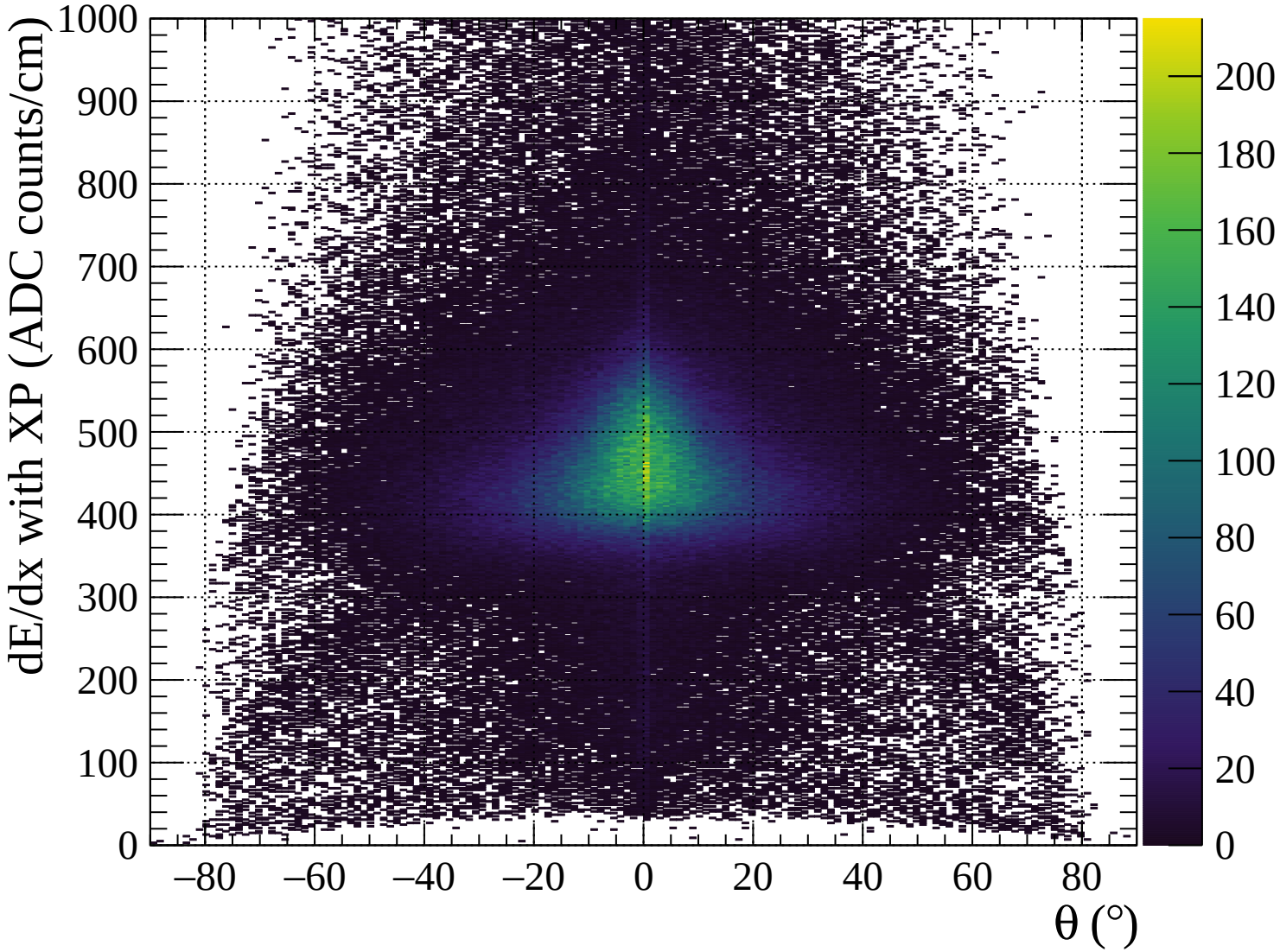


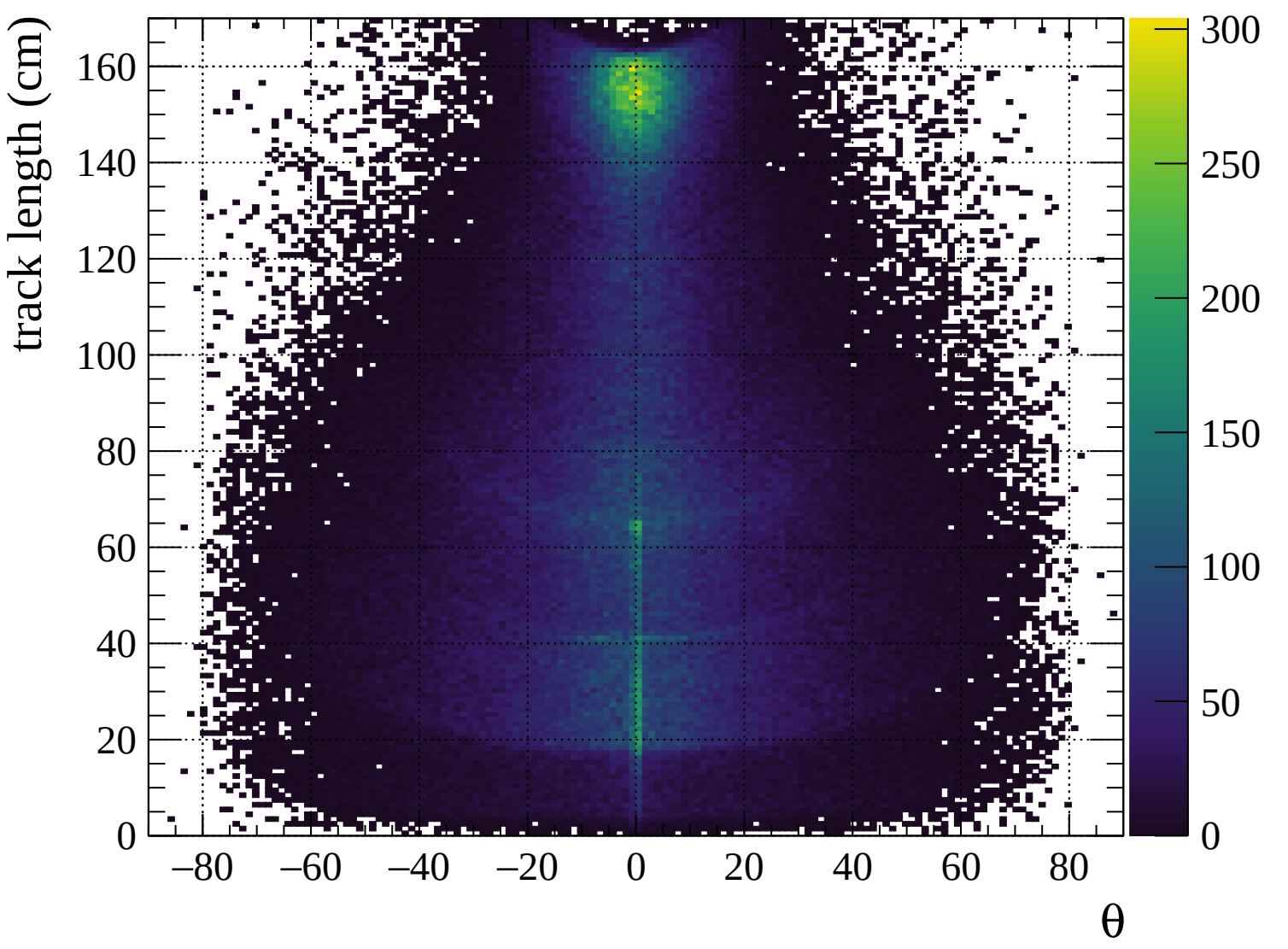


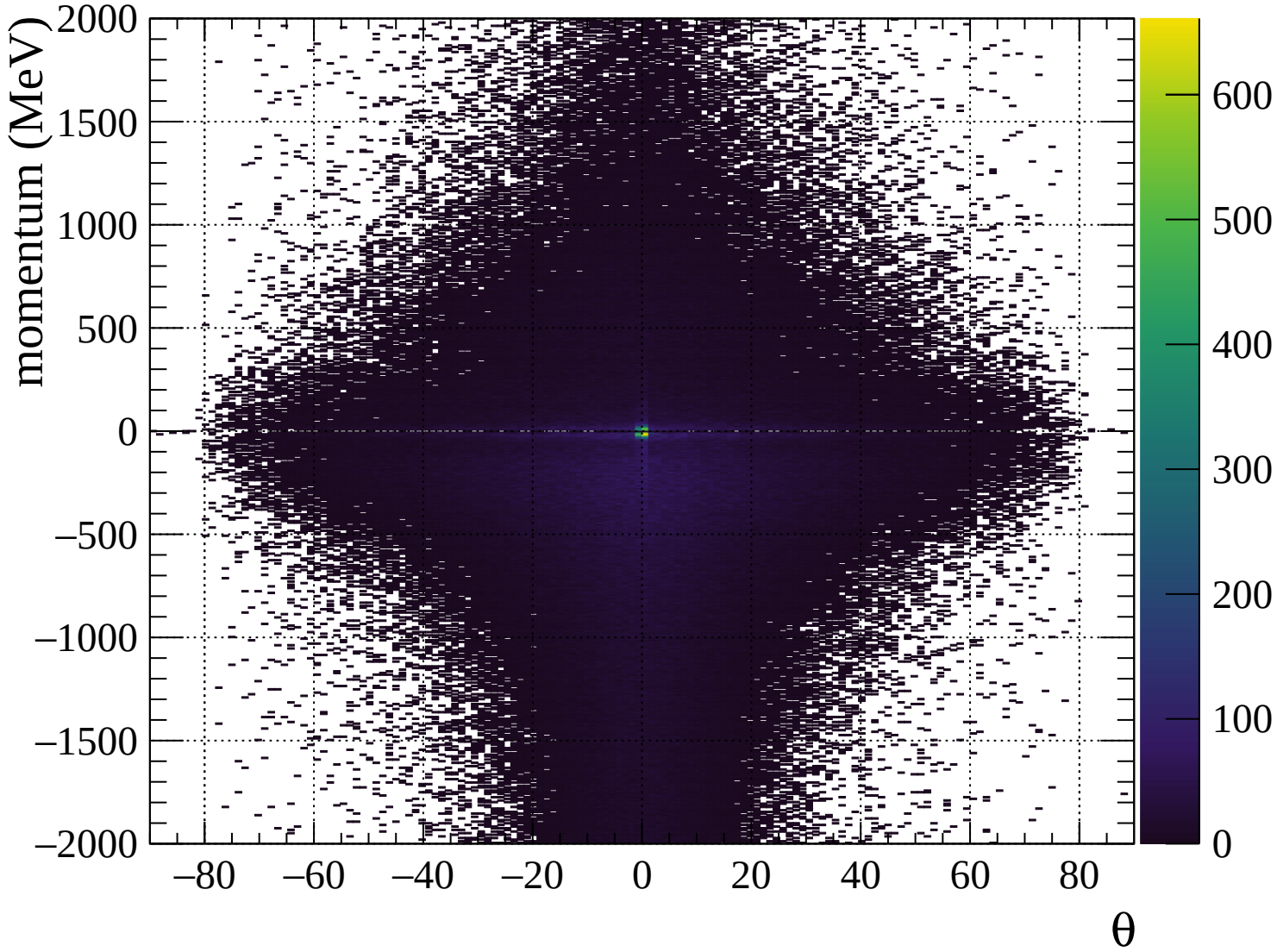


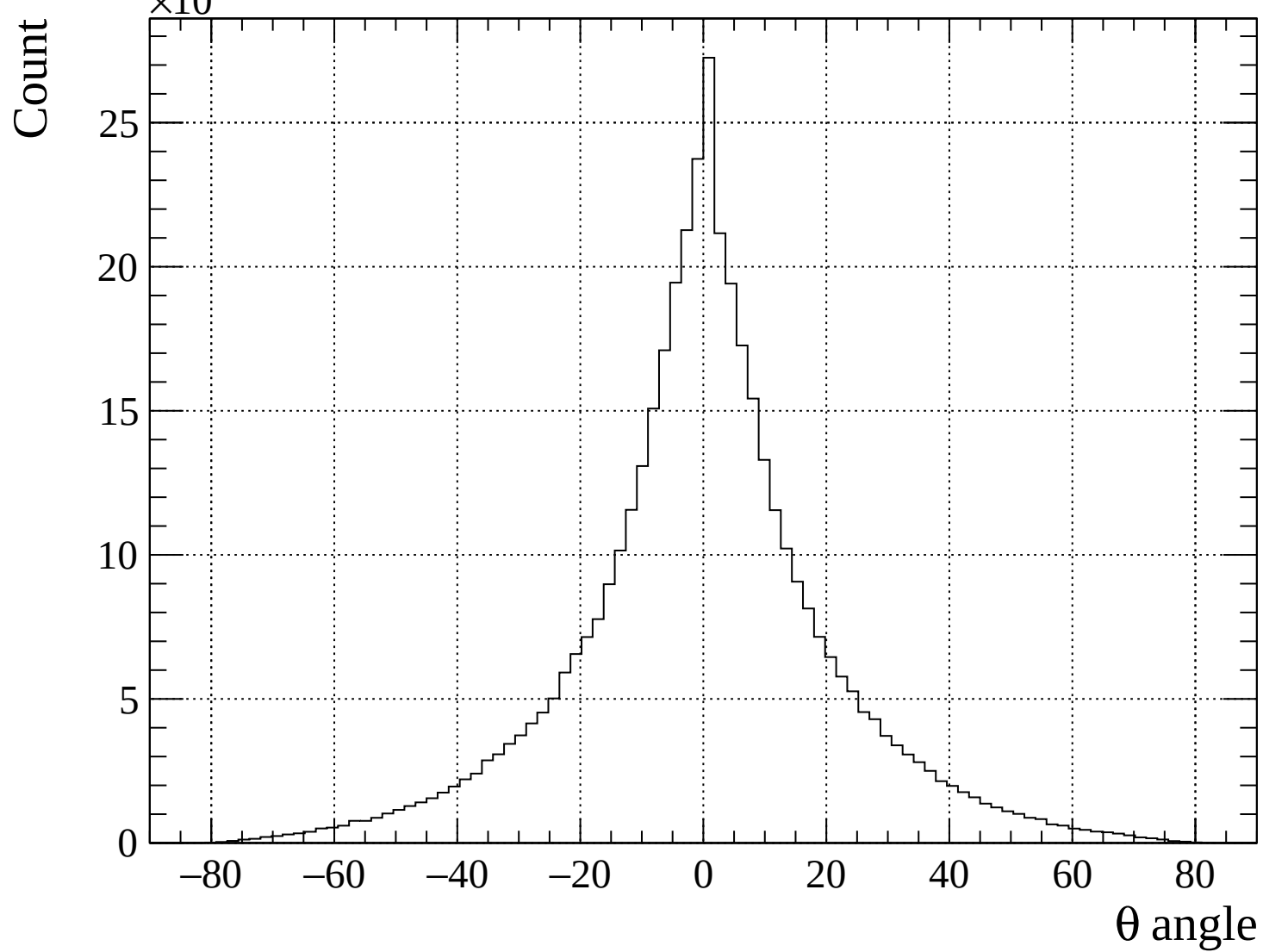


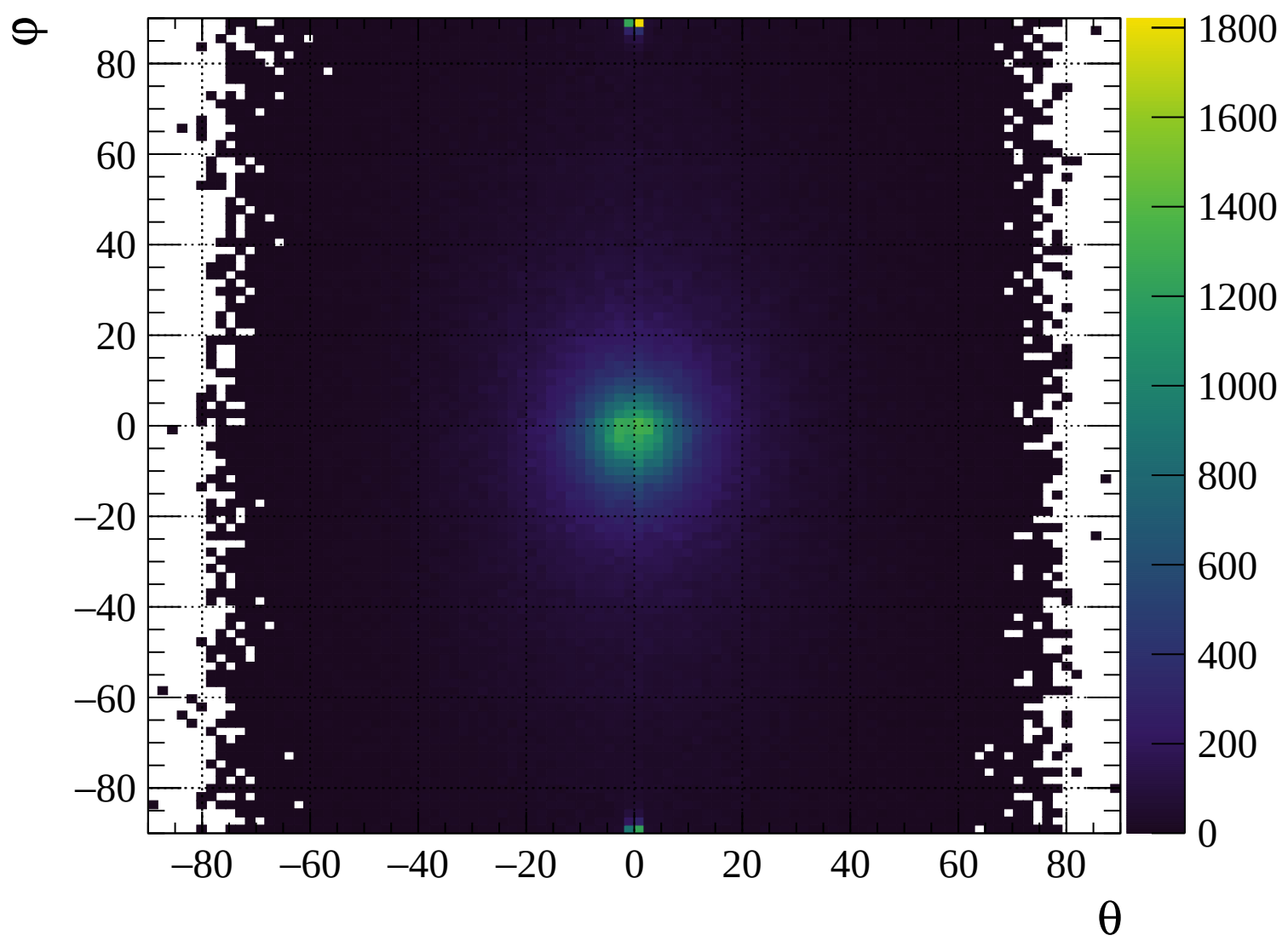


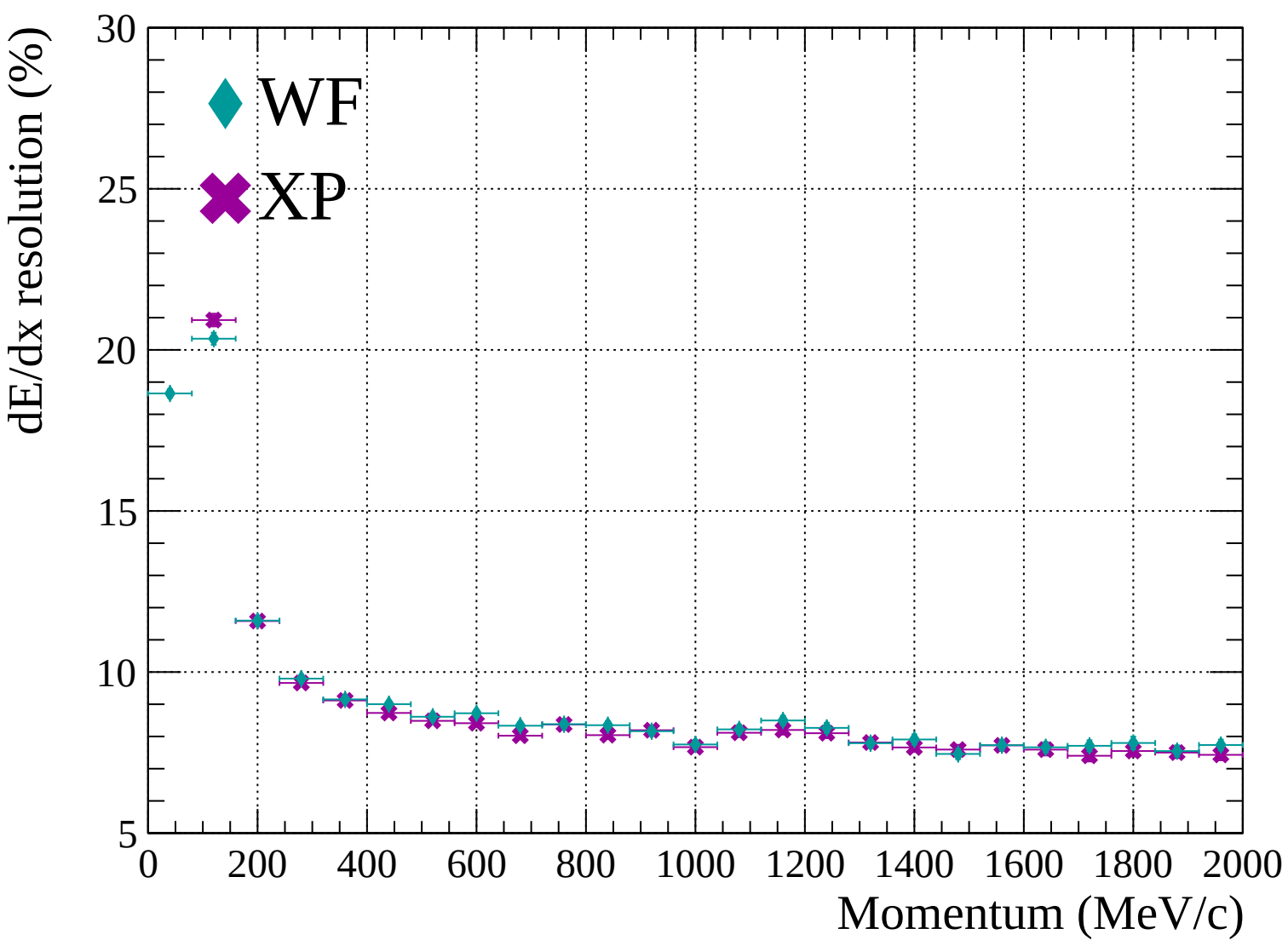




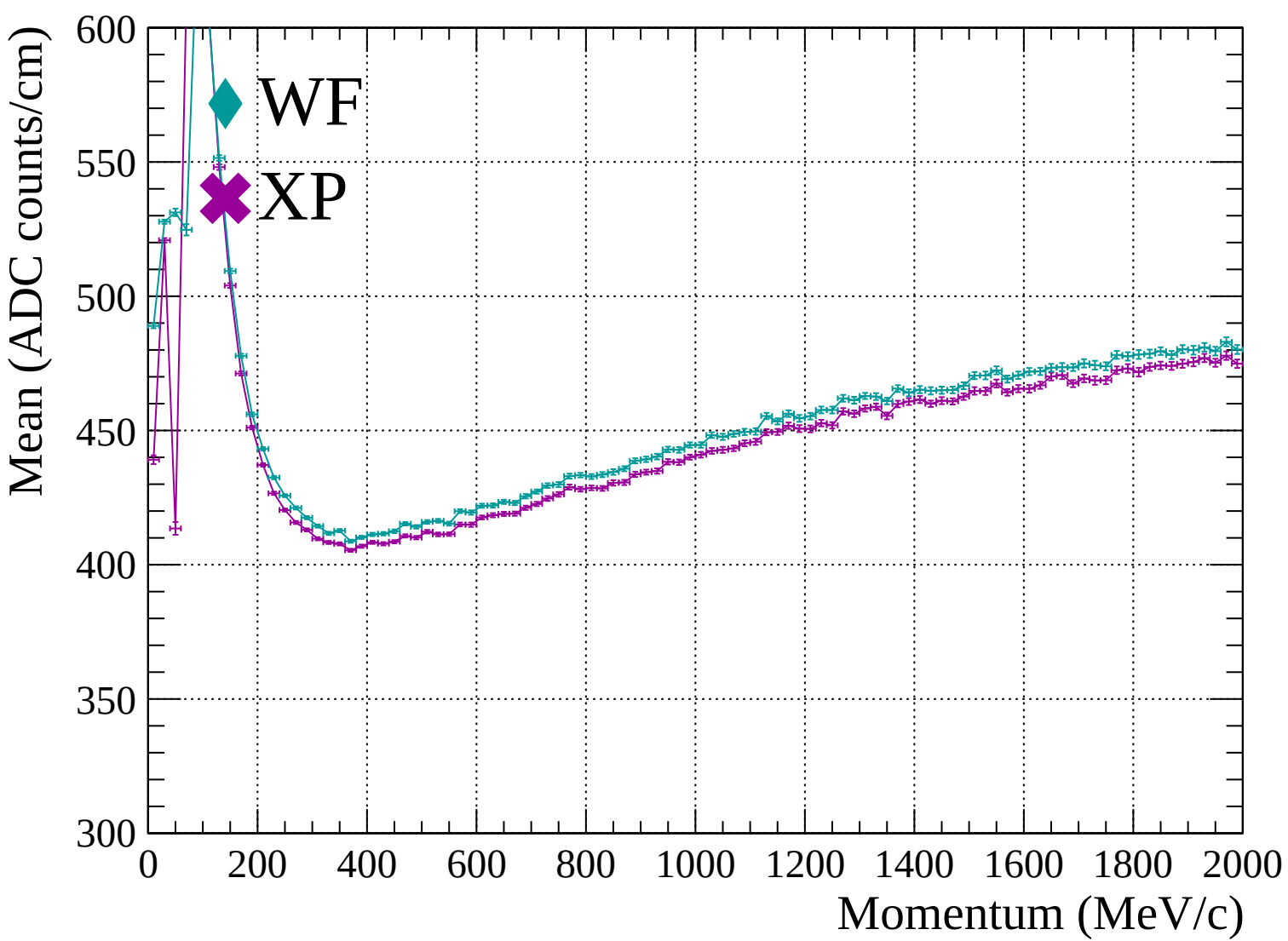


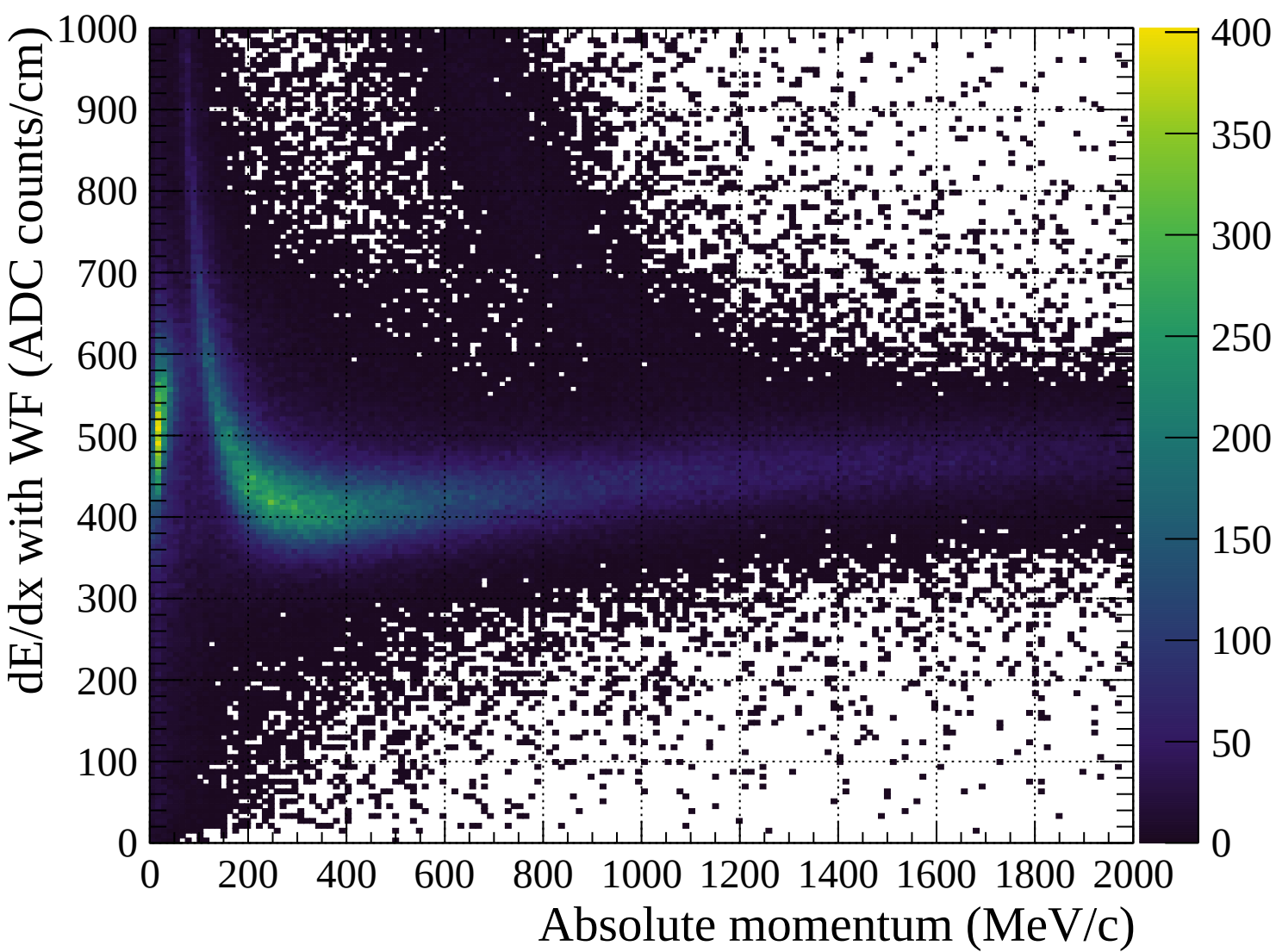


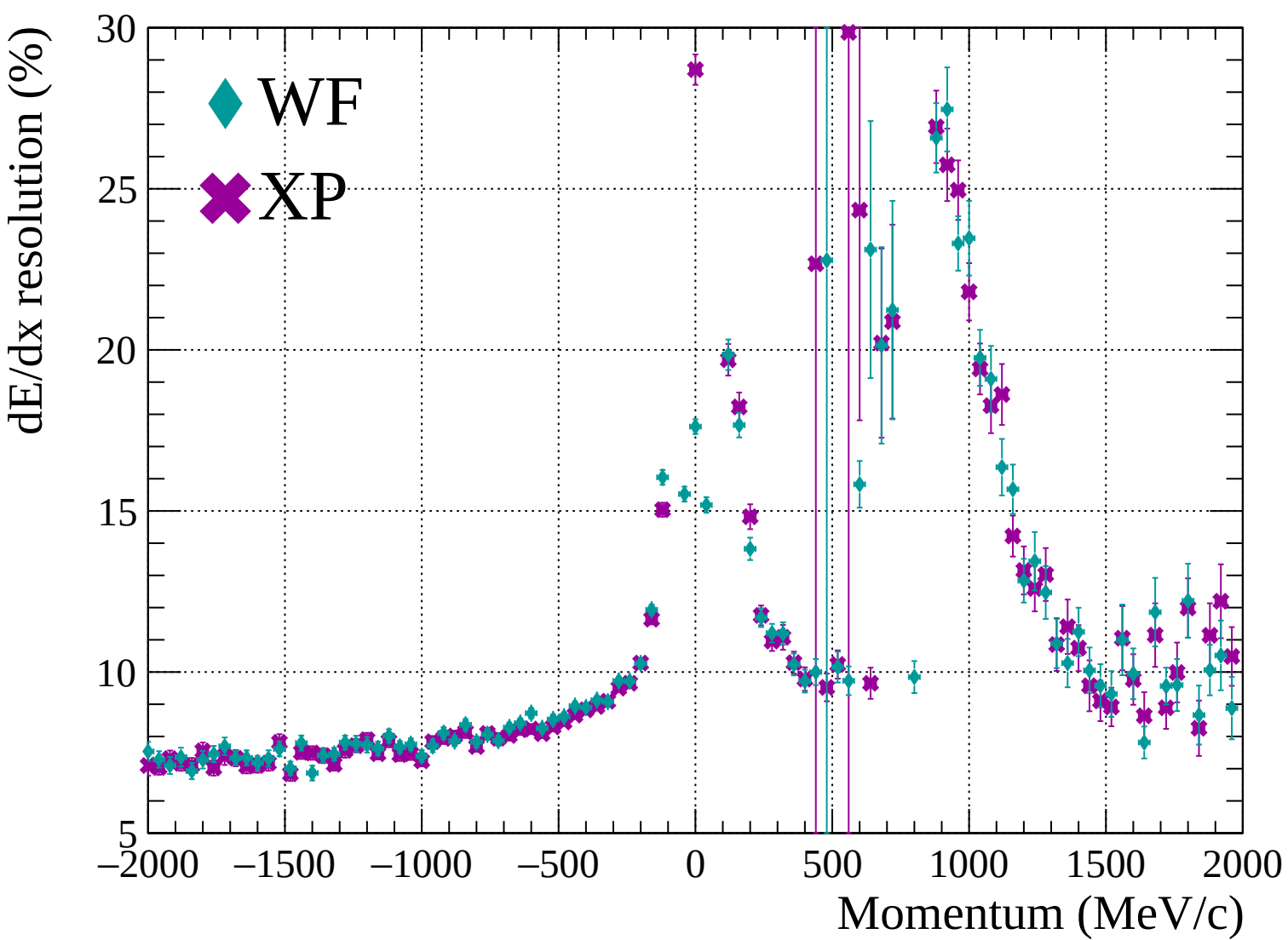


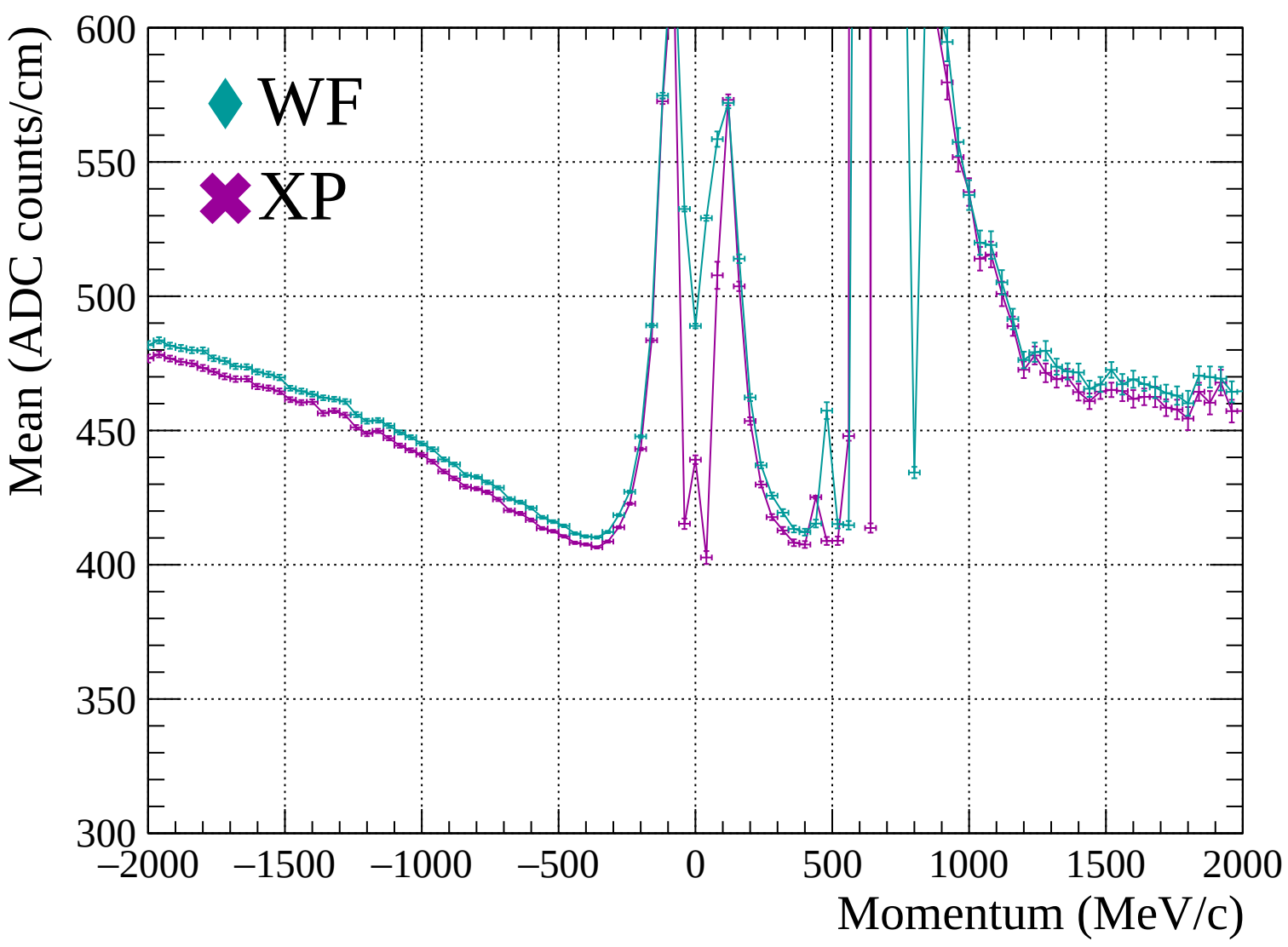


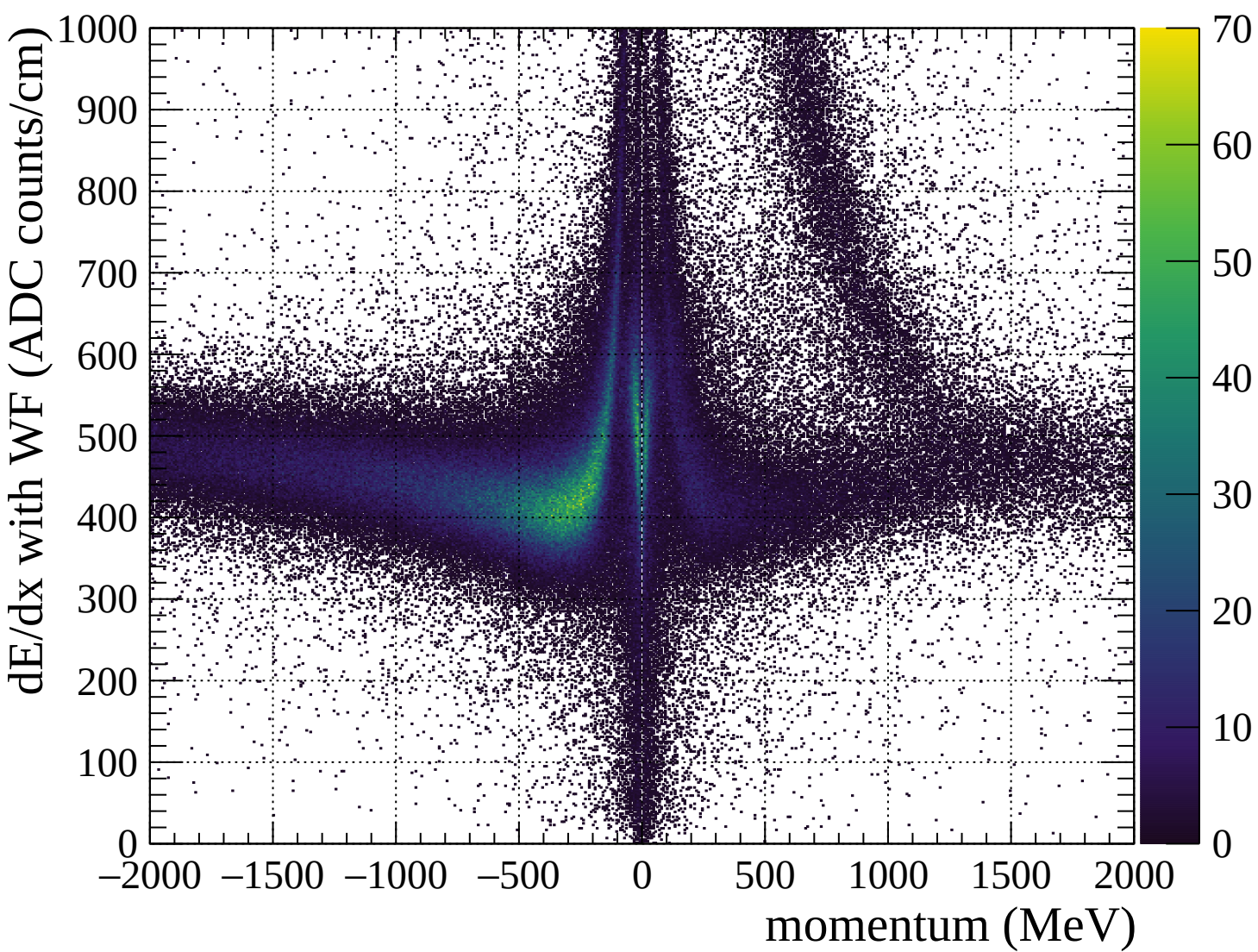


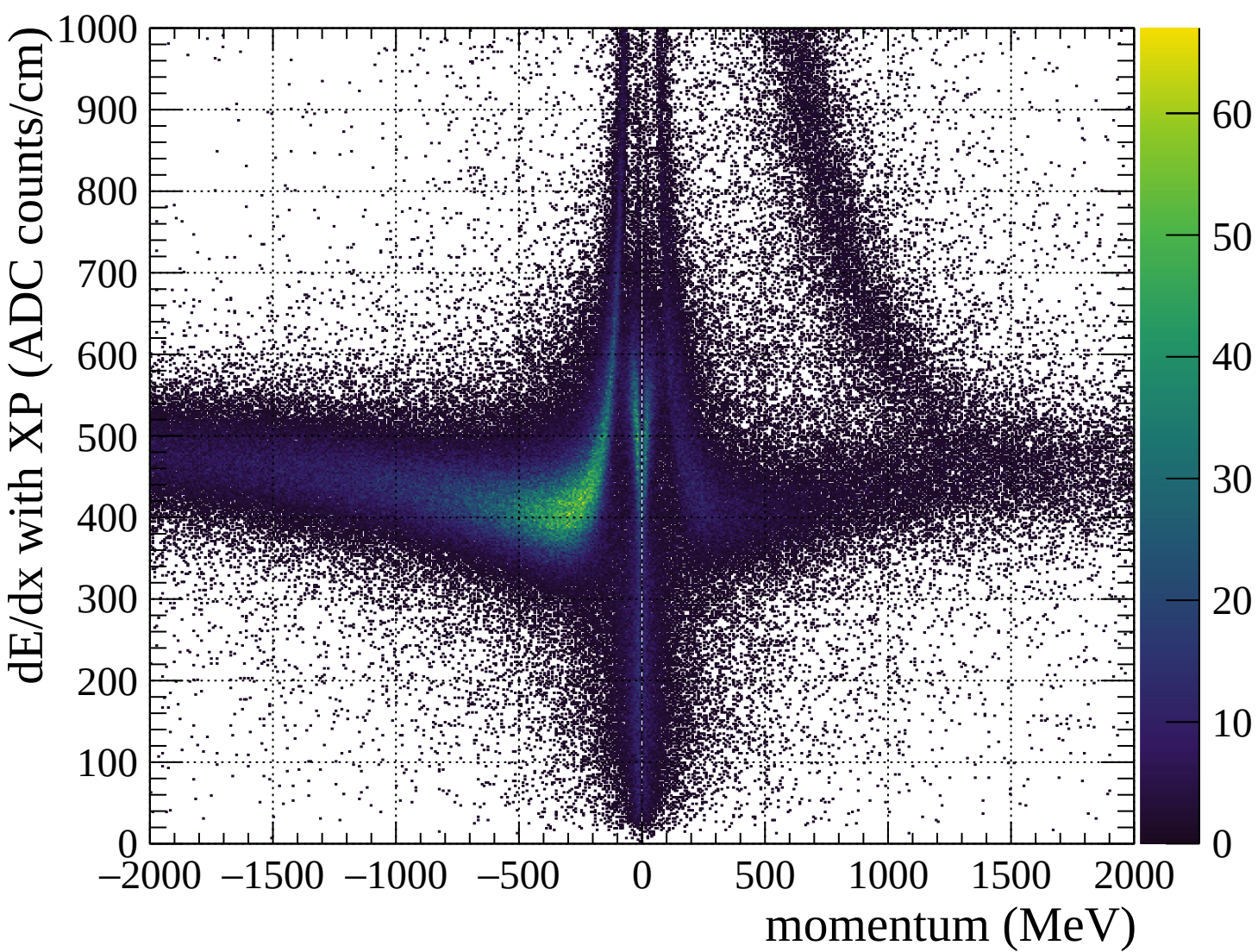


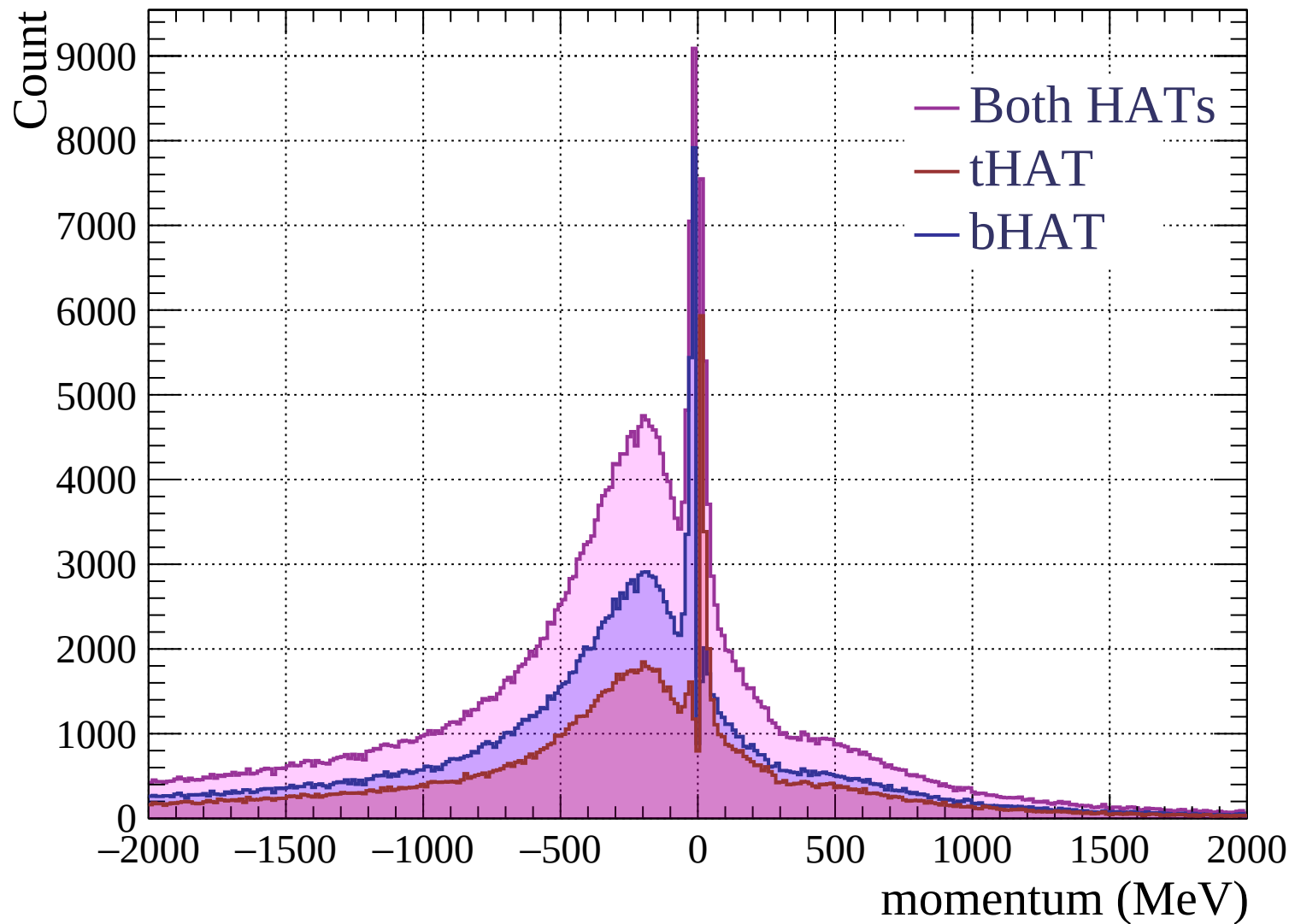




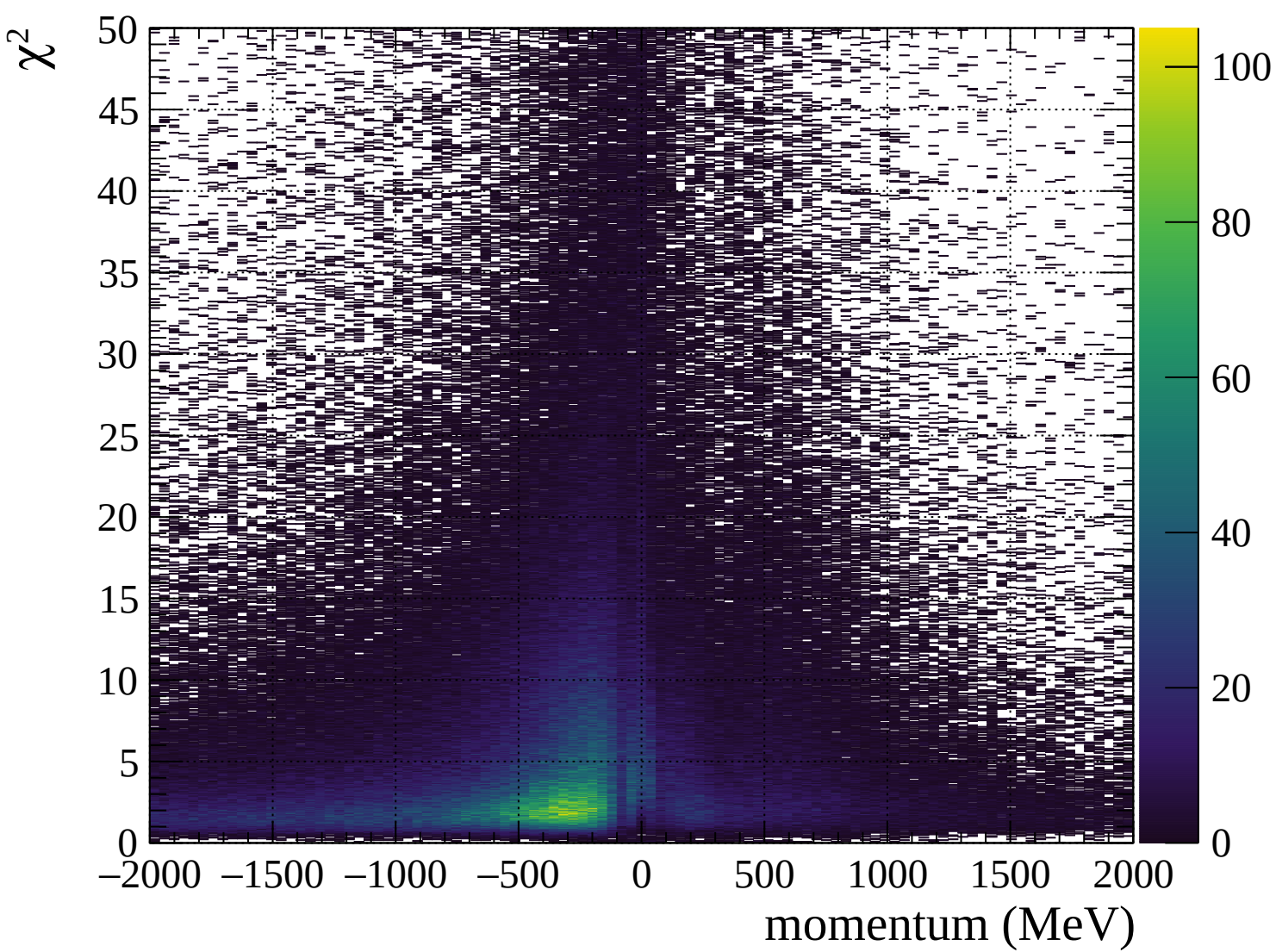






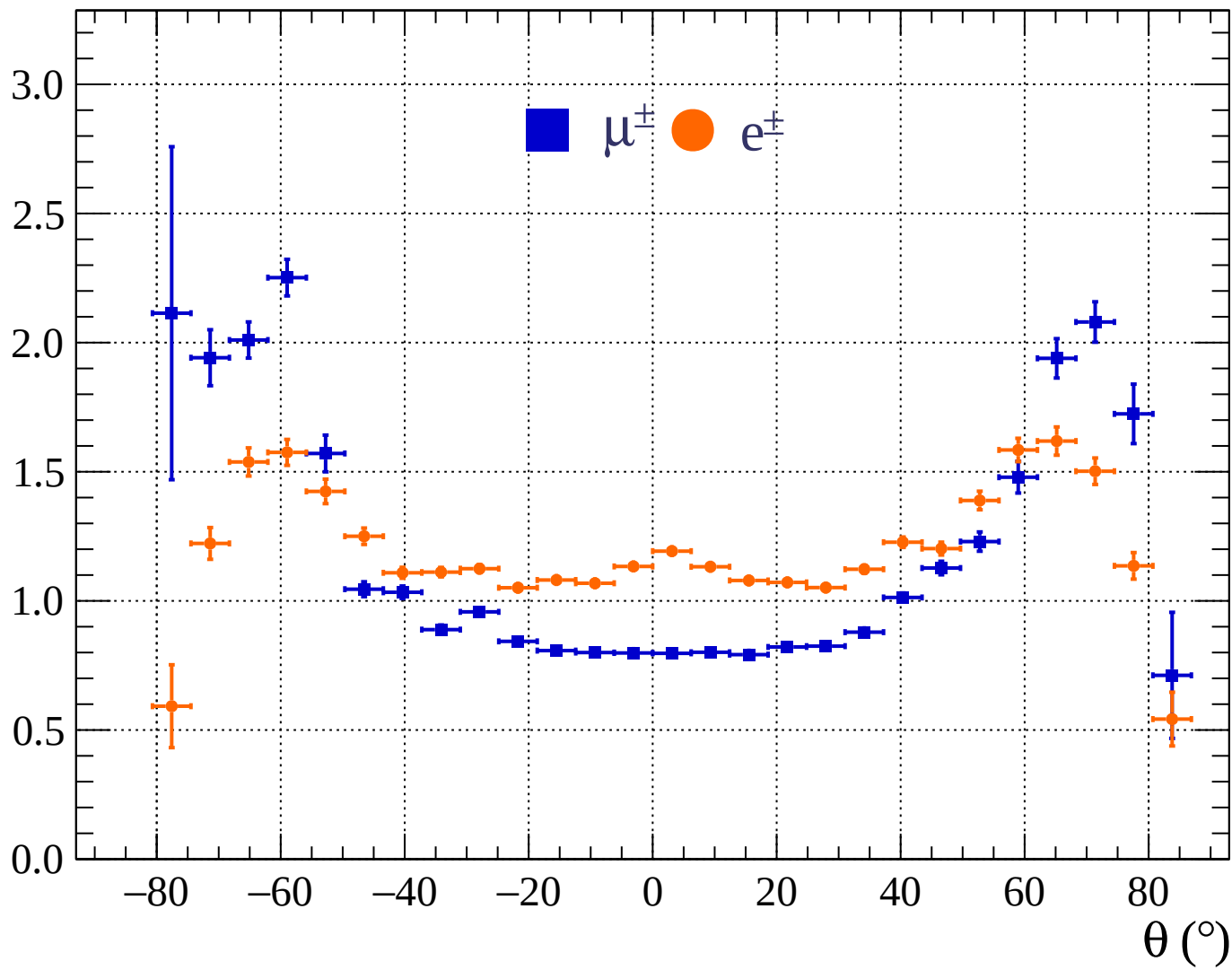




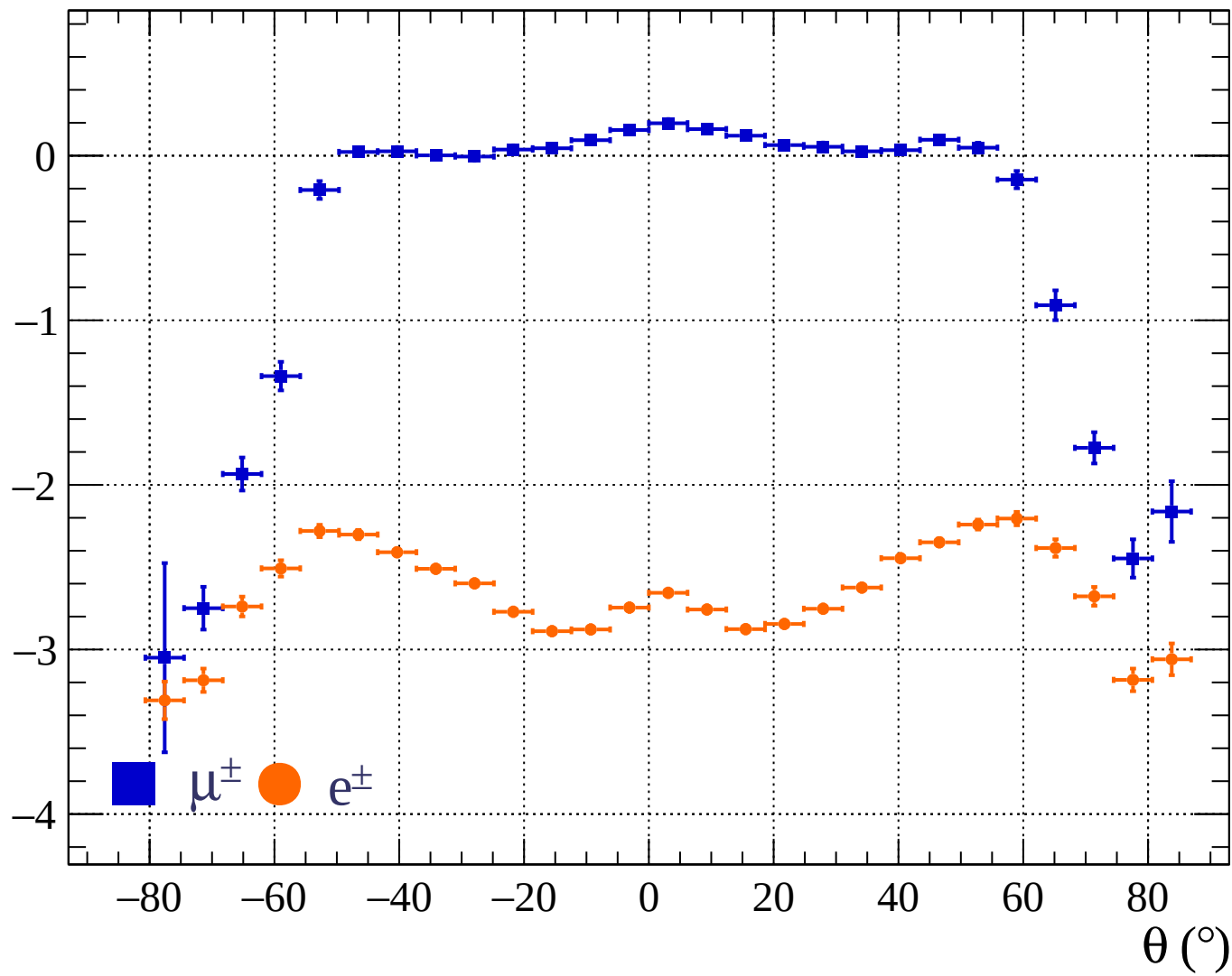




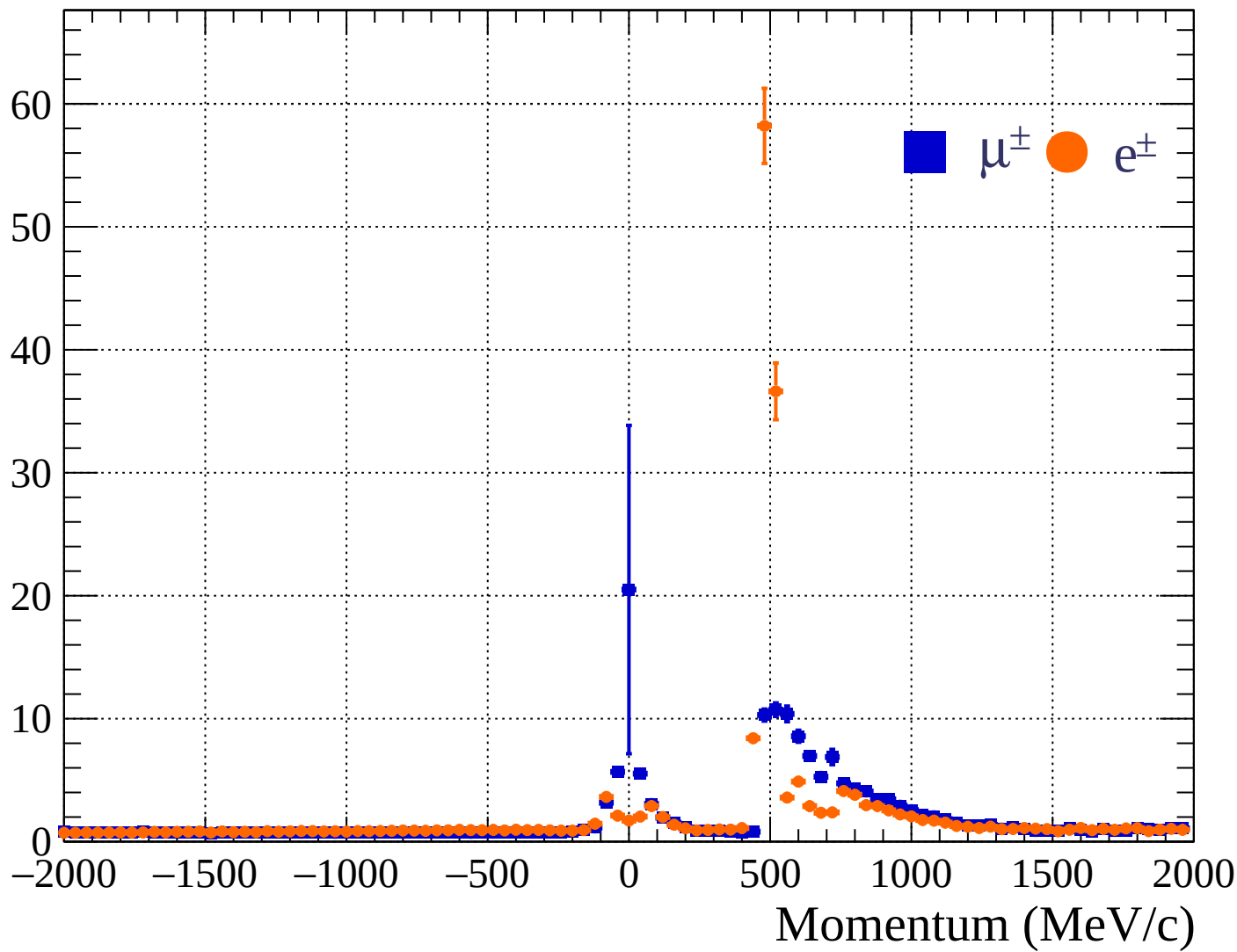
Pulls standard deviation

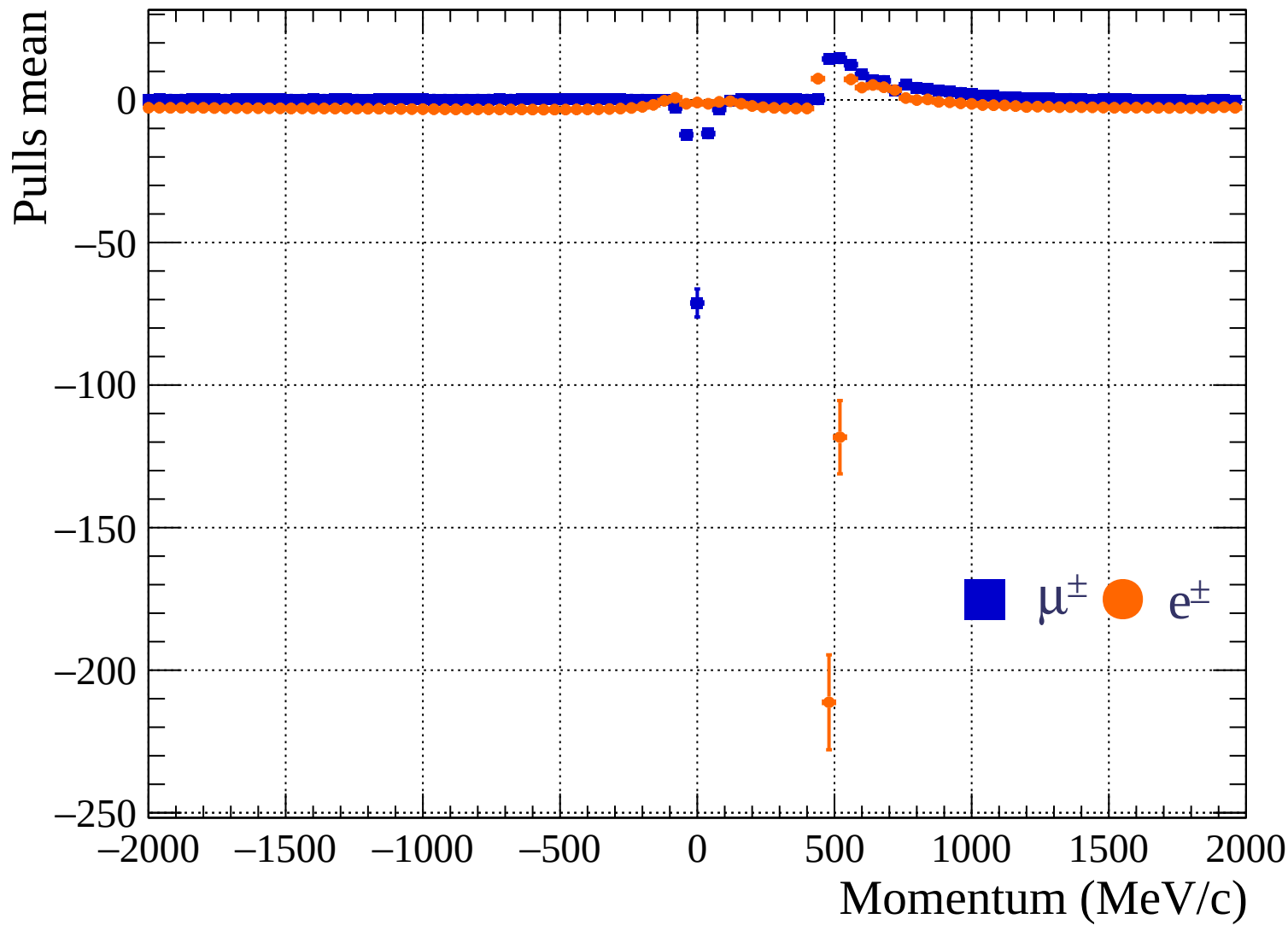


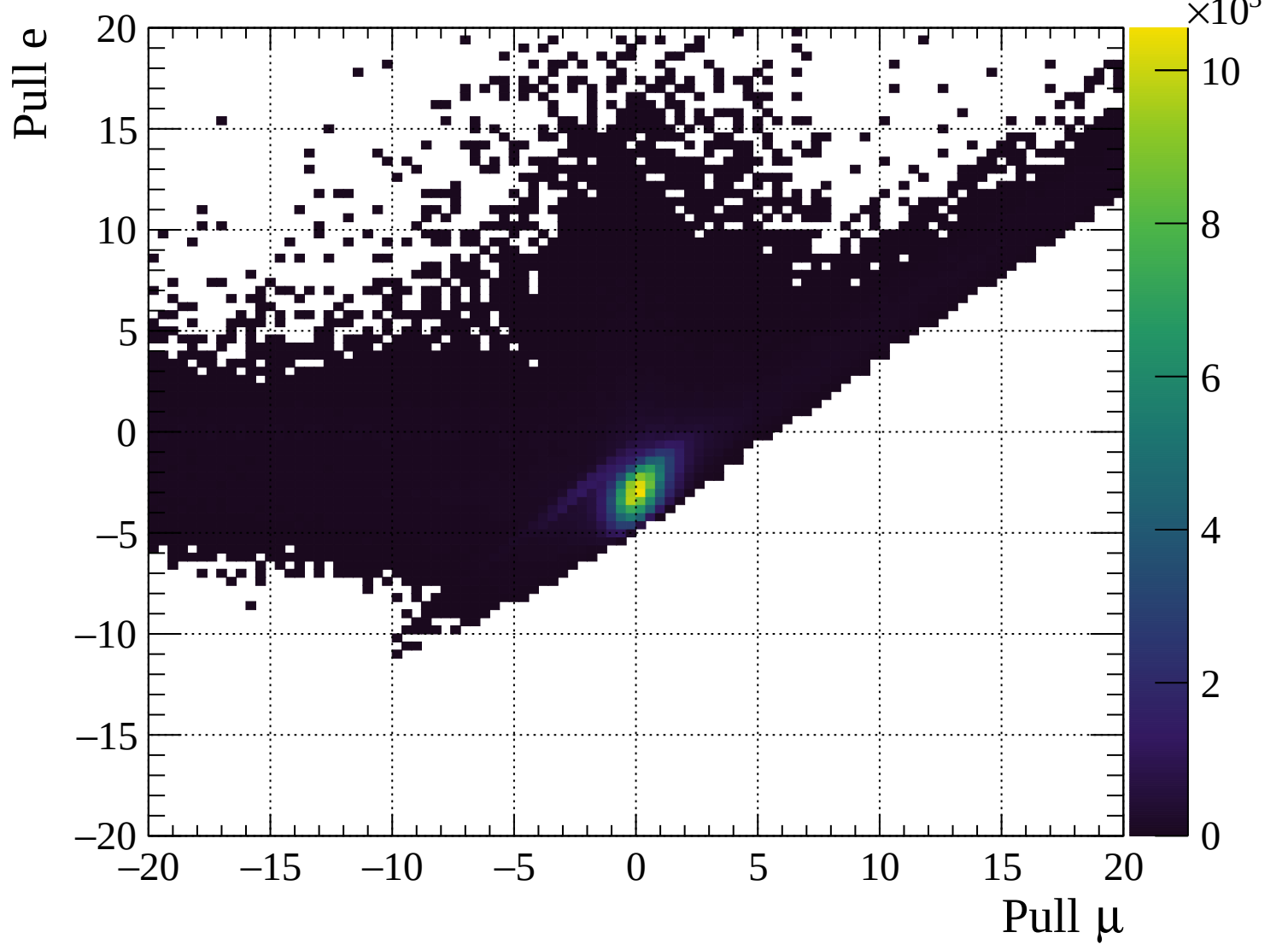
Pulls mean

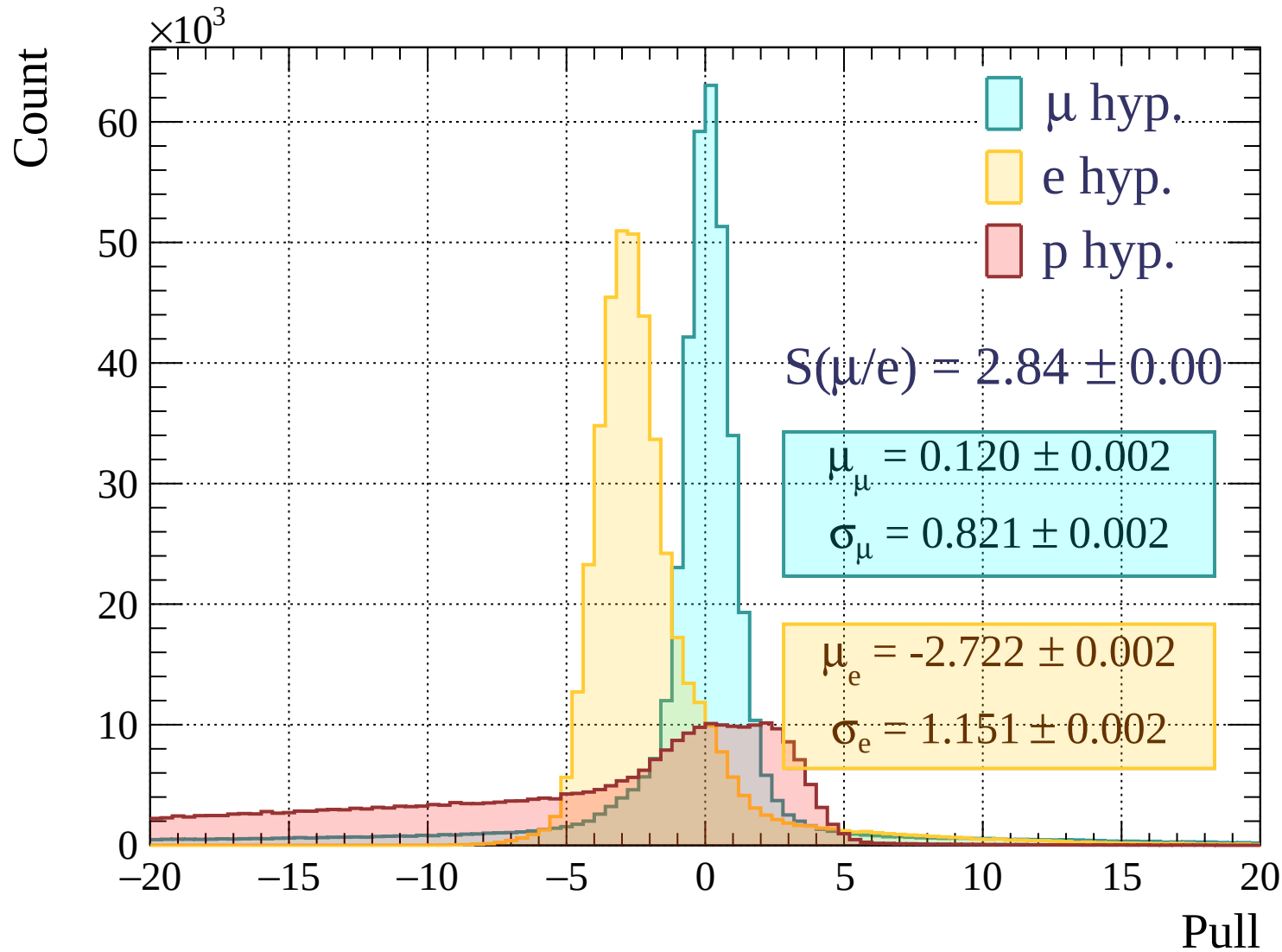


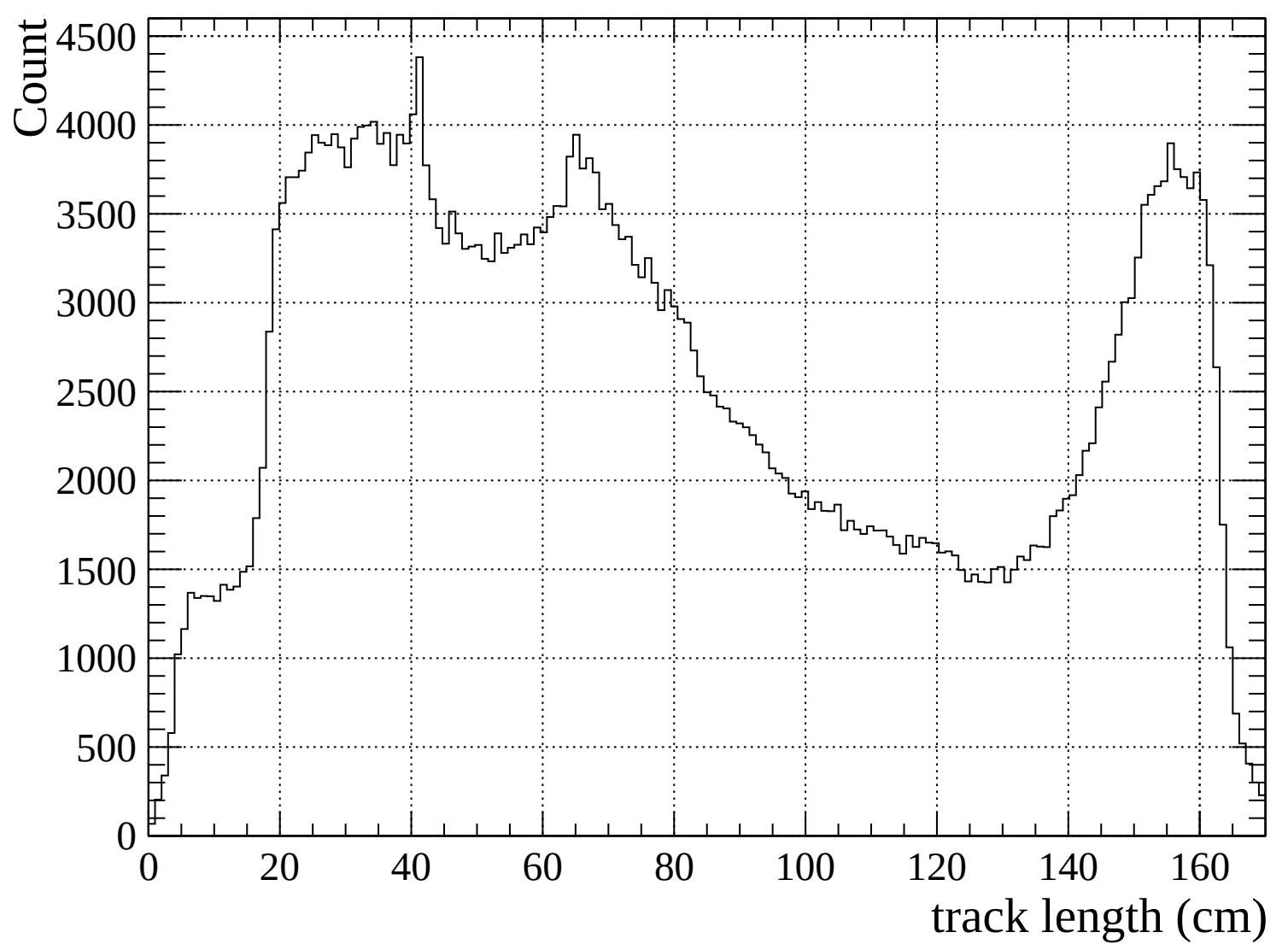
Pulls standard deviation

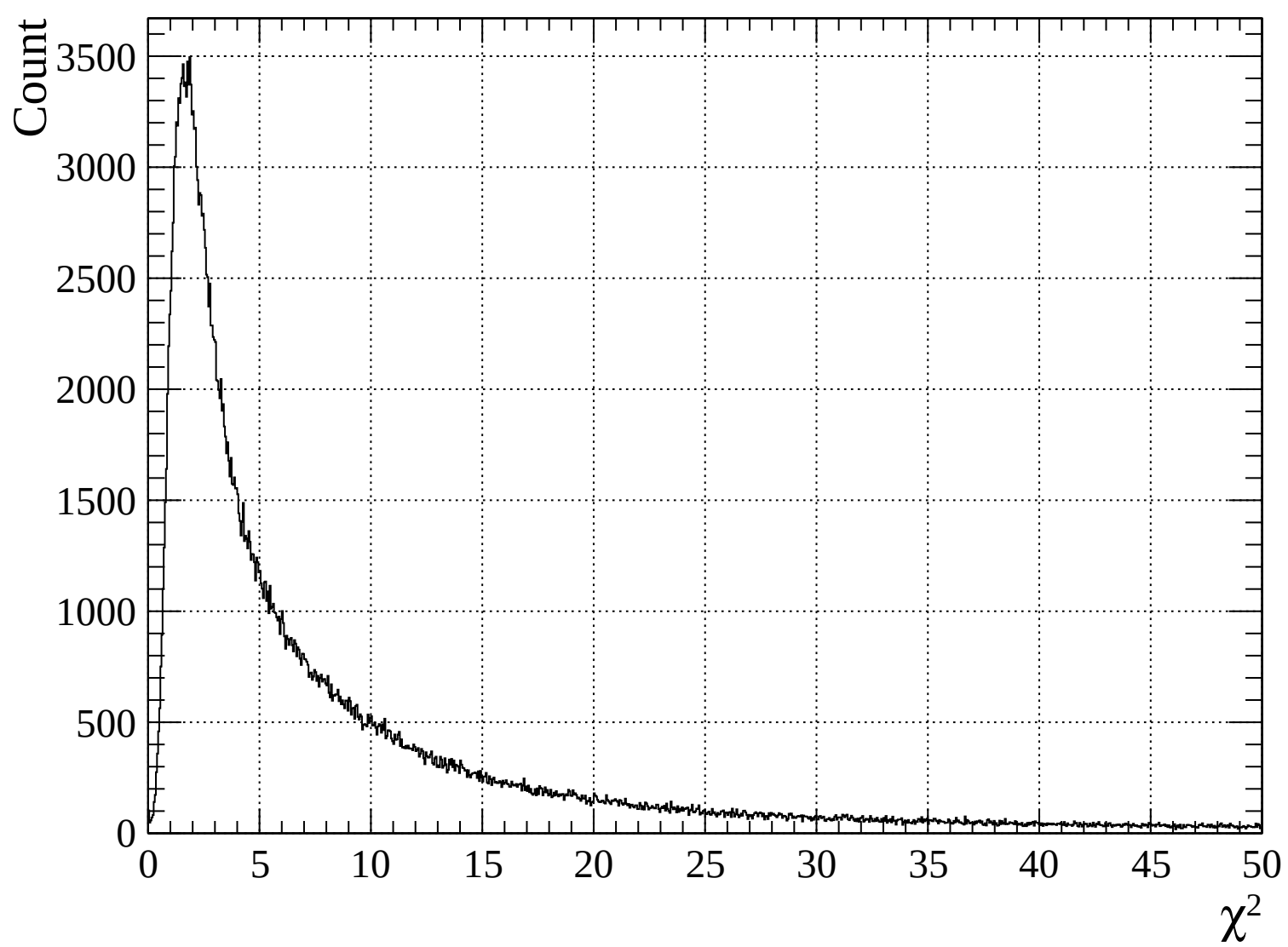




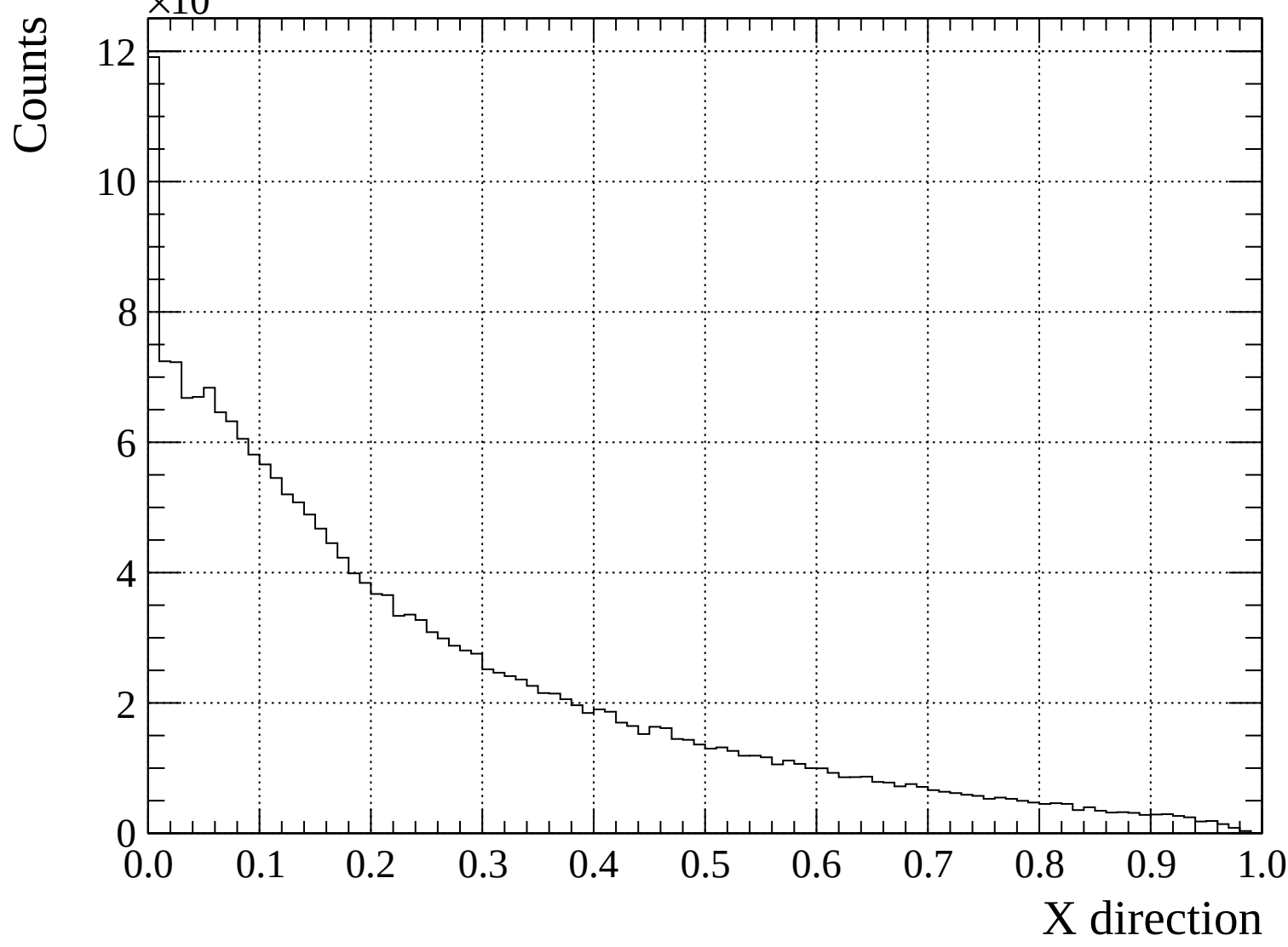


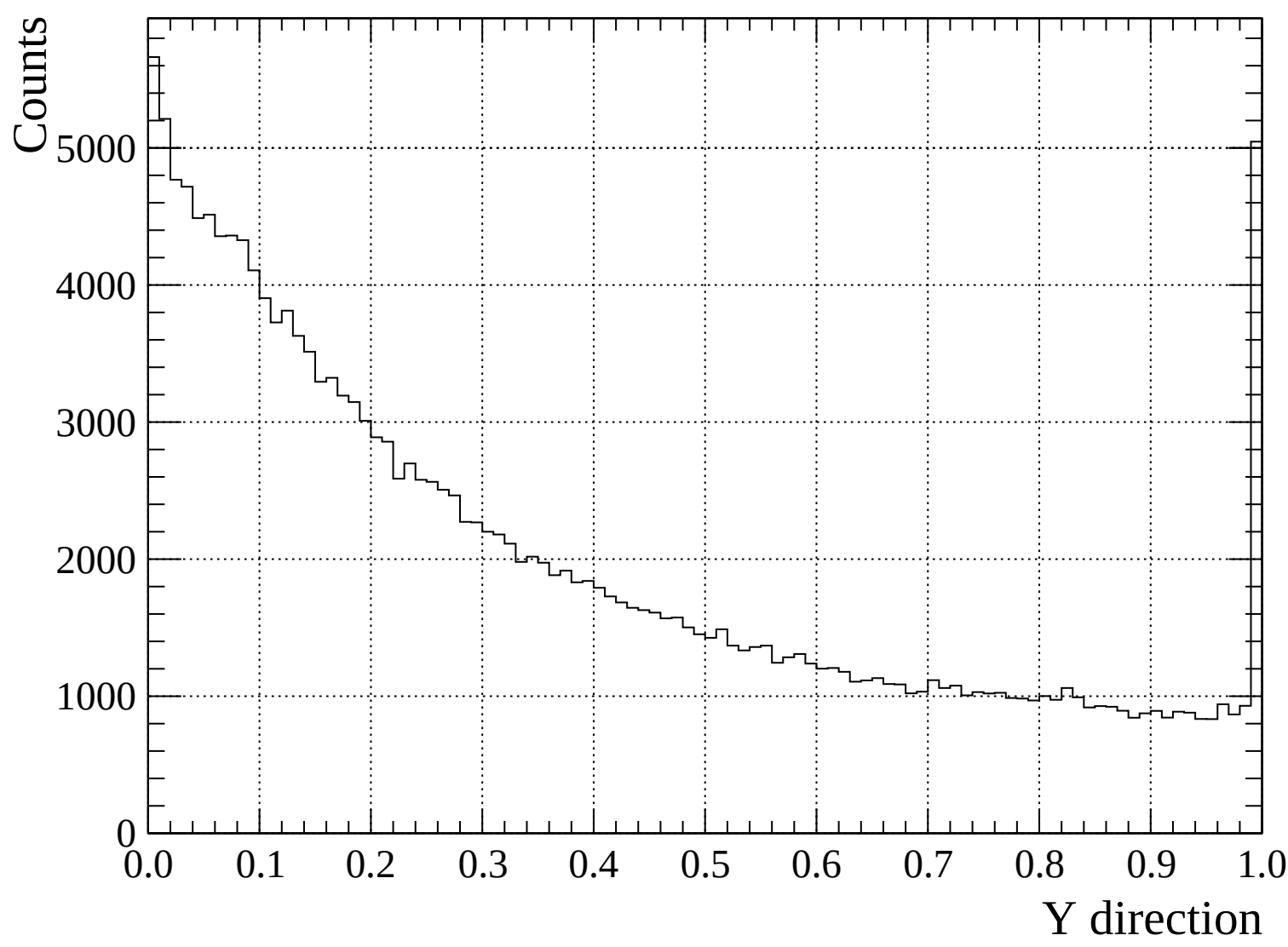


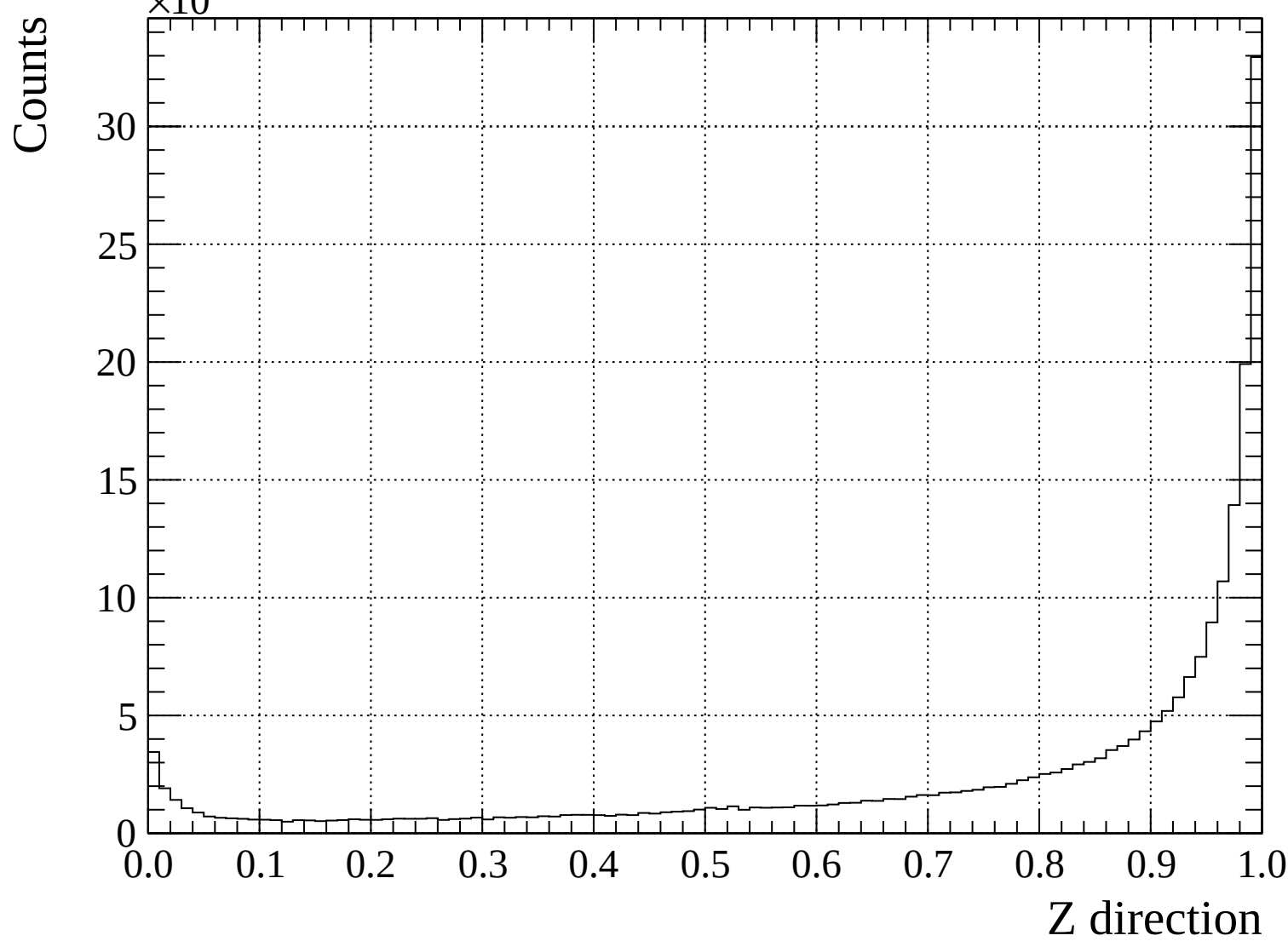


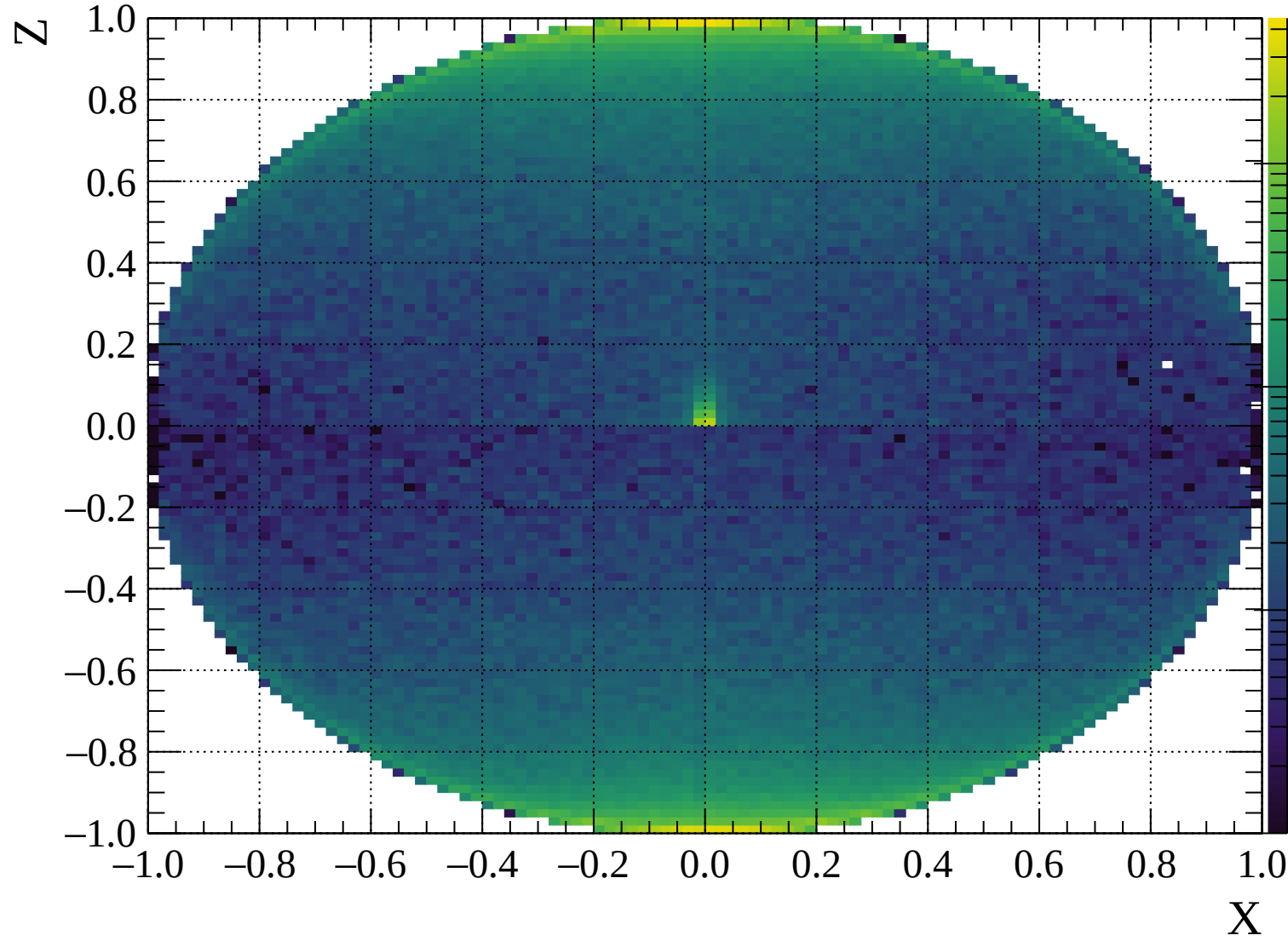


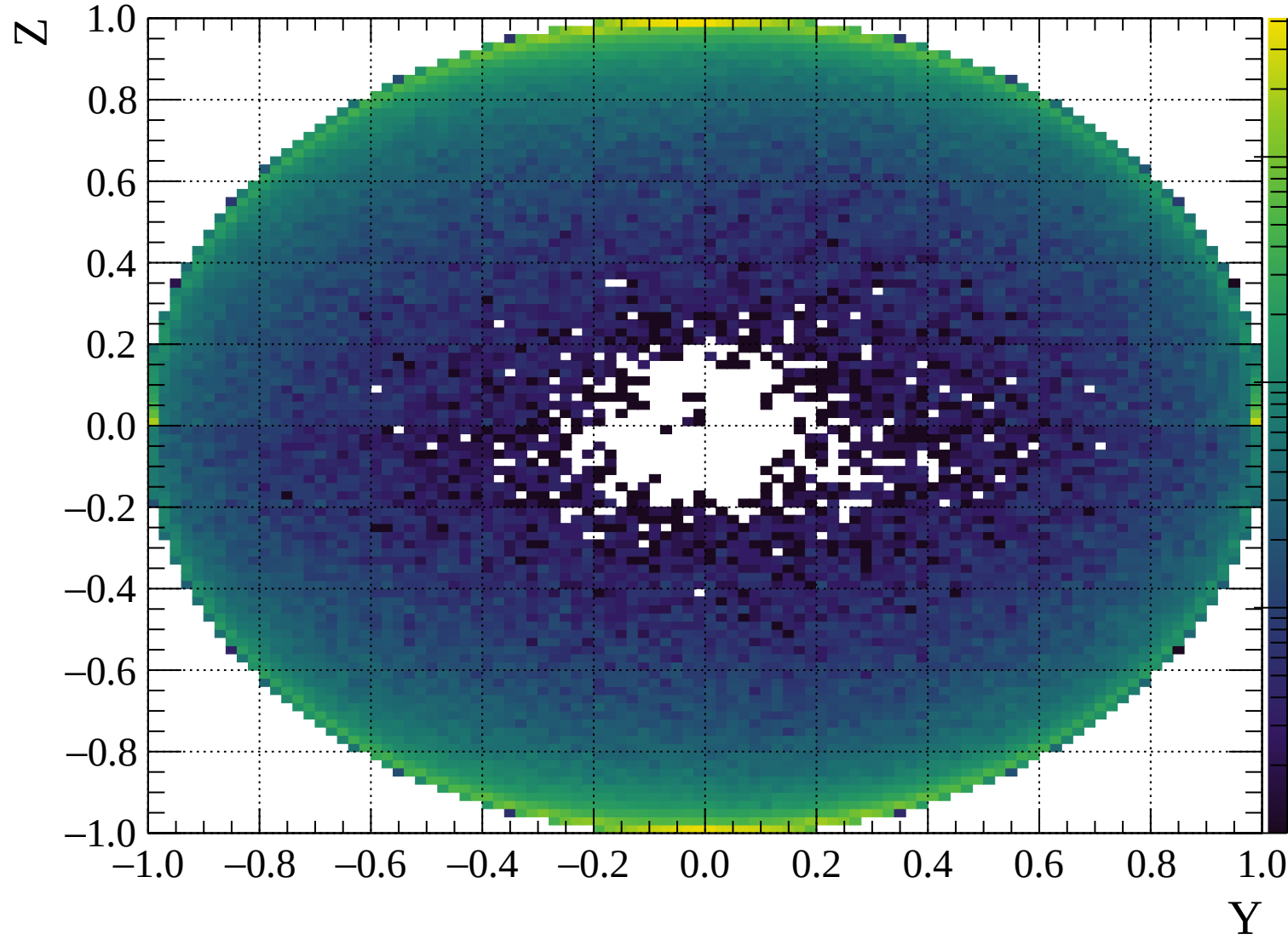


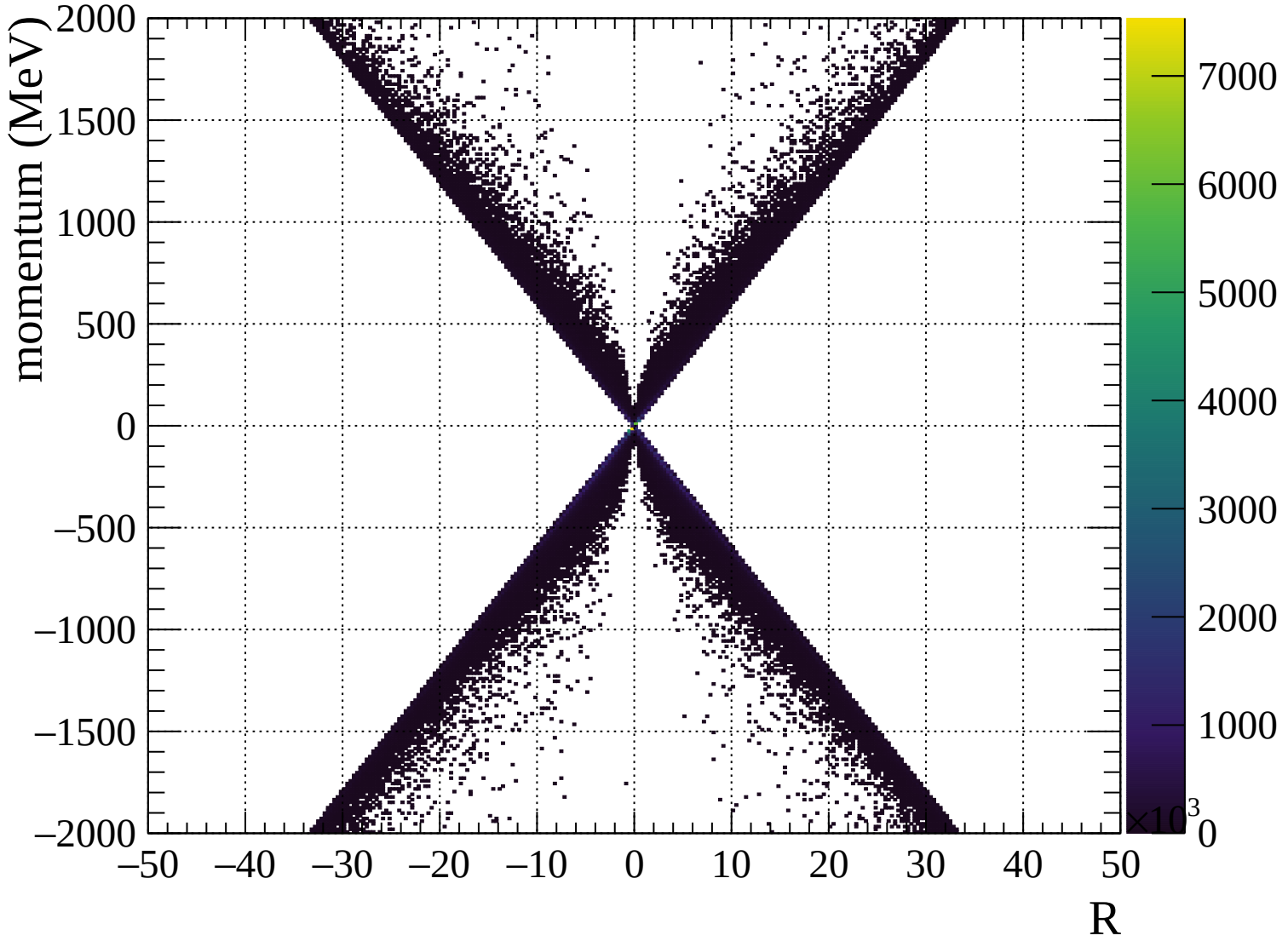


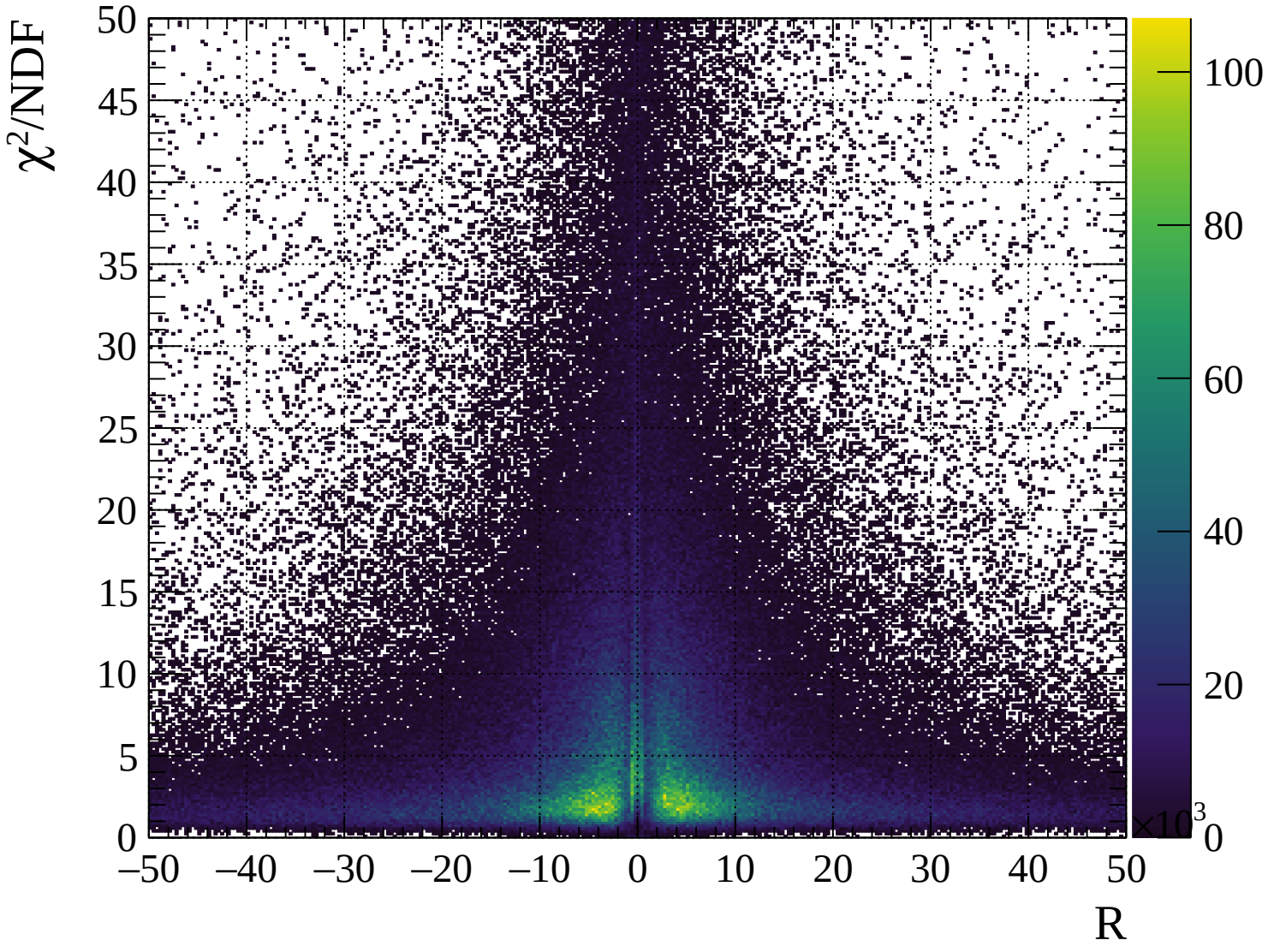




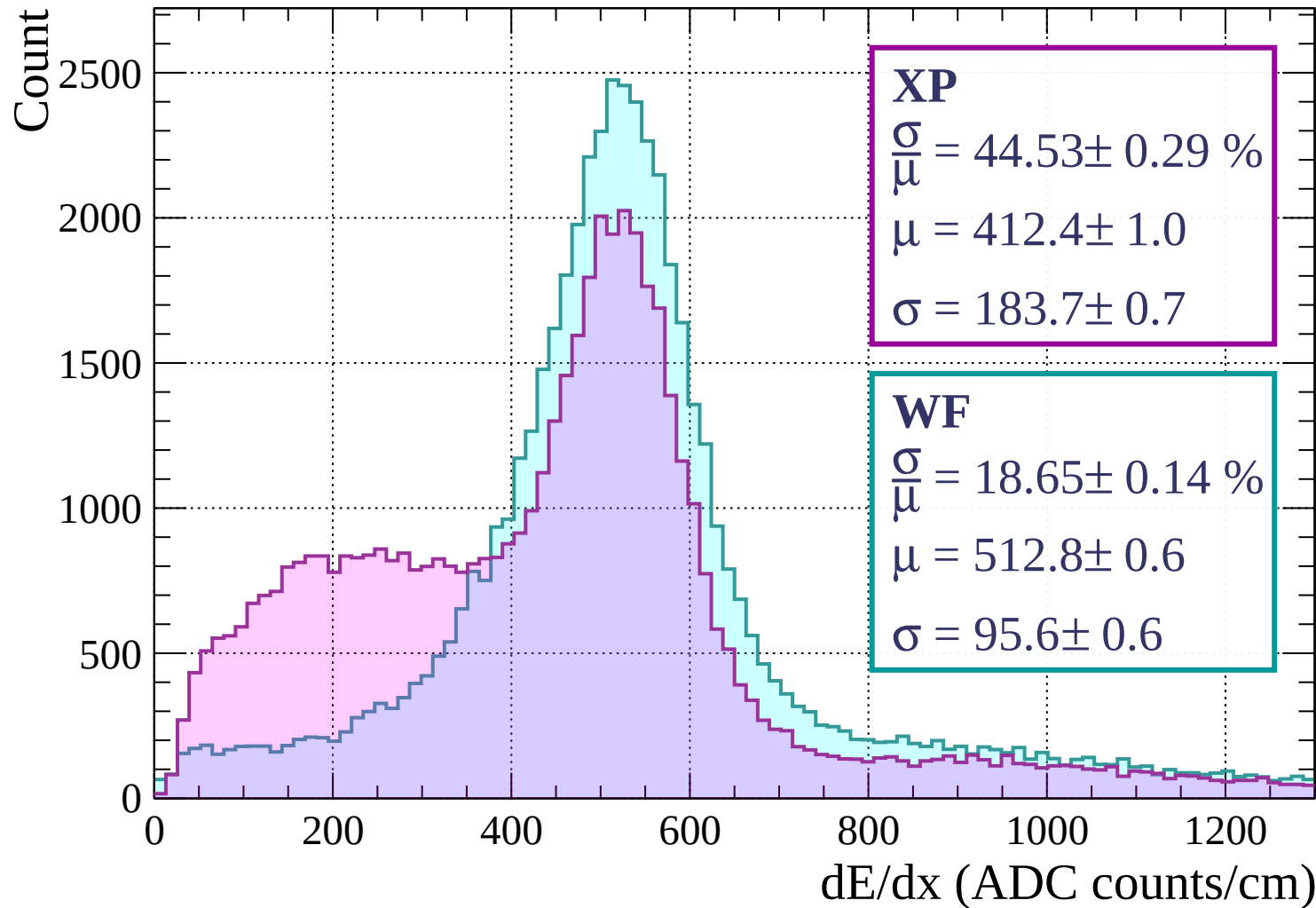






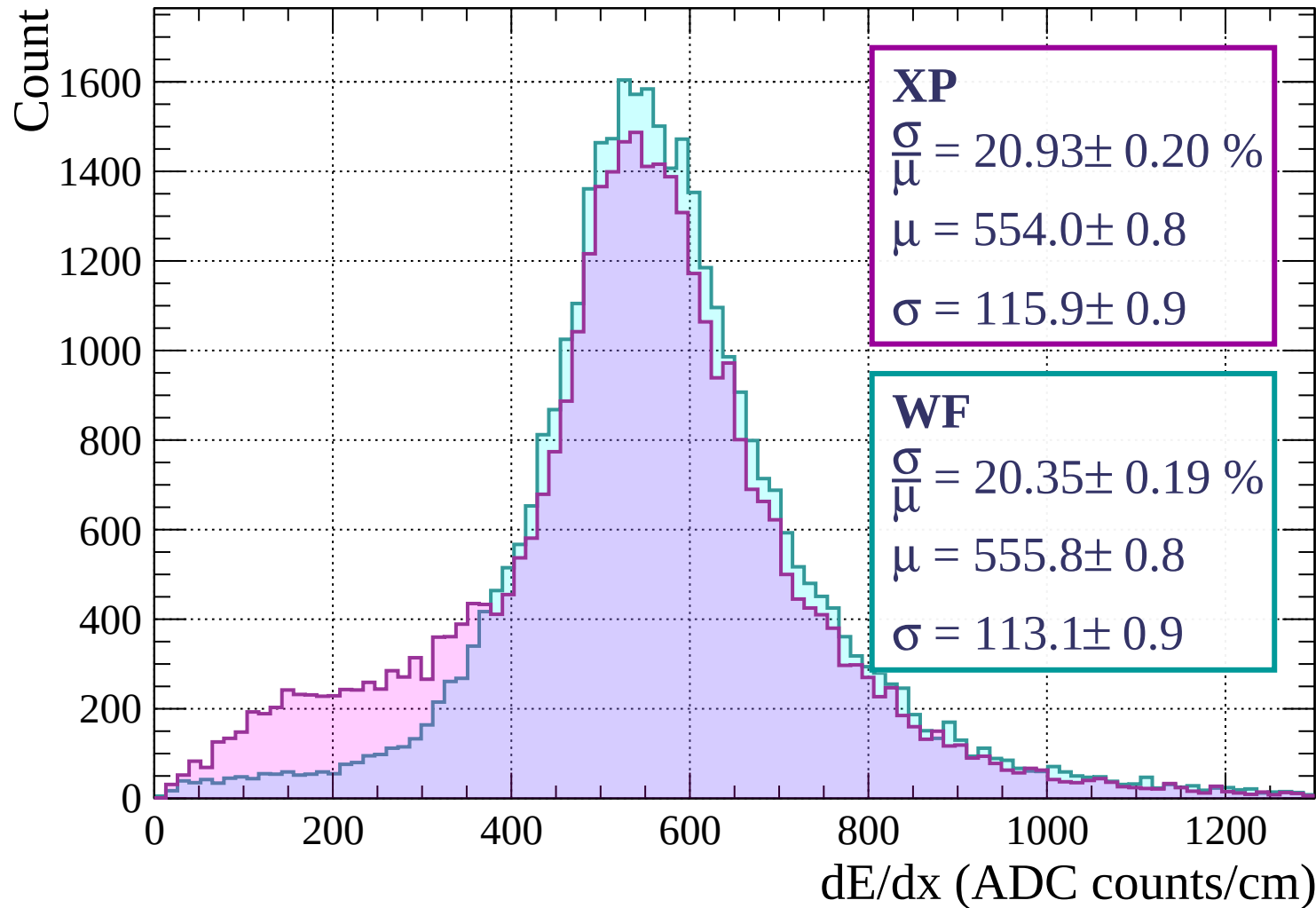


# Energy loss | $0 < p < 80$

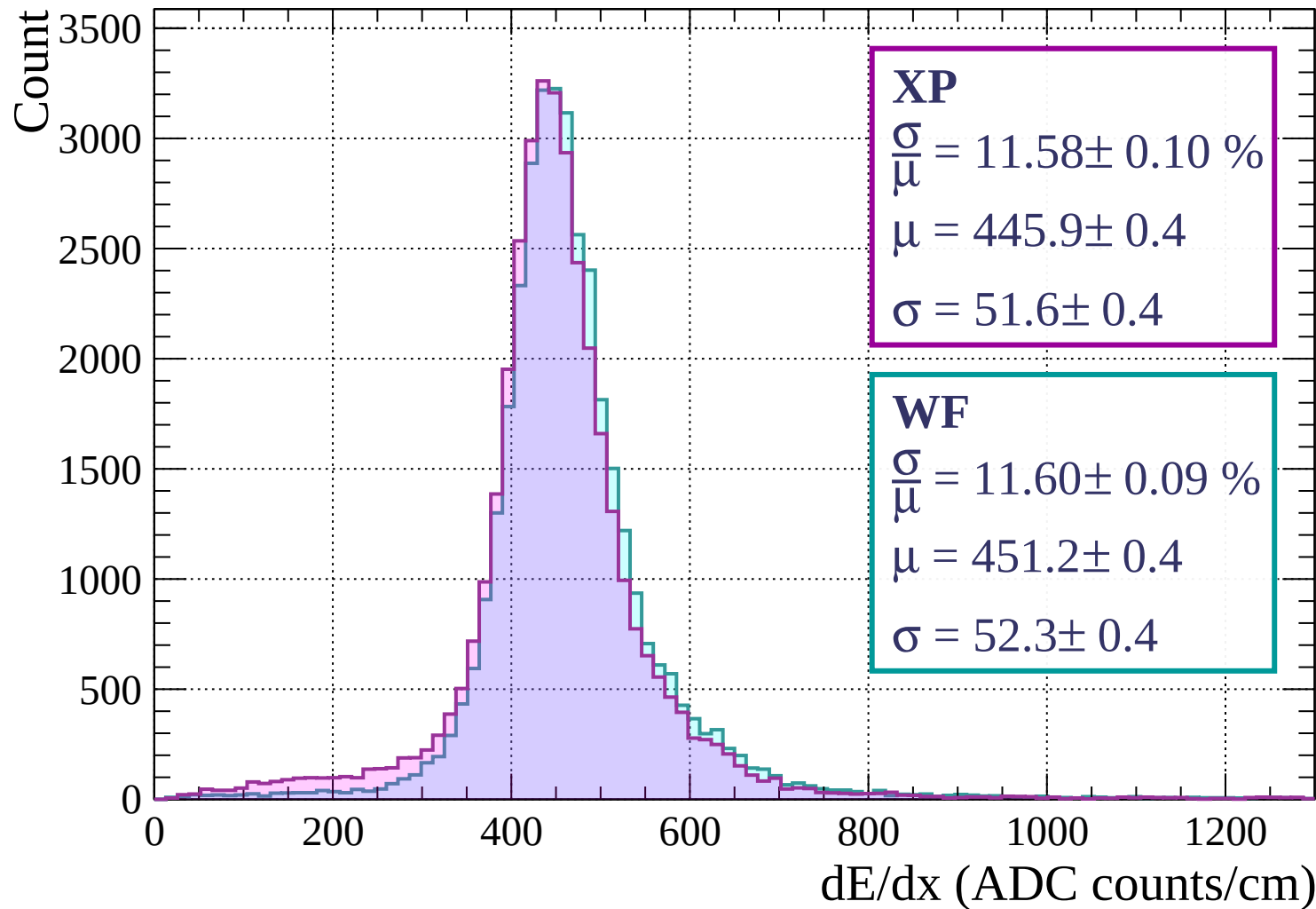




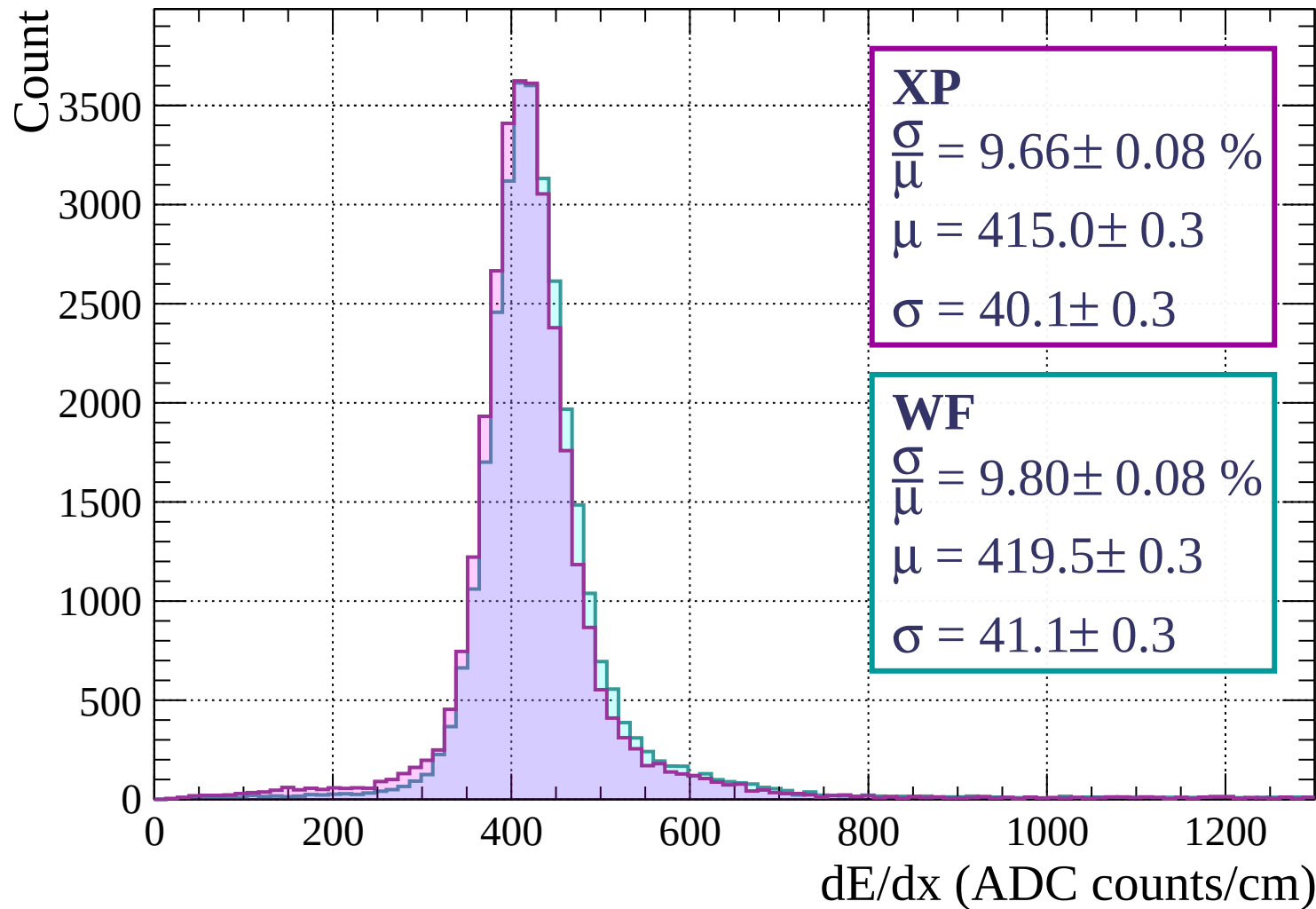
# Energy loss | $80 < p < 160$



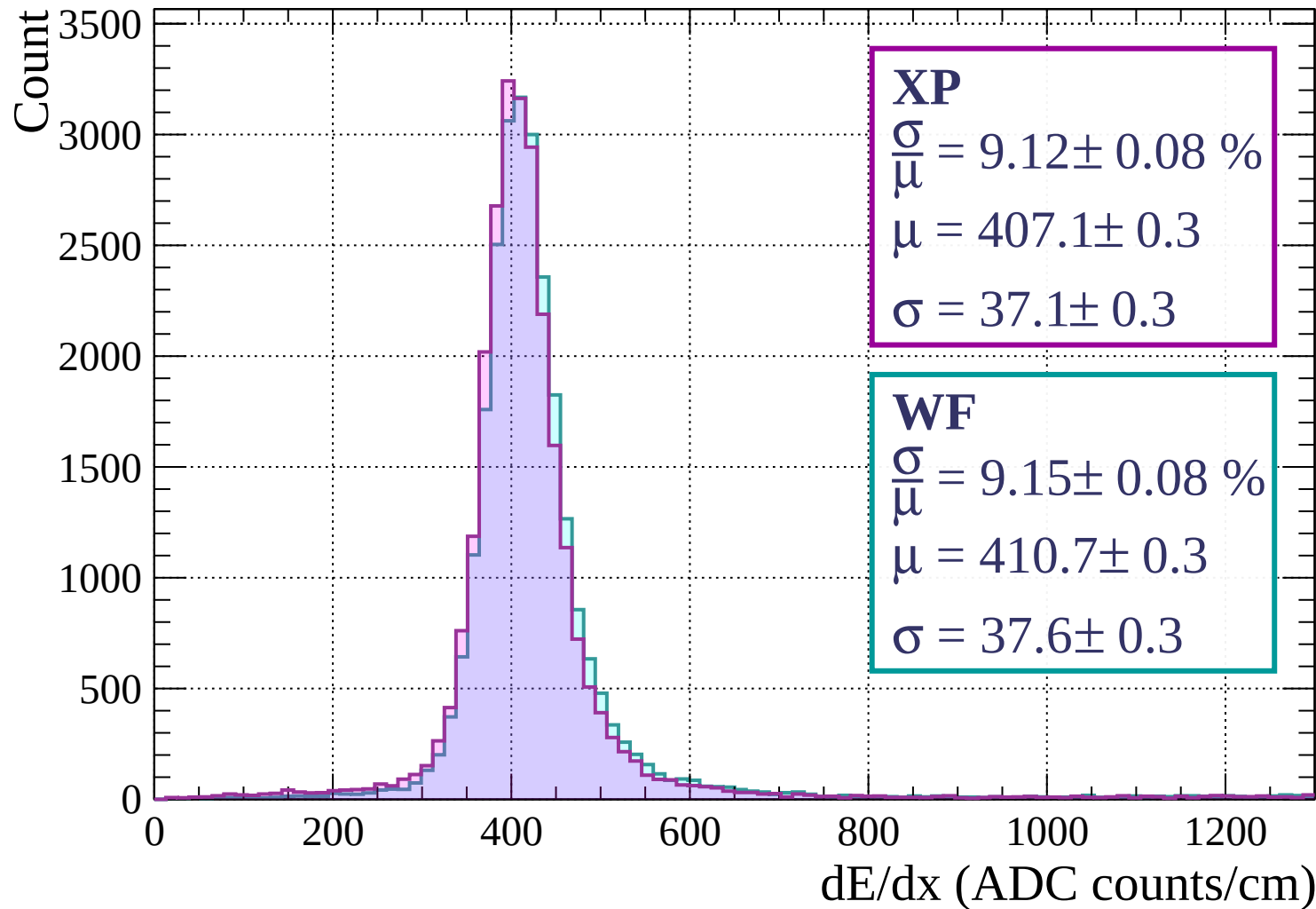
Energy loss |  $160 < p < 240$



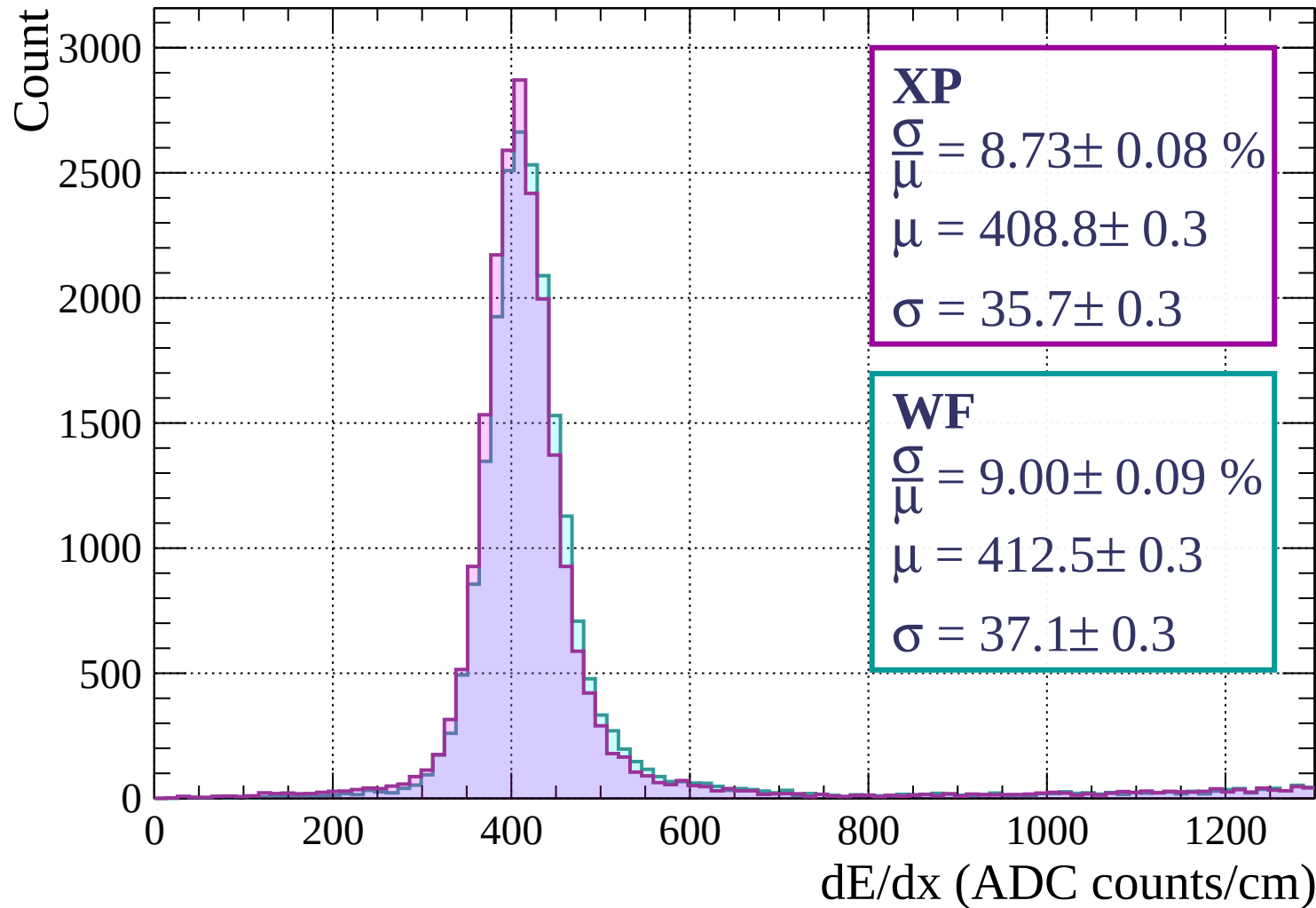
Energy loss |  $240 < p < 320$



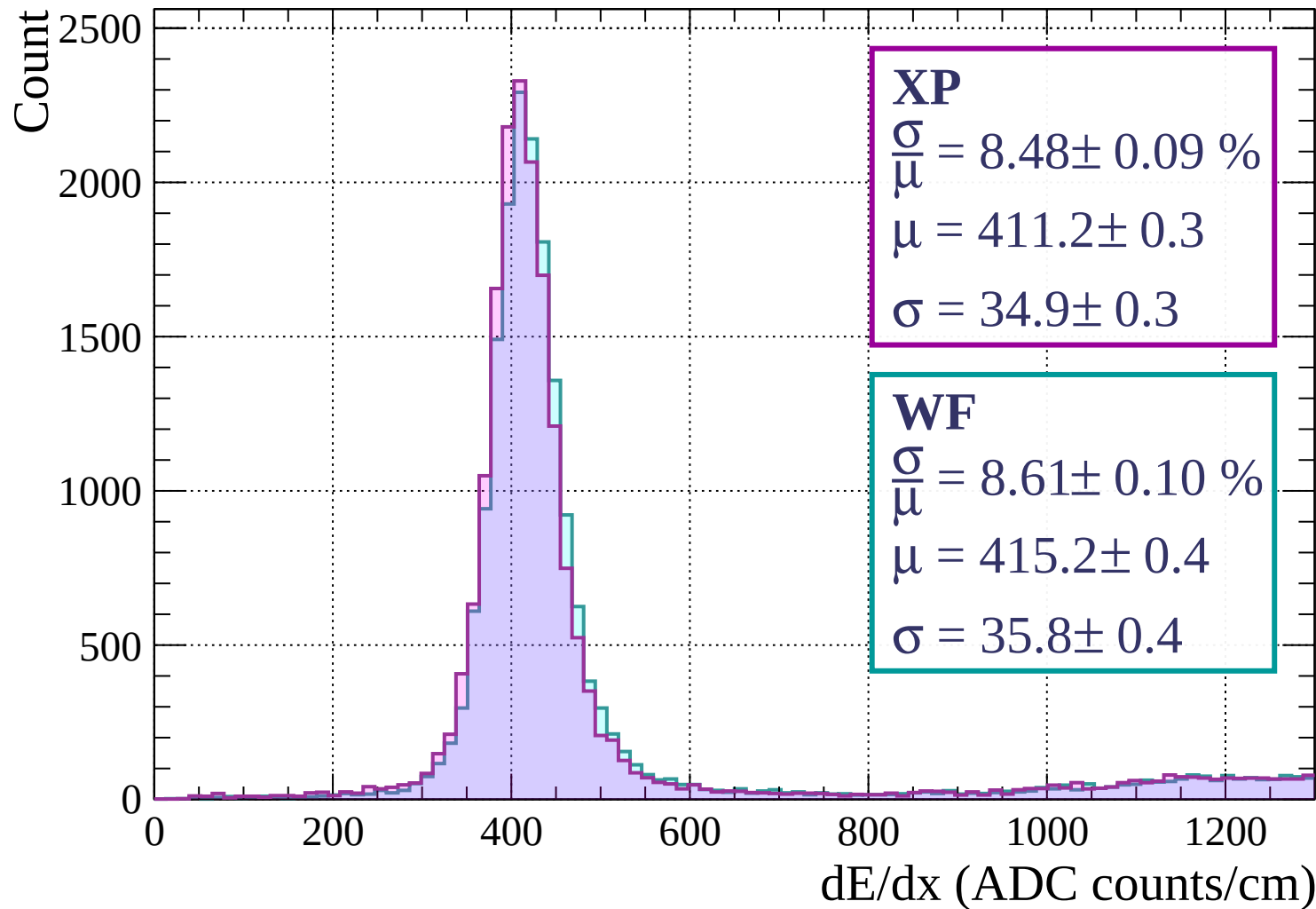
Energy loss |  $320 < p < 400$



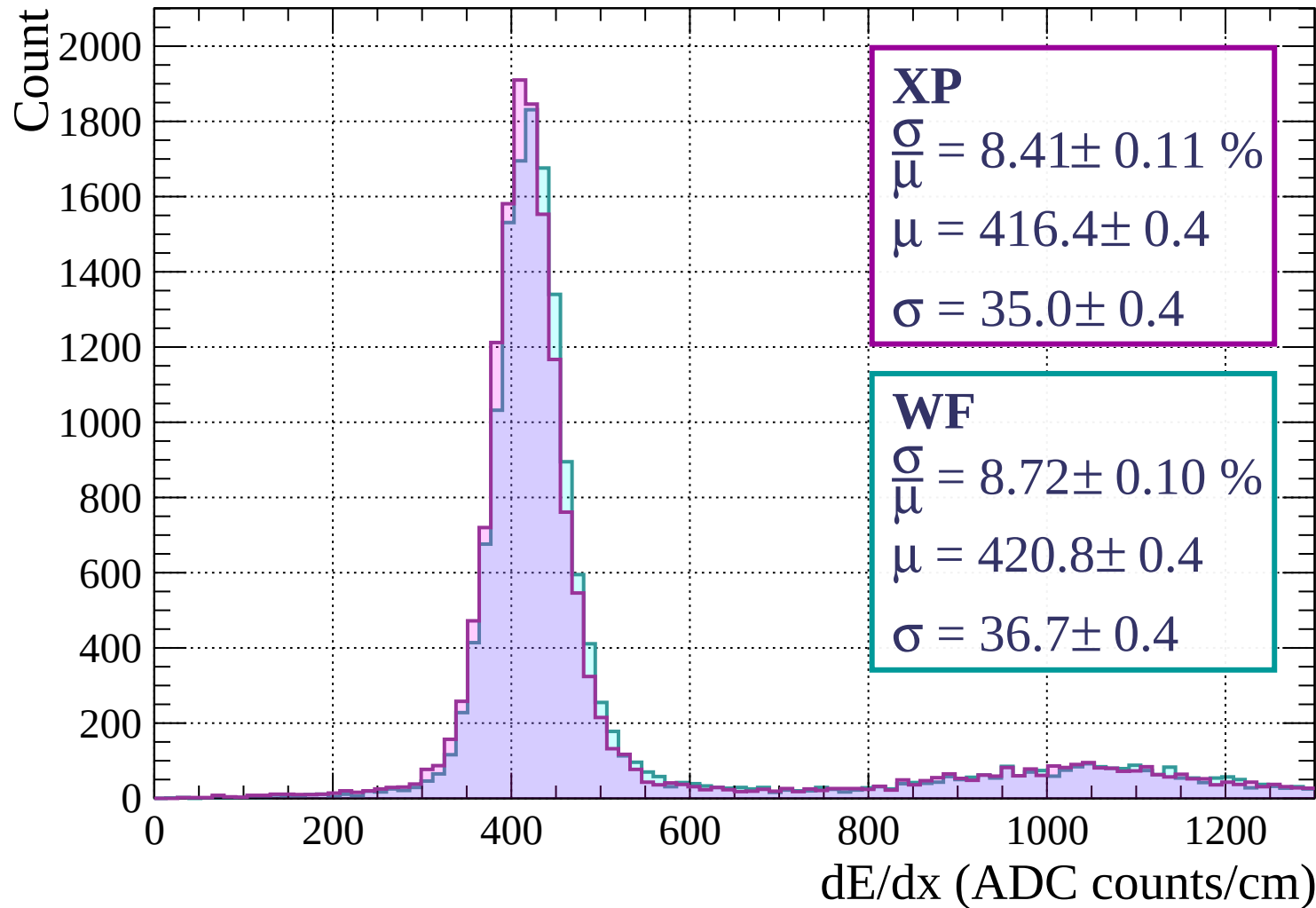
Energy loss |  $400 < p < 480$



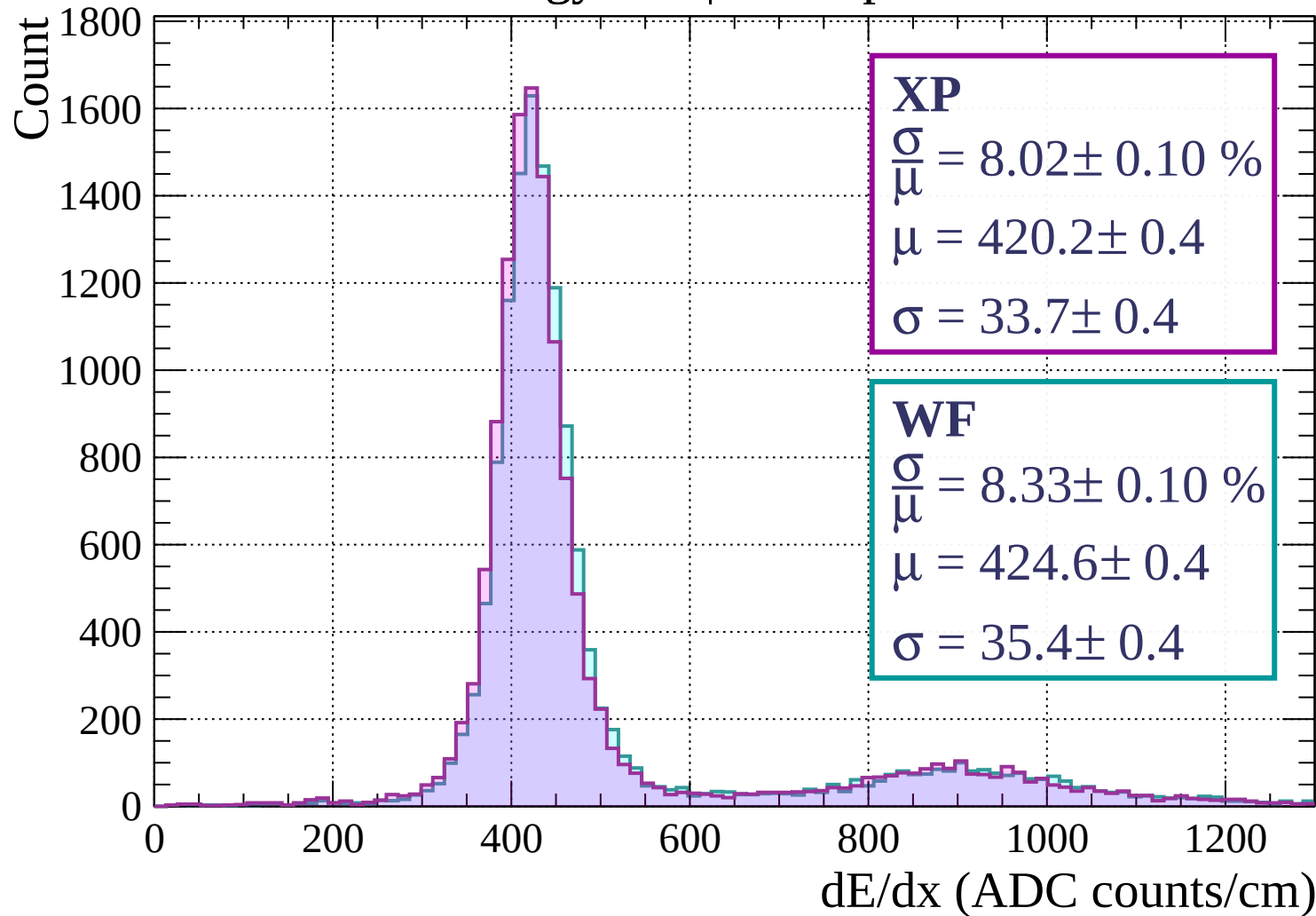
Energy loss |  $480 < p < 560$



Energy loss |  $560 < p < 640$

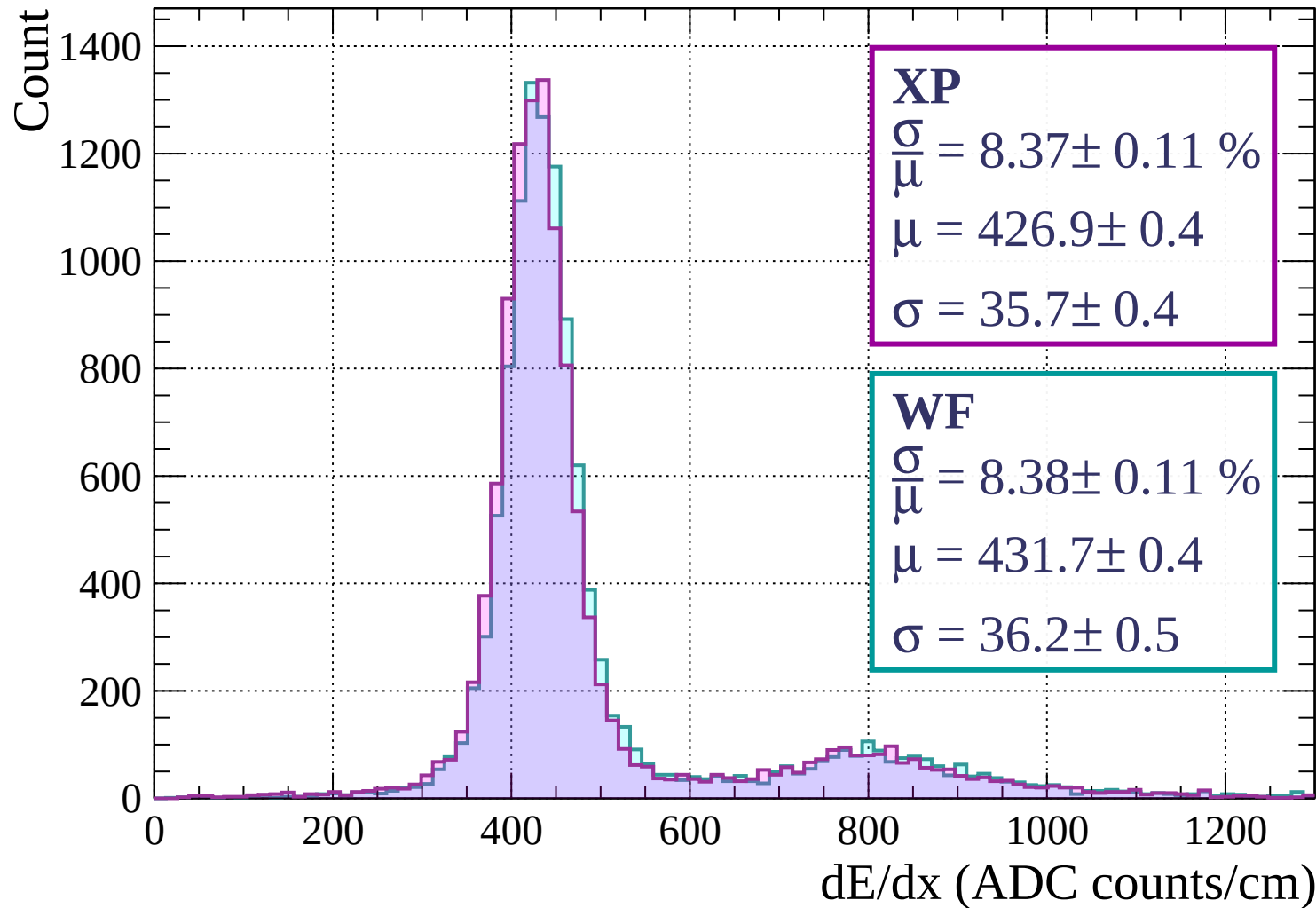


Energy loss |  $640 < p < 720$

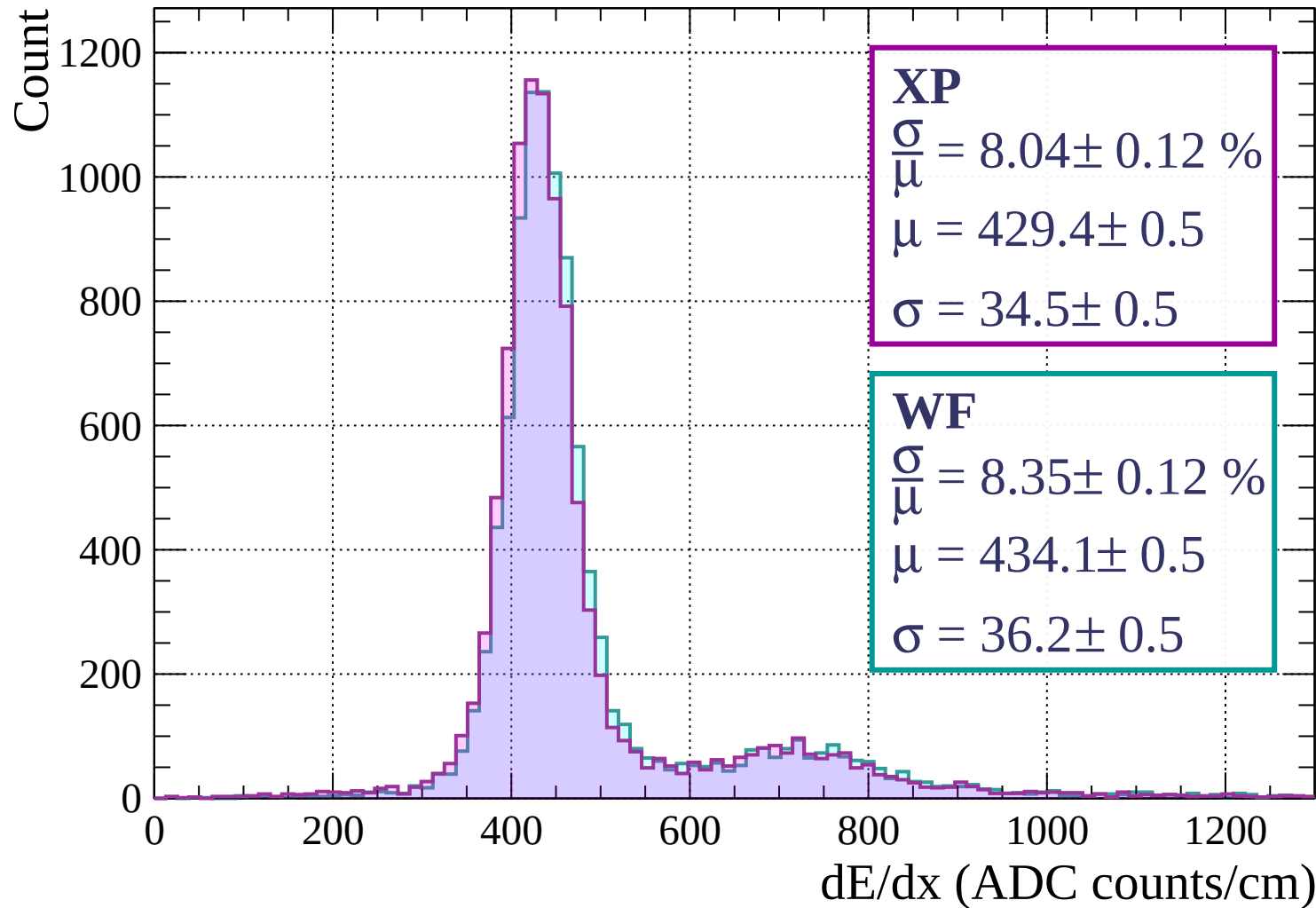




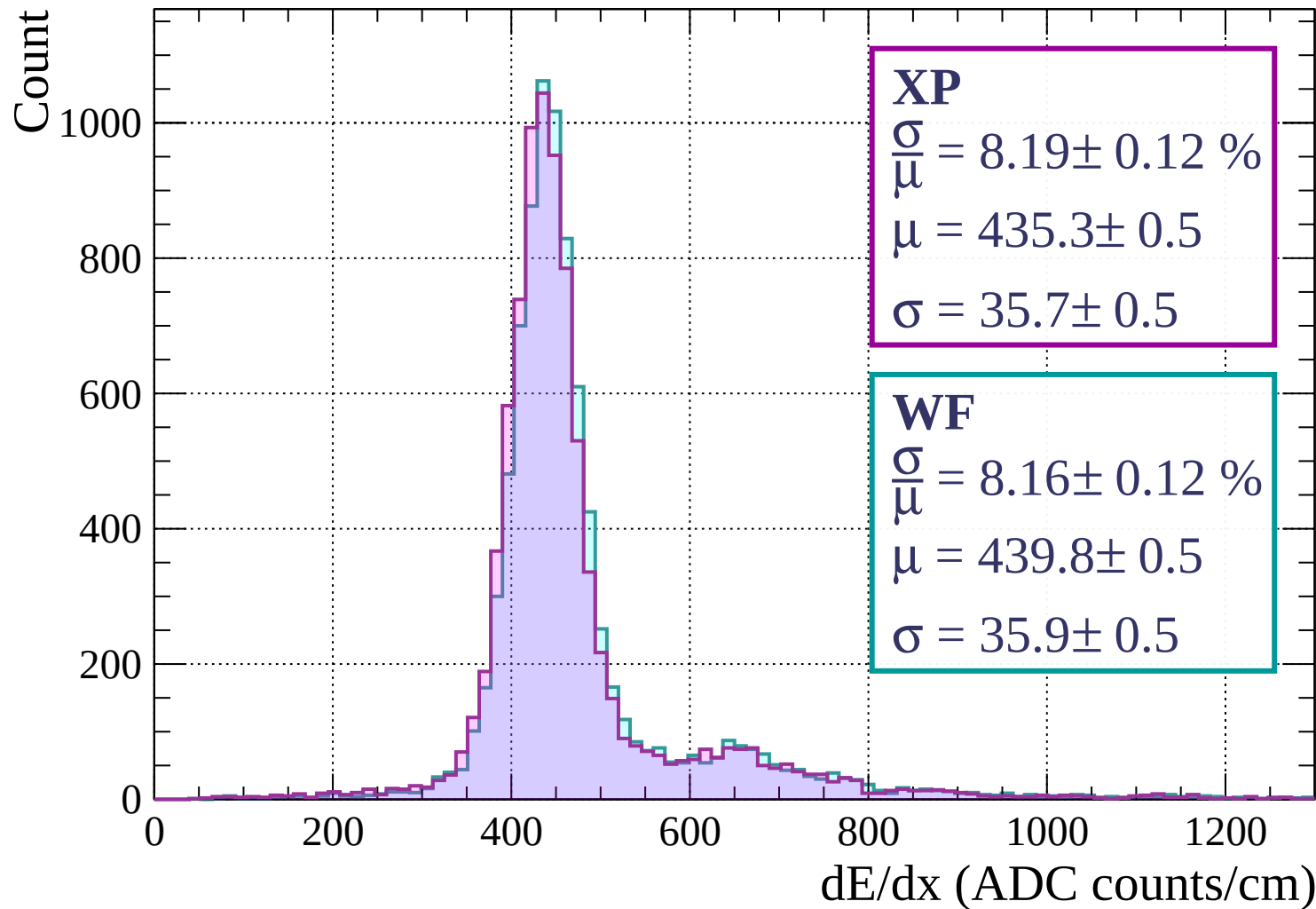
Energy loss |  $720 < p < 800$



Energy loss |  $800 < p < 880$

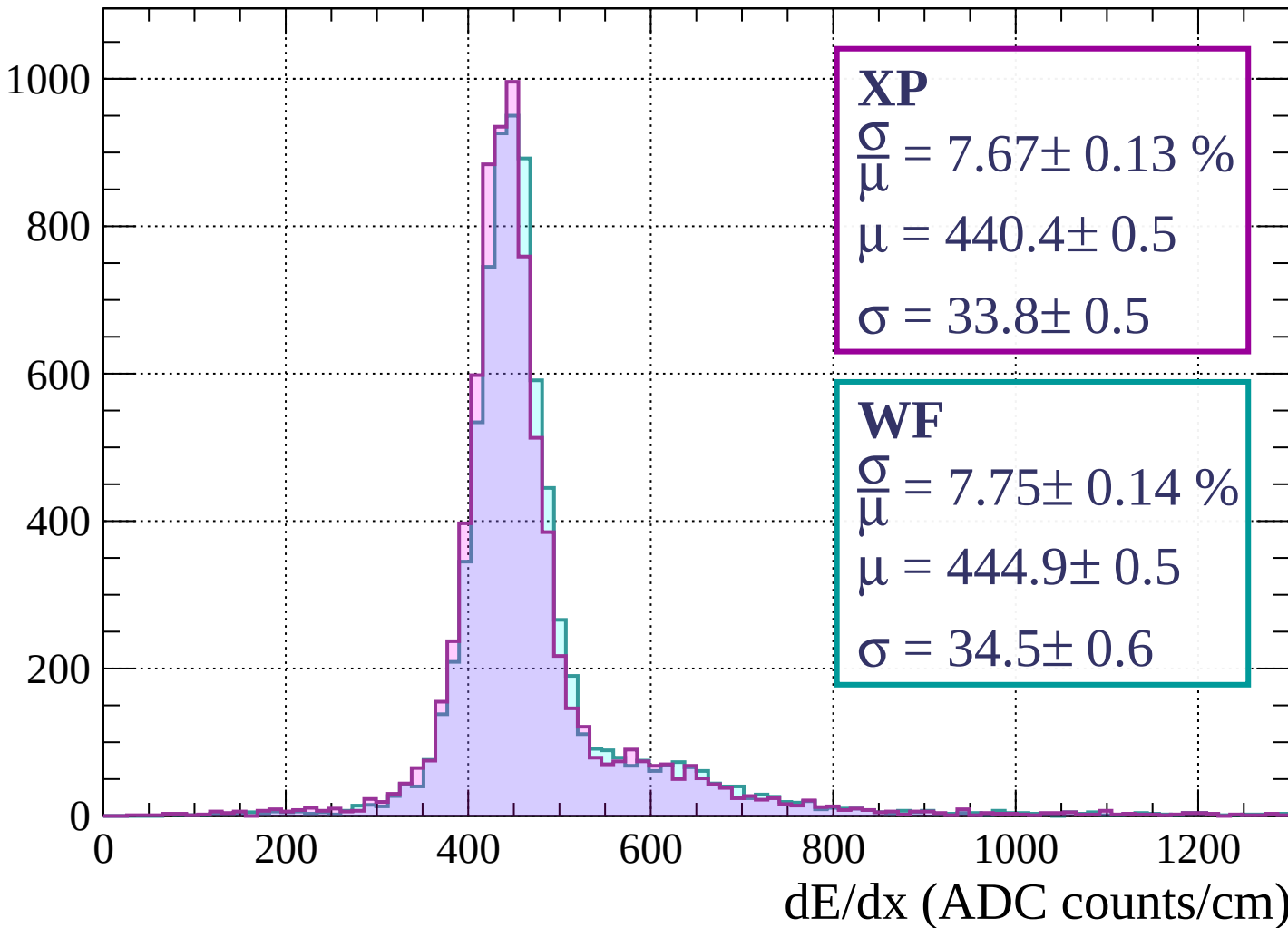


Energy loss |  $880 < p < 960$



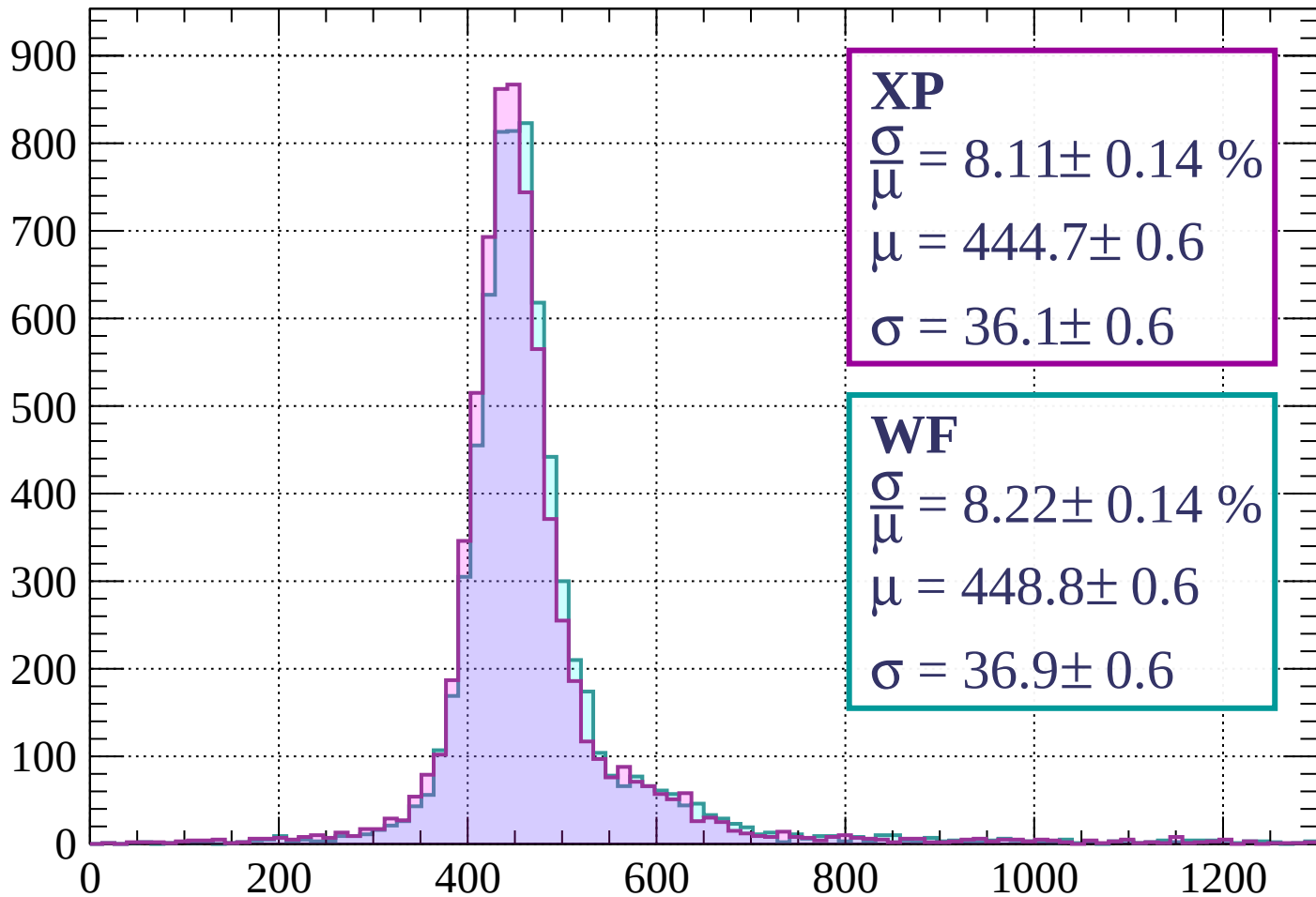
Energy loss |  $960 < p < 1040$

Count



Energy loss |  $1040 < p < 1120$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.11 \pm 0.14 \%$$

$$\mu = 444.7 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.22 \pm 0.14 \%$$

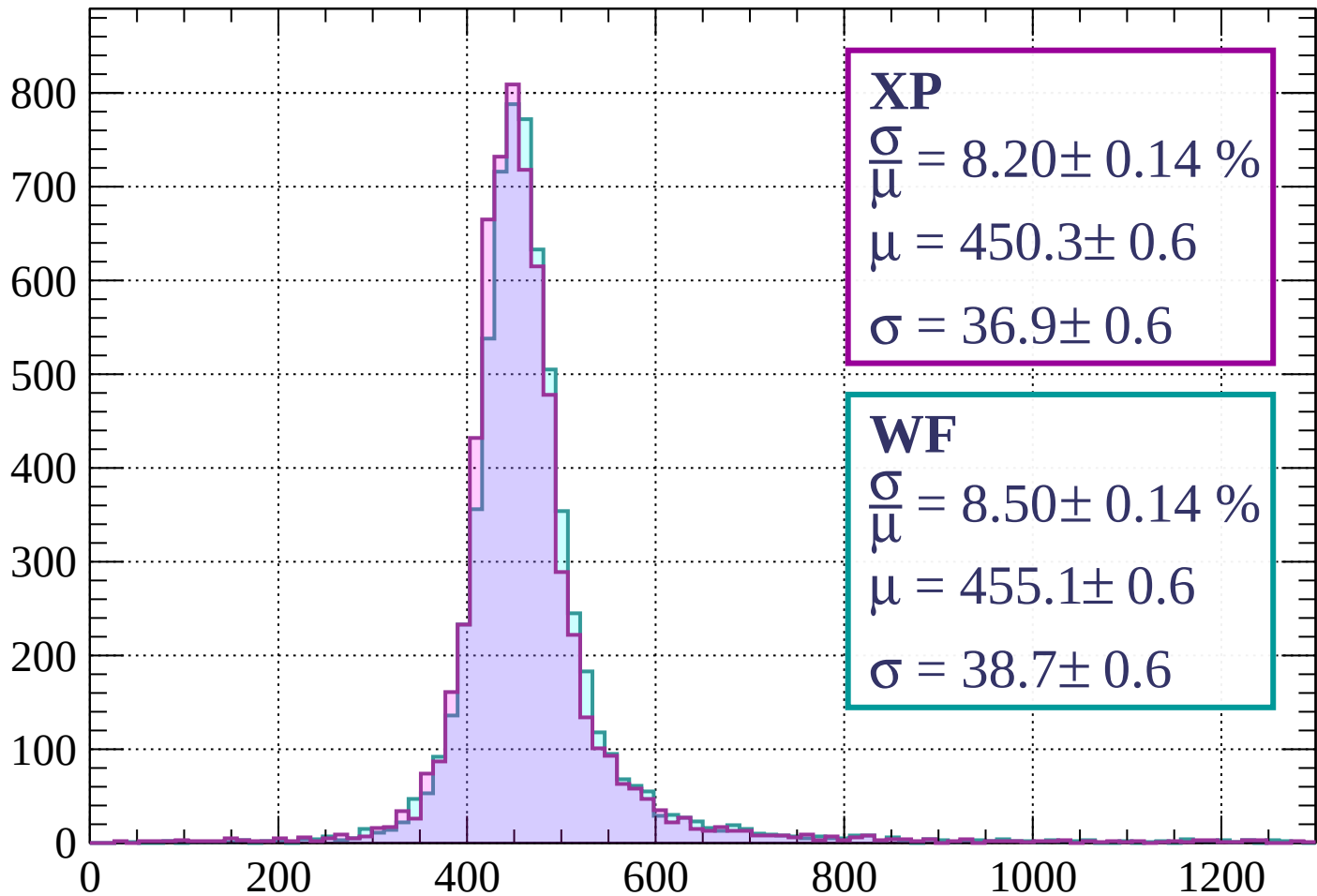
$$\mu = 448.8 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $1120 < p < 1200$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.20 \pm 0.14 \%$$

$$\mu = 450.3 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.50 \pm 0.14 \%$$

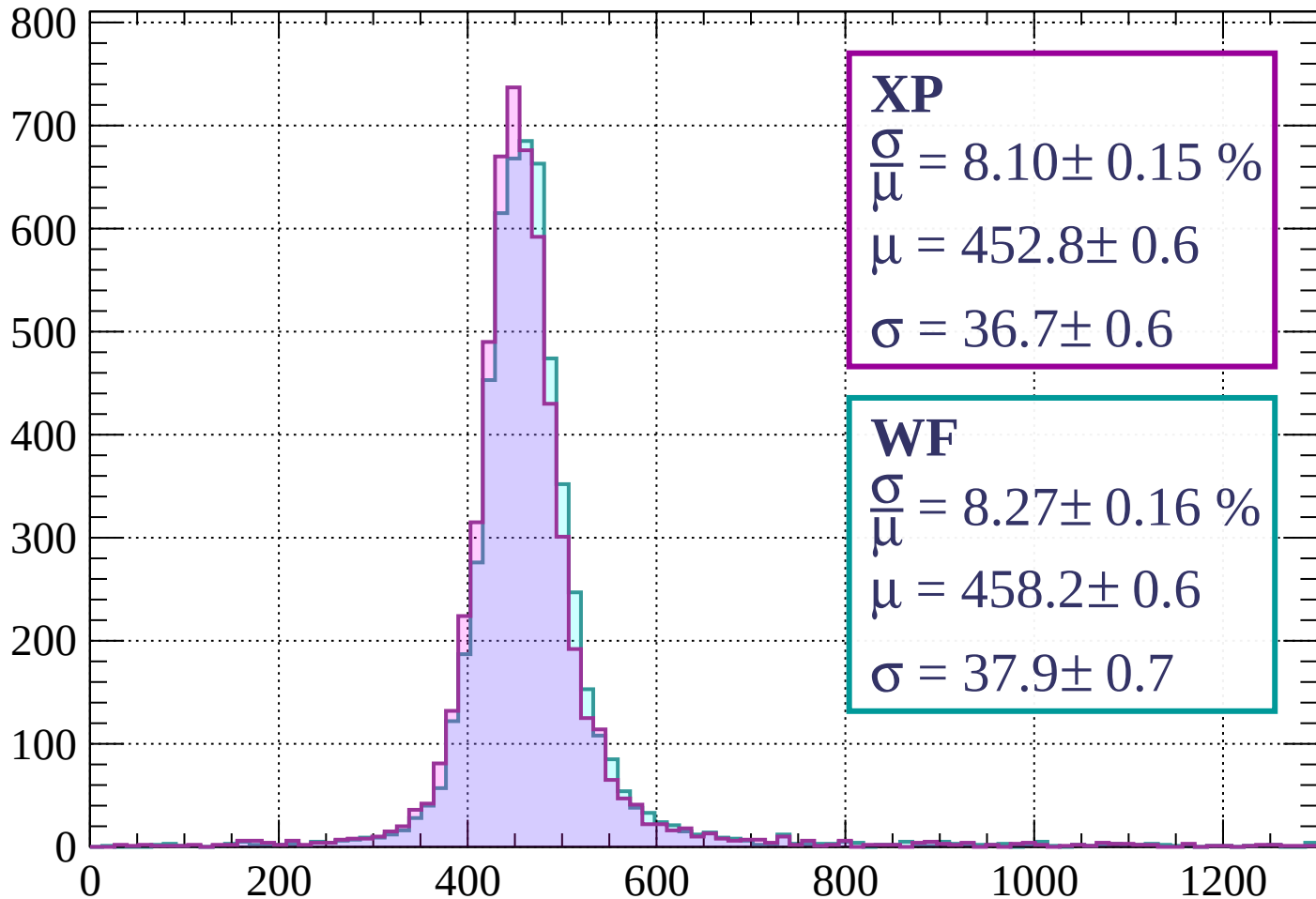
$$\mu = 455.1 \pm 0.6$$

$$\sigma = 38.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $1200 < p < 1280$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.10 \pm 0.15 \%$$

$$\mu = 452.8 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

**WF**

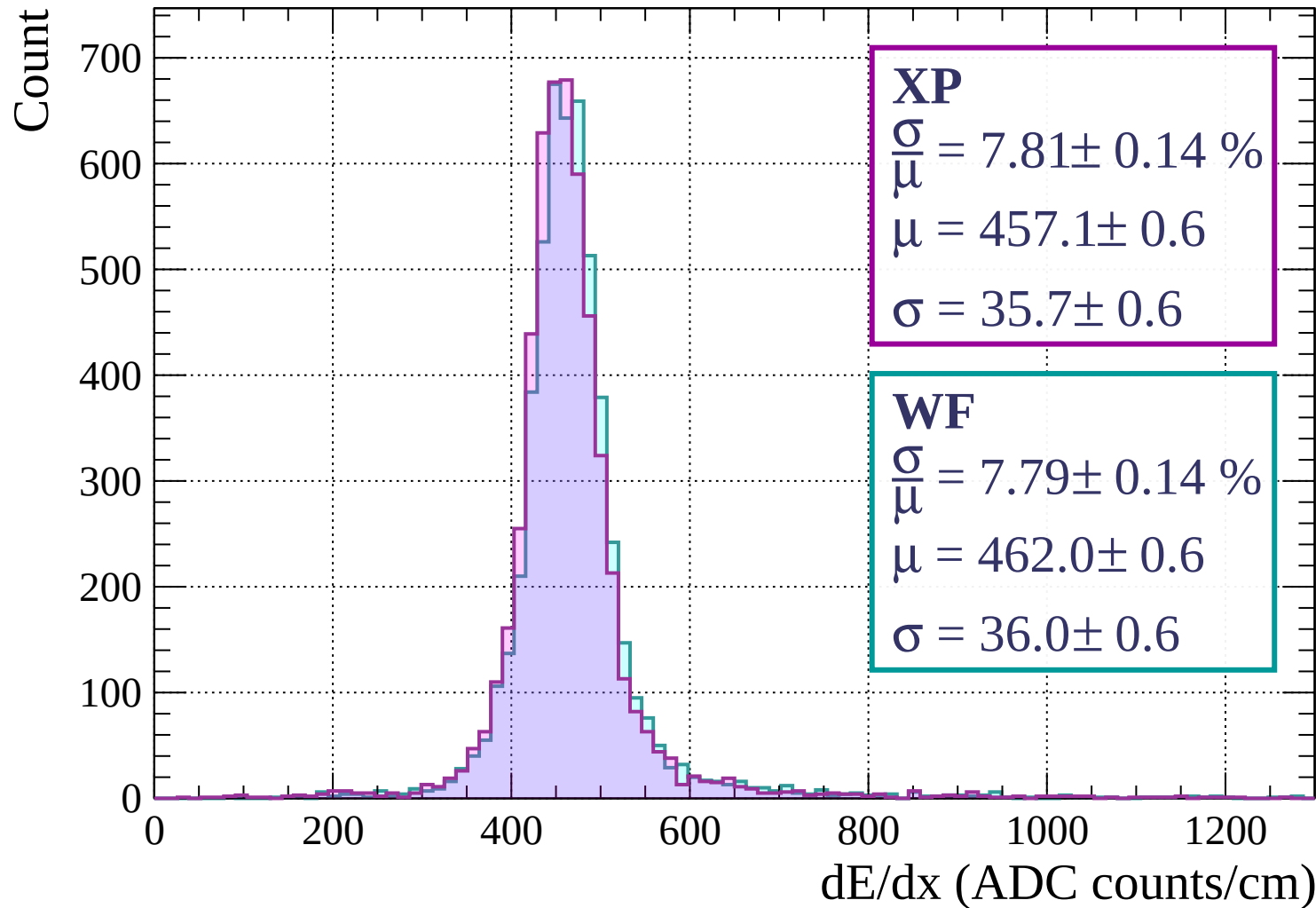
$$\frac{\sigma}{\mu} = 8.27 \pm 0.16 \%$$

$$\mu = 458.2 \pm 0.6$$

$$\sigma = 37.9 \pm 0.7$$

dE/dx (ADC counts/cm)

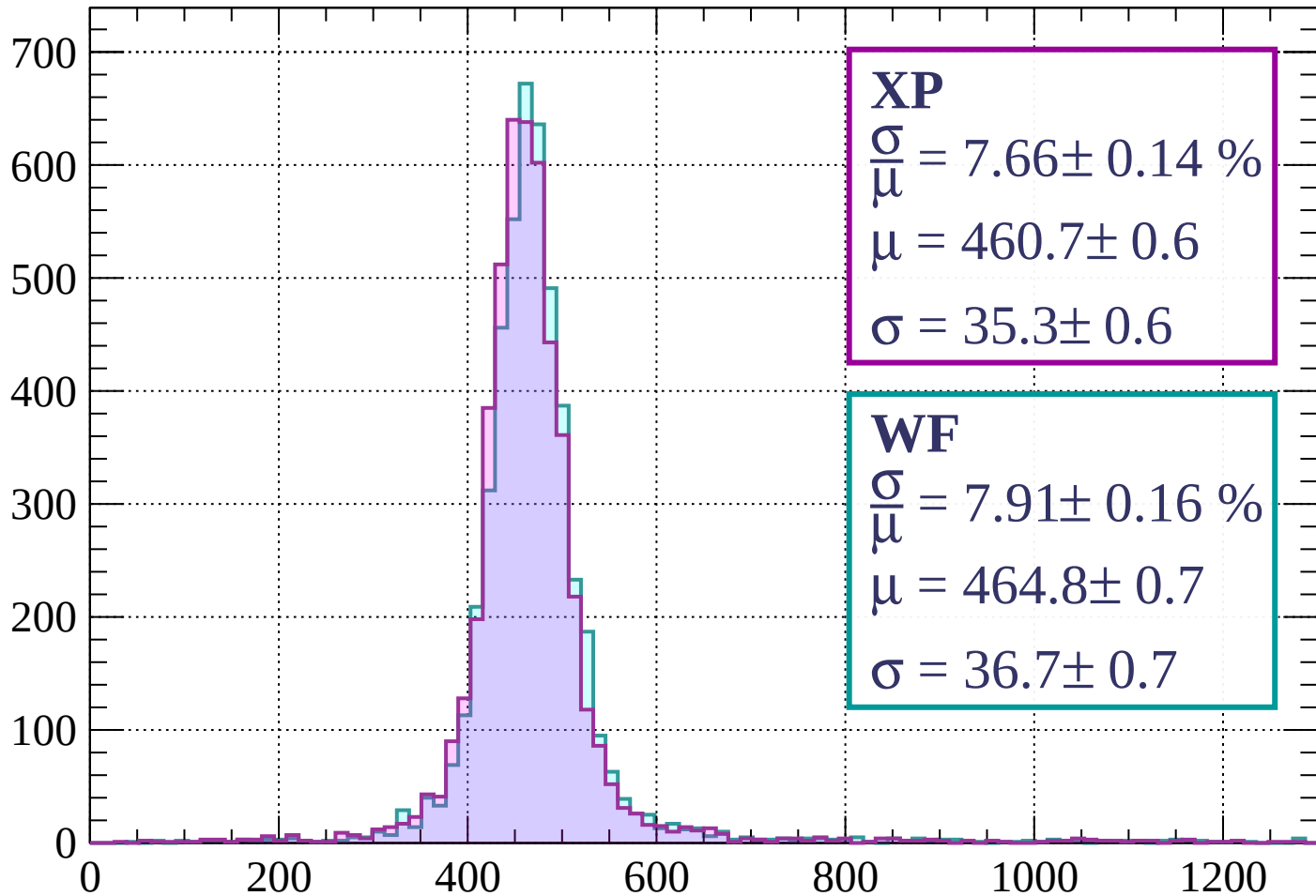
Energy loss |  $1280 < p < 1360$





Energy loss |  $1360 < p < 1440$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.66 \pm 0.14 \%$$

$$\mu = 460.7 \pm 0.6$$

$$\sigma = 35.3 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 7.91 \pm 0.16 \%$$

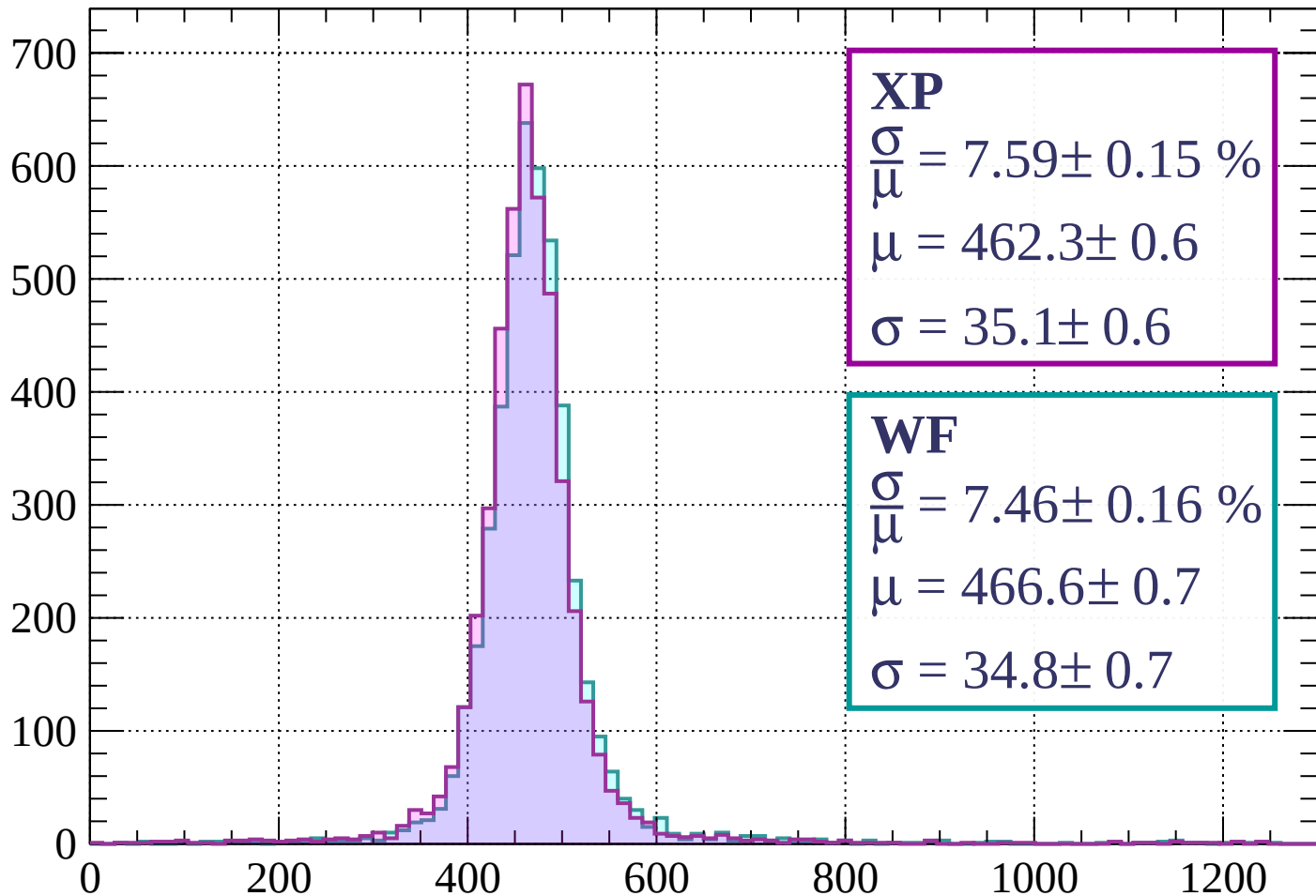
$$\mu = 464.8 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $1440 < p < 1520$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.59 \pm 0.15 \%$$

$$\mu = 462.3 \pm 0.6$$

$$\sigma = 35.1 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 7.46 \pm 0.16 \%$$

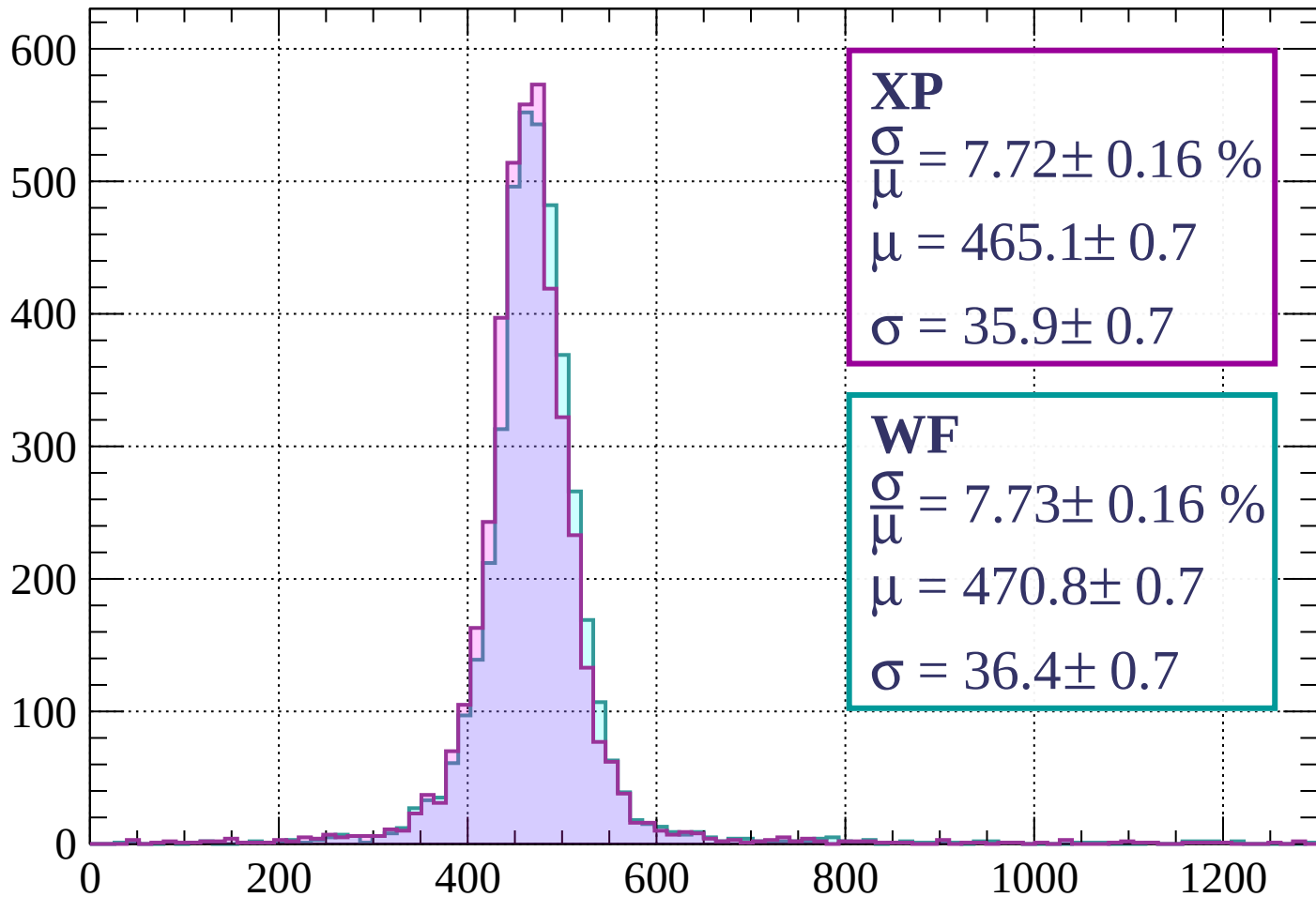
$$\mu = 466.6 \pm 0.7$$

$$\sigma = 34.8 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1520 < p < 1600$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.72 \pm 0.16 \%$$

$$\mu = 465.1 \pm 0.7$$

$$\sigma = 35.9 \pm 0.7$$

**WF**

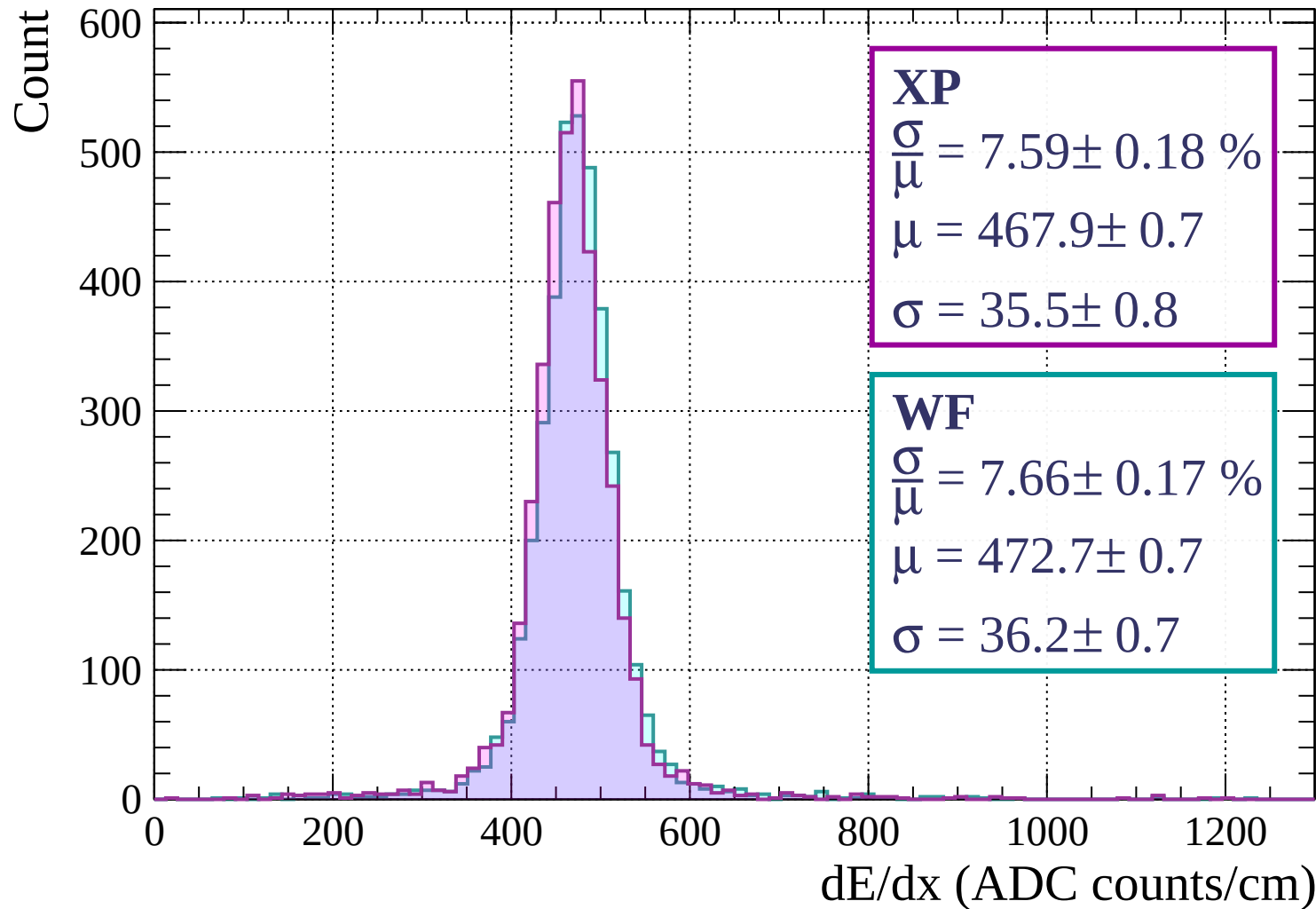
$$\frac{\sigma}{\mu} = 7.73 \pm 0.16 \%$$

$$\mu = 470.8 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

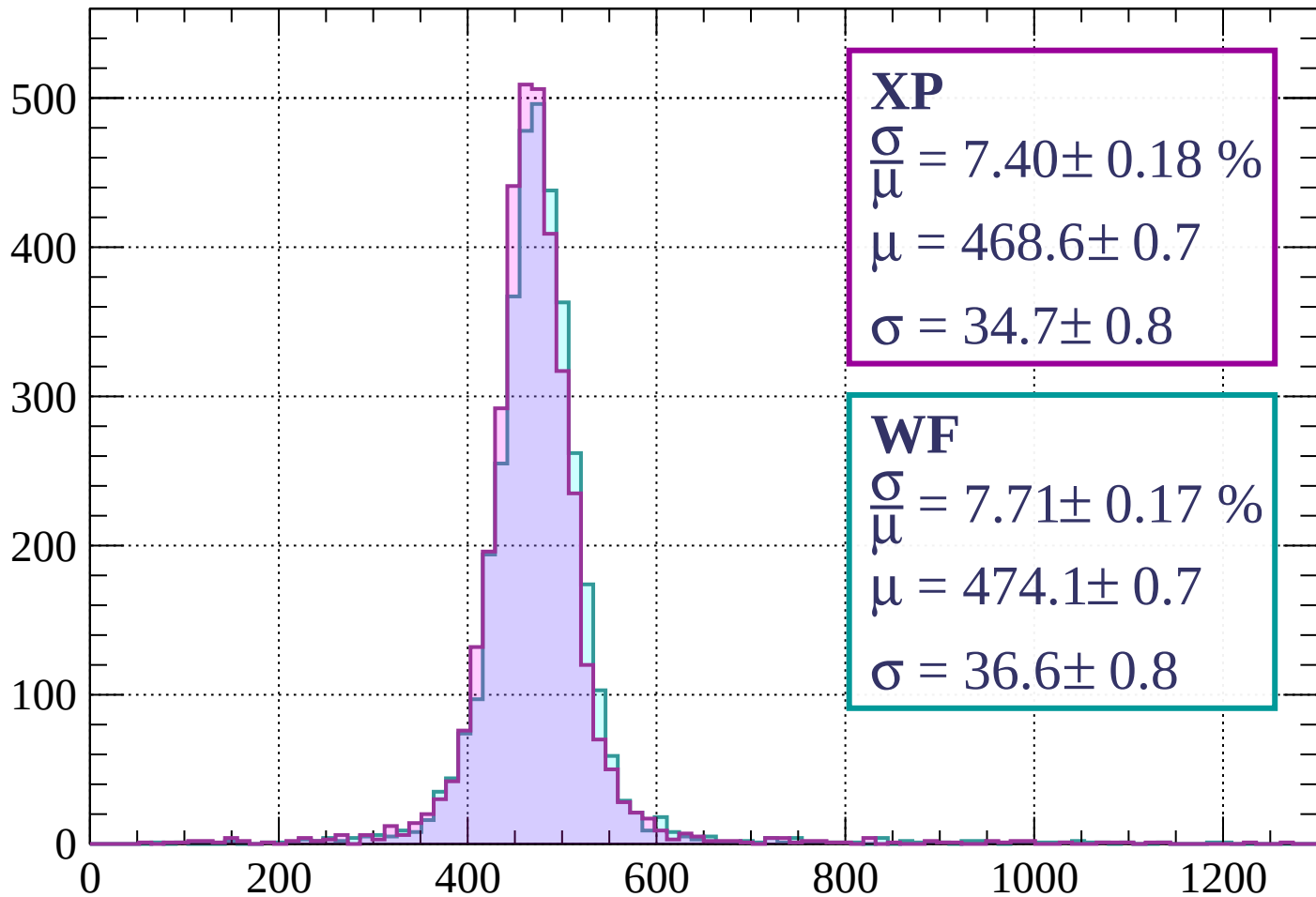
dE/dx (ADC counts/cm)

Energy loss |  $1600 < p < 1680$



Energy loss |  $1680 < p < 1760$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.40 \pm 0.18 \%$$

$$\mu = 468.6 \pm 0.7$$

$$\sigma = 34.7 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.71 \pm 0.17 \%$$

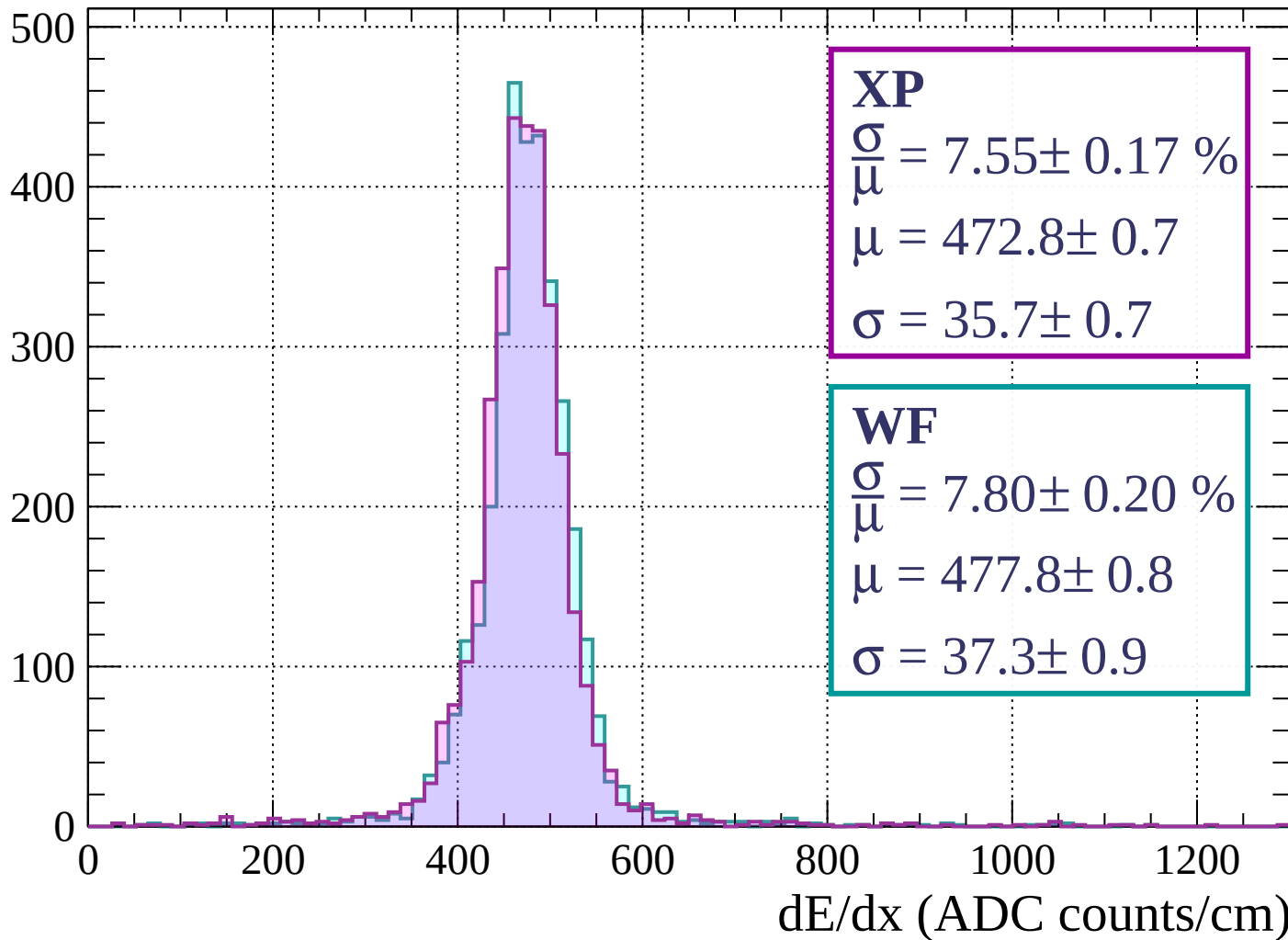
$$\mu = 474.1 \pm 0.7$$

$$\sigma = 36.6 \pm 0.8$$

dE/dx (ADC counts/cm)

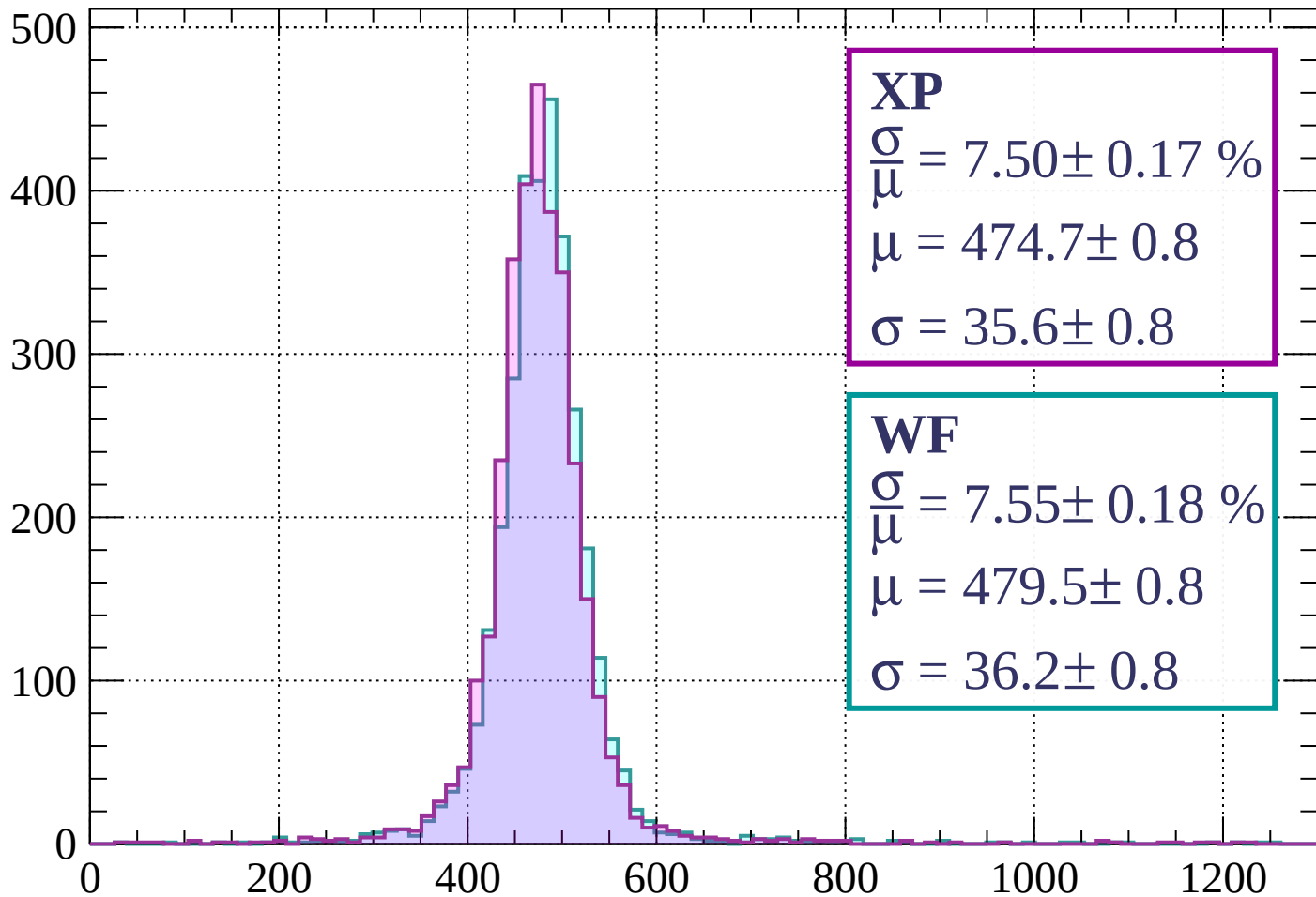
Energy loss |  $1760 < p < 1840$

Count



Energy loss |  $1840 < p < 1920$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.50 \pm 0.17 \%$$

$$\mu = 474.7 \pm 0.8$$

$$\sigma = 35.6 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.55 \pm 0.18 \%$$

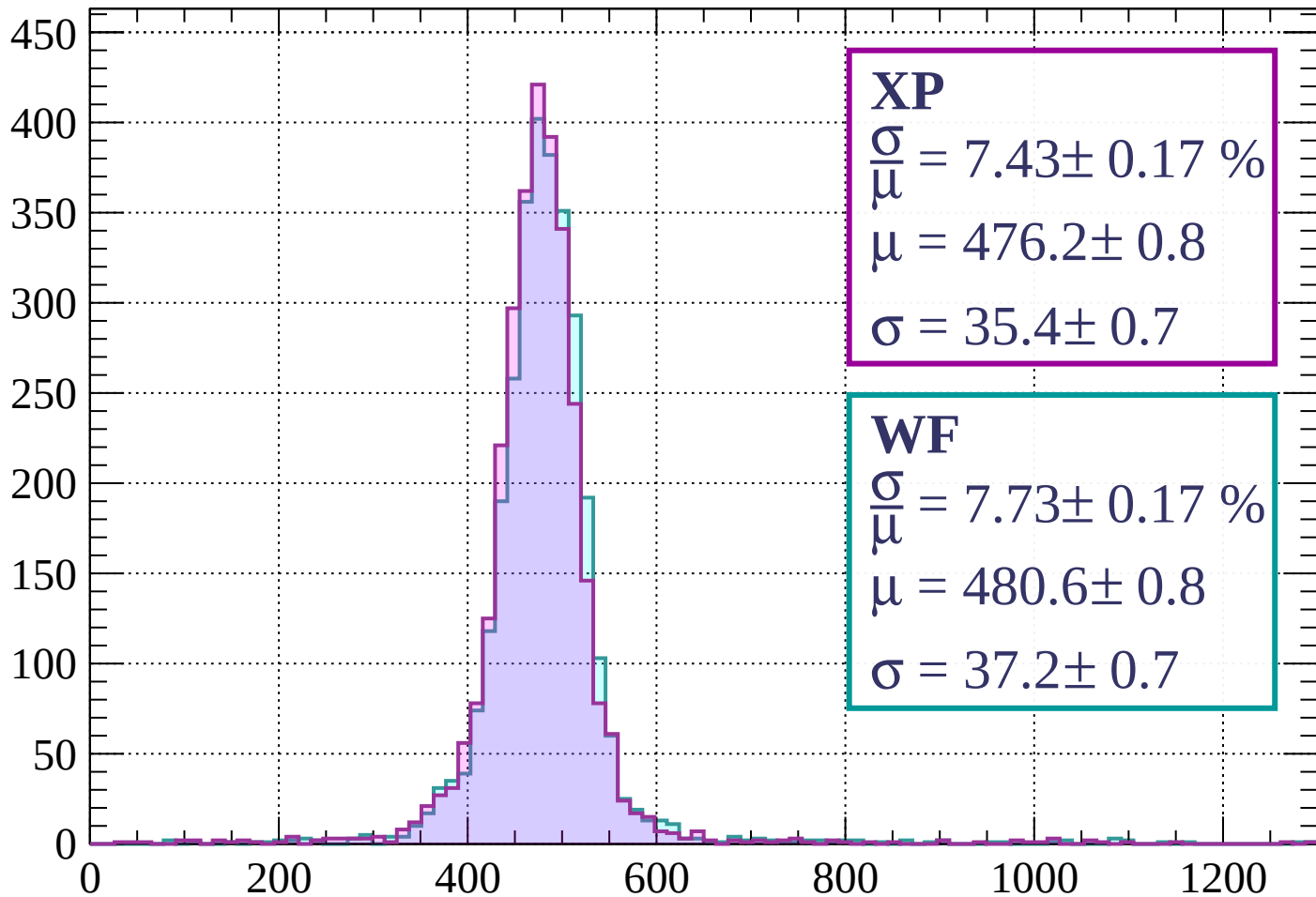
$$\mu = 479.5 \pm 0.8$$

$$\sigma = 36.2 \pm 0.8$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1920 < p < 2000$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.43 \pm 0.17 \%$$

$$\mu = 476.2 \pm 0.8$$

$$\sigma = 35.4 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 7.73 \pm 0.17 \%$$

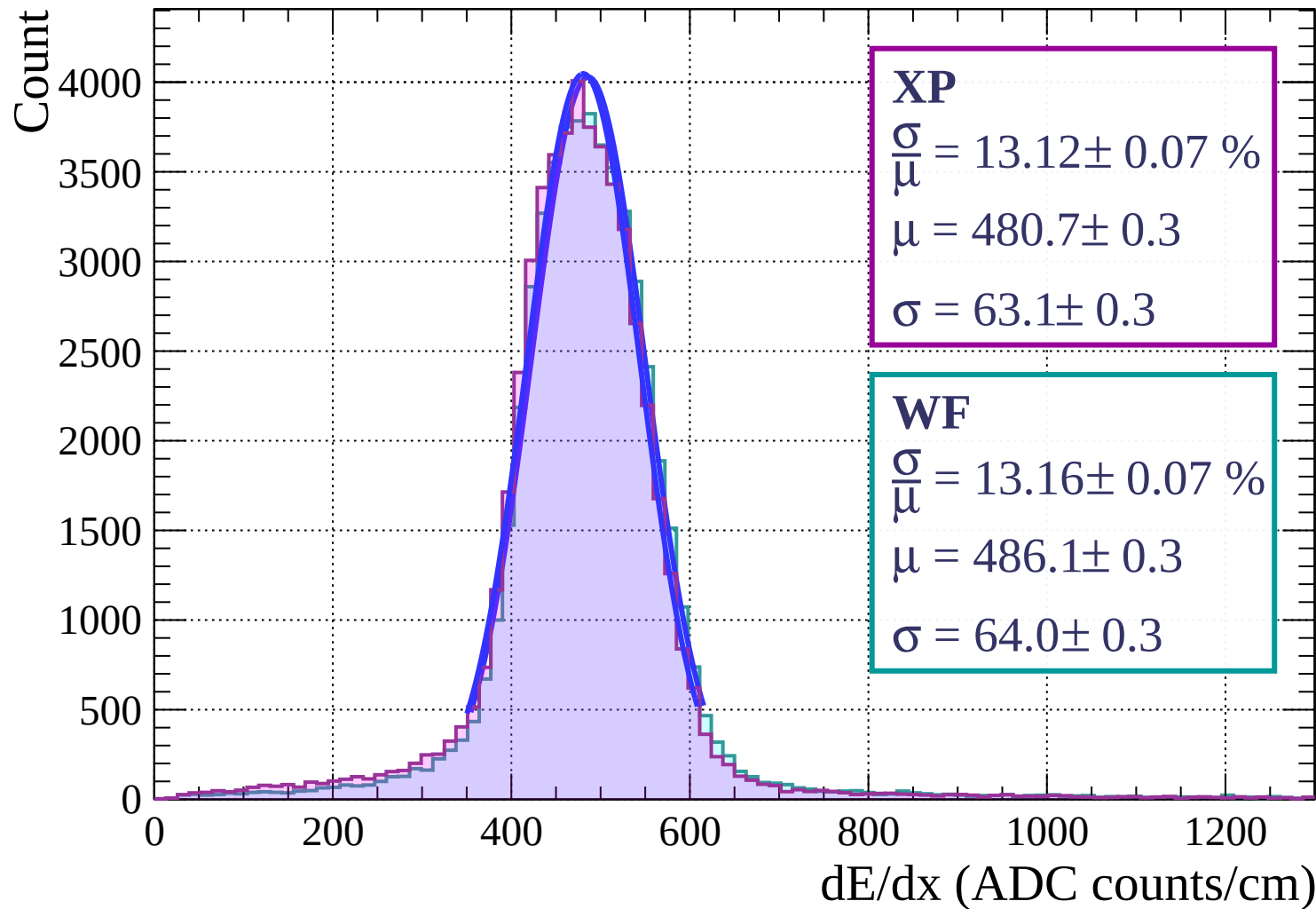
$$\mu = 480.6 \pm 0.8$$

$$\sigma = 37.2 \pm 0.7$$

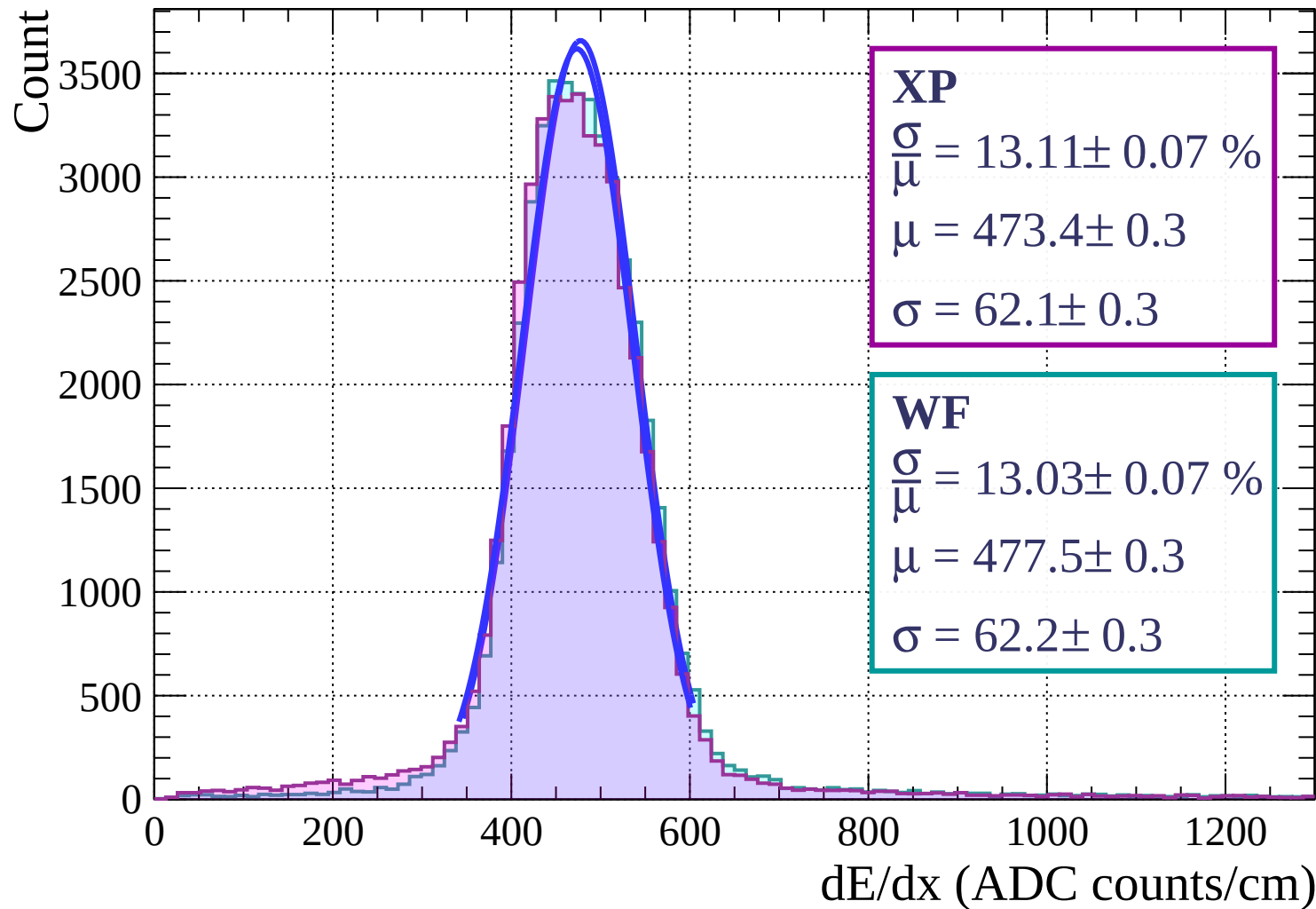
dE/dx (ADC counts/cm)



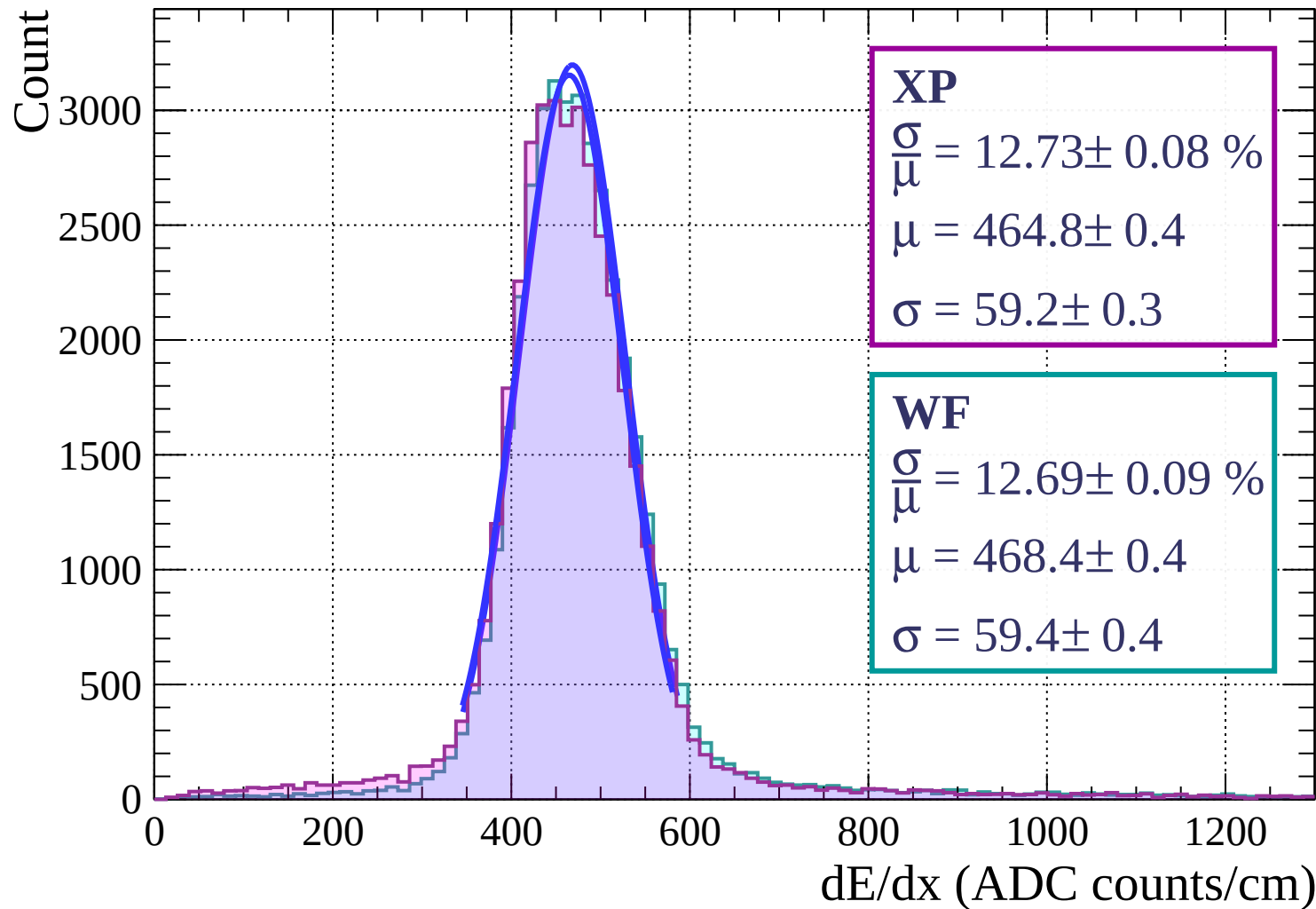
# Energy loss | $0 < \phi < 3$



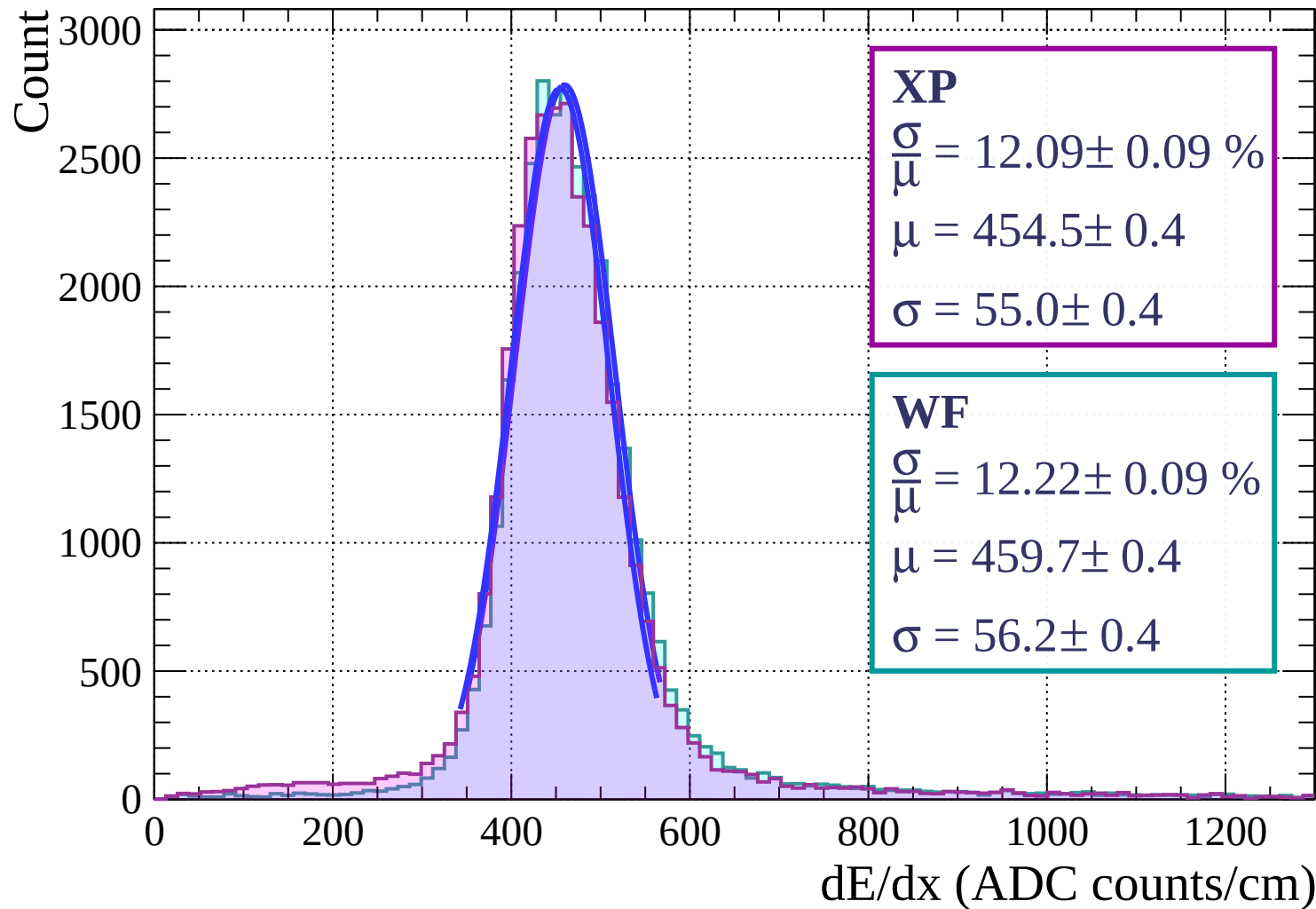
# Energy loss | $3 < \phi < 6$



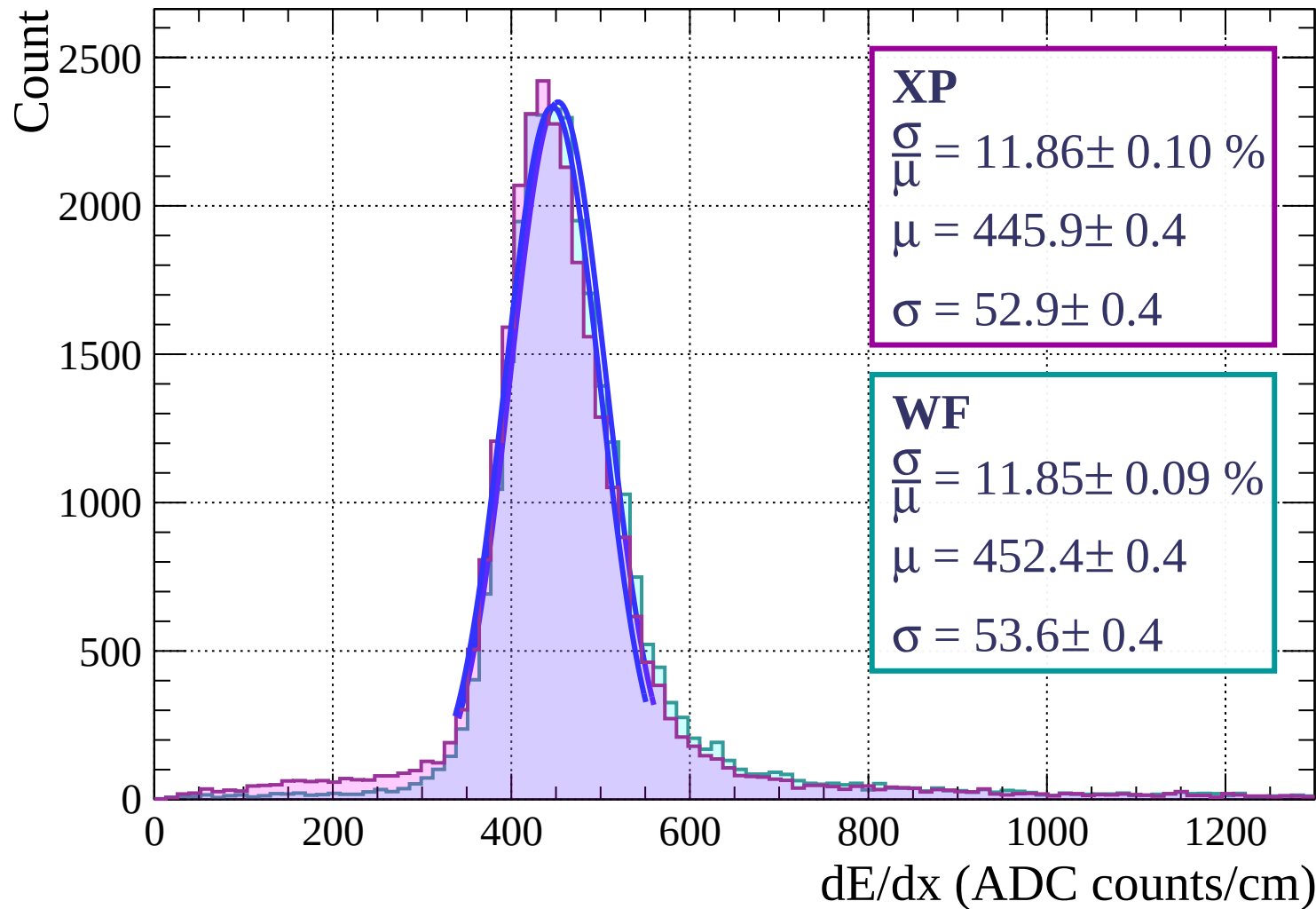
# Energy loss | $6 < \phi < 9$



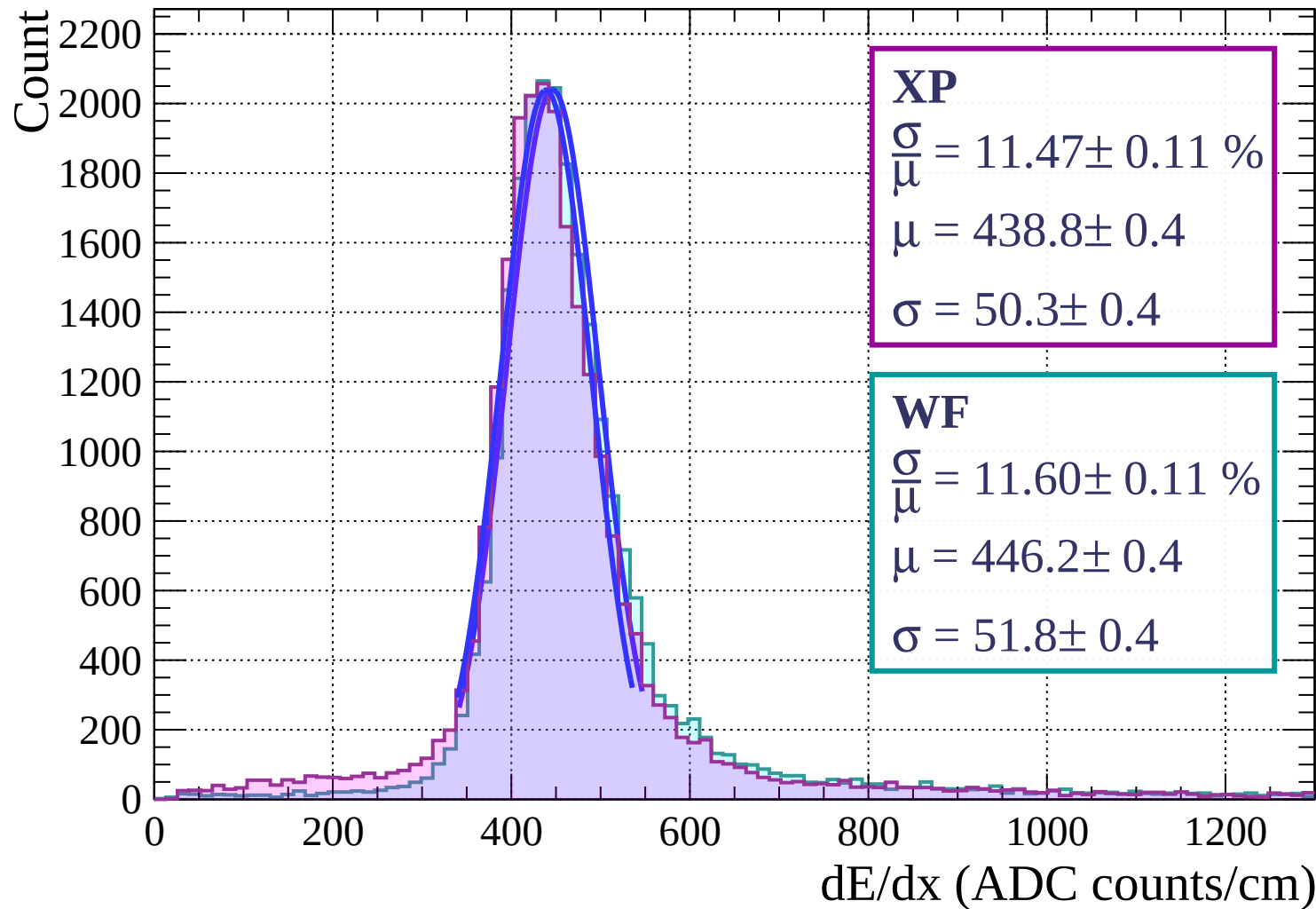
# Energy loss | $9 < \phi < 12$



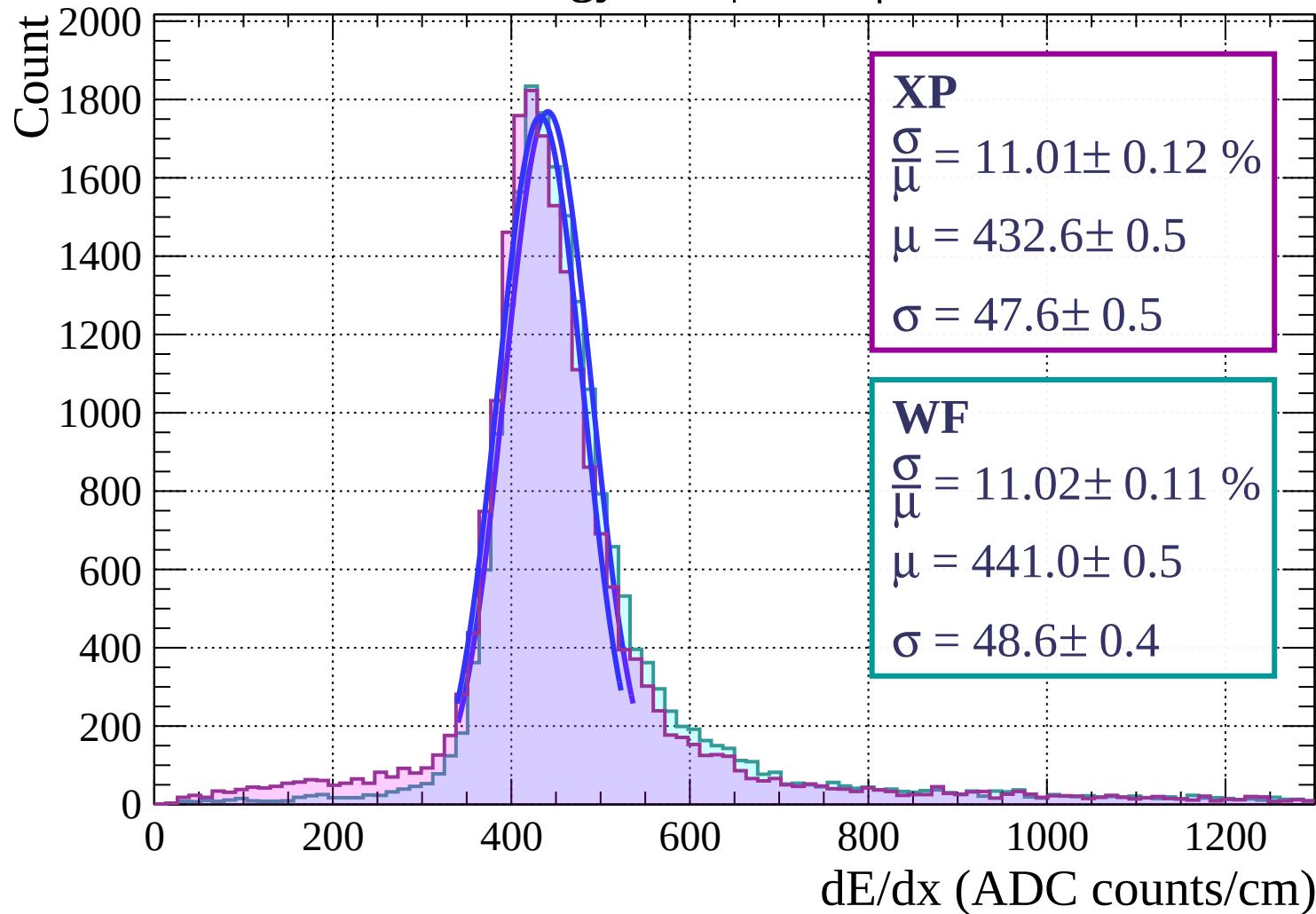
# Energy loss | $12 < \phi < 15$



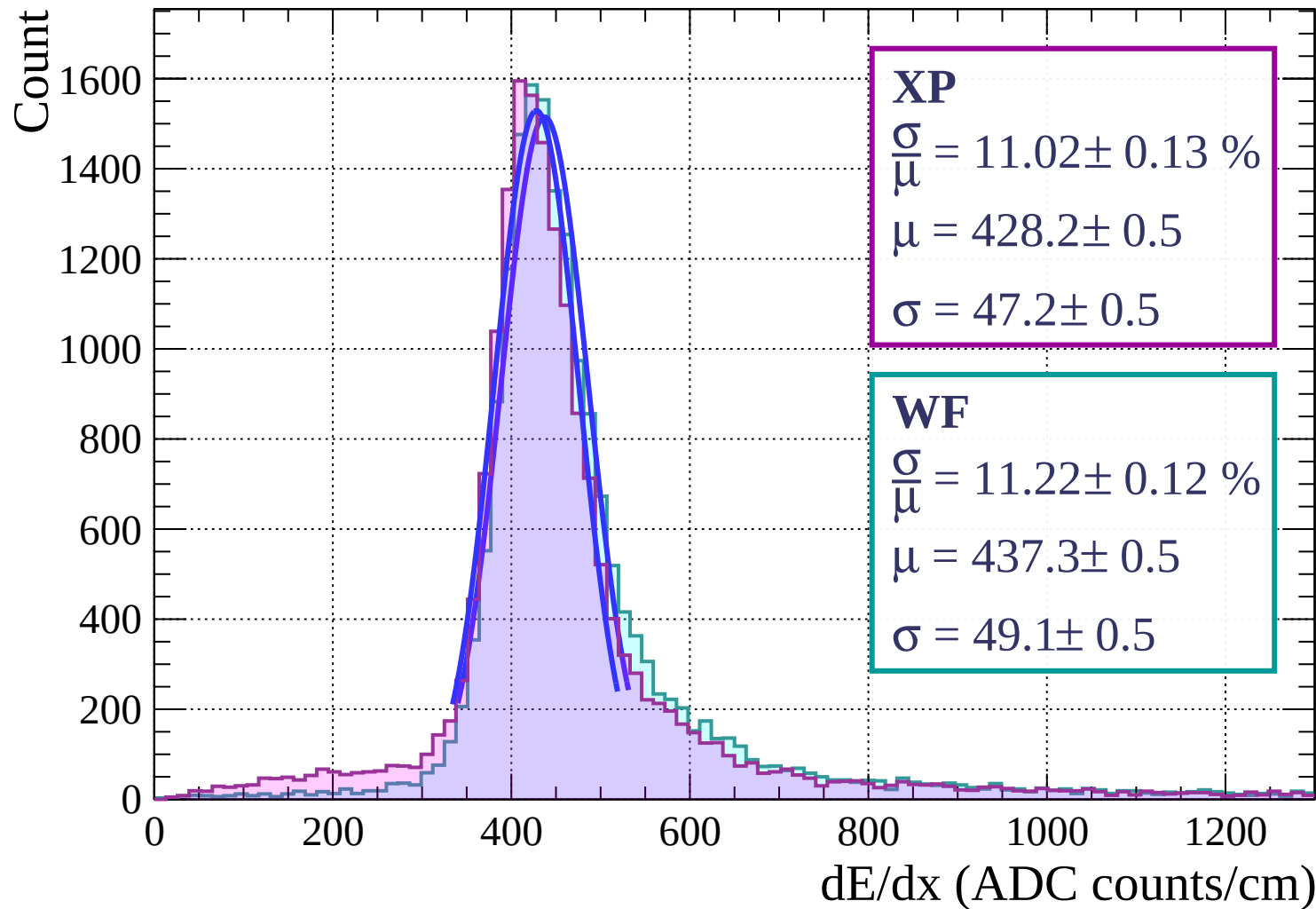
# Energy loss | $15 < \phi < 18$



# Energy loss | $18 < \phi < 21$

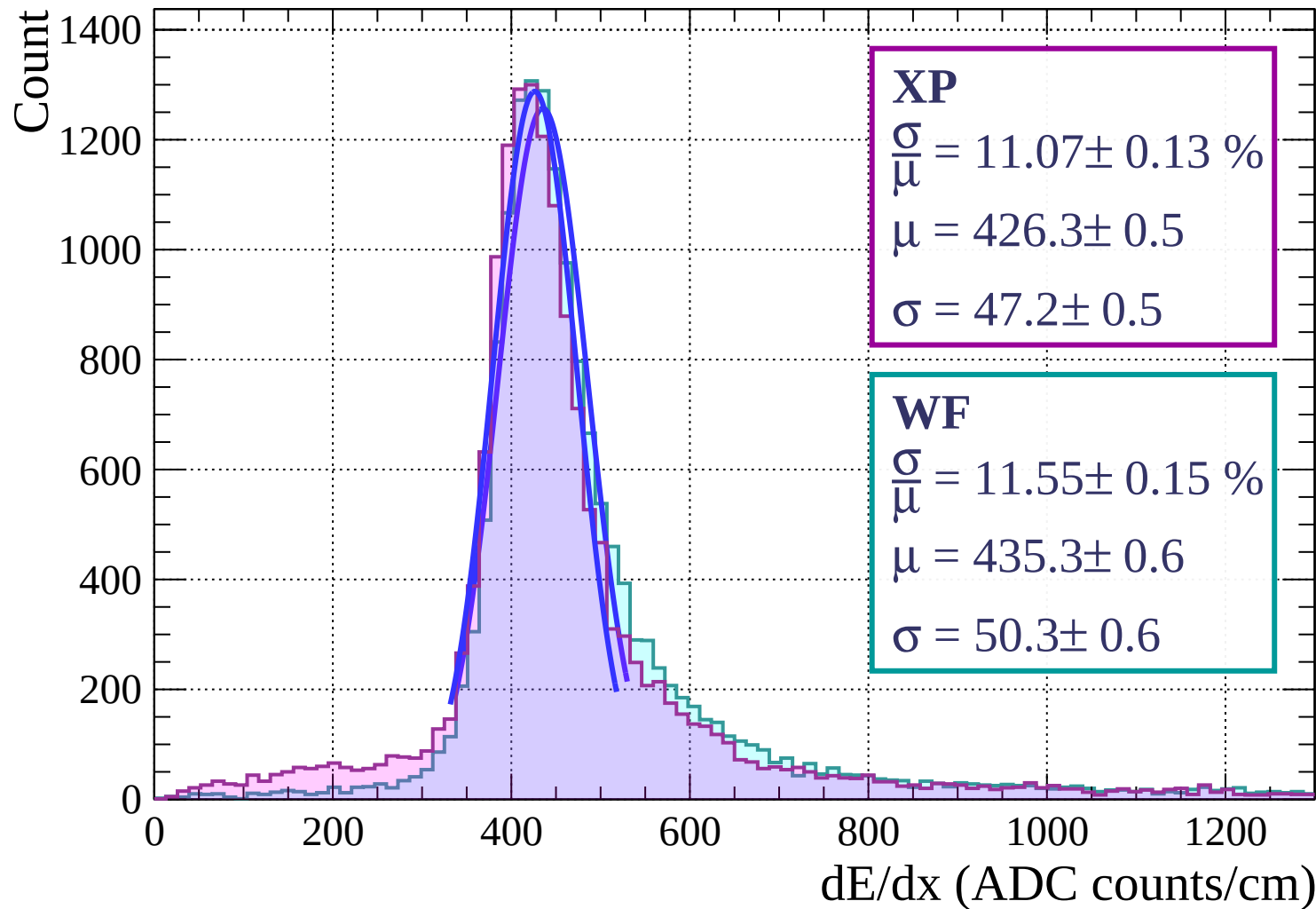


Energy loss |  $21 < \phi < 24$

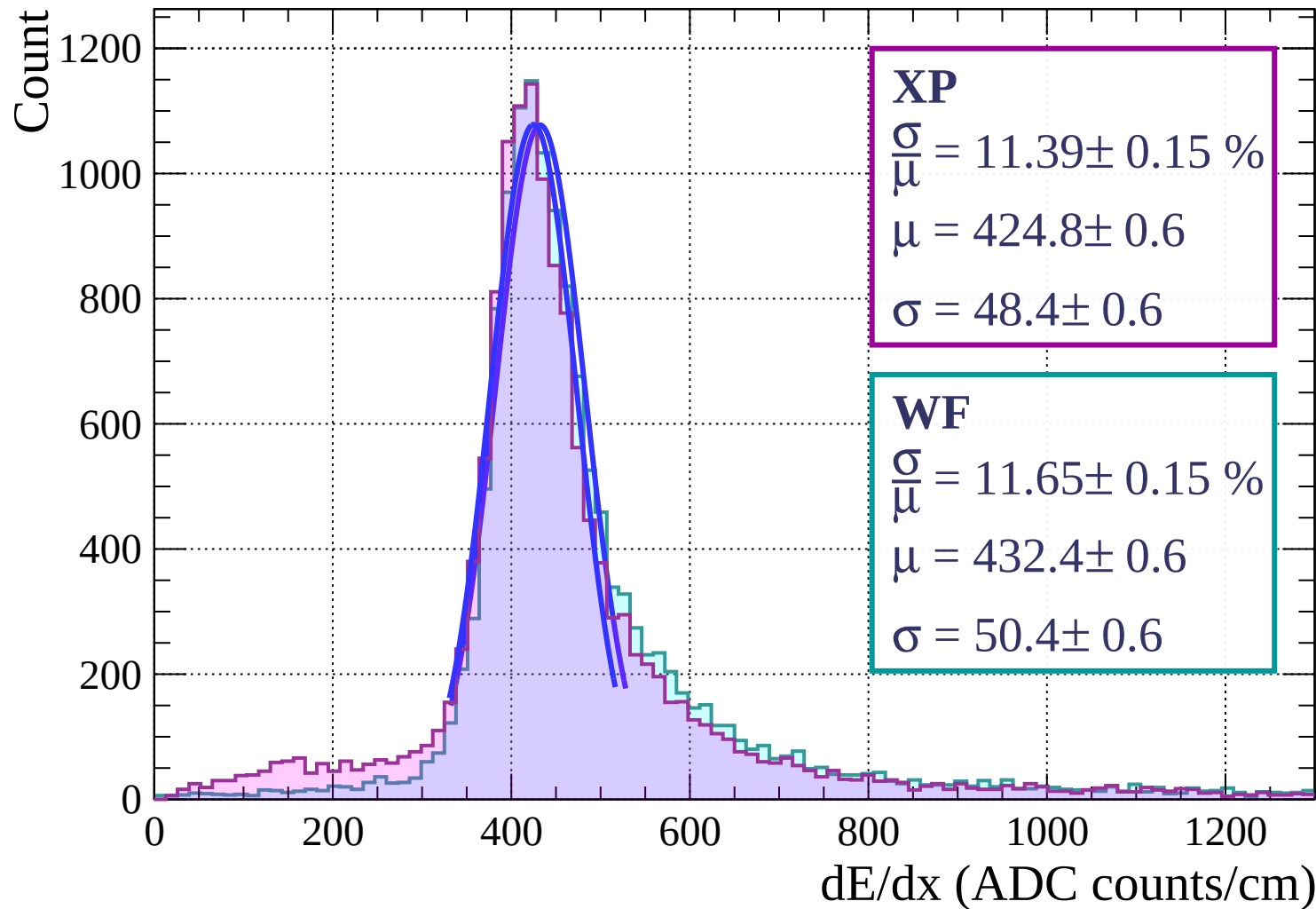




Energy loss |  $24 < \phi < 27$

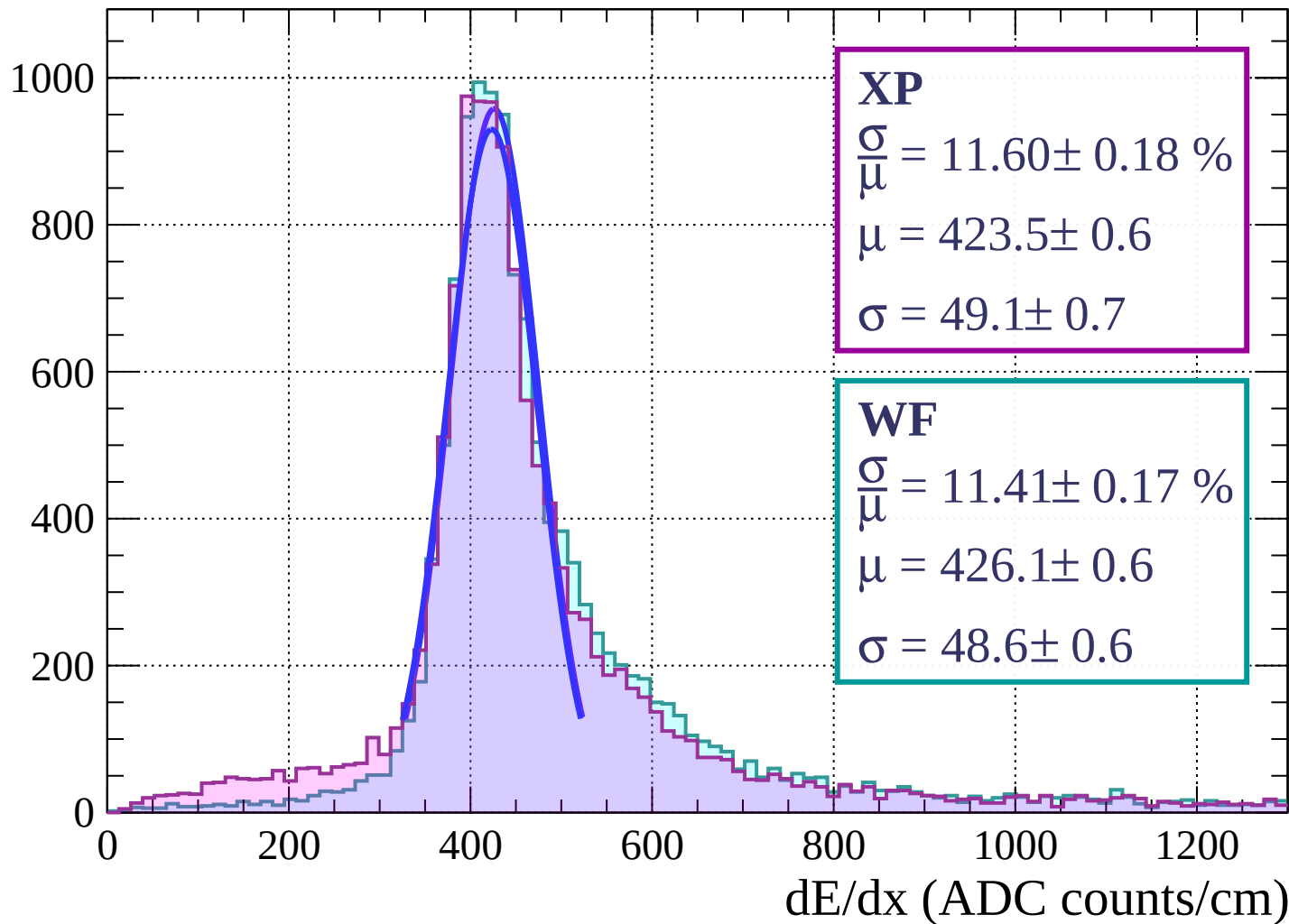


Energy loss |  $27 < \phi < 30$

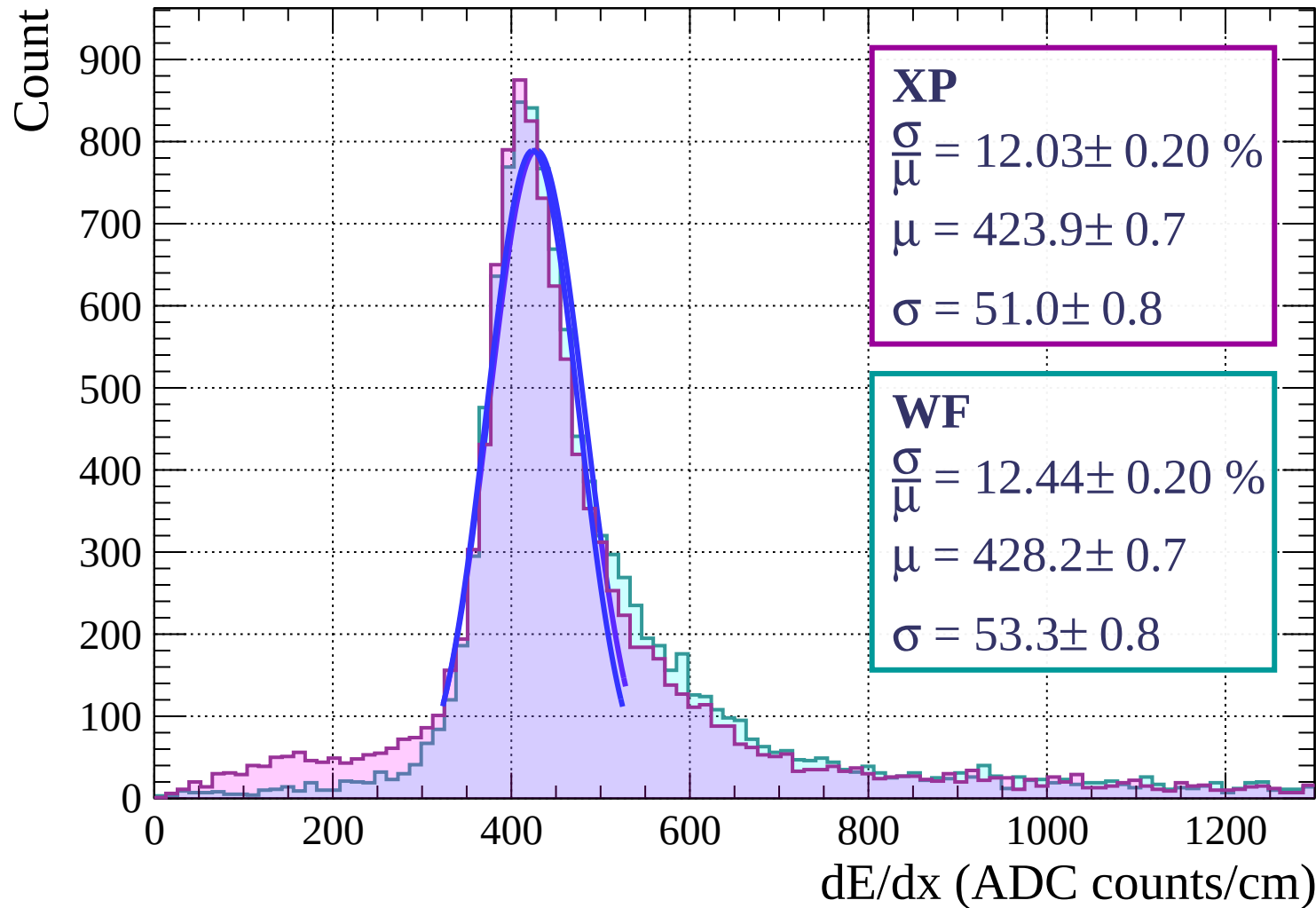


Energy loss |  $30 < \phi < 33$

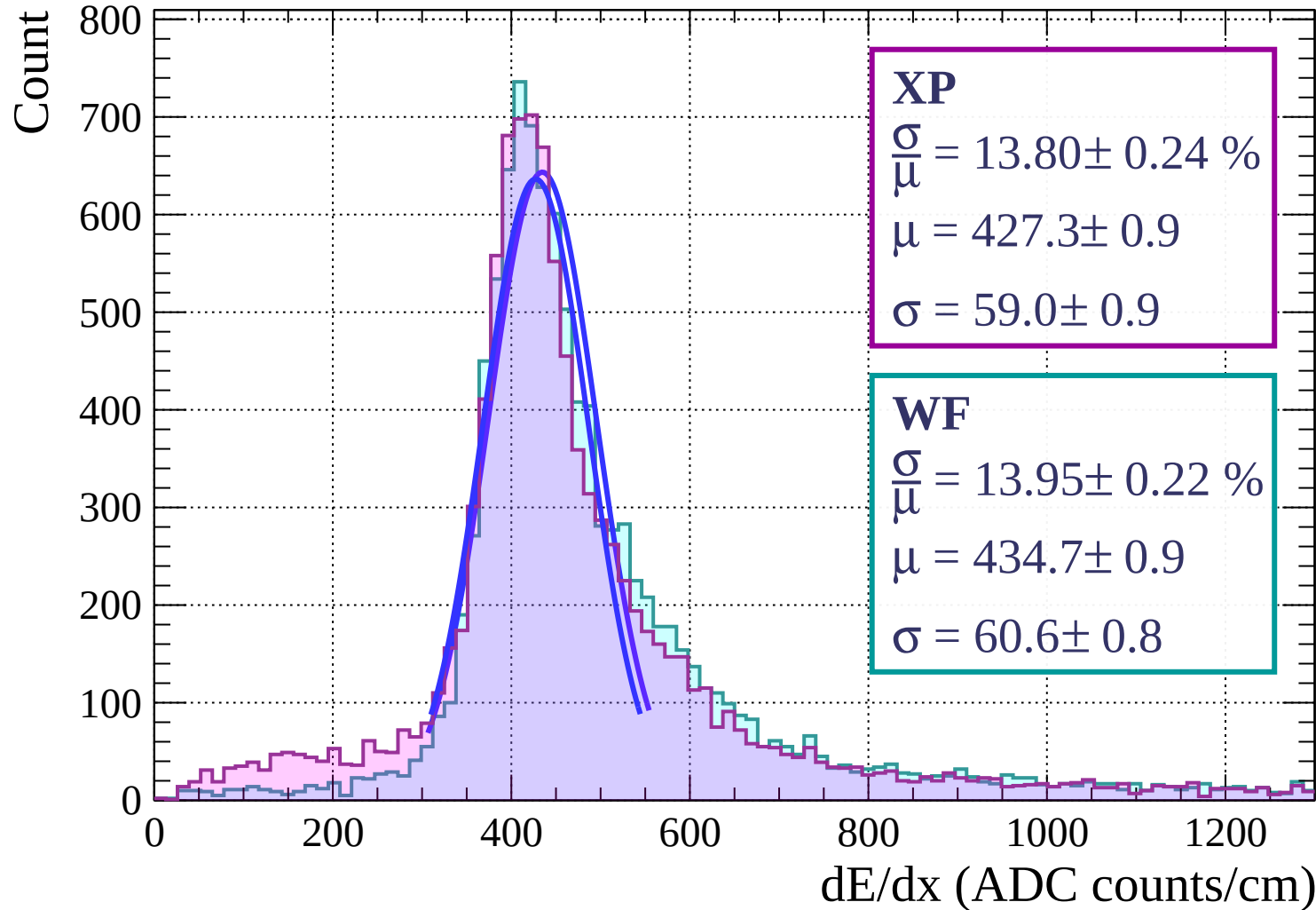
Count



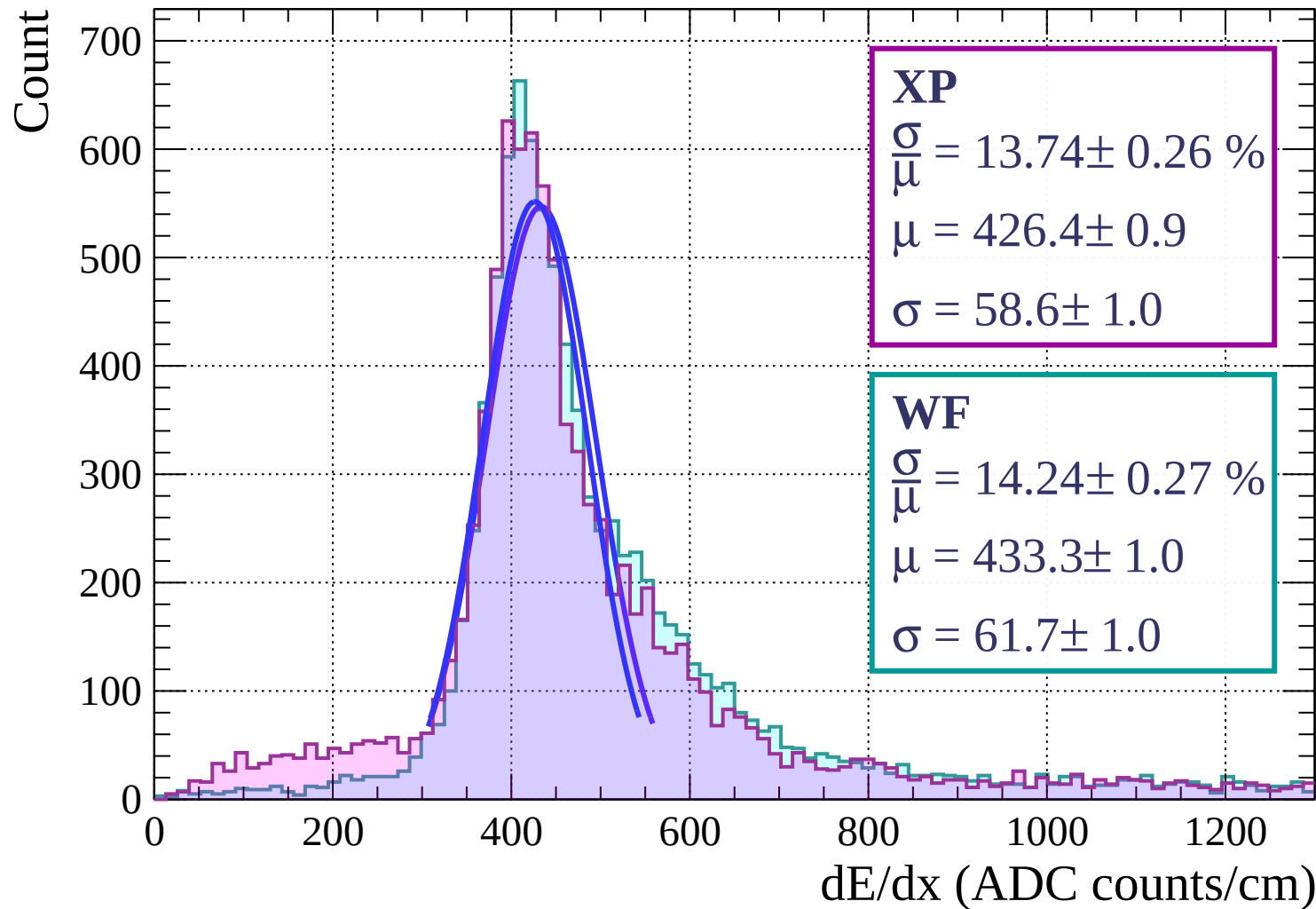
# Energy loss | $33 < \phi < 36$



Energy loss |  $36 < \phi < 39$

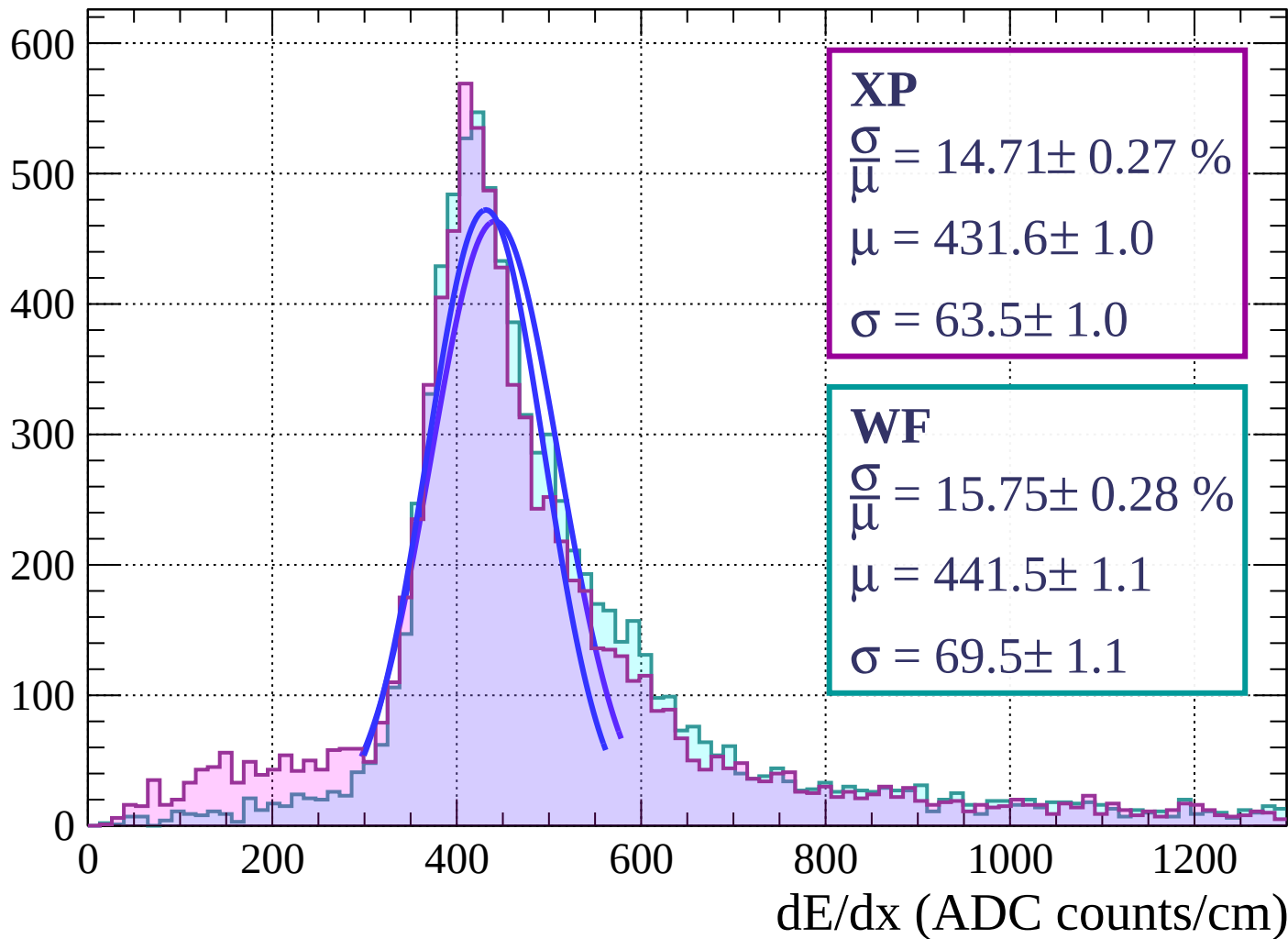


Energy loss |  $39 < \phi < 42$



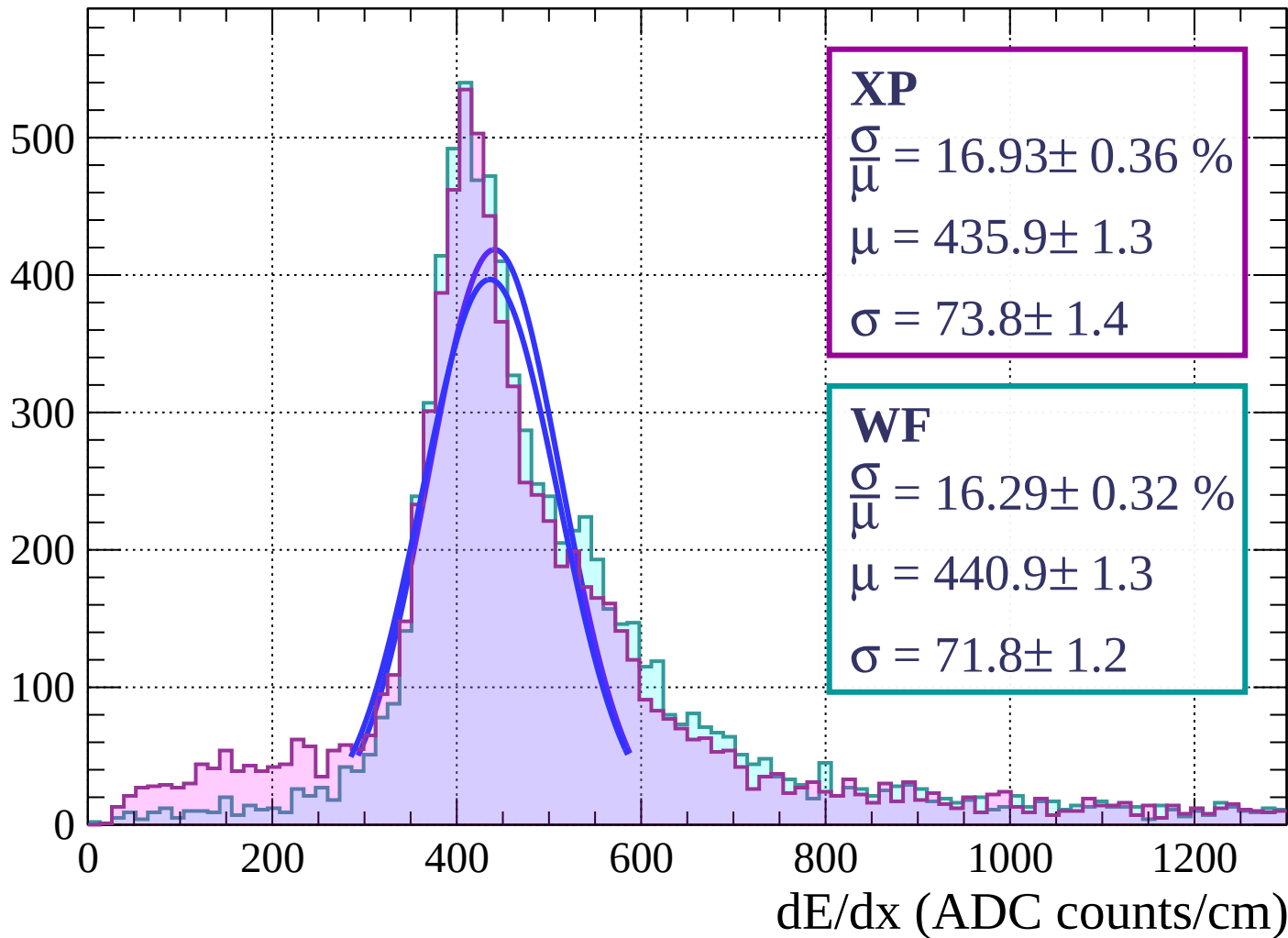
Energy loss |  $42 < \phi < 45$

Count



Energy loss |  $45 < \phi < 48$

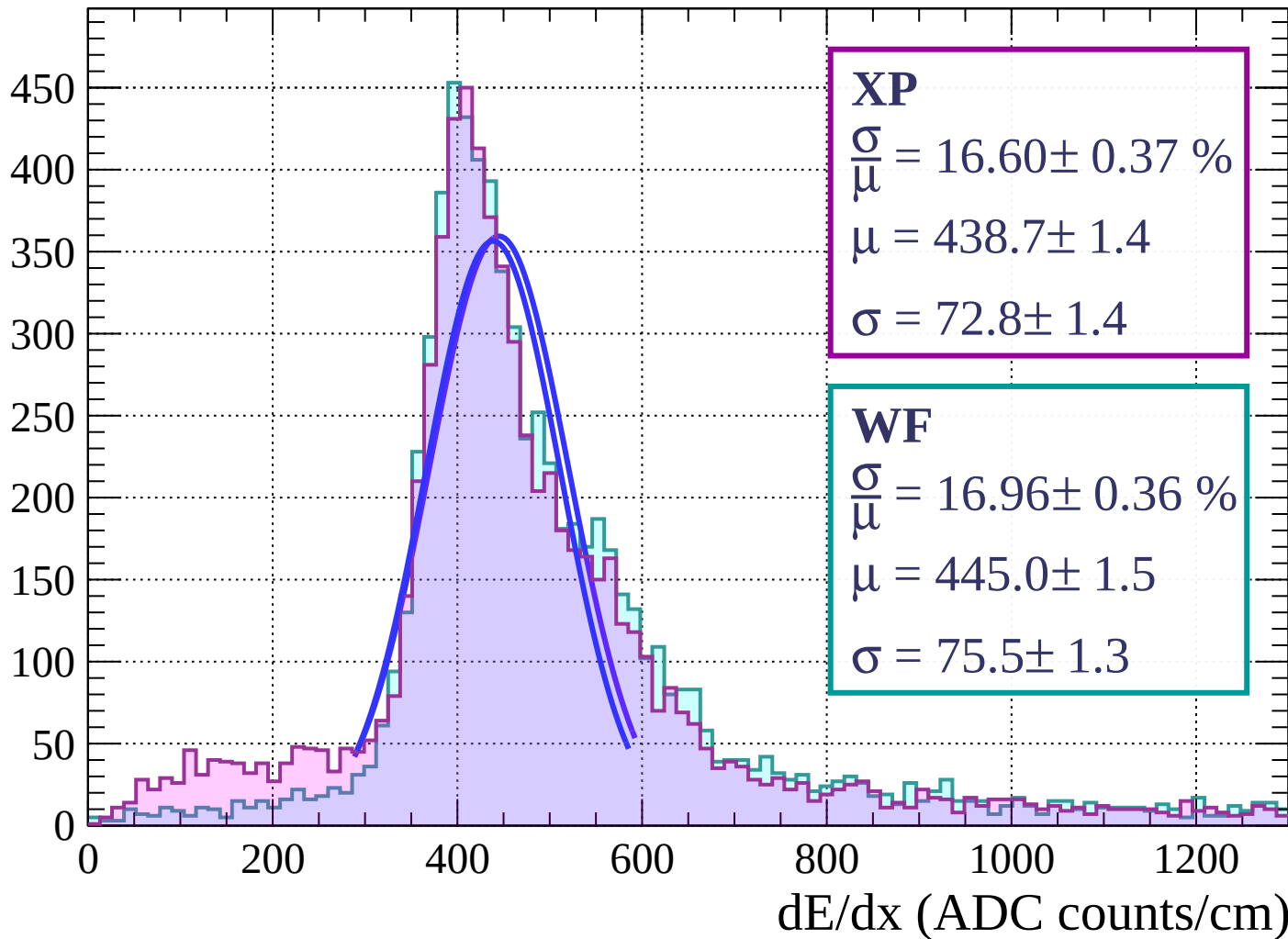
Count





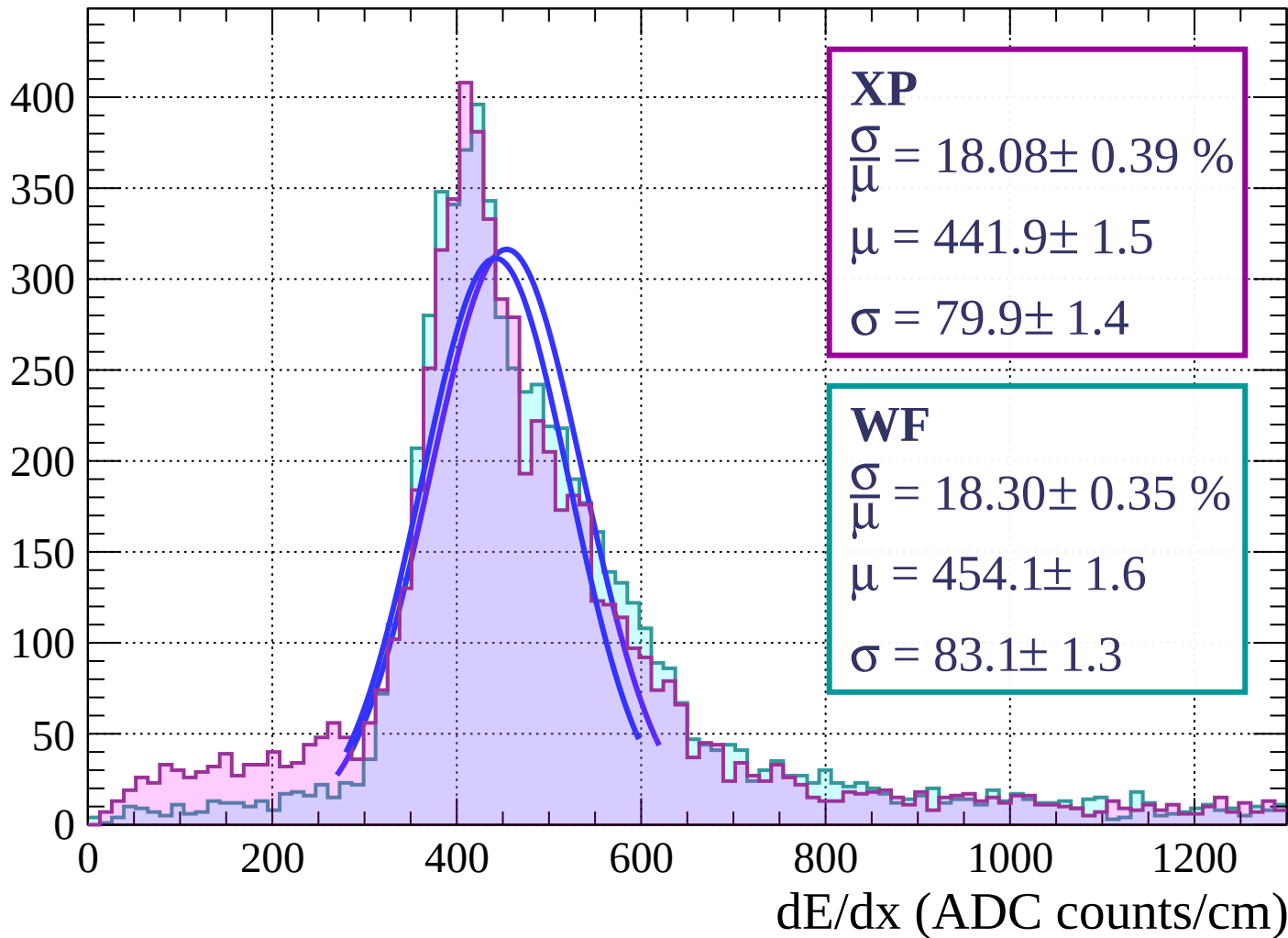
Energy loss |  $48 < \phi < 51$

Count



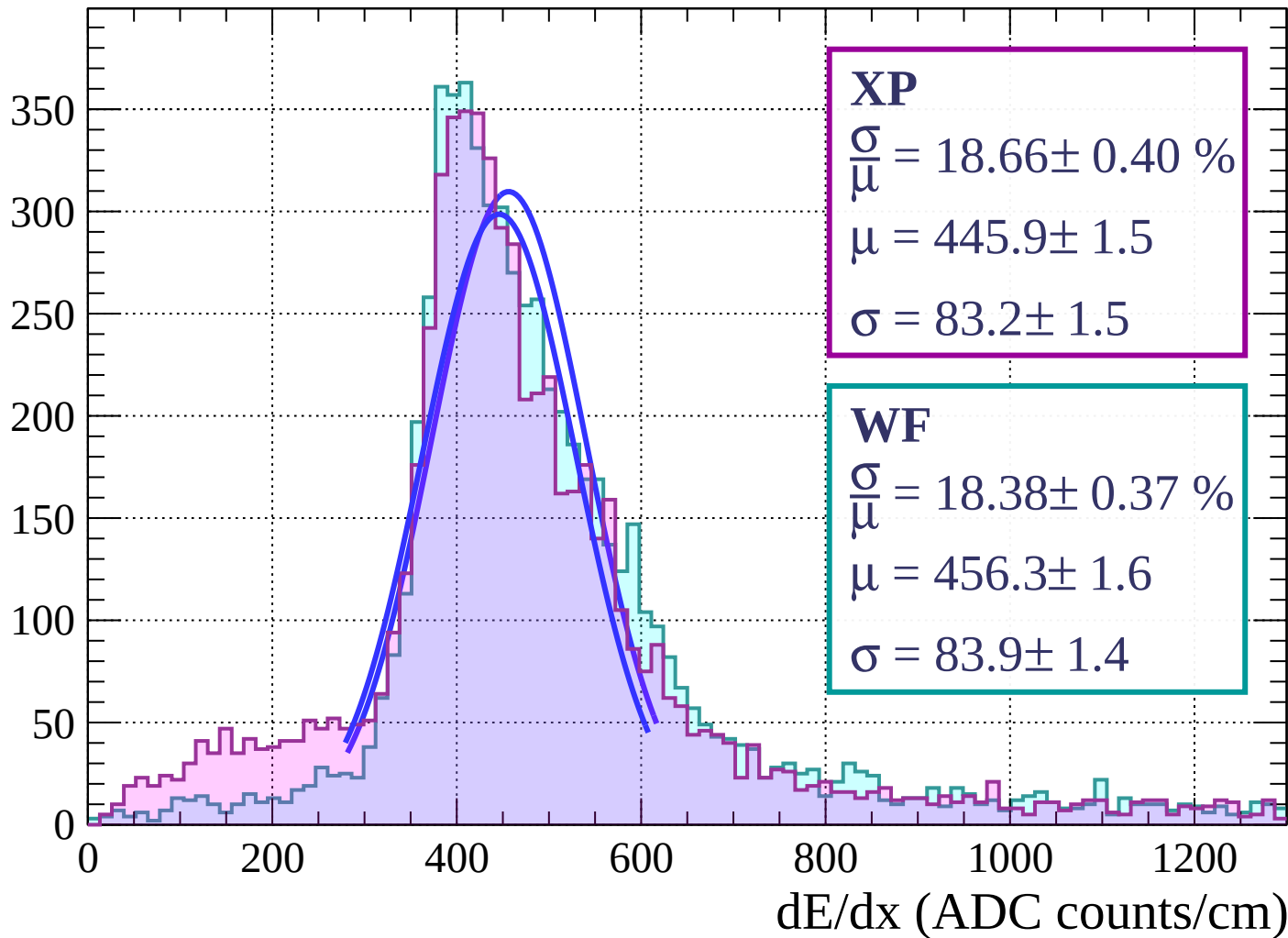
# Energy loss | $51 < \phi < 54$

Count

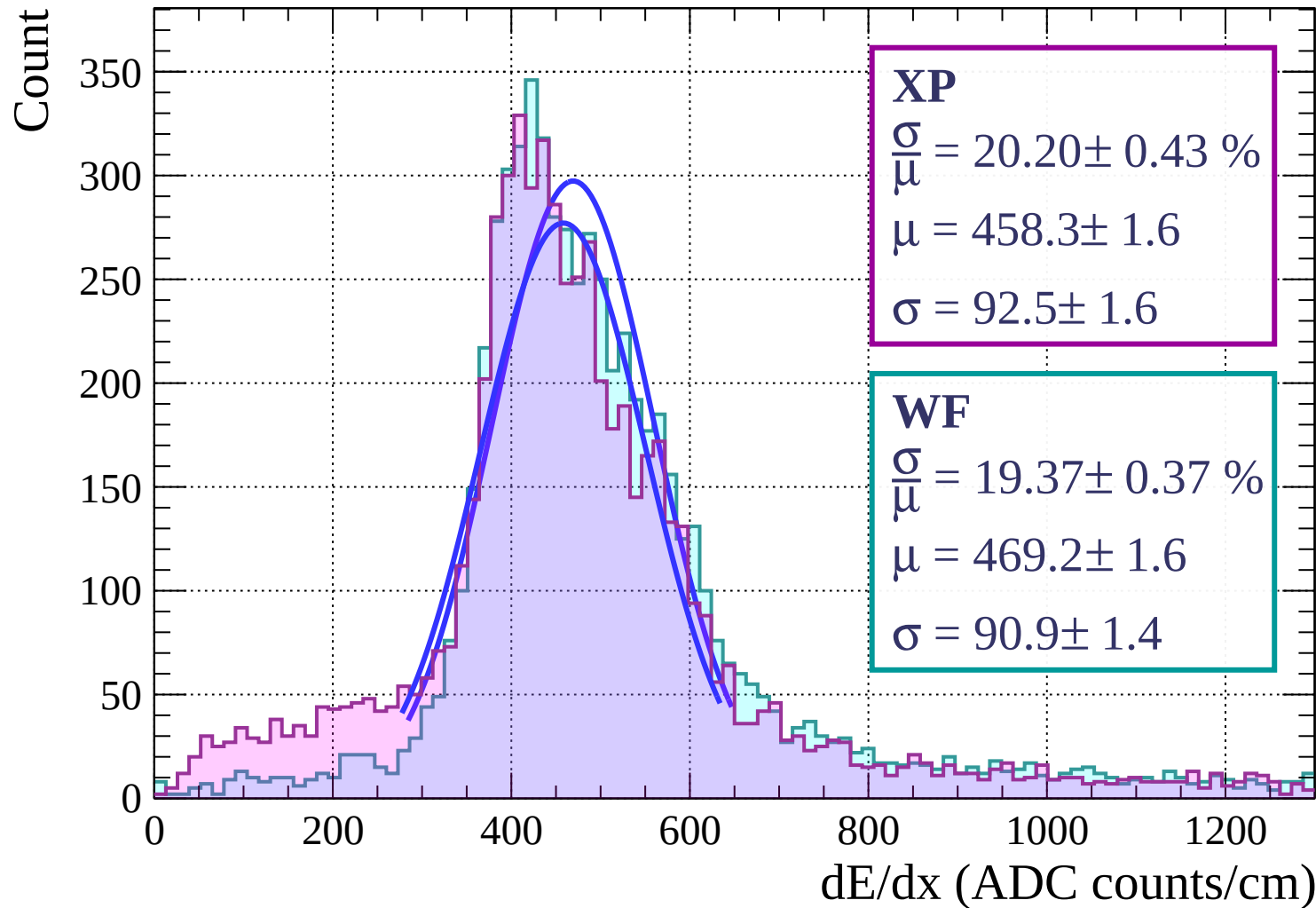


Energy loss |  $54 < \phi < 57$

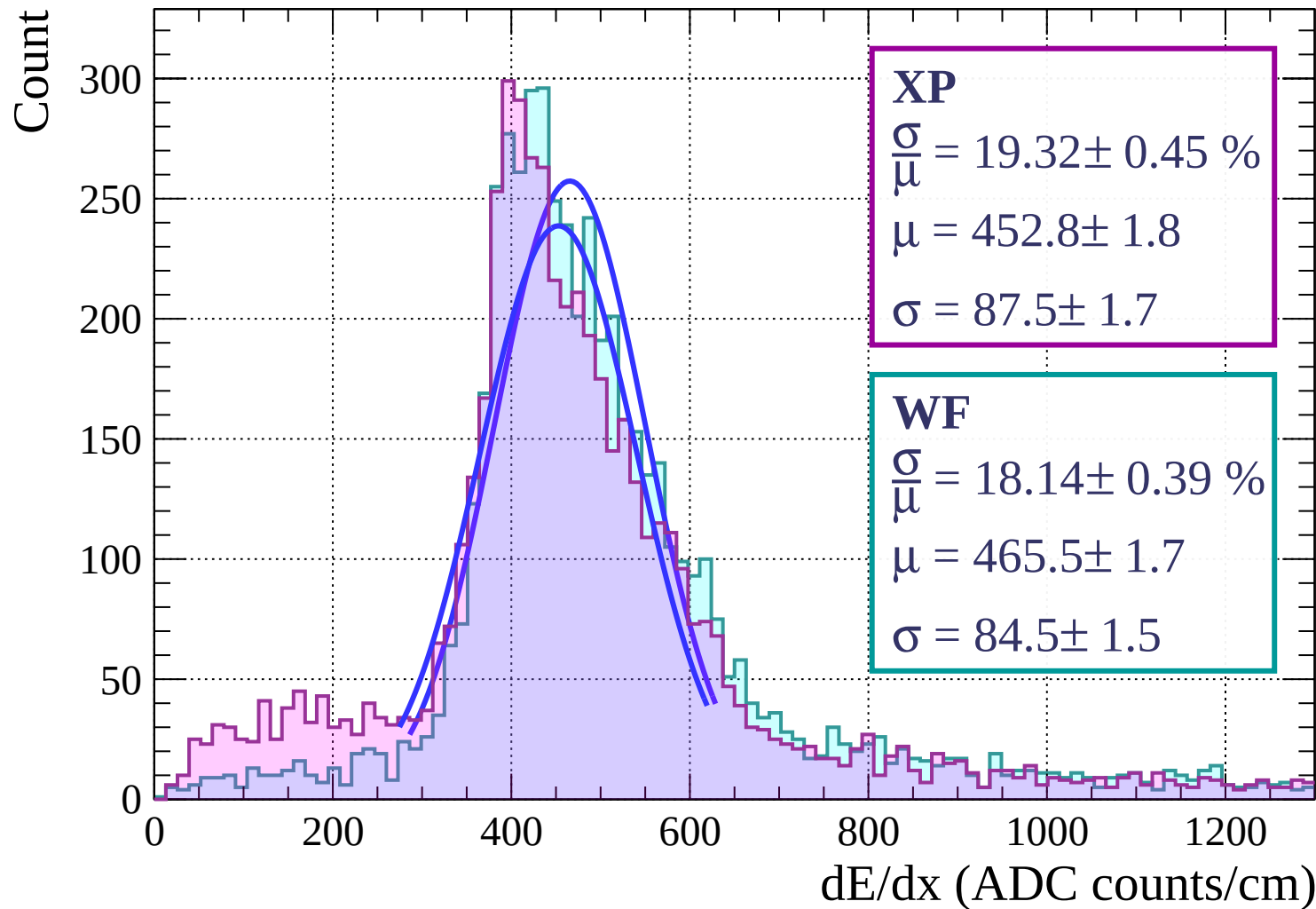
Count



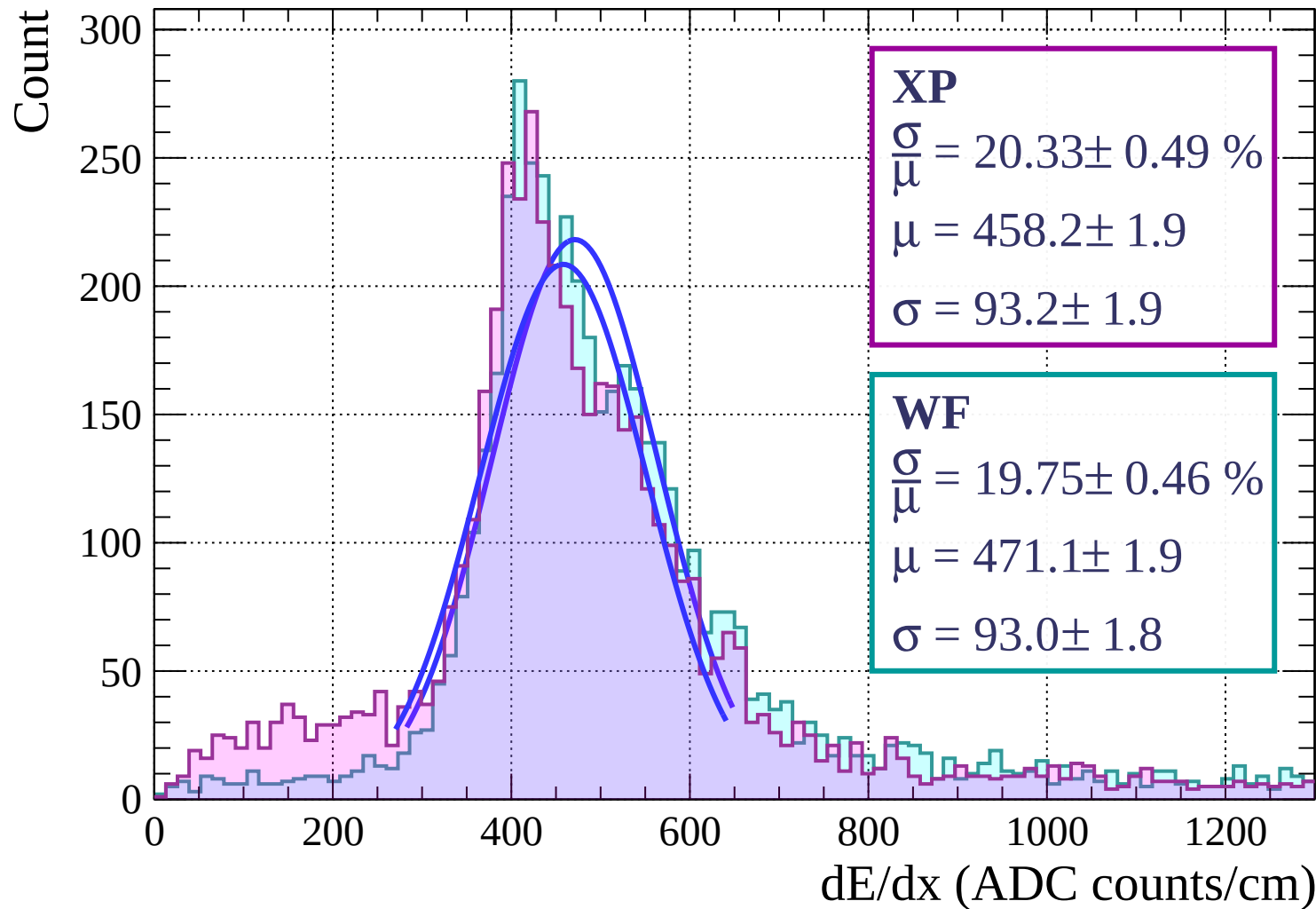
# Energy loss | $57 < \phi < 60$



# Energy loss | $60 < \phi < 63$

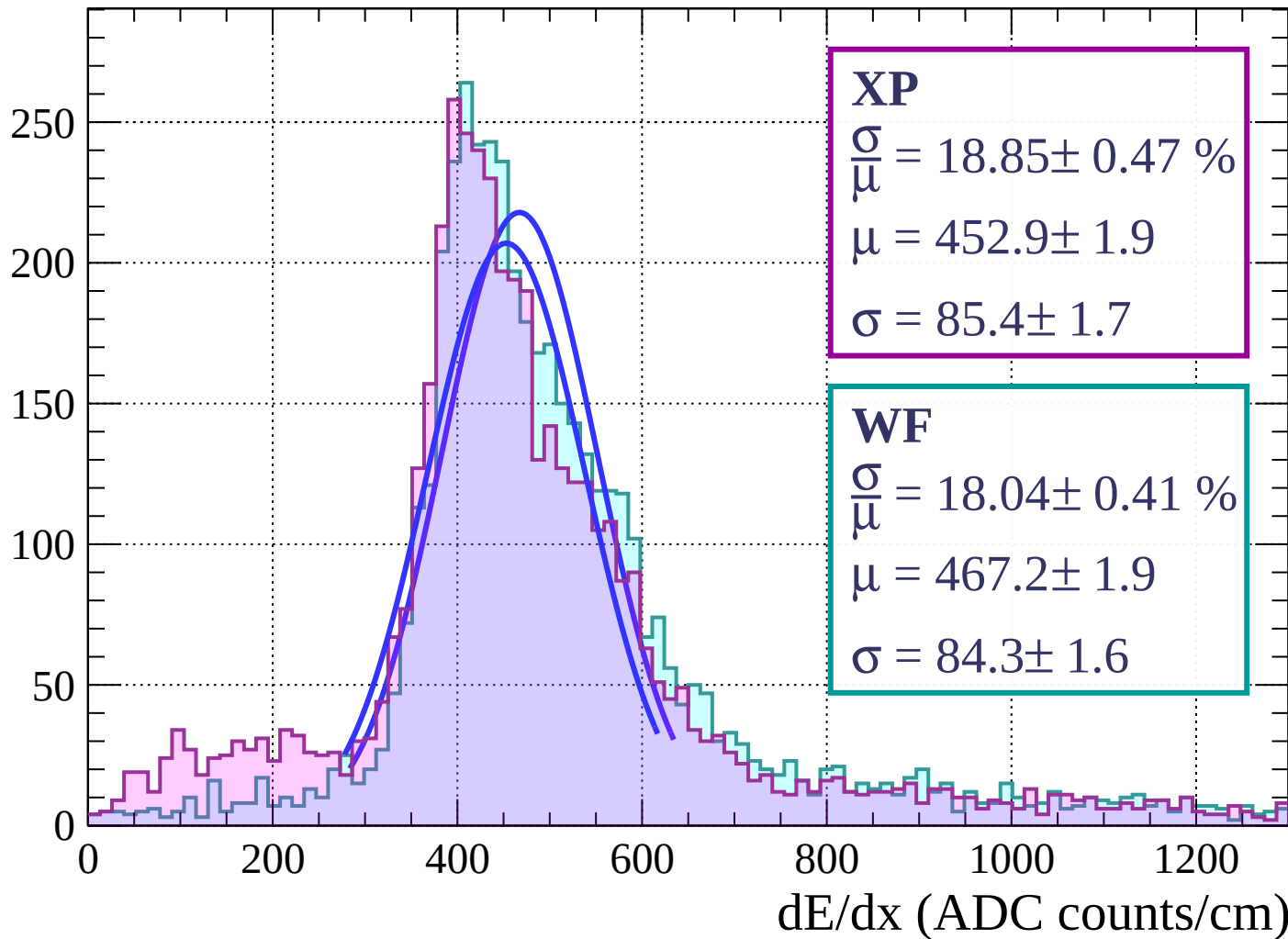


Energy loss |  $63 < \phi < 66$

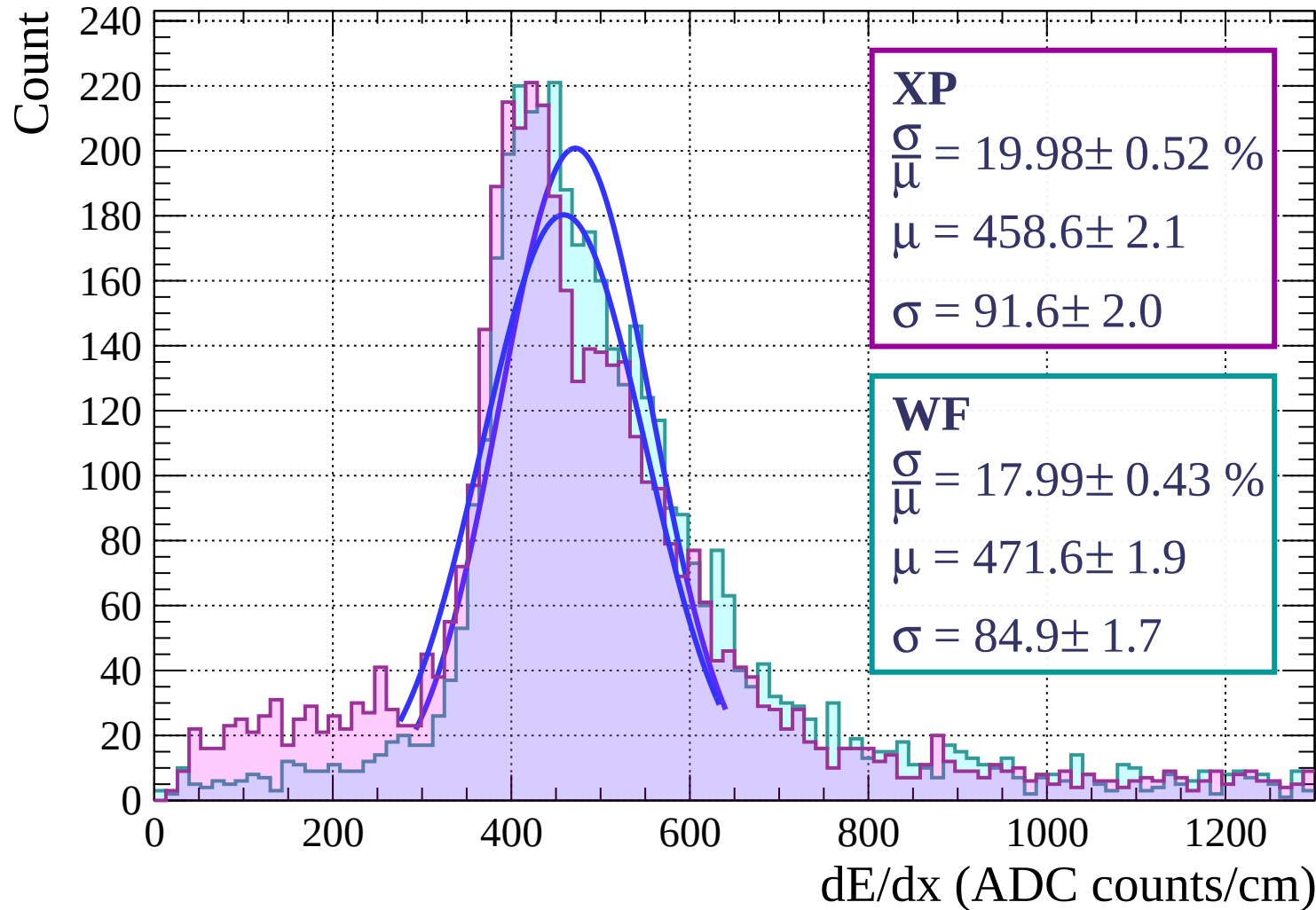


Energy loss |  $66 < \phi < 69$

Count

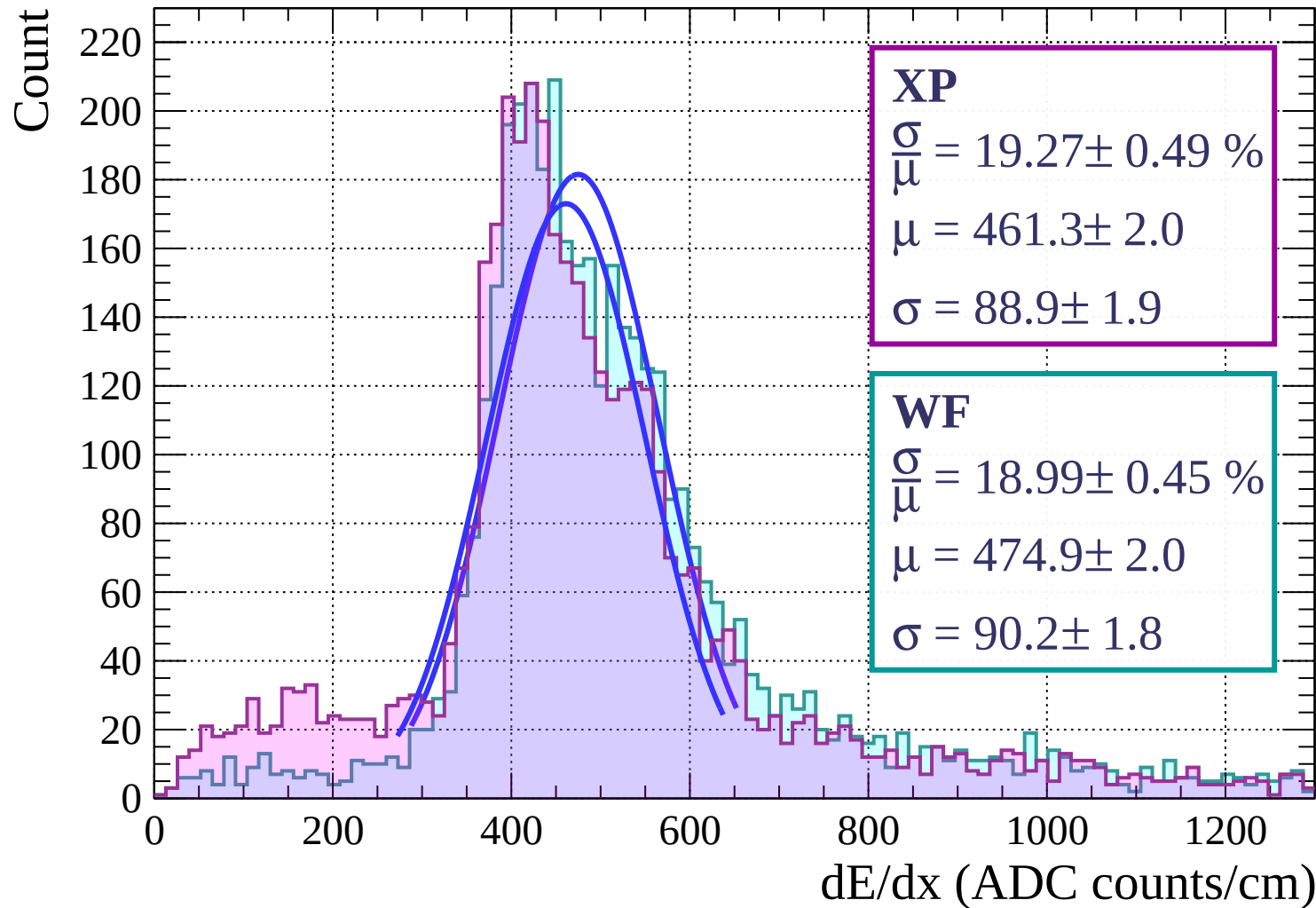


# Energy loss | $69 < \phi < 72$



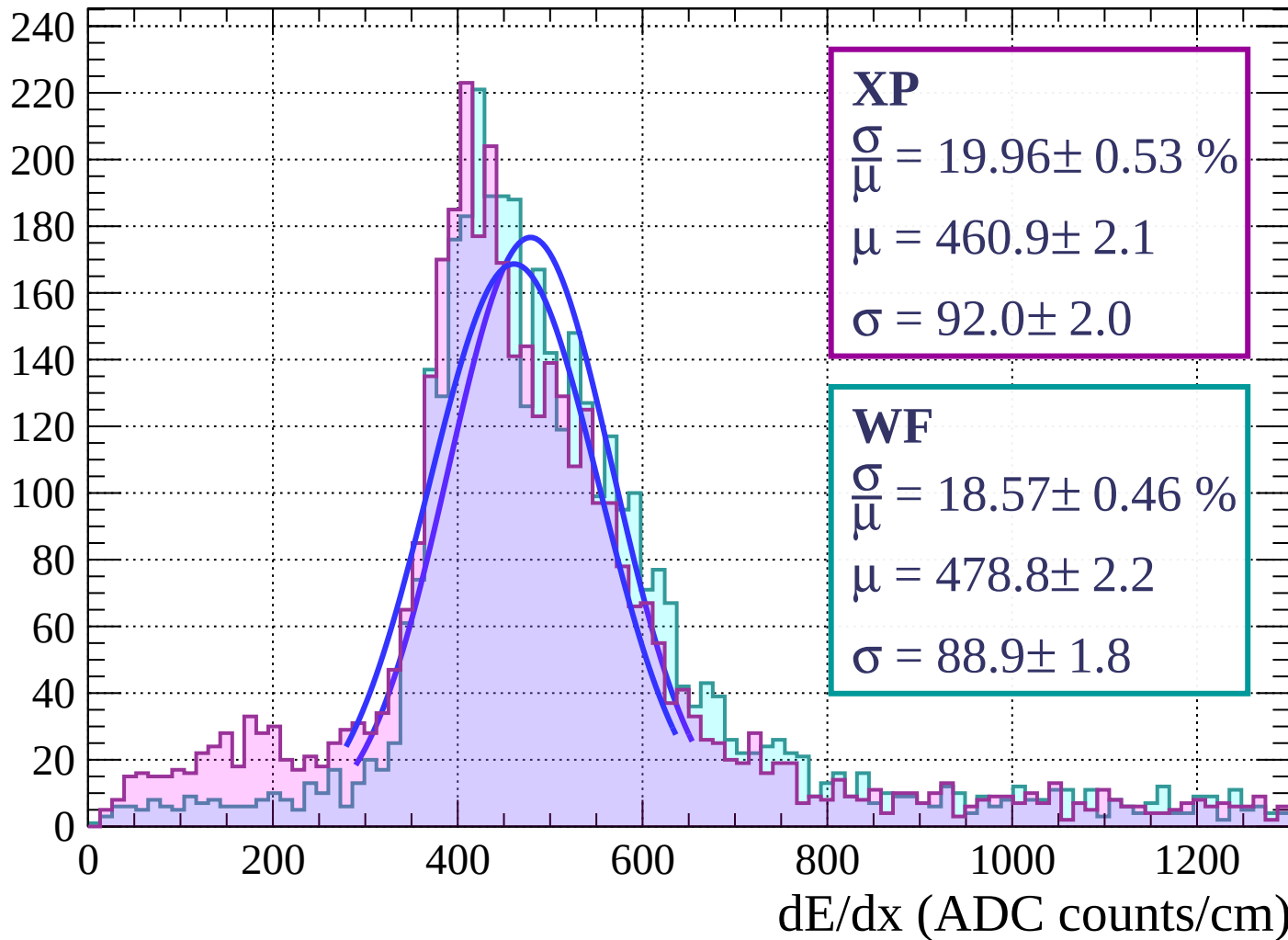


# Energy loss | $72 < \phi < 75$



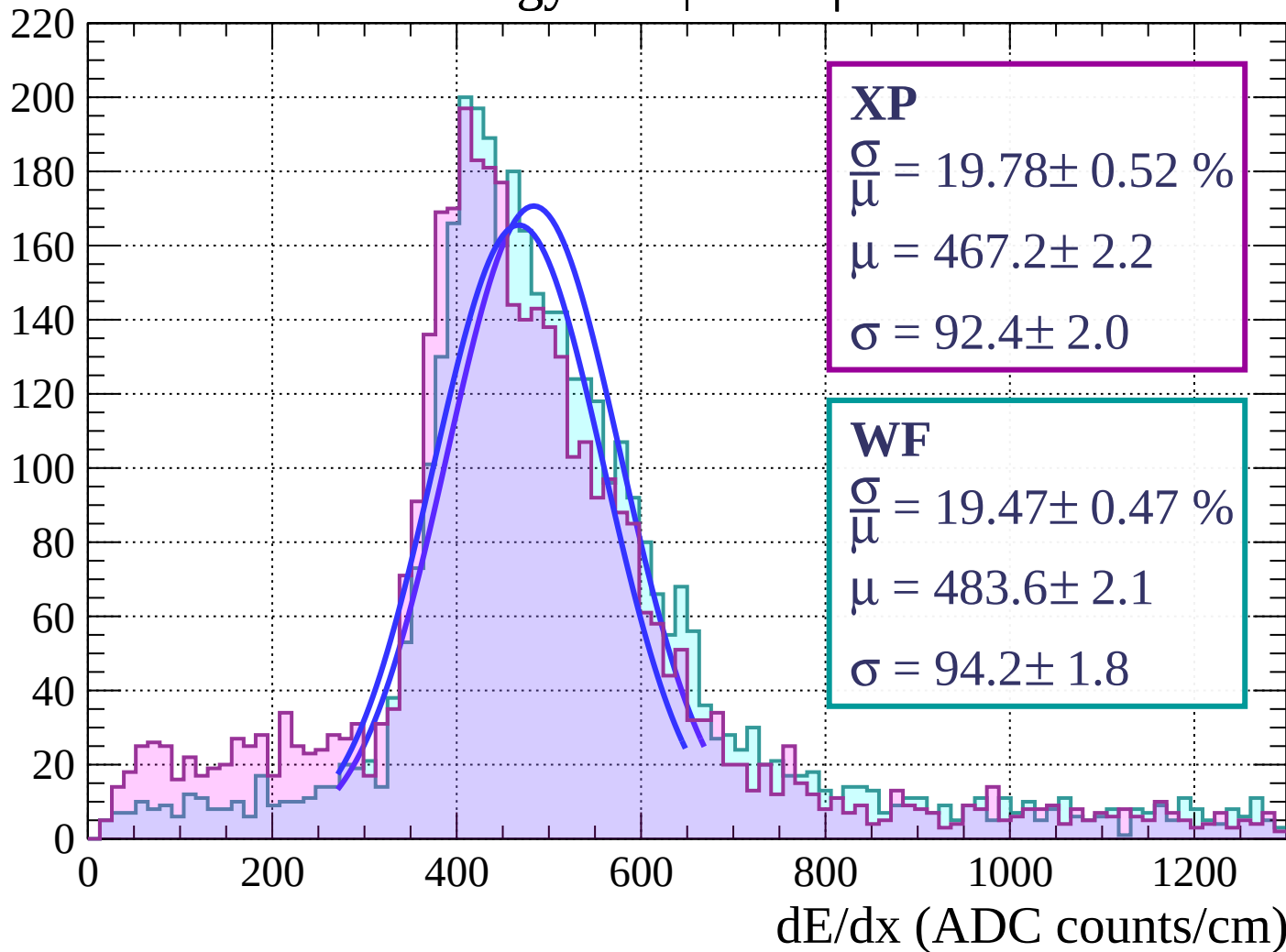
Energy loss |  $75 < \phi < 78$

Count



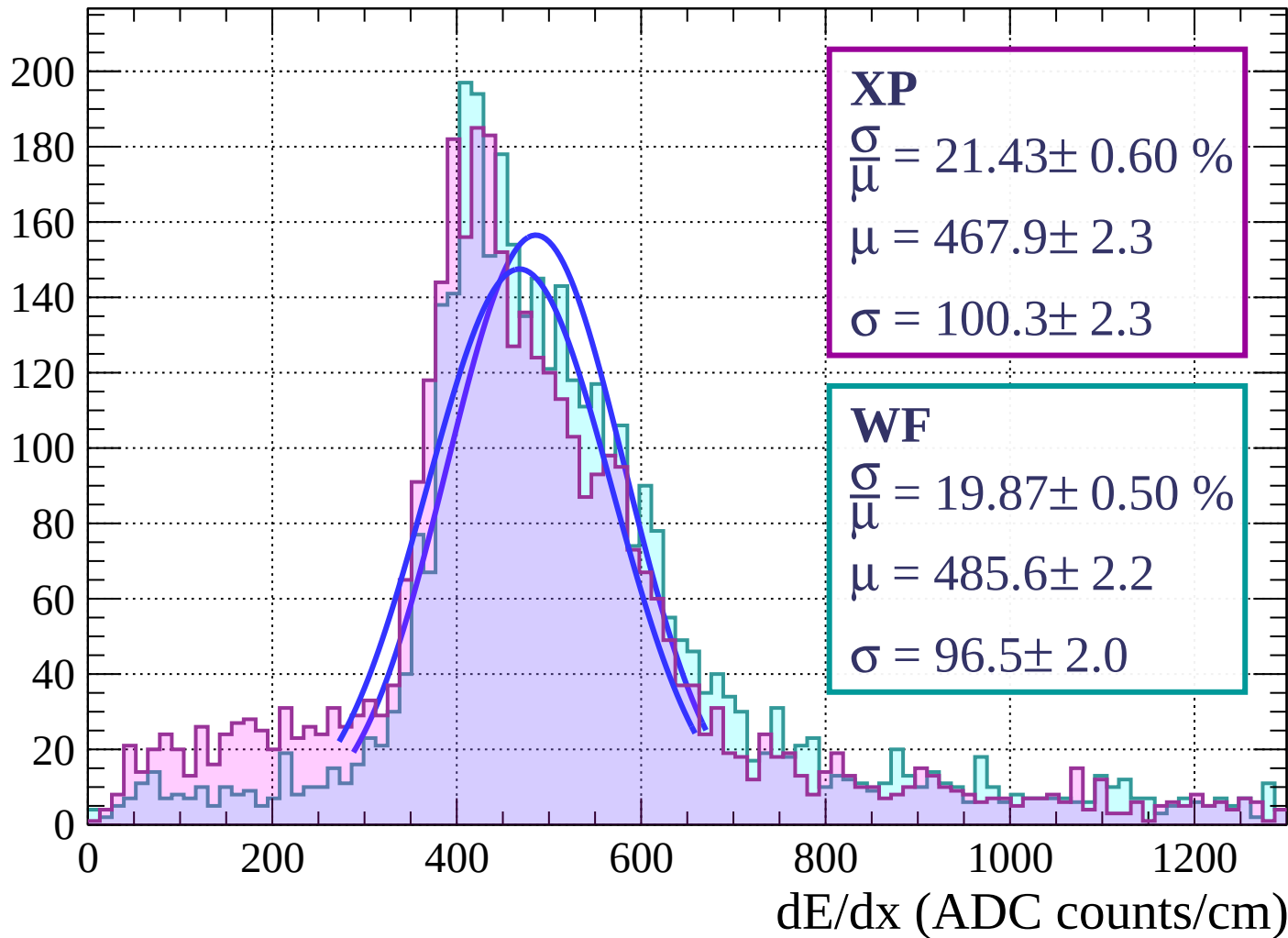
Energy loss |  $78 < \phi < 81$

Count

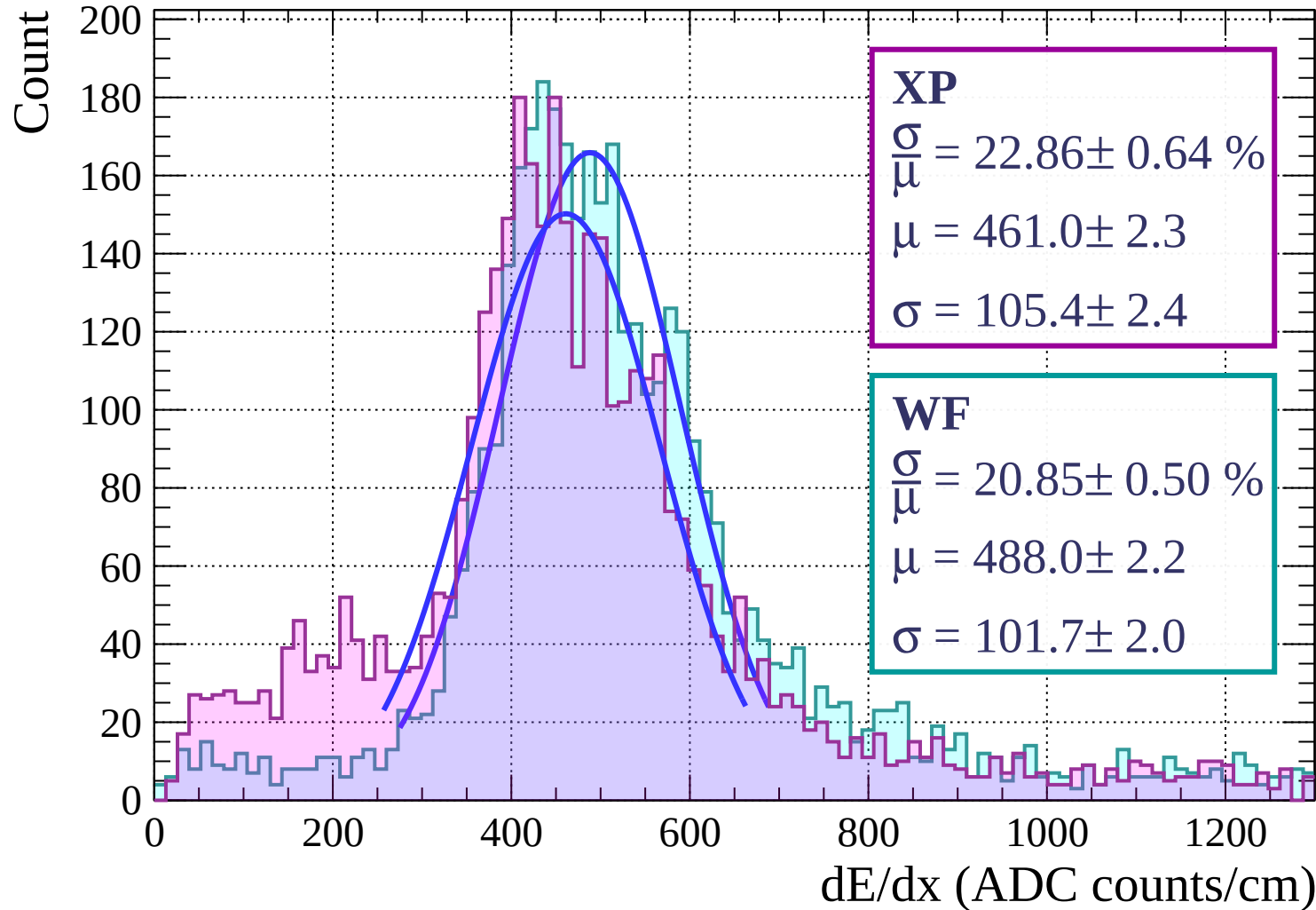


Energy loss |  $81 < \phi < 84$

Count

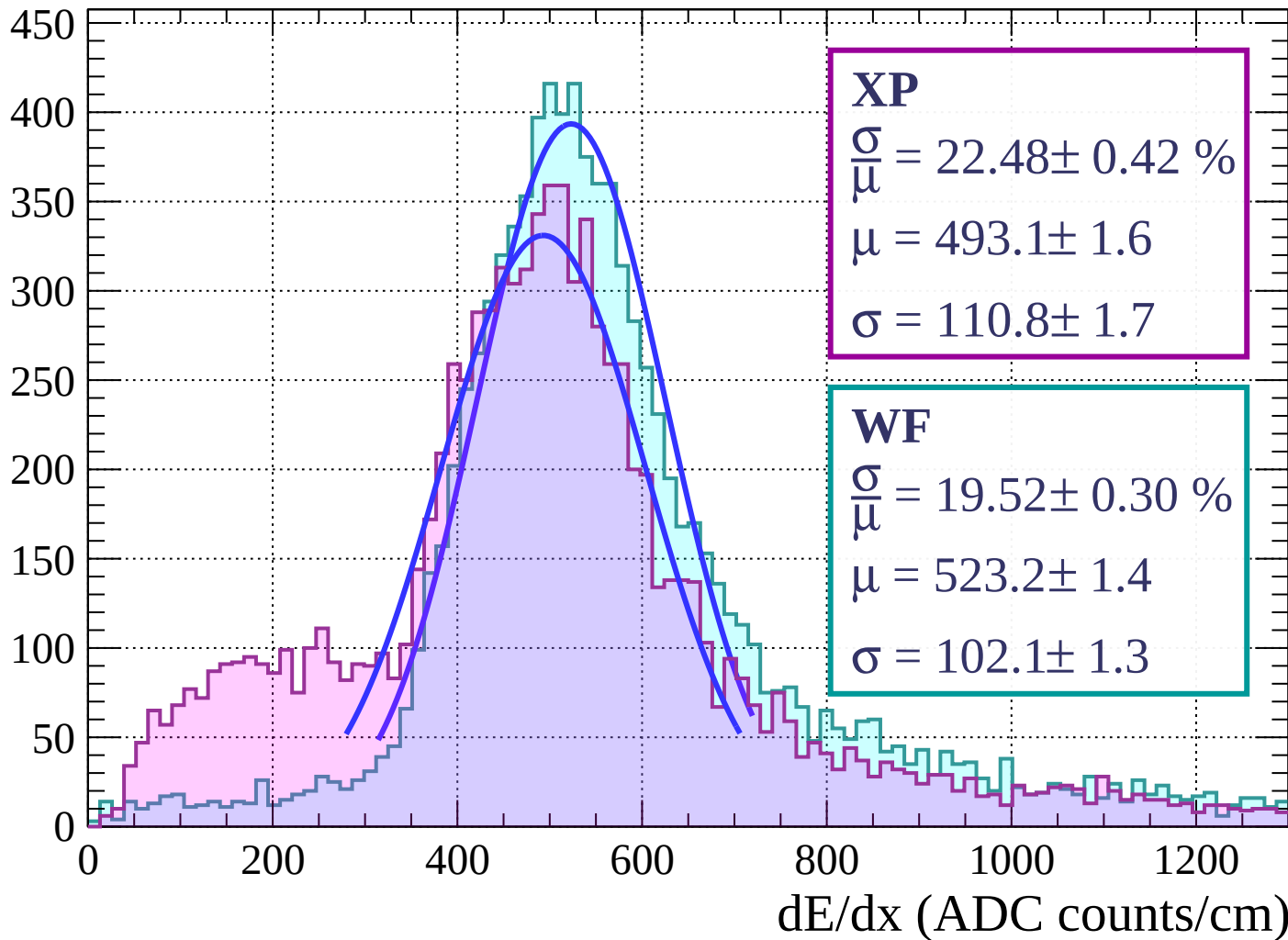


Energy loss |  $84 < \phi < 87$

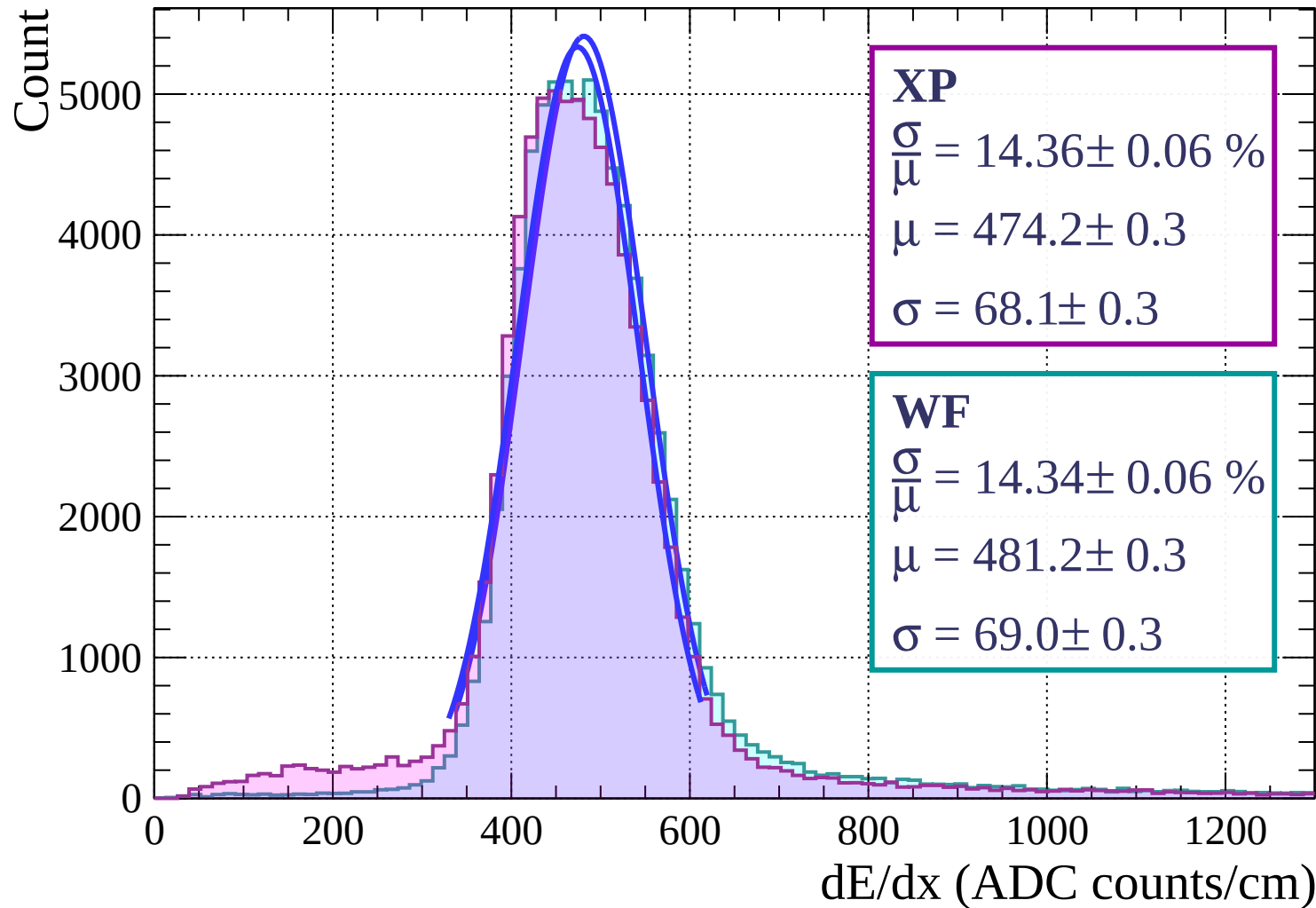


# Energy loss | $87 < \phi < 90$

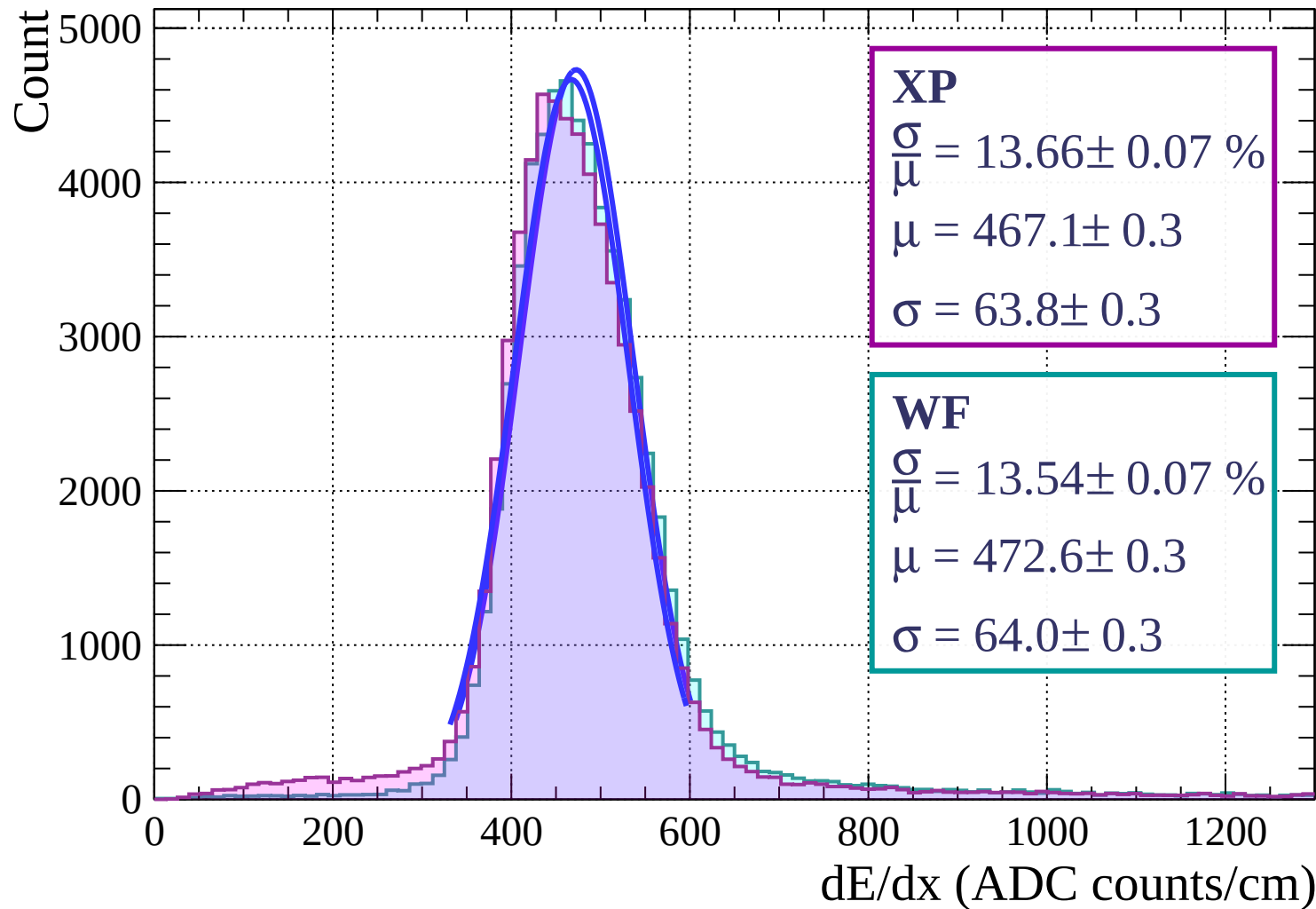
Count



Energy loss |  $0 < \theta < 3$

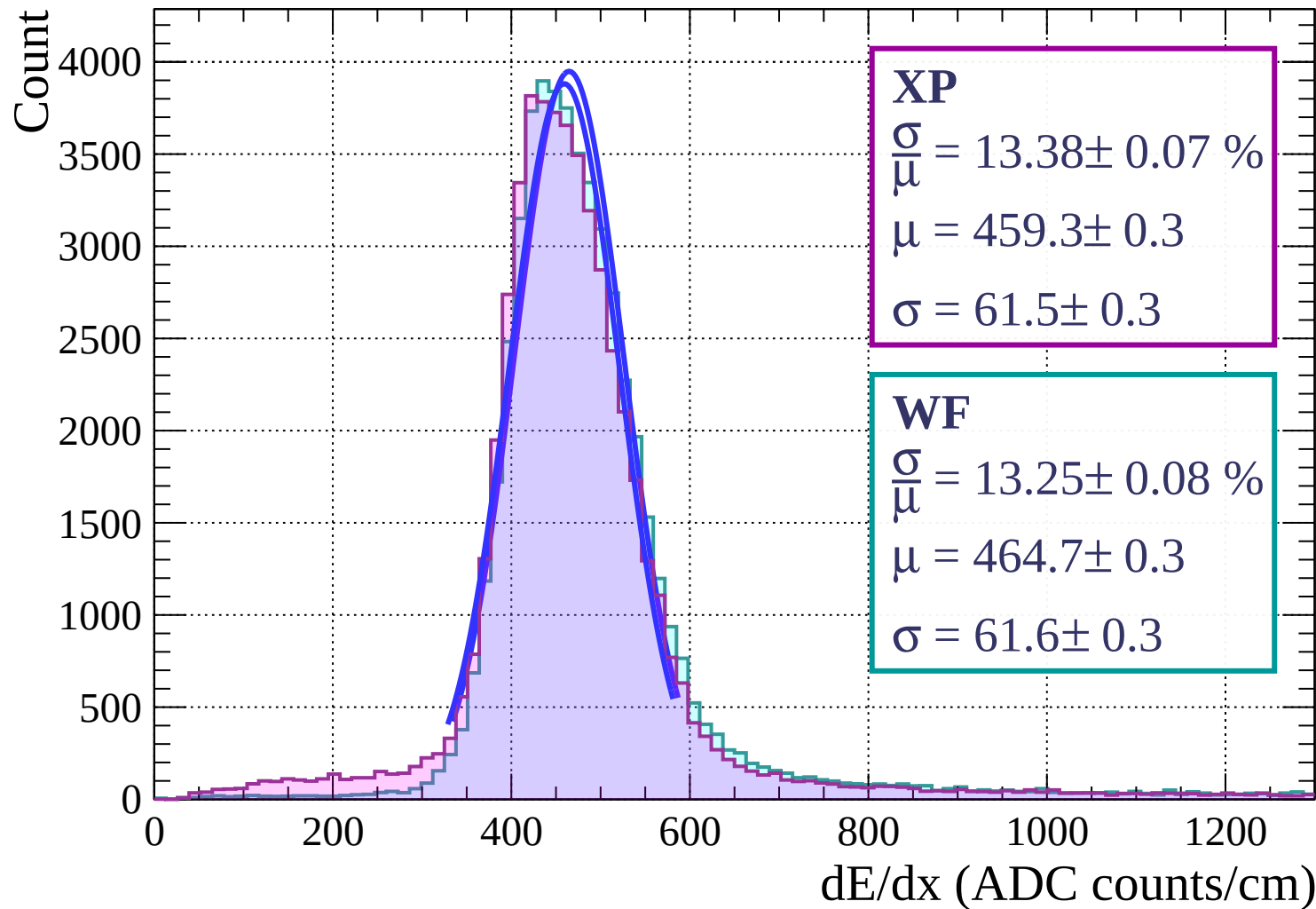


# Energy loss | $3 < \theta < 6$

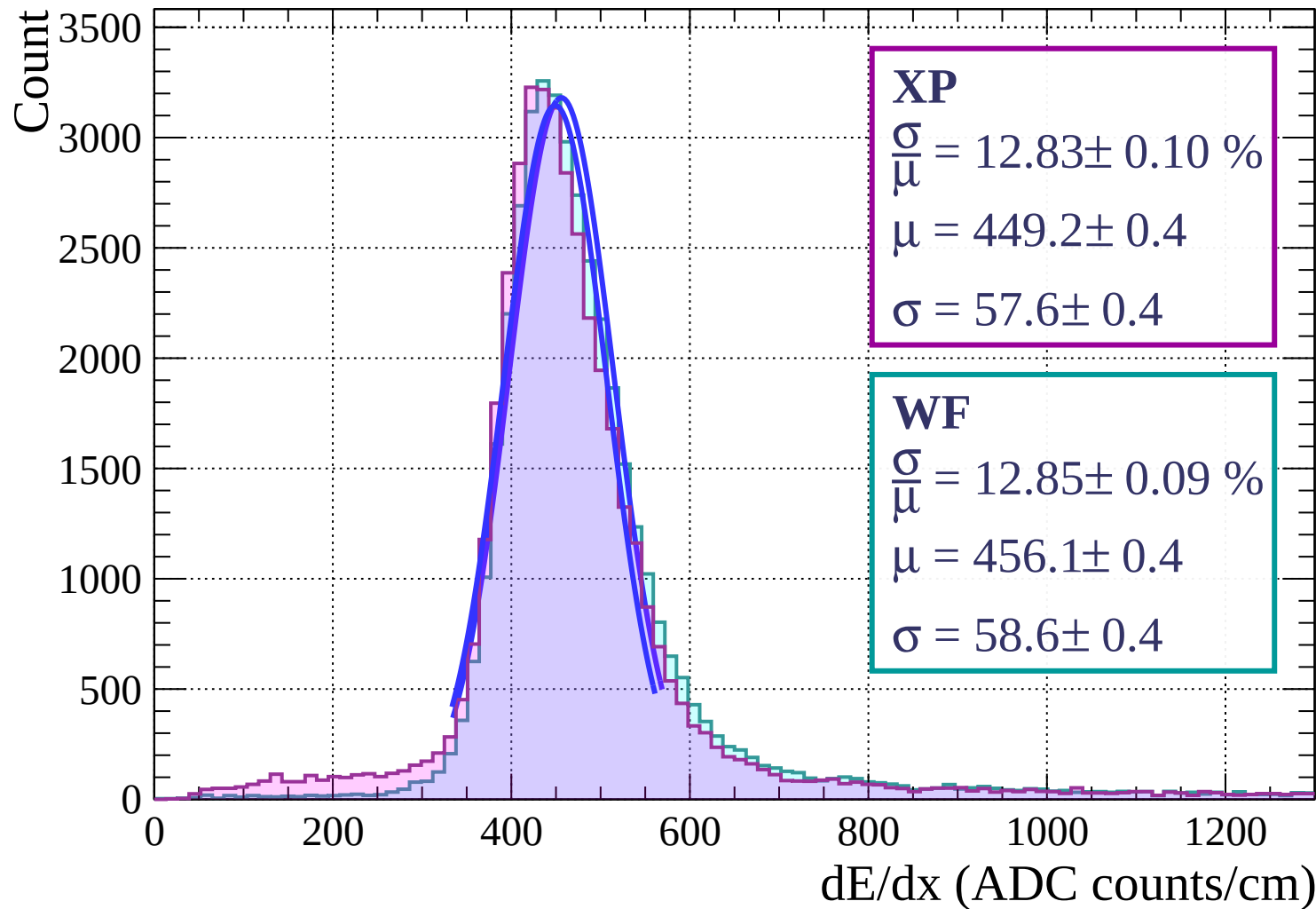




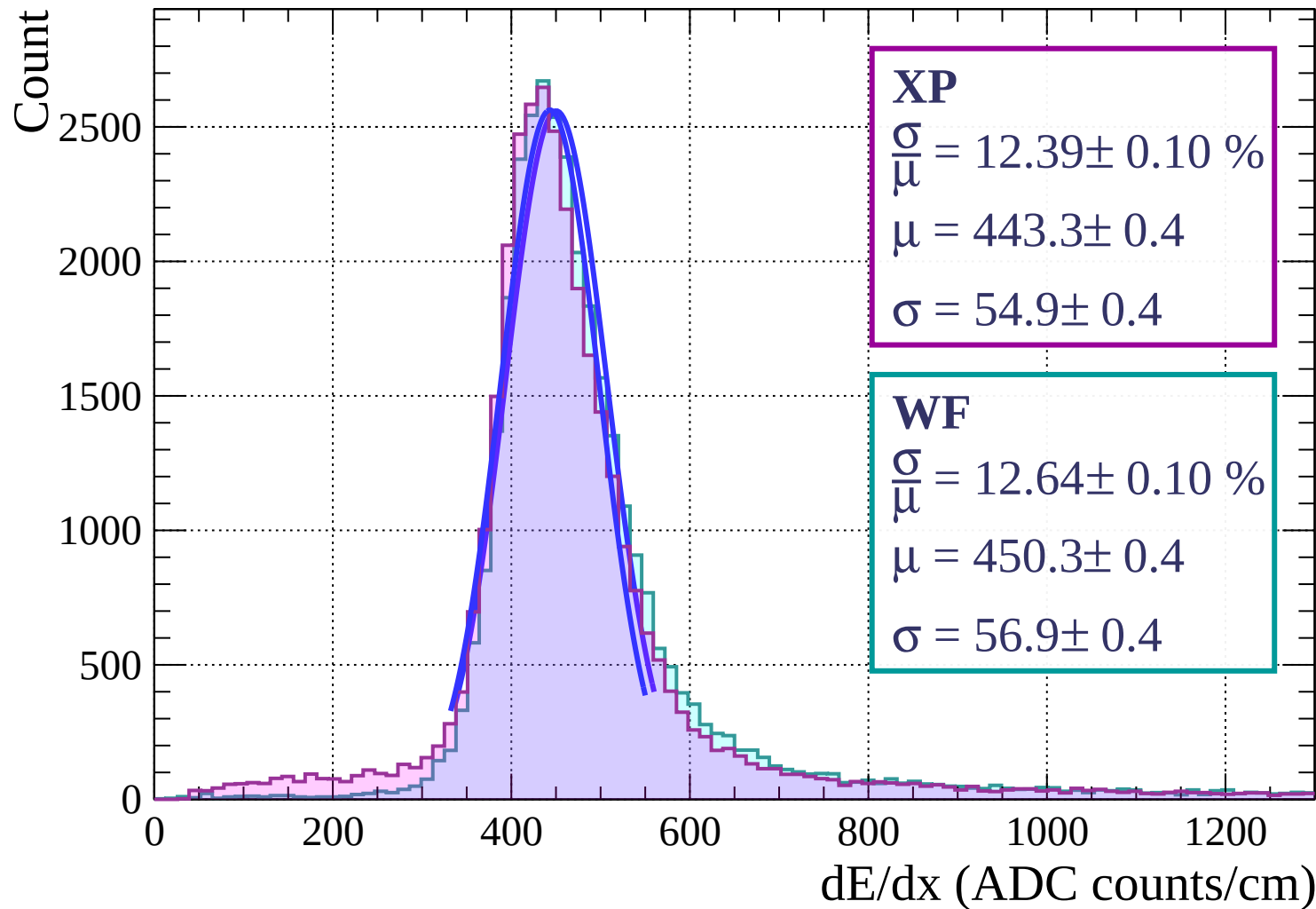
# Energy loss | $6 < \theta < 9$



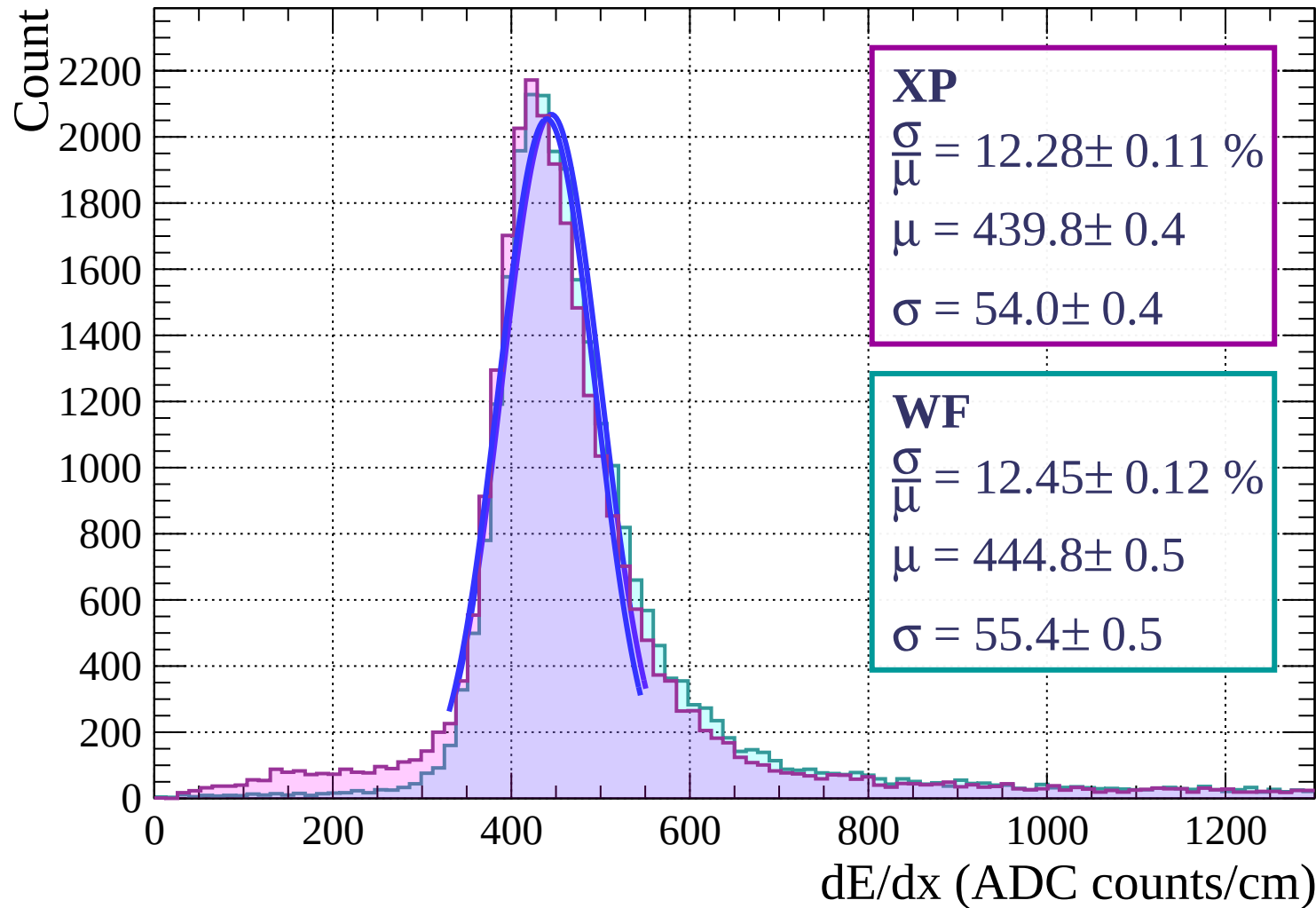
Energy loss |  $9 < \theta < 12$



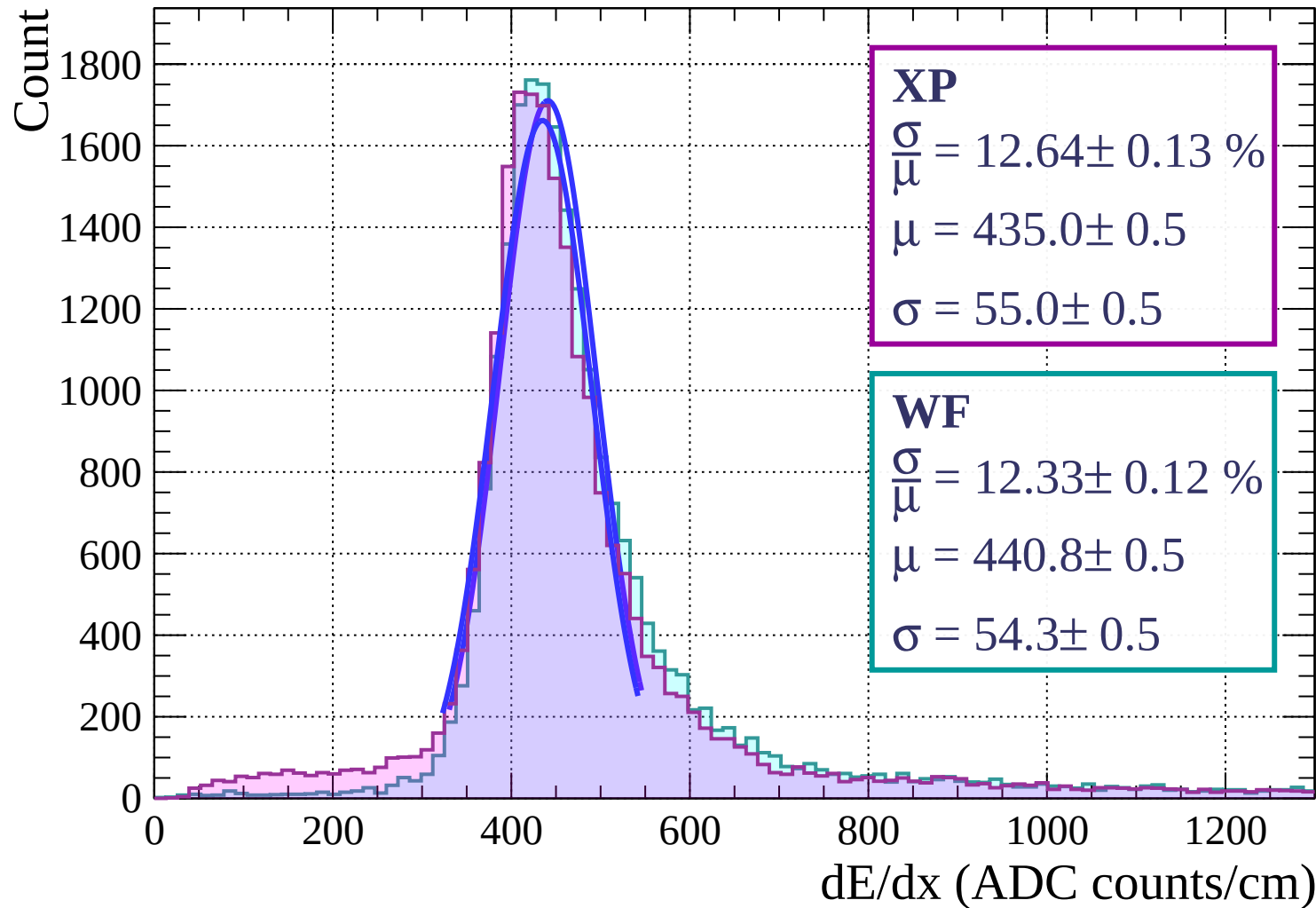
# Energy loss | $12 < \theta < 15$



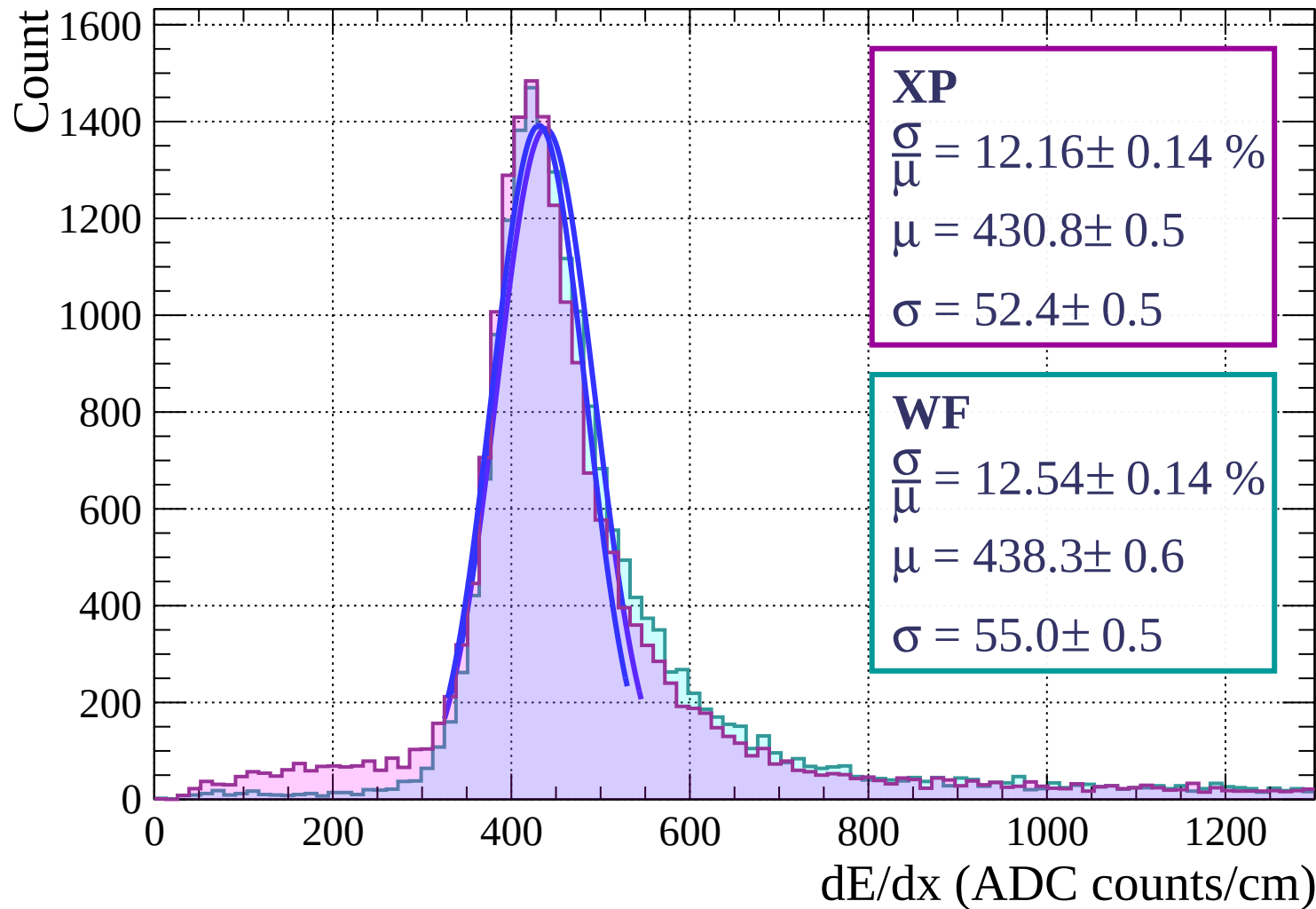
Energy loss |  $15 < \theta < 18$



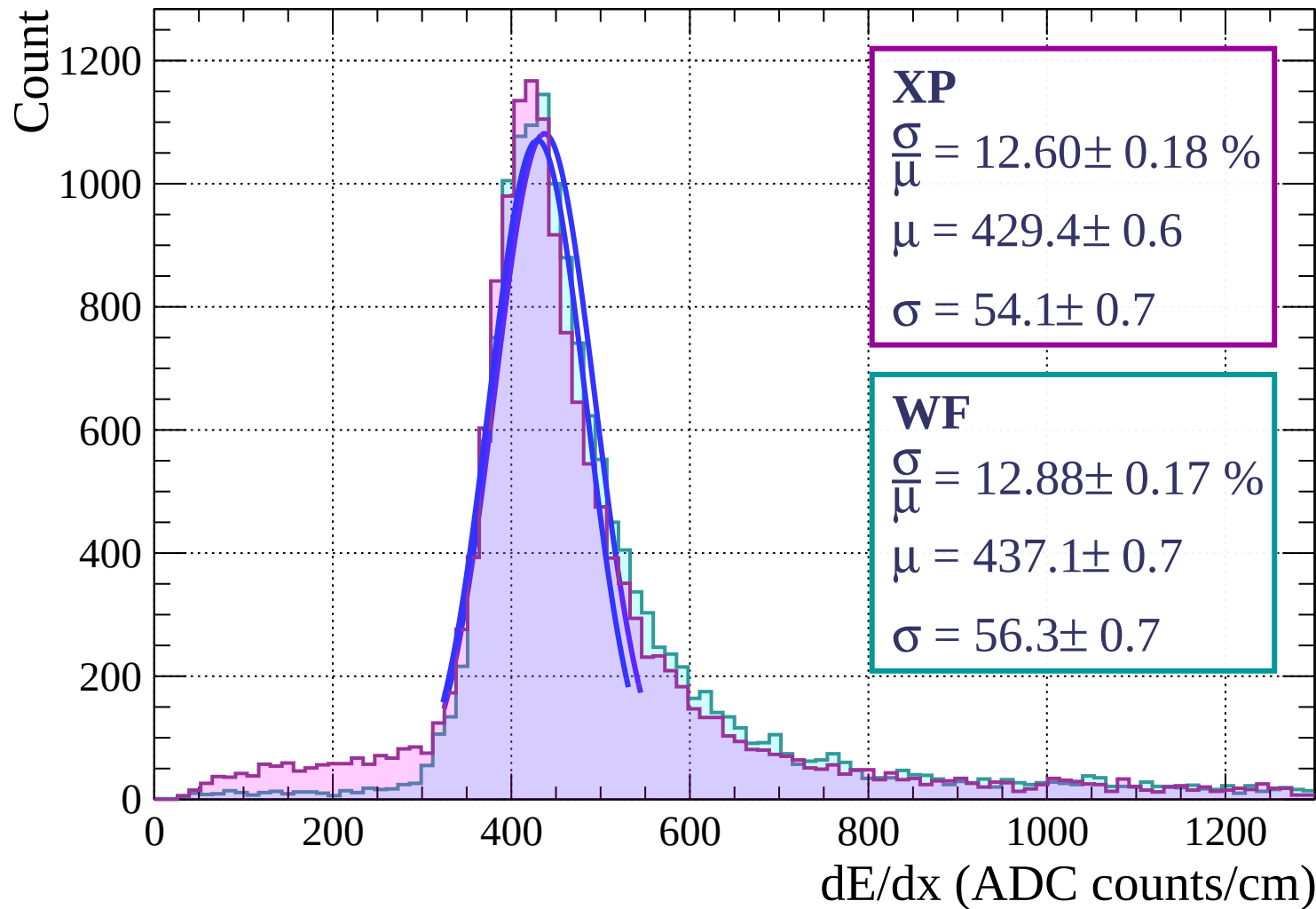
Energy loss |  $18 < \theta < 21$



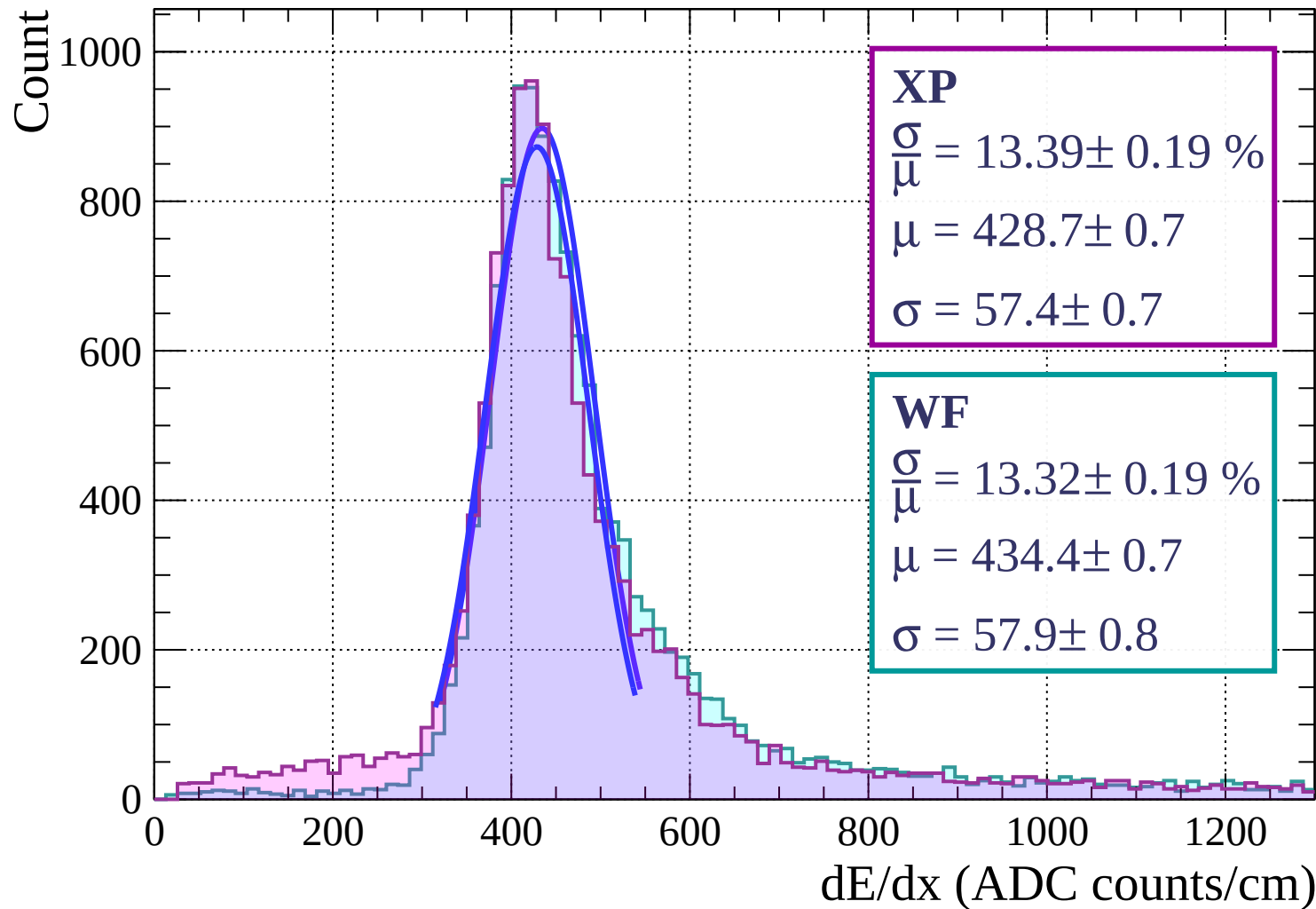
Energy loss |  $21 < \theta < 24$



Energy loss |  $24 < \theta < 27$



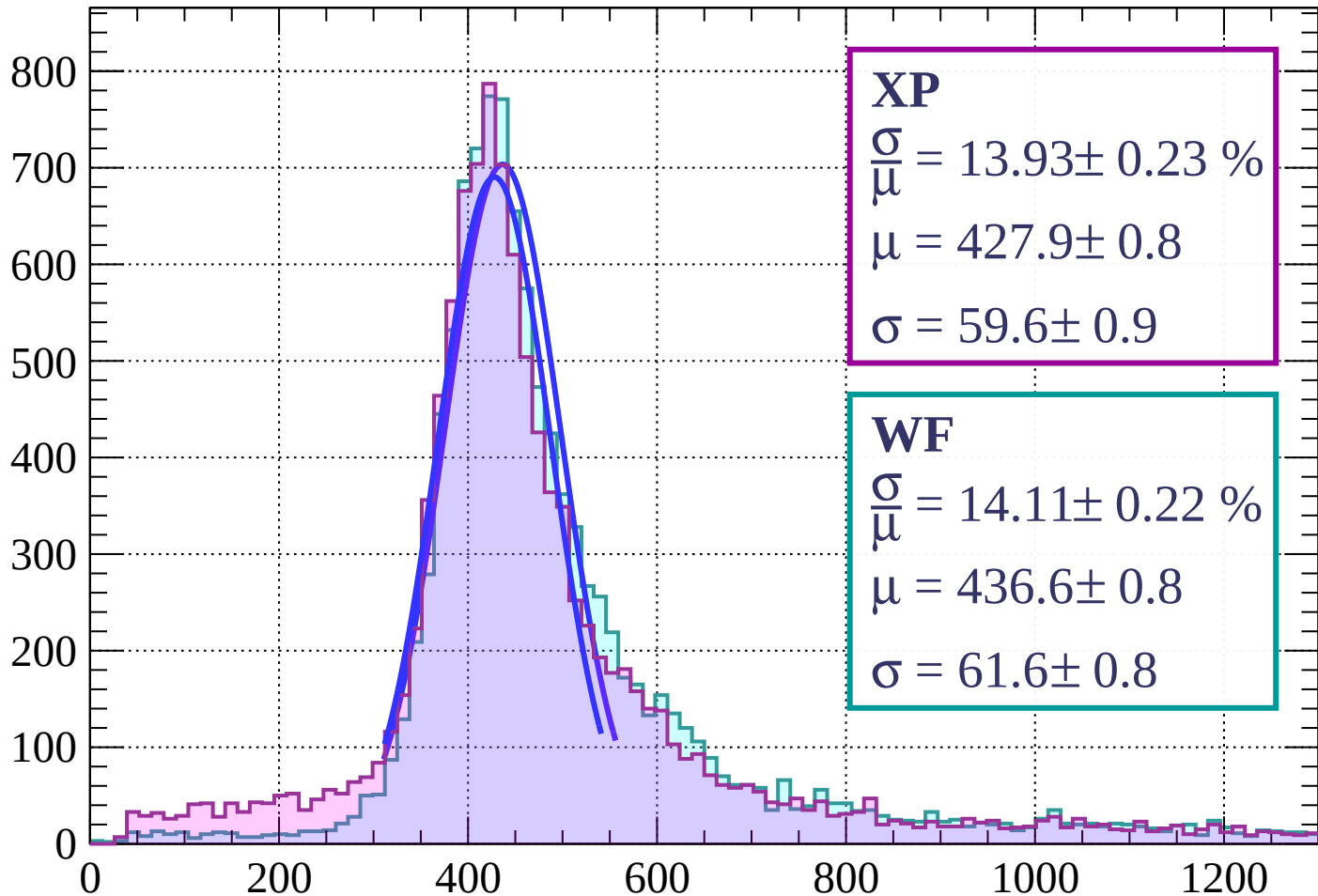
# Energy loss | $27 < \theta < 30$





Energy loss |  $30 < \theta < 33$

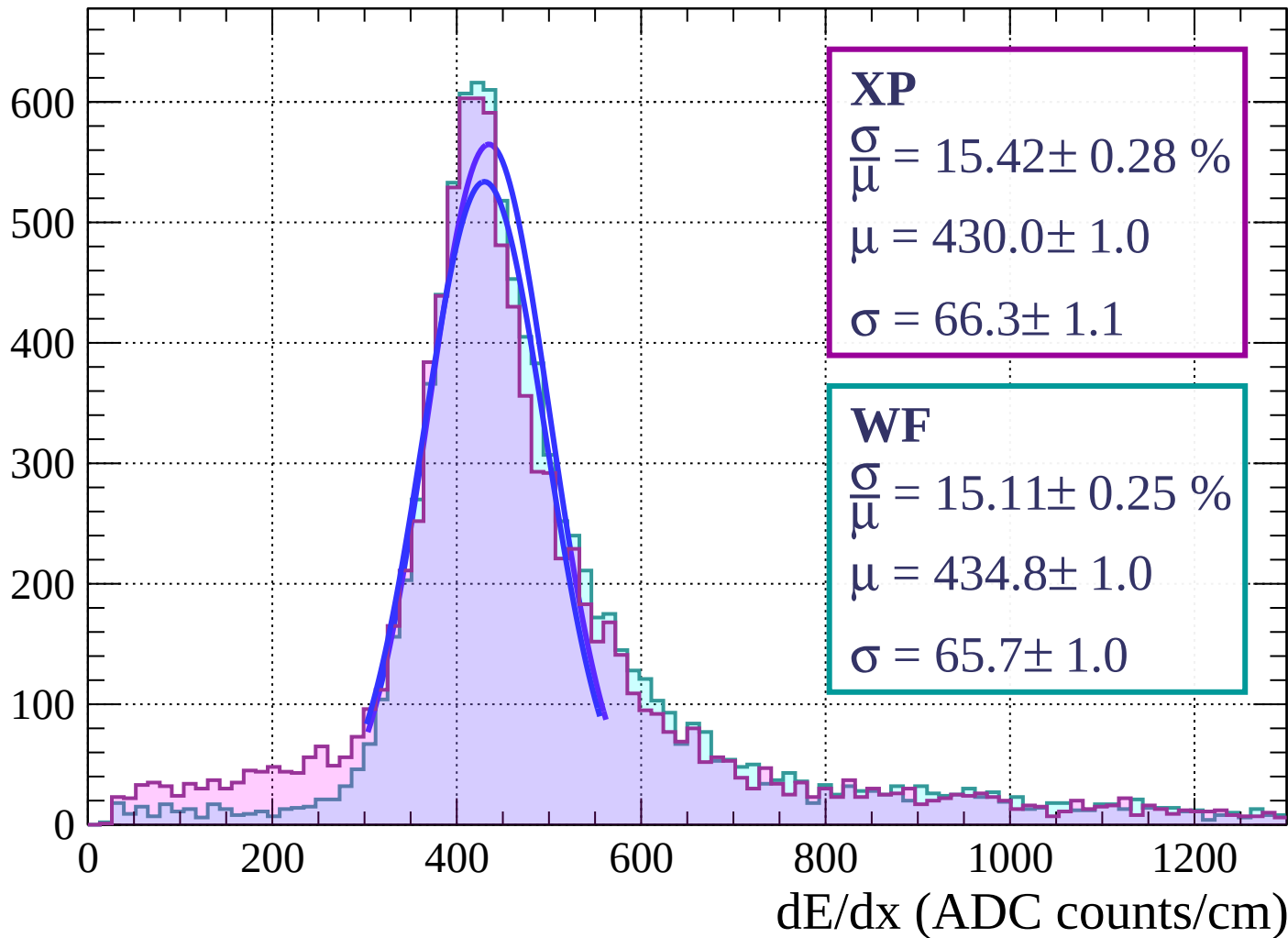
Count



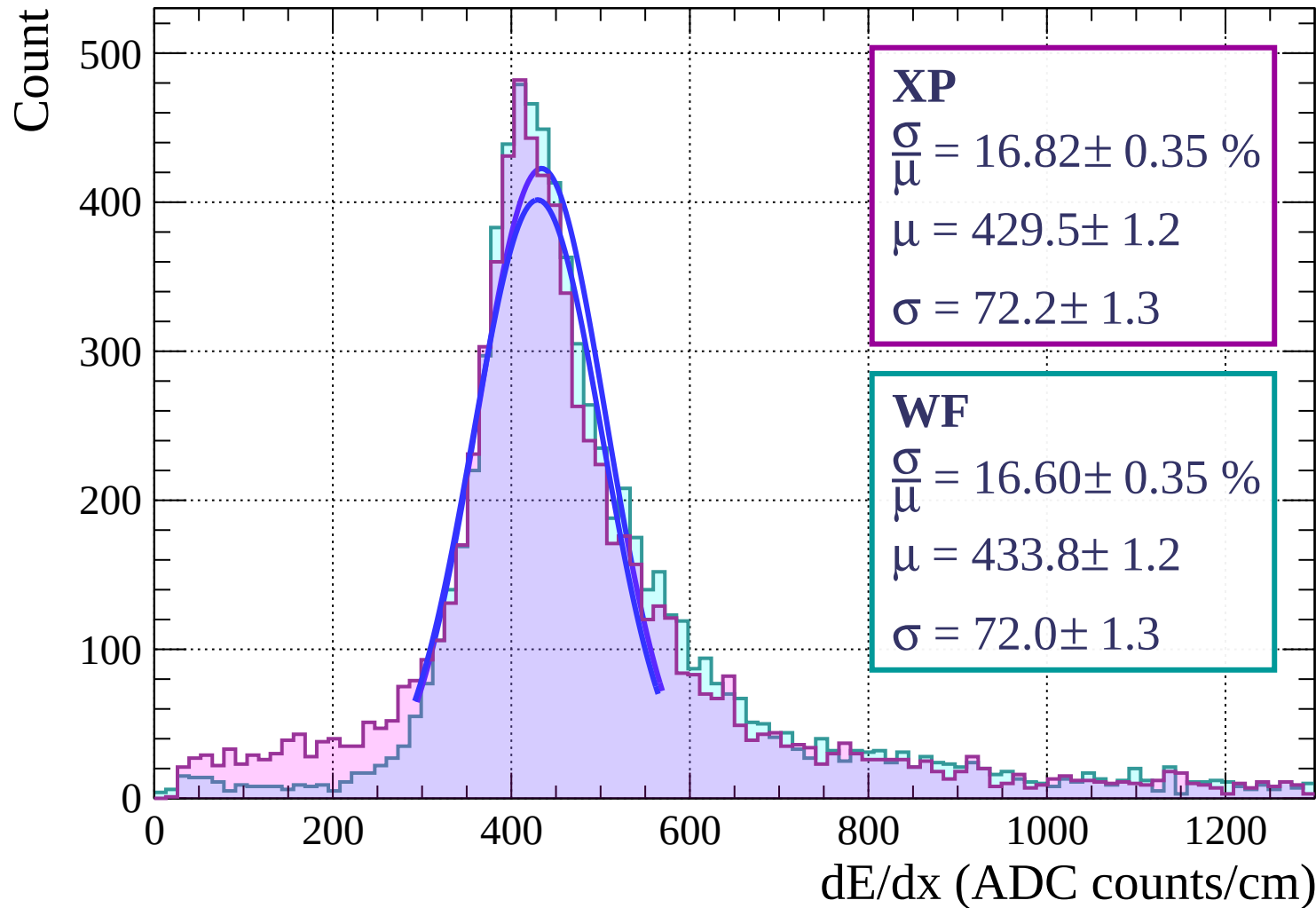
dE/dx (ADC counts/cm)

Energy loss |  $33 < \theta < 36$

Count

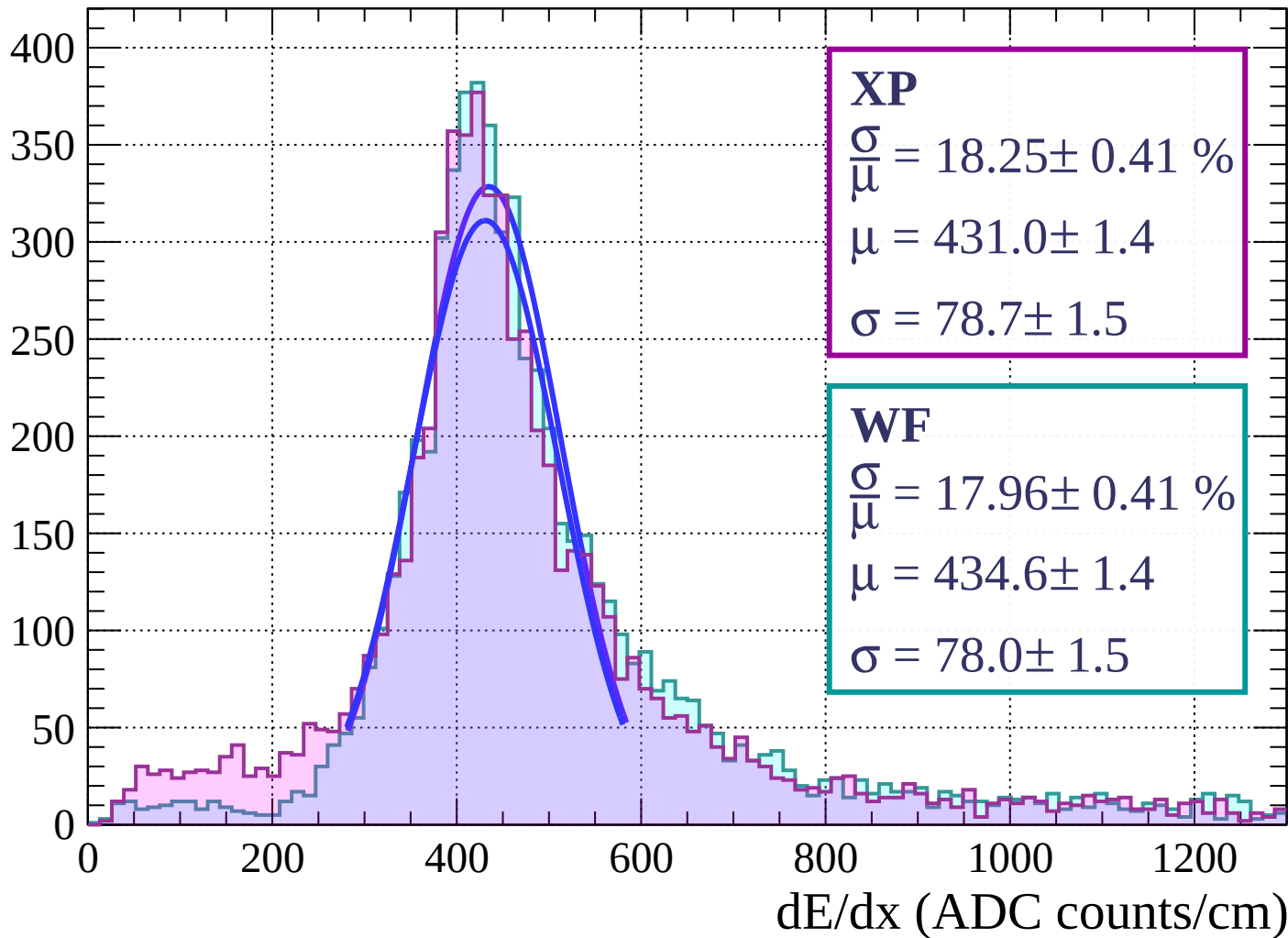


Energy loss |  $36 < \theta < 39$



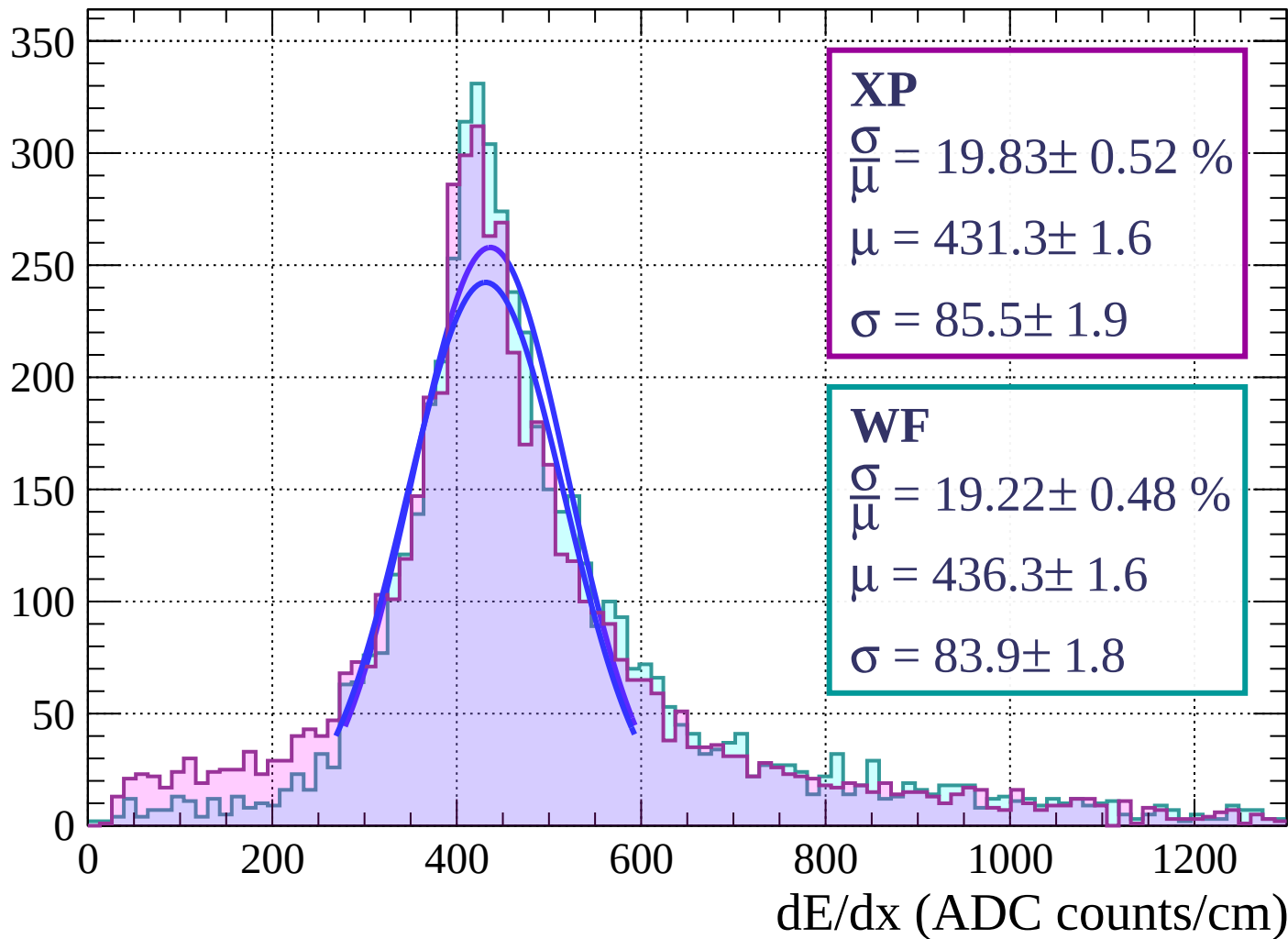
Energy loss |  $39 < \theta < 42$

Count



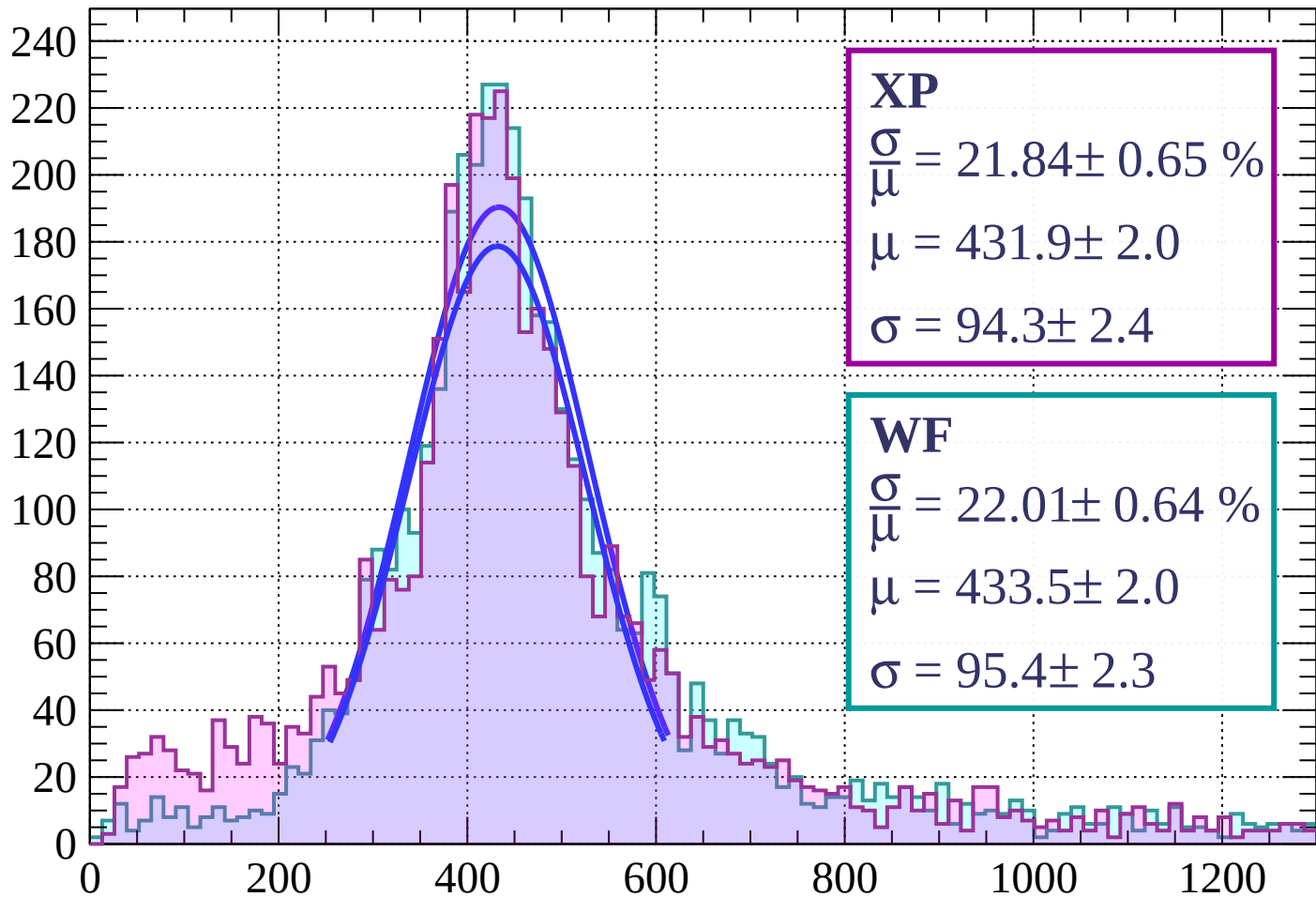
Energy loss |  $42 < \theta < 45$

Count



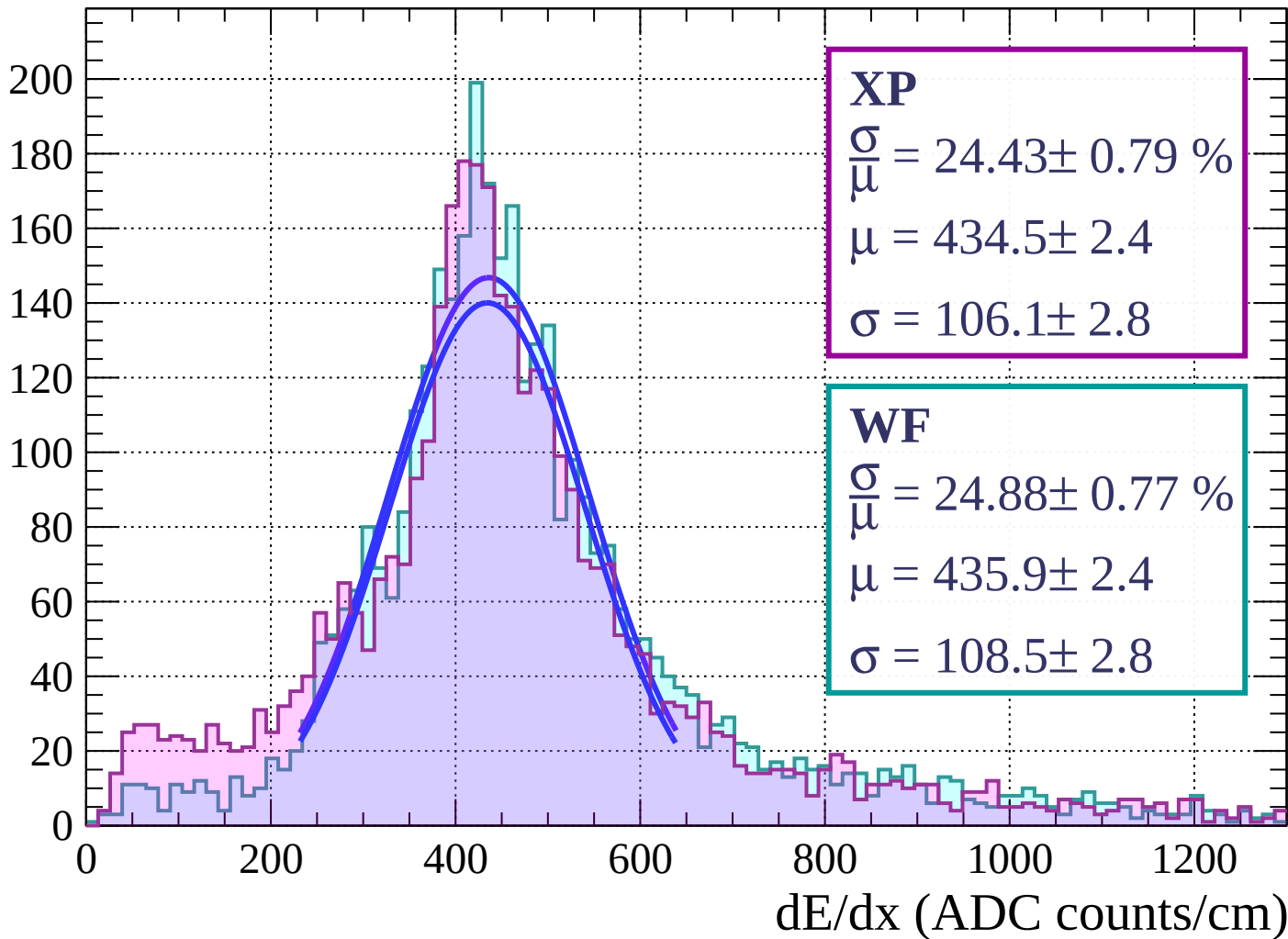
Energy loss |  $45 < \theta < 48$

Count



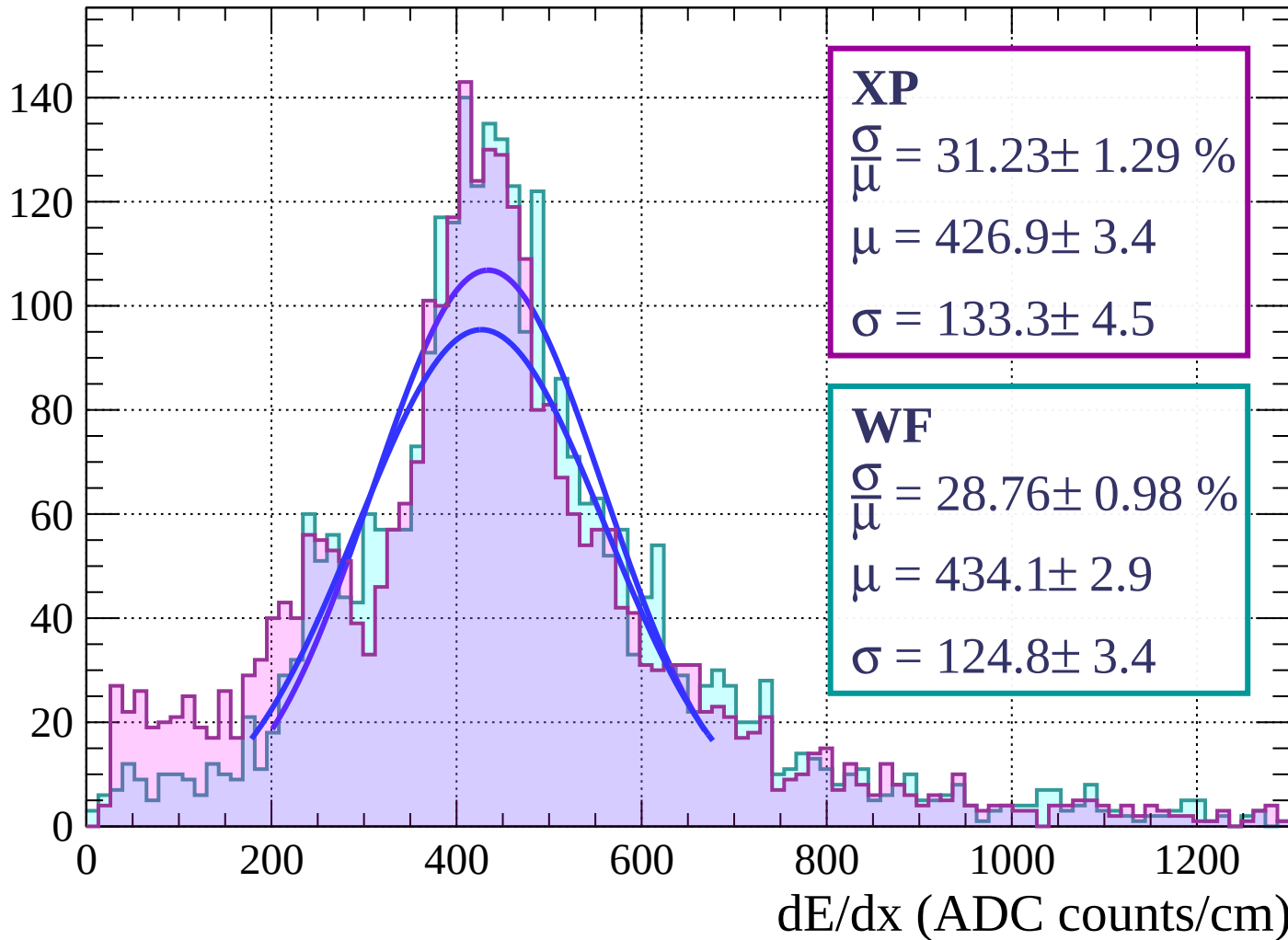
Energy loss |  $48 < \theta < 51$

Count



Energy loss |  $51 < \theta < 54$

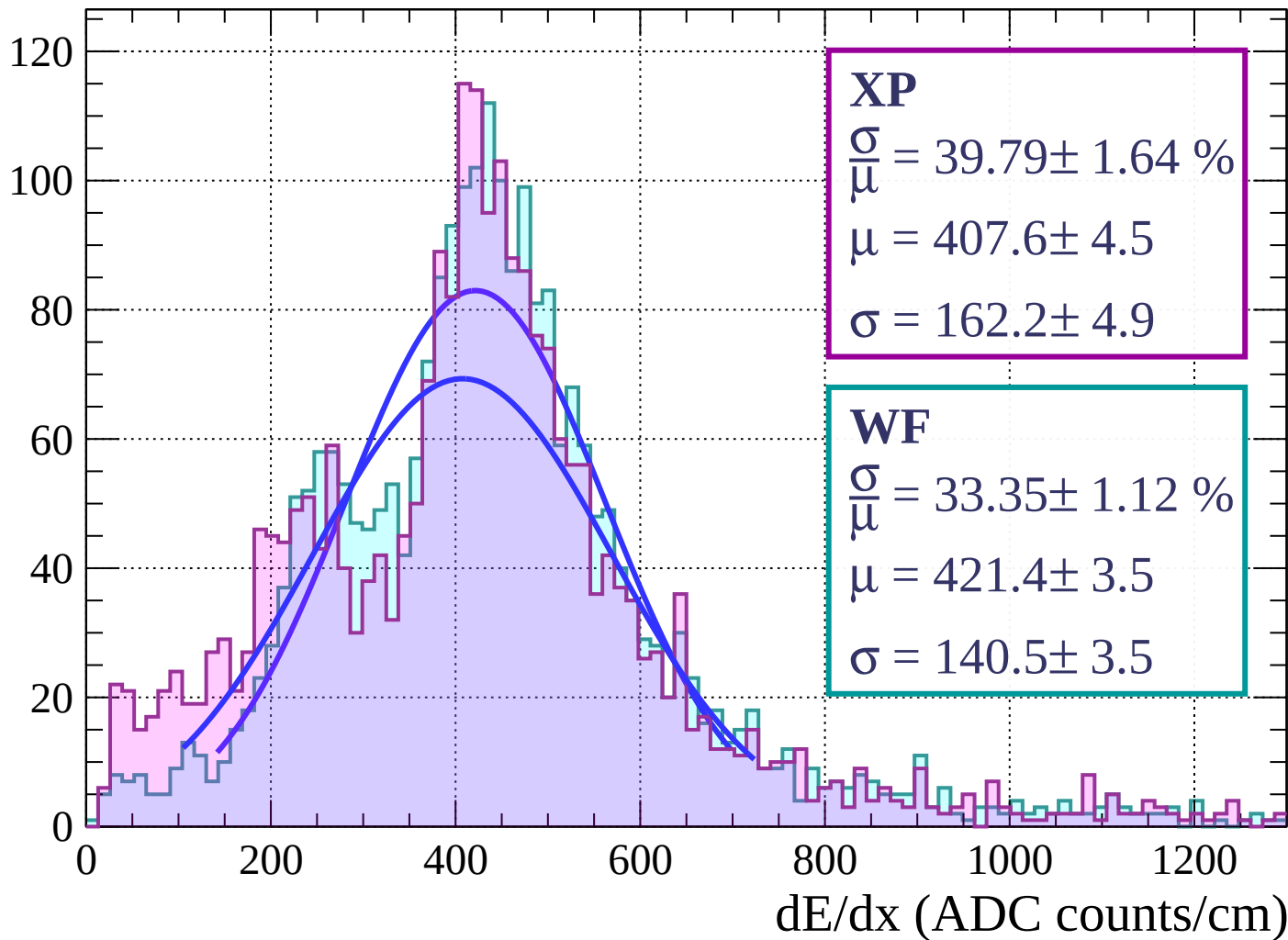
Count





Energy loss |  $54 < \theta < 57$

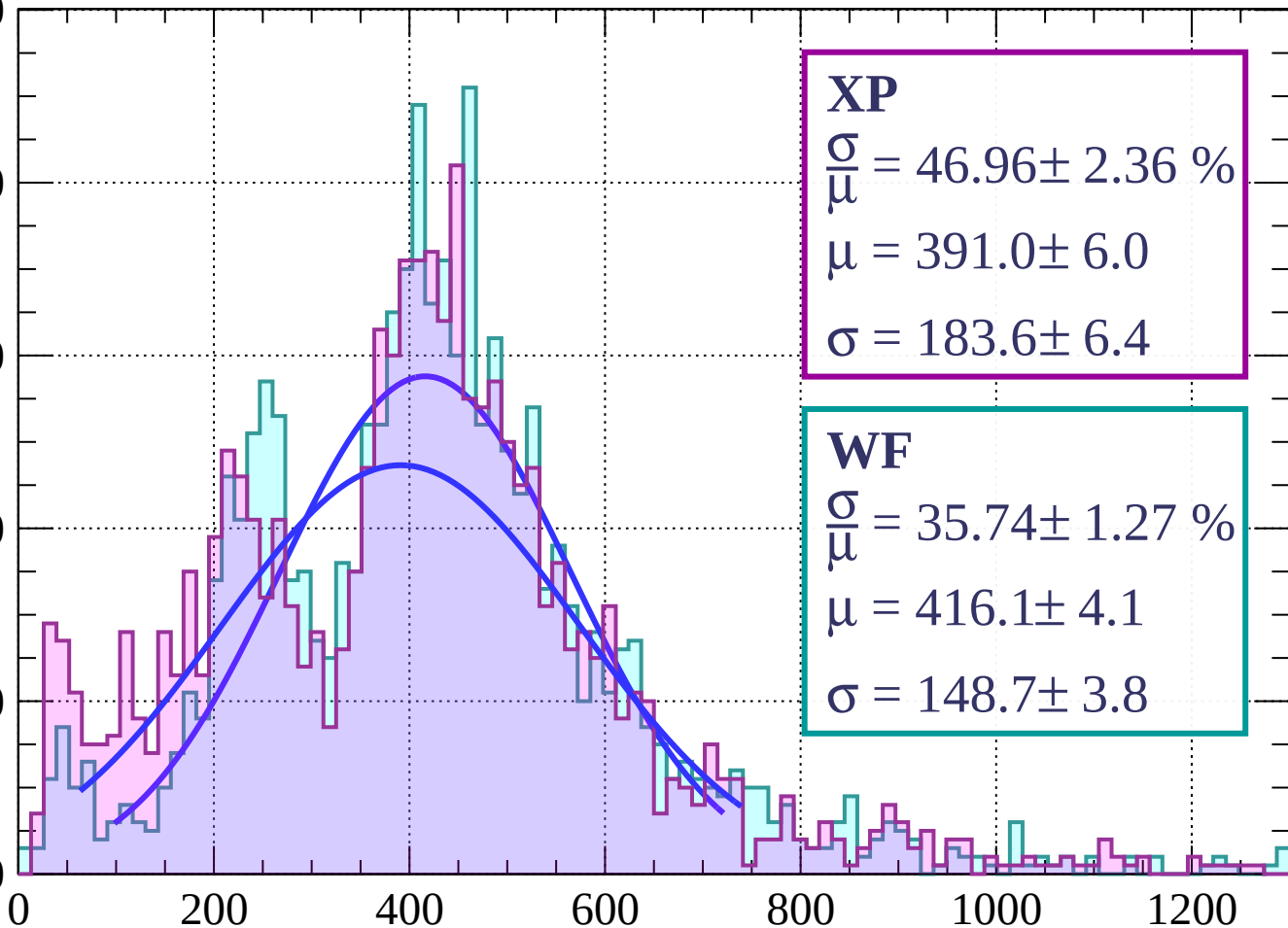
Count



Energy loss |  $57 < \theta < 60$

Count

100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 46.96 \pm 2.36 \%$$

$$\mu = 391.0 \pm 6.0$$

$$\sigma = 183.6 \pm 6.4$$

**WF**

$$\frac{\sigma}{\mu} = 35.74 \pm 1.27 \%$$

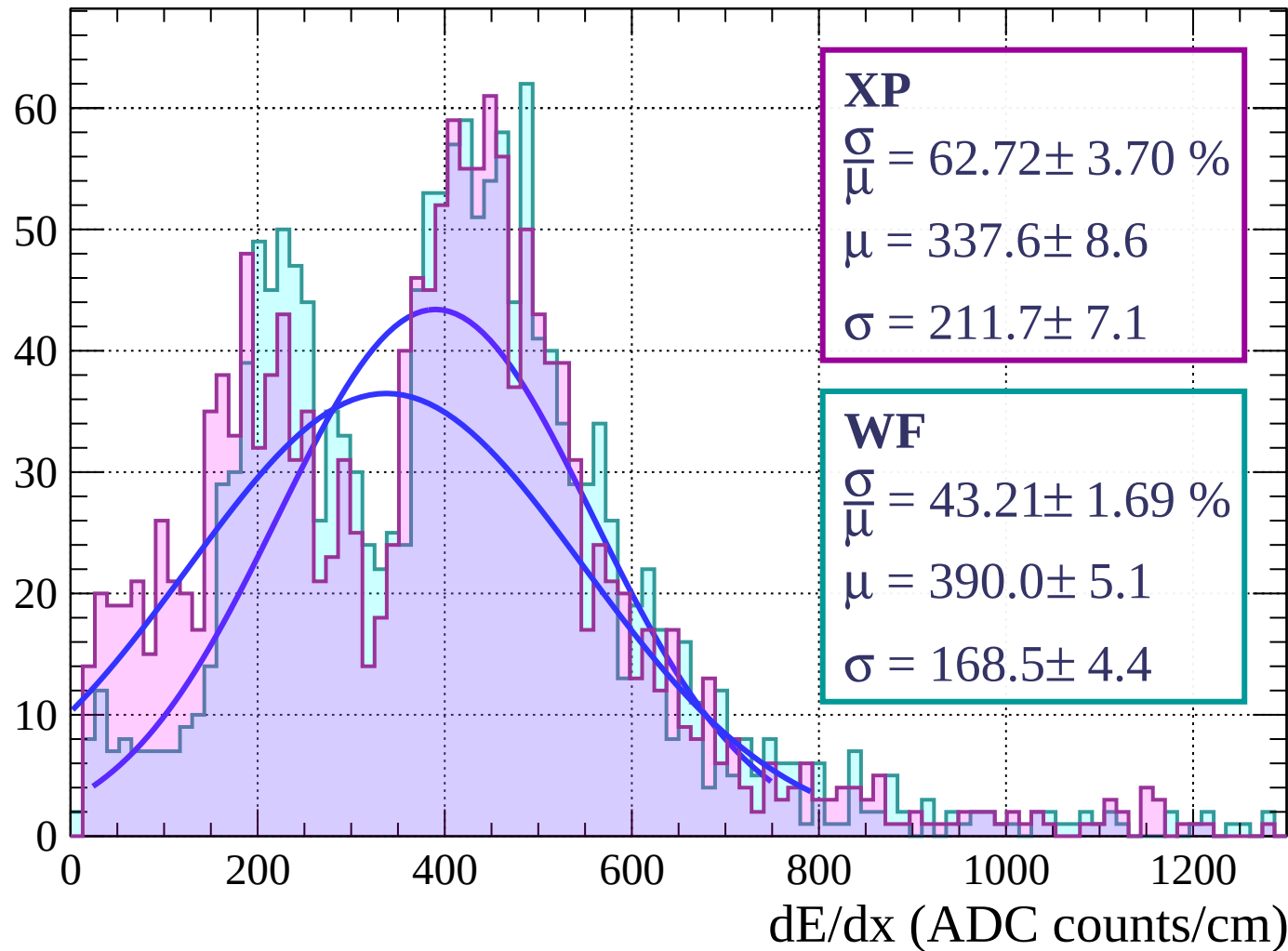
$$\mu = 416.1 \pm 4.1$$

$$\sigma = 148.7 \pm 3.8$$

$dE/dx$  (ADC counts/cm)

# Energy loss | $60 < \theta < 63$

Count



Energy loss |  $63 < \theta < 66$

Count

**XP**

$$\frac{\sigma}{\mu} = 56.04 \pm 2.96 \%$$

$$\mu = 347.5 \pm 7.5$$

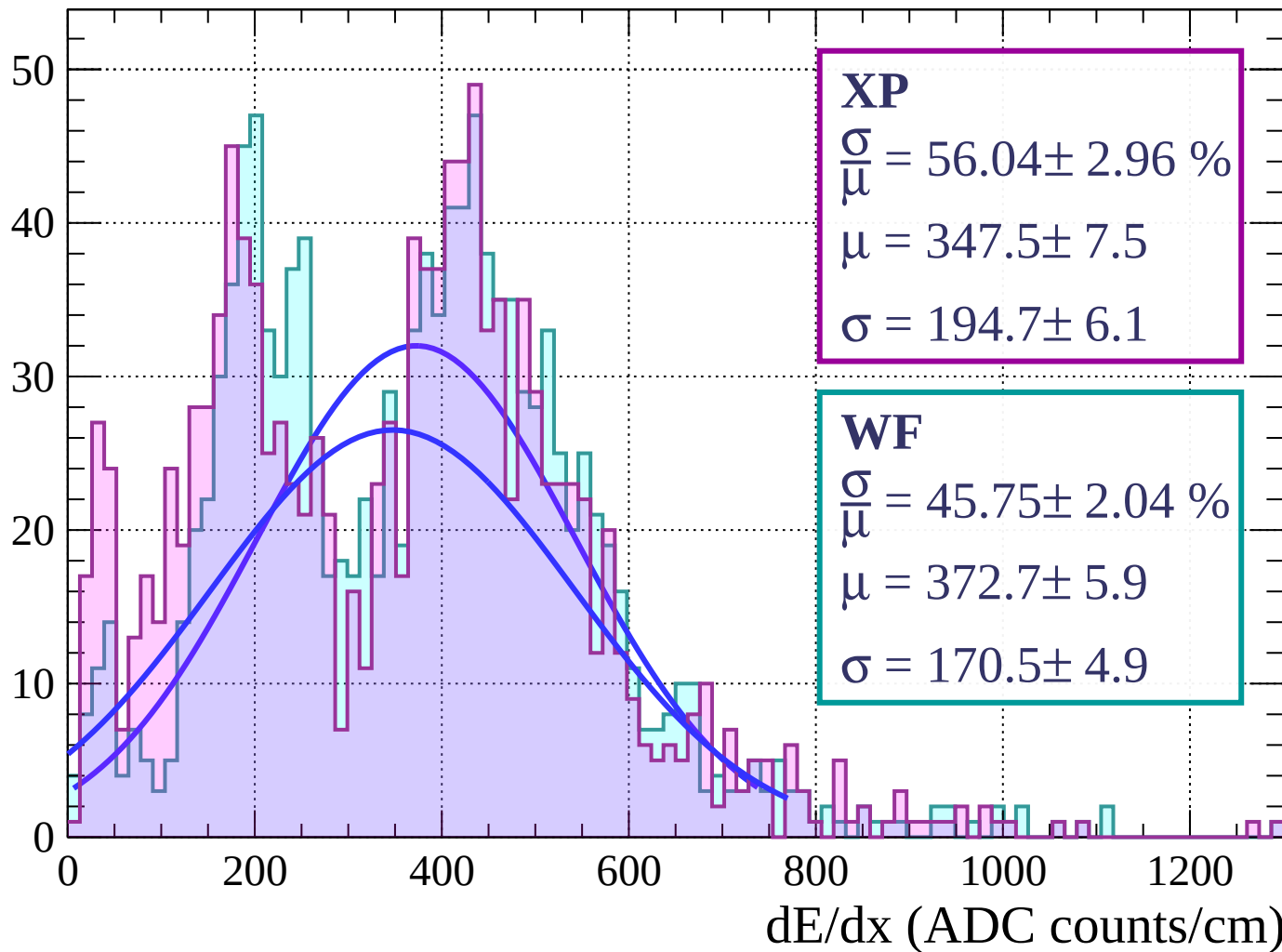
$$\sigma = 194.7 \pm 6.1$$

**WF**

$$\frac{\sigma}{\mu} = 45.75 \pm 2.04 \%$$

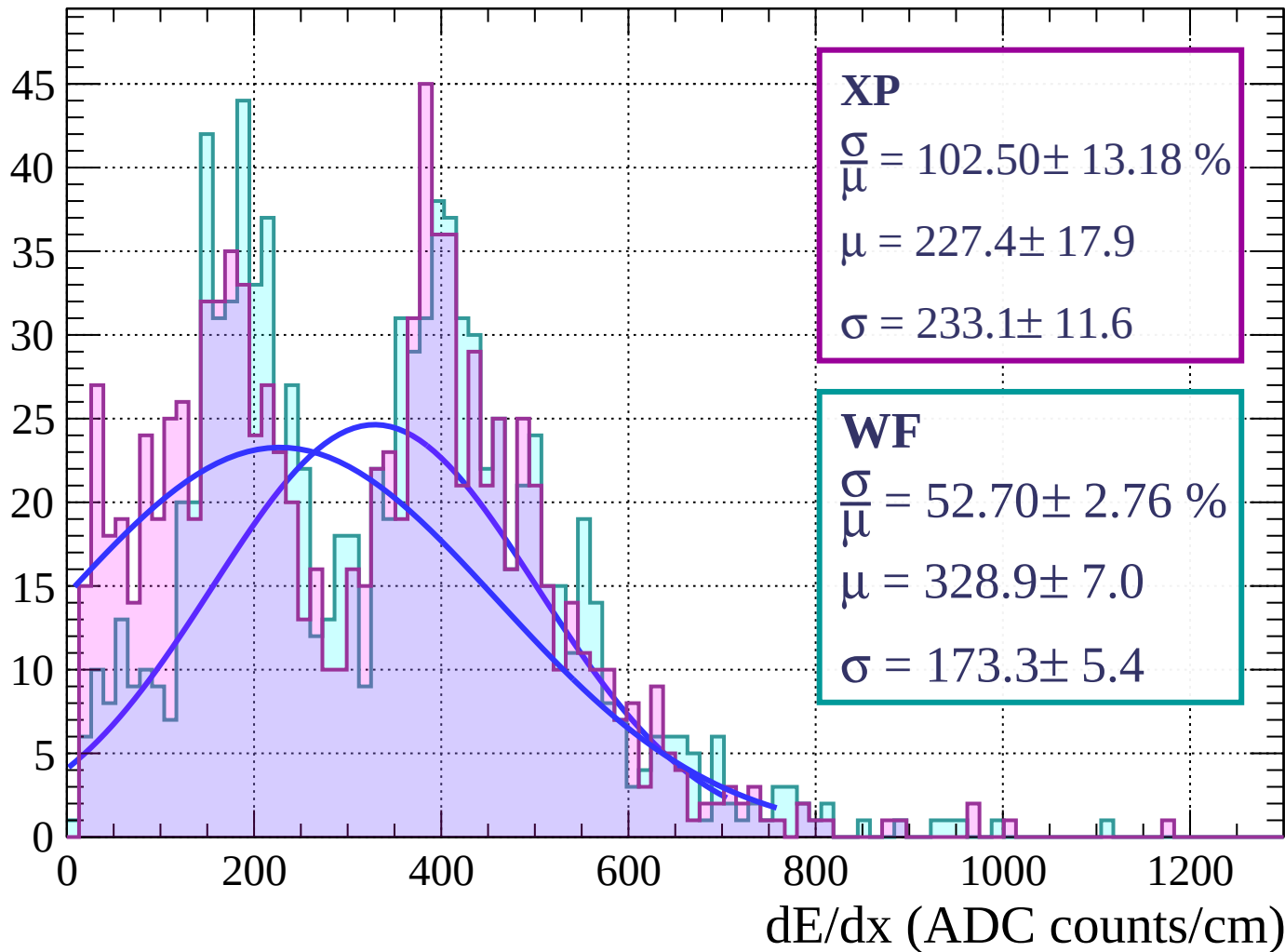
$$\mu = 372.7 \pm 5.9$$

$$\sigma = 170.5 \pm 4.9$$



# Energy loss | $66 < \theta < 69$

Count



Energy loss |  $69 < \theta < 72$

Count

**XP**

$$\frac{\sigma}{\mu} = 176.57 \pm 54.22 \%$$

$$\mu = 141.9 \pm 33.0$$

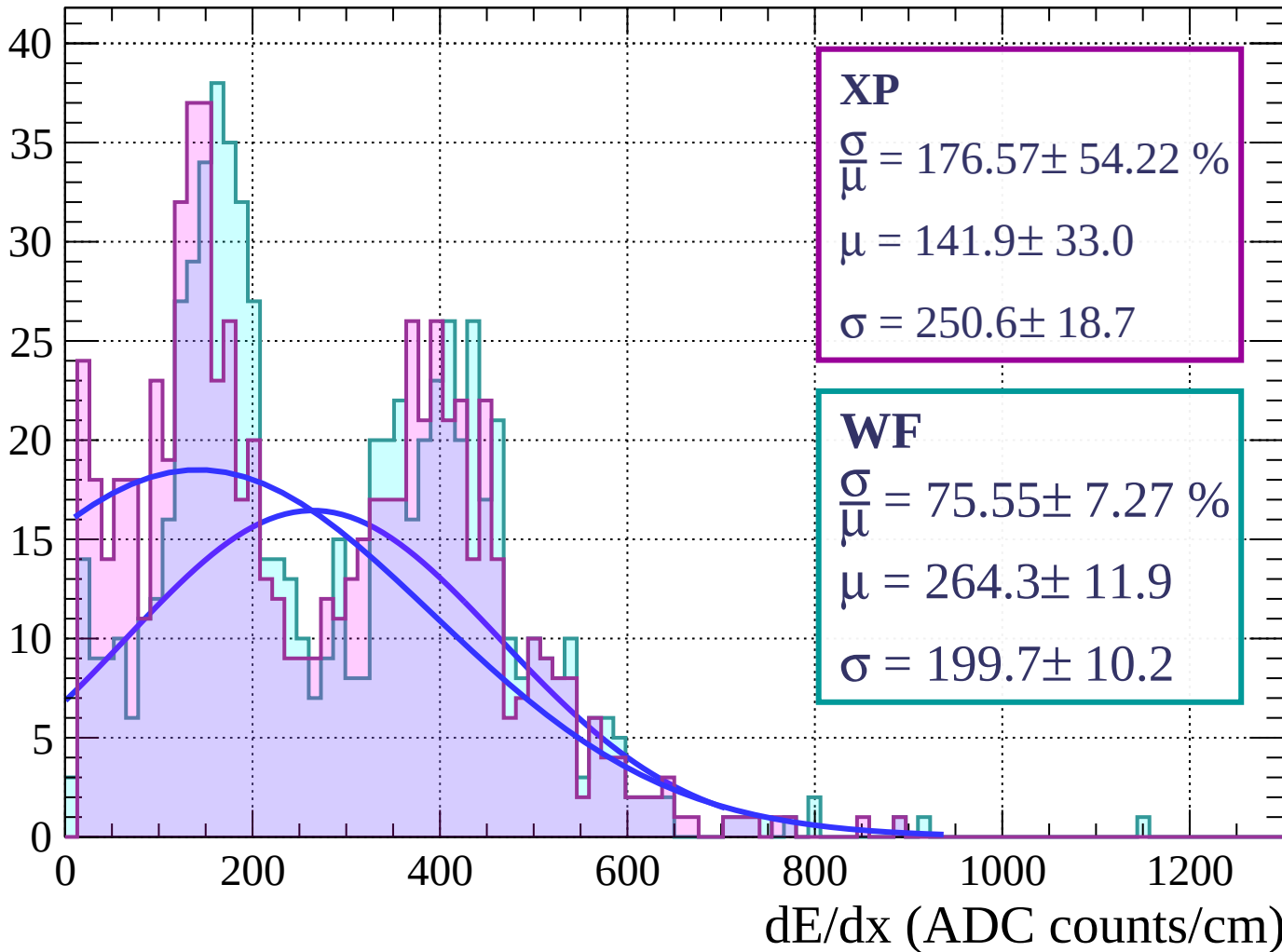
$$\sigma = 250.6 \pm 18.7$$

**WF**

$$\frac{\sigma}{\mu} = 75.55 \pm 7.27 \%$$

$$\mu = 264.3 \pm 11.9$$

$$\sigma = 199.7 \pm 10.2$$



Energy loss |  $72 < \theta < 75$

Count

**XP**

$$\frac{\sigma}{\mu} = 107.17 \pm 18.39 \%$$

$$\mu = 166.6 \pm 15.7$$

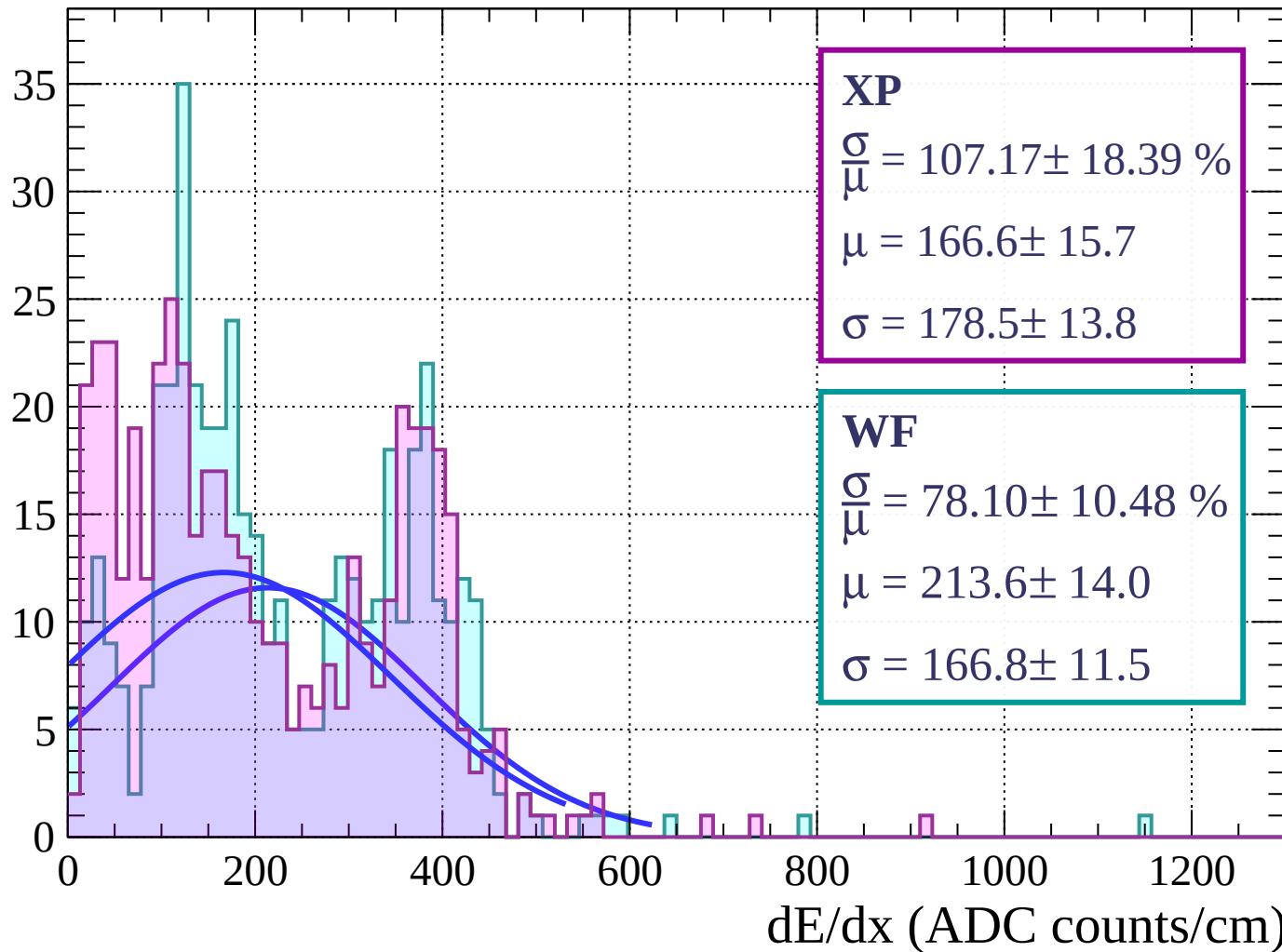
$$\sigma = 178.5 \pm 13.8$$

**WF**

$$\frac{\sigma}{\mu} = 78.10 \pm 10.48 \%$$

$$\mu = 213.6 \pm 14.0$$

$$\sigma = 166.8 \pm 11.5$$



Energy loss |  $75 < \theta < 78$

Count

**XP**

$$\frac{\sigma}{\mu} = 145.49 \pm 67.96 \%$$

$$\mu = 61.2 \pm 16.9$$

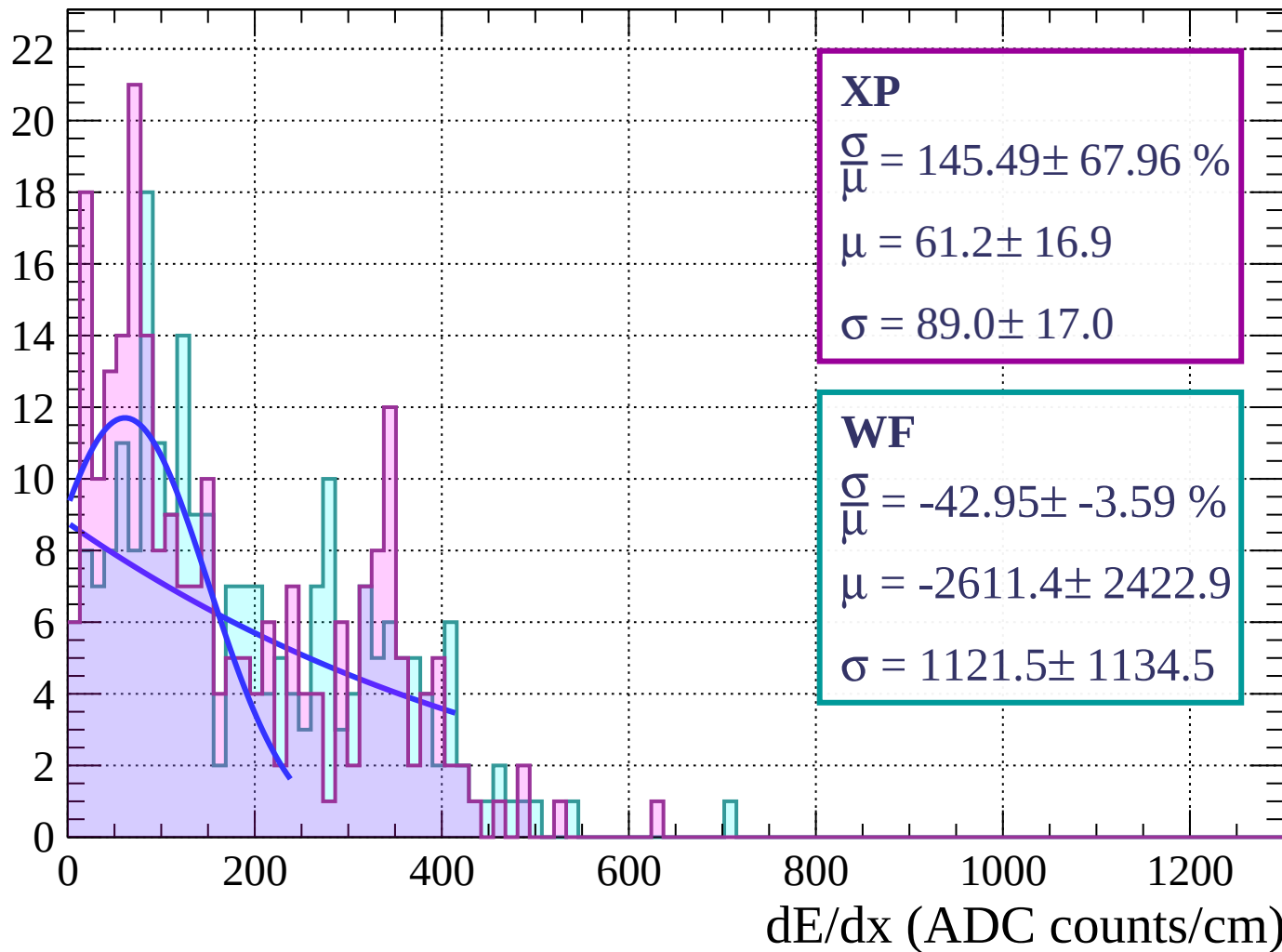
$$\sigma = 89.0 \pm 17.0$$

**WF**

$$\frac{\sigma}{\mu} = -42.95 \pm -3.59 \%$$

$$\mu = -2611.4 \pm 2422.9$$

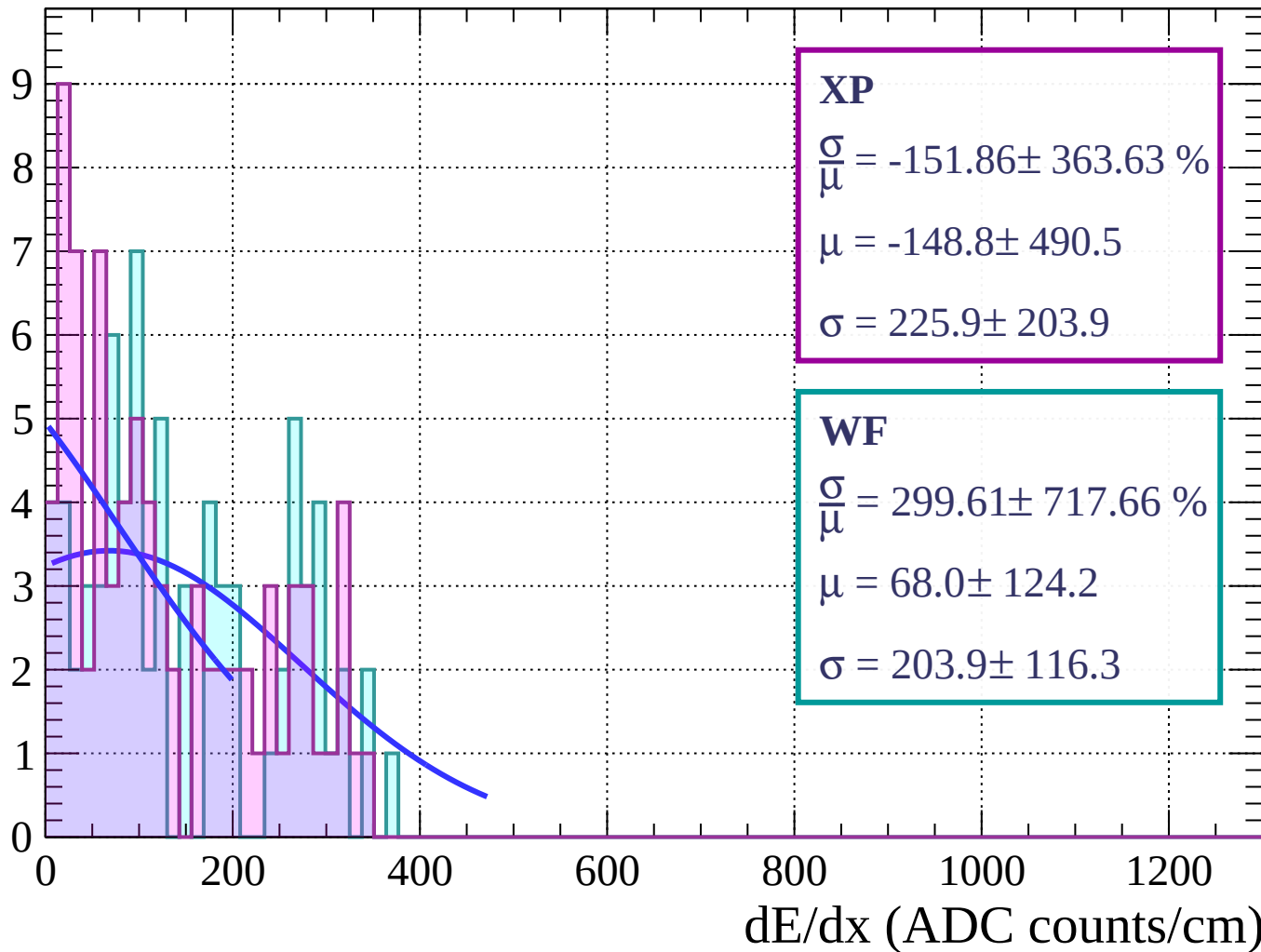
$$\sigma = 1121.5 \pm 1134.5$$





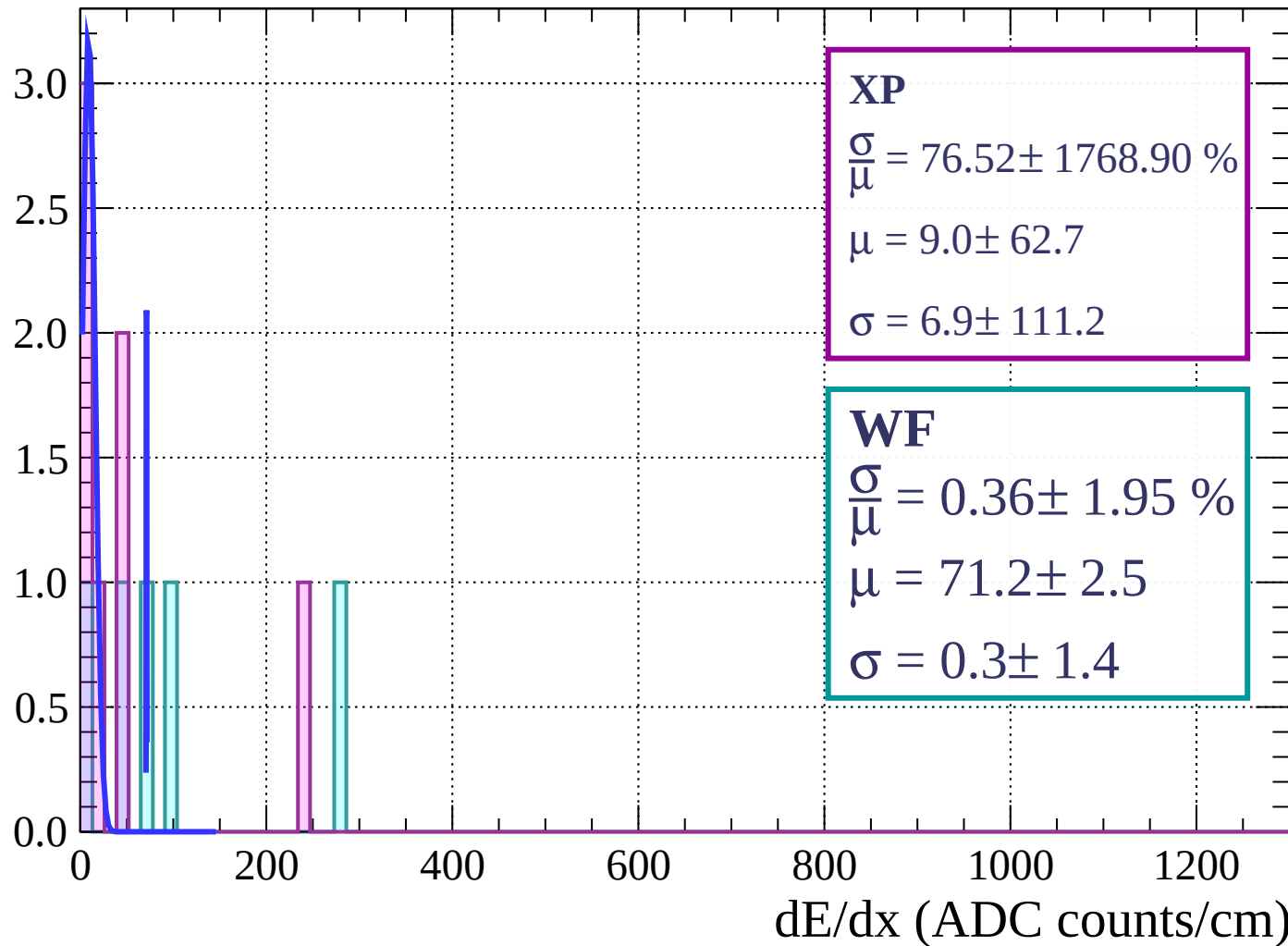
Energy loss |  $78 < \theta < 81$

Count

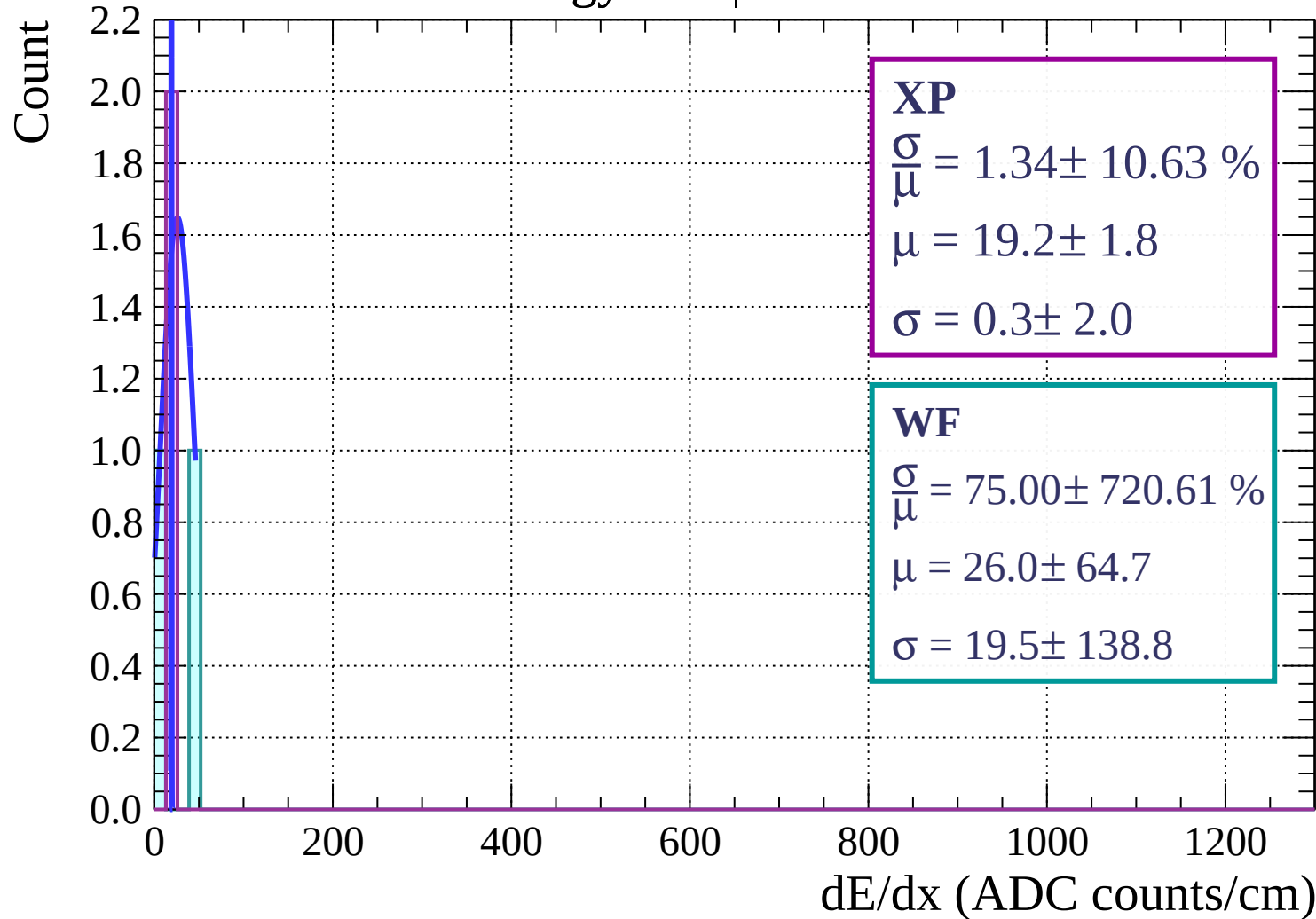


Energy loss |  $81 < \theta < 84$

Count

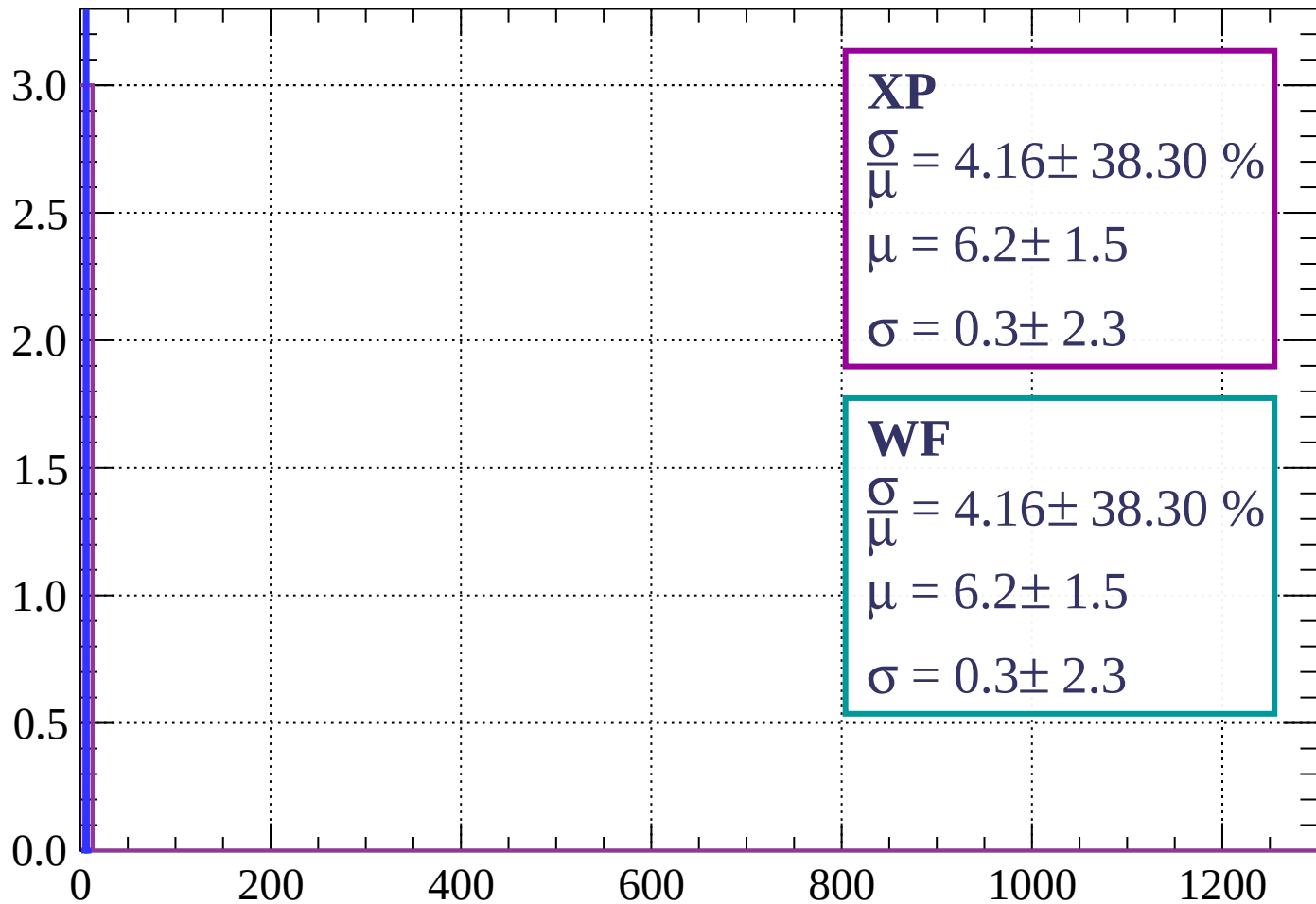


Energy loss |  $84 < \theta < 87$

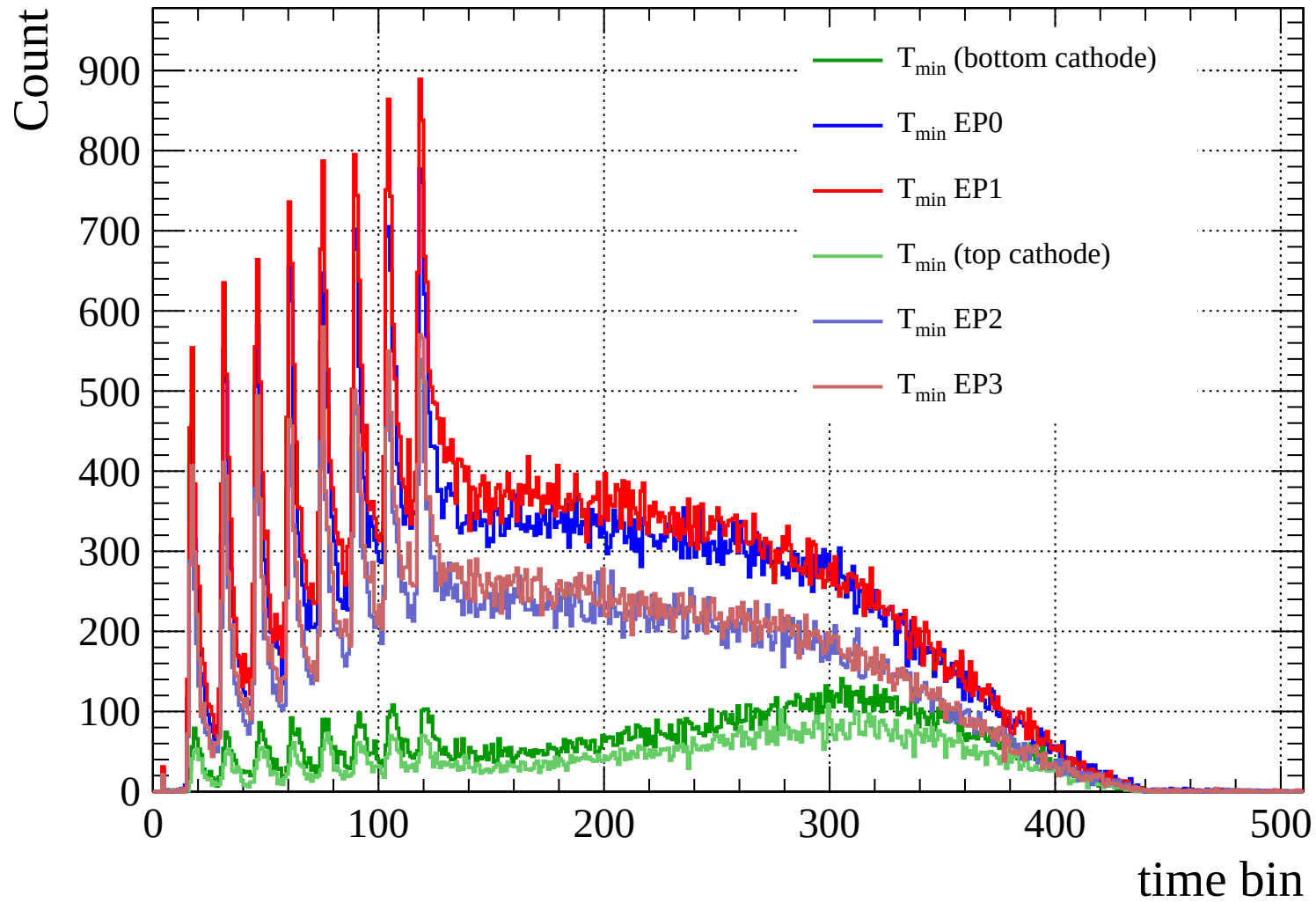


Energy loss |  $87 < \theta < 90$

Count



$dE/dx$  (ADC counts/cm)



Count

